

codex-windows / 019f2174-3a94-7cc3-9a94-76a03359fbad

Metadata

```
- Source: `codex-windows`  
- Kind: `codex`  
- Source file: `C:\Users\avidu\codex\archived_sessions\rollout-2026-07-02T11-41-33-019f2174-3a94-7cc3-9a94-76a03359fbad.jsonl`  
- SHA-256: `9137f4c3fc3bdd246cbeaa1d9cd2c5f3b73c86a27804838e3a4512b151dd7015`  
- Source modified: `2026-07-02T06:11:33+00:00`  
- Imported at: `2026-07-05T16:48:16+00:00`  
- cli_version: `0.142.5`  
- cwd: `C:\Users\avidu\Projects\badminton-highlight-indexer`  
- model_provider: `openai`  
- session_id: `019f2174-3a94-7cc3-9a94-76a03359fbad`  
- source: `vscode`
```

Transcript

1. user (2026-07-02T06:11:33.143Z)

Hi. Can you run that skill which I added today regarding github repos at session start? I want you to "pick up from you left off" last time and I have a starter project for you which is to plumb a very rich "observability output per processed video" feature in the highlights generation tool and also the data collector tool. i.e. once the data collector or sports highlighter ingests / processes a new video, they should also emit a very rich side output which captures all the stages the video went through and the details eg. fps, resolution, frames extracted, rallies etc. all the details assuming it will be used to catch some inconsistencies and evaluate output differences as well. Show me a plan for both and then I want you to ensure that you iterate on the quality of the report by evaluating it as if "someone who is reading the README file and some FAQs is running the tool and he should be able to make sense of the report and do an initial debugging if required" as a stretch goal, we should also ensure we produce a "customer friendly video report" where we would remove all the system processing details and the user will know metadata about the video and some meta level highlights emitted by the tools processings.

I am happy to clarify any questions you have and make sure you plumb it well enough that this 'house of cards' doesn't fall!

2. assistant (2026-07-02T06:11:33.143Z)

I'll start by running the session-start skill you added today, then read my handoff/memory to pick up where we left off.

3. assistant (2026-07-02T06:11:33.143Z)

```
[external_agent_tool_call: Skill]  
input: {"skill":"anthropic-skills:new-session-hello"}  
[/external_agent_tool_call]
```

4. assistant (2026-07-02T06:11:33.143Z)

```
[external_agent_tool_result]  
Launching skill: anthropic-skills:new-session-hello  
[/external_agent_tool_result]
```

5. assistant (2026-07-02T06:11:33.143Z)

```
[external_agent_tool_call: Read]  
file: C:\Users\avidu\AppData\Roaming\Claude\local-agent-mode-sessions\skills-  
plugin\b4180250-5146-4b86-9001-c9f3dd74258c\87f748ce-e72b-44f3-8ef0-78e3212a99c3\skills\new-  
session-hello\SKILL.md  
[/external_agent_tool_call]
```

6. assistant (2026-07-02T06:11:33.143Z)

```
[external_agent_tool_result]
1      ---
2      name: new-session-hello
3      description: "I want claude to read the github repos linked and have some context when I
start a new coding session"
4      ---
5
6      Read the github folders linked in the session and build a context based on it and mostly
there could be a code map file which you write to index the code so that re-read is faster and
cheaper. Always look to build context from where I had left off, all this only when the session
is initiated with >=1 github repositories.
7
[/external_agent_tool_result]
```

7. assistant (2026-07-02T06:11:33.143Z)

Skill loaded. Per its instructions and my memory's "? Start here" pointers, let me ramp up from where I left off. Reading the live checkpoint, handoff, and code-map pointers in parallel.

8. assistant (2026-07-02T06:11:33.143Z)

```
[external_agent_tool_call: Read]
file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

9. assistant (2026-07-02T06:11:33.143Z)

```
[external_agent_tool_result]
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-07 · ? **SESSION SUNSET.** master clean (PRs #25?#35 merged; 209
tests green; only
13     feat/khelacharya-shot-intelligence retained; no open PRs). Audio E1 done ? classical
onset = ~2.2× ceiling, NOT
14     adopted (detector kept, PR #35); quantitative findings + lessons in
[[audio-e1-findings]] (+ plan §10). VLC
15     annotator = standalone repo github.com/avidullu/rally-annotator (MIT, multi-sport).
16     **?? RESUME NEXT SESSION on EITHER:** (1) **GOLDEN EVALS** ? owner self-rates clips in
the VLC annotator ?
17     `import-local`/--labels` ? unblocks distilled-model LOVO (n?3) + human-gold E1 re-
confirm = the highest-leverage
18     unblock; OR (2) **AUDIO E2** ? learned timbre classifier (needs labeled hits). Also
queued: pose/stance #2, owner's
19     "other hunch", Option B (owned end-to-end model). Read [[audio-e1-findings]], [[code-
map-artifact]] (docs/CODE_MAP.md),
20     docs/RALLY_DETECTION_RESEARCH.md, ADR-007 to ramp.
21     ? **PR #36 (docs/AUDIO_E1_FINDINGS.md) MERGED to master 2026-06-07** (owner restored the
branch after an accidental
22     close; reopened + squash-merged). Record now in-repo + [[audio-e1-findings]] (memory) +
plan §10. master fully clean, no open PRs.
23
```

```

24      --- (prior checkpoint) CLEAN SLATE. Indexer PRs #25?#32 MERGED; master 202 tests green,
tree clean, branches pruned
25      (kept feat/khelacharya-shot-intelligence). Audio decided (ADR-007). VLC annotator ?
standalone public repo
26      github.com/avidullu/rally-annotator (MIT, multi-sport). **ALL PRs #25?#34 MERGED; slate
FULLY clean** (master only + retained feat/khelacharya-shot-intelligence; no open
27      PRs; tree clean). #33 removed the stale annotator copy; #34 expanded ADR-007 (fusion +
Option B) + repointed the
28      ref ? both merged safely. **PARKED: modeling waits on golden labels (user self-rates via
annotator); next code
29      lever = AUDIO (E1 go/no-go probe).**
30      **ARCHITECTURE DECISION captured (ADR-007 $Integration architecture, PR #34):** audio
fuses in OUR rally layer
31      (WASB stays frozen black box ? feature-level into distill_local + decision-level recall-
recovery); **Option B = the
32      intended end-state** ? once a healthy golden corpus exists, train an OWNED end-to-end
multimodal model (frames+audio+
33      pose+??rallies) fine-tuned on our data, dropping WASB-weight dependence + recall
ceiling. Modular cues now (audio?
34      stance?other) are how we earn Option B (each generates the labels it needs). See
resumption runbook ?.
35
36      ## DISTILLED MODEL BUILT (2026-06-06) ? user wants this approach; honest result = needs
more data
37      - **ALL PRs #27-#31 MERGED to master** (0c6f2c1). master now has: code-map,
gate-A/rally_gate, gemini-refine
38      (+provider hardening + extract_video_chunk scale_width), calibrate_local,
distill_local. This session's 5
39      branches PRUNED (local+remote). Retained: feat/khelacharya-shot-intelligence (PR #11).
Pre-existing merged
40      stragglers still on origin (left, optional prune): docs/ending-reason-forced-
unforced(#19), feat/rally-slicer(#20),
41      feat/wasb-substrate(#21).
42      - **User will RATE the clips ? cached artifacts LEFT IN PLACE** (do NOT delete):
Temp\{adarsh,testlarge}_frames +
43      trajectories (~clips + Temp), Temp\verylarge_frames (partial 12246/46080).
Adarsh+TestLargeVideo trajectories
44      ready ? when user import-locals golden labels, calibrate_local/distill_local use them
with zero re-extraction.
45      (#30 stranded calibrate_local on gate branch ? RECOVERED in PR #31.)
46      - `backend/eval/distill_local.py` (PR #31, OPEN): learned model ? WASB recall-oriented
candidates (0.005,1.0,0.5)
47      ? features [duration,...
[/external_agent_tool_result]

```

10. assistant (2026-07-02T06:11:33.143Z)

```

[external_agent_tool_call: Read]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\session-handoff.md
[/external_agent_tool_call]

```

11. assistant (2026-07-02T06:11:33.143Z)

```

[external_agent_tool_result]
1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff for this project ? current state, next
steps, open decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key
decision/ADR, phase shift, session wrap); \"update the handoff\" forces an immediate full
rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: eb4b7b53-2f77-4c4d-83e1-4fa5e77f3e3f

```

```

8      ---
9
10     # Session handoff ? badminton-highlight-indexer (+ siblings: sports-data-collector,
rally-annotator)
11
12     **Last updated:** 2026-06-07 . **Refresh policy: AUTO-refresh at meaningful
checkpoints** ? PR merged,
13     key decision/ADR, phase shift, session wrap-up (NOT on-request-only). "update the
handoff" forces an
14     immediate full rewrite. **Staleness tripwire:** if ?3 substantive [[current-state]]
checkpoints ? or any
15     PR-merge / new ADR / phase shift ? have landed since the "Last updated" date above,
refresh this handoff now.
16     Pointer, not a duplicate. **For the freshest blow-by-blow state, read [[current-state]]
first**
17     (updated every micro-checkpoint); this handoff is the slower ~1-page orientation.
18
19     ## You are here (2026-06-07)
20     - **Rally-detection stack matured & all merged.** WASB shuttle-trajectory substrate is
**resumable/
21     checkpointed** (md5-ish cache, incremental detector flush, tracker re-run). On top: a
rally layer =
22     windowing thresholds + a **net-crossing "gate A"** + a **distilled classifier**
(`distill_local`).
23     Eval harness: `calibrate_wasb` / `calibration.py` (LOVO) / `metrics.py`.
24     - **Honest core finding:** WASB-only threshold tuning **ceilings ~F1 0.44** and the
learned distilled
25     model **underperforms the baseline at n=2 videos**. The blocker is **golden data +
WASB candidate
26     recall**, not the method. (Gemini-as-teacher gave us the first real measurement.)
27     - **Strategy decided (ADR-007):** Gemini = **offline teacher only**; **AUDIO** hit-
detection is the next
28     local cue, **fused in our rally layer** (WASB stays a frozen black box) ? Option A.
**Pose/stance =
29     parked #2.** **Option B = the eventual end-state**: one OWNED end-to-end multimodal
model
30     (frames+audio+pose?rallies) fine-tuned on our data, dropping WASB-weight dependence +
the recall ceiling.
31     - **Golden data path:** owner **self-rates** clips with the new **standalone VLC
annotator repo**
32     (github.com/avidullu/rally-annotator, MIT, multi-sport) ? CSV ? in-repo `--labels`
loaders / collector
33     `import-local`. Human labels > the Gemini silver labels the model was starving on.
34     - **? SESSION SUNSET (2026-06-07).** Indexer PRs **25?#35 merged**, master green (**209
tests**), tree clean
35     (only `feat/khelacharya-shot-intelligence` retained; **no open PRs**). Annotator repo
public/clean. **Audio E1
36     ran ? classical onset = ~2.2x discrimination ceiling, NOT adopted for fusion**
(detector kept; quantitative
37     gains/losses + reusable lessons in [[audio-e1-findings]]). Modeling **PARKED** pending
golden labels.
38     ? PR #36 (`docs/AUDIO_E1_FINDINGS.md`) MERGED to master (2026-06-07); record also in
[[audio-e1-findings]] + plan §10.
39
40     ## Next steps / open threads
41     1. **Owner:** rate a few clips in VLC (annotator v1.2 ? needs a live multi-sport smoke
test), and/or merge
42     nothing pending (none open).
43     2. **Audio E1 ? DONE ? PR #35. Verdict: classical-onset audio is too weak (~2.2x
ceiling).** Built
44     `backend/audio/hit_detector.py` + `audio_e1_probe.py` (numpy/ffmpeg, no scipy, 209
tests). On existing GoPro
45     audio vs Gemini silver: audio fires in 100% of rallies + recovers 100% of WASB-missed
rallies (real recall
46     signal), BUT discrimination only ~2.2x and audio-only F1 < WASB. **Band-pass (Option

```

A) tried ? does NOT help**
47 (HF bands worse; ceiling holds; NOT a warm-up artifact ? between-point
tosses/footwork/talk fool a classical
48 onset detector). **Next (user decides):** E2 learned TIMBRE classifier (needs
labelle...
[/external_agent_tool_result]

12. assistant (2026-07-02T06:11:33.143Z)

[external_agent_tool_call: Read]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\code-map-artifact.md
[/external_agent_tool_call]

13. assistant (2026-07-02T06:11:33.143Z)

[external_agent_tool_result]
1 ---
2 name: code-map-artifact
3 description: Where the dense cold-start CODE MAP of the indexer lives ? read it FIRST to
ramp up on codebase structure instead of crawling files
4 metadata:
5 node_type: memory
6 type: reference
7 originSessionId: 42211f80-2169-4025-ad9d-99c1aa29ff70
8 ---
9
10 For **code/architecture ramp-up** on `badminton-highlight-indexer`, read
`docs/CODE\_MAP.md` in
11 the repo FIRST ? a dense, anchor-rich (`@path::symbol`), machine-first map: pipeline
(runtime +
12 calibration/eval), per-subsystem file/symbol index, canonical data contracts, WASB WSL
boundary,
13 cross-repo collector contract, cross-cutting gotchas, "to do X -> open Y" ramp paths,
and a
14 test->subsystem map. It replaces a DFS/BFS crawl of 40+ files.
15
16 - Landed via PR #27 (`docs/code-map`).
17 - **Regenerate** after any registry/contract/WSL-interface change by rerunning the
`build-code-map`
18 workflow (script saved under the session's `workflows/scripts/`): parallel subsystem
readers ->
19 synthesize -> adversarial verify (the verify phase re-checks every anchor against real
files).
20 - Caveat baked into the doc: Sections 4 (WASB-SBDT) & 5 (collector) describe EXTERNAL
repos and are
21 unverified against their source ? only the repo-side callers were checked.
22 - This is for CODE structure (which is repo-derivable, hence a doc not a memory). Non-
code project
23 state stays in [[current-state]] / [[session-handoff]].
24
[/external_agent_tool_result]

14. assistant (2026-07-02T06:11:33.143Z)

Memory ramped me up well. I'm picking up from a clean master (PRs #25?#36 merged, 209 tests
green, modeling parked pending golden labels). The new task ? rich per-video observability
output in **both** the indexer (highlights tool) and the collector ? is greenfield. Let me read
the code map and survey both repos' pipelines + their README/FAQ docs (since the report must be
debuggable by a README-reader).

15. assistant (2026-07-02T06:11:33.144Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md
[/external_agent_tool_call]

16. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_result]
1      # CODE_MAP.md
2
3      **Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-
hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a
segmenter-agnostic eval/calibration stack.
4
5      **Regenerate:** rerun the `build-code-map` workflow (parallel subsystem readers ? this
doc). Re-run after any registry/contract/WSL-interface change.
6
7      **LEGEND**
8      - `@path` = file (always repo-relative)
9      - `::sym` = symbol (class/func/const) in the nearest preceding `@path`
10     - `->` = dataflow / call / "produces"
11     - `$` = data contract (canonical shape)
12     - `?` = gotcha / footgun
13     - `?` = registry / plugin extension point
14     - `WSL` = runs inside WSL2 conda env, NOT the Windows Python process
15
16     ---
17
18     ## 1. PIPELINE
19
20     ### 1A. Runtime: video file -> highlight reel
21     ```
22     config.json (json.load; absent -> {})
23     -> video_id = MD5(whole file) @backend/main.py::get_file_md5 (? chunk size
varies per endpoint: 64KB main, 1MB run_process, 8KB run_eval, 64KB phase3 ? all hash WHOLE
file so values agree)
24     -> validate
@backend/validators/base.py::registry.run_all(video_path, config)
25     FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity
-> SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see
@backend/validators/__init__.py)
26     -> db.add_video(video_id, filepath, status, [r.dict()]) @backend/database.py
27     -> seg = segmenter_registry.get(name)(db, config) ?
@backend/indexer/base.py::segmenter_registry
28     -> segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)
29     [STOP POINT: segments-only ? predictions now in DB; eval/regression operate here
without stitching]
30     -> select segment ids -> [(start,end)] (+ stitching padding)
31     -> VideoCompiler.compile_highlights(filepath, time_pairs, out_name)
@backend/stitcher/compiler.py
32     per-clip ffmpeg re-encode (libx264/superfast/crf22) -> concat -c copy ->
output/<name>.mp4
33     ```
34     Hybrid segmenter internal 4-phase (yolo_hybrid / tracknet_hybrid / wasb_hybrid ? same
shape):
35     ```
36     P1 chunk video (yolo only; tracknet/wasb = whole-video single pass, chunk_index=0)
37     -> P2 candidate "action windows" from motion/trajectory [STOP: candidates persisted
via db.add_candidate_segment]
38     -> P3 pad clip (ai_handoff_padding) -> ThreadPoolExecutor ->
provider.analyze_video(clip, prompt) ? provider_registry
39     -> P4 db.add_play_segment + db.link_candidate_to_segment
40     ```
41     Pure-AI path `gemini`: no P2; chunk-or-single -> provider -> heavy validate
(clamp/dedup) -> add_play_segment.
42     Legacy `motion`: pixel-diff -> segments + **fabricated** metadata/events (? not ground
truth).
```

```

43
44     ### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
45     ```
46     WasbHybridSegmenter.process_video                                @backend/indexer/wasb_hybrid.py
47     -> WasbRunner.healthcheck() (gate; not ok -> return [])
@backend/detectors/wasb_runner.py
48     -> WasbRunner.run_predict(video, out_dir)
49     stage video + wasb_infer.py into WSL fs -> `wsl bash -c 'source conda.sh && python
wasb_infer.py ...'`
50     -> @backend/detectors/wasb_infer.py::run (WSL): sliding windows -> GPU detector
pass (CHECKPOINTED det_raw.jsonl) -> CPU tracker (always full re-run) -> trajectory CSV
51     -> TrackNetRunner.parse_trajectory_csv(csv)                    @backend/detectors/tracknet_runner.py
(shared, re-exported by WasbRunner)
52     -> TrackNetRunner.trajectory_to_action_windows(points, fps, frame_width, chunk_start,
velocity_thresh, merge_gap, min_window_duration, chunk_index) [the model-agnostic WINDOWING
BRAIN]
53     ```
54     (`tracknet_hybrid` identical, swapping WasbRunner->TrackNetRunner; `_probe_fps_width`
fallback (30.0, 1280.0).)
55
56     ### 1C. Calibration / eval flow (NO pipeline run unless told)
57     ```
58     GT CSV -> annotations.load_annotations / calibrate_wasb.load_golden_gts ->
List[Interval]
59     DB predicti...
[/external_agent_tool_result]

```

17. assistant (2026-07-02T06:11:33.144Z)

```

[external_agent_tool_call: Glob]
input: {"pattern": "**/*.md", "path": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer"}
[/external_agent_tool_call]

```

18. assistant (2026-07-02T06:11:33.144Z)

```

[external_agent_tool_result]
.pytest_cache\\README.md
TEST_RUN_SUMMARY.md
docs\\EVALUATION.md
docs\\MONETIZATION_AUDIT.md
docs\\TRACKNET_WSL_SETUP.md
README.md
docs\\GOLDEN_SET_IMPLEMENTATION.md
docs\\ROADMAP.md
docs\\GOLDEN_SET_PHASE1_VIDEOS.md
docs\\GOLDEN_SET_VENDORING.md
docs\\HOW_RALLY_DETECTION_WORKS.md
docs\\CODE_MAP.md
docs\\RALLY_DETECTION_RESEARCH.md
docs\\VIDEO_RECORDING_GUIDELINES.md
docs\\DECISIONS.md
docs\\AUDIO_RALLY_DETECTION_PLAN.md
docs\\BACKLOG.md
docs\\AUDIO_E1_FINDINGS.md
[/external_agent_tool_result]

```

19. assistant (2026-07-02T06:11:33.144Z)

```

[external_agent_tool_call: Glob]
input: {"pattern": "**/*.md", "path": "C:\\Users\\avidu\\Projects\\sports-data-collector"}
[/external_agent_tool_call]

```

20. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_result]
C:\Users\avidu\Projects\sports-data-collector\.pytest_cache\README.md
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\NDA_TEMPLATE.md
C:\Users\avidu\Projects\sports-data-collector\docs\DECISIONS.md
C:\Users\avidu\Projects\sports-data-collector\README.md
C:\Users\avidu\Projects\sports-data-collector\docs\PIPELINE_ROLE.md
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ANNOTATION_GUIDE.md
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\R00RA_BRIEF.md
C:\Users\avidu\Projects\sports-data-collector\data\validation\pilot_badminton_01_report.md
C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md
[/external_agent_tool_result]
```

21. assistant (2026-07-02T06:11:33.144Z)

Indexer architecture is clear. Now I need the collector's pipeline + both READMEs (to calibrate the "README-reader debuggability" bar). Let me read those and fan out an Explore agent to map the collector's processing stages and current output-emission points.

22. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

23. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\README.md
[/external_agent_tool_call]
```

24. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\docs\PIPELINE_ROLE.md
[/external_agent_tool_call]
```

25. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_result]
1      # Badminton Highlight & Rally Indexer
2
3      A local, AI-powered web application and CLI utility designed to process long-form
badminton recordings (e.g., from GoPro), run comprehensive pluggable validations (quality,
integrity, and relevance checks), index all rallies into a relational SQLite database, and
dynamically compile custom highlight reels using high-speed FFmpeg stitching.
4
5      The pipeline is built around three pluggable registries ? validators,
segmenters, and AI providers ? plus a sport profile registry so the same engine can
target other racket/court sports in the future. Badminton is the default profile.
6
7      > Where this is heading: the project is evolving from an AI-API-dependent prototype
into a self-improving tool that learns rally detection offline from a commercially-licensed,
annotated corpus. See [docs/ROADMAP.md](docs/ROADMAP.md) for the four-phase plan and
[docs/DECISIONS.md](docs/DECISIONS.md) (ADR-002?004) for the rationale.
8
9      ---
10
11     ## ? Key Features
12
13     * Pluggable Video Validation Framework: Easily extendable base-class architecture
that automatically runs checks on:
14     * Static Format & Resolution: Verifies container is MP4, resolution >= 720p, and
framerate >= 30 FPS.
15     * PTS Timeline Continuity: Scans keyframe timestamps for timeline gaps, jumps,
```


or editing splices to detect tampering or splices.

```
16      *   *Scene Cut Density Check*: Measures sub-sampled HSV frame histogram correlation
to verify visual cuts. A continuous raw camera stream should have near 0 cuts.
17      *   *Blurriness Check*: Calculates average frame Laplacian variance to reject out-
of-focus footage.
18      *   *Relevance (Is it Badminton?): Uses high-performance HSV color court matching
to identify court layouts.
19      *   **Three Pluggable Rally Segmenters** (`backend/indexer/`):
20      *   `yolo_hybrid` *(default)* ? two-stage pipeline. Stage 1 runs **torchvision
person detection** (Faster R-CNN, BSD-licensed) with **ByteTrack** tracking (supervision, MIT-
licensed) to cheaply filter the video down to a small set of high-motion candidate windows.
Stage 2 hands each candidate (with a small padding) to the AI provider for high-precision
semantic boundaries and shot counts. Massively cheaper than running the LLM on the full video.
*(The registry key remains `yolo_hybrid` for back-compat; the AGPL-licensed YOLOv8 was replaced
with permissive components ? see [docs/MONETIZATION_AUDIT.md](docs/MONETIZATION_AUDIT.md) and
ADR-005.)*
21      *   `gemini` ? sends the entire video (or a `chunk_duration` slice) to Google Gemini
directly. Highest precision per call, highest cost.
22      *   `motion` ? zero-dependency OpenCV pixel-difference heuristic. No API key
required. Useful as a baseline or for offline/air-gapped use.
23      *   **Pluggable AI Provider Registry** (`backend/providers/`): The Gemini provider ships
in-tree; new providers (e.g. an OpenAI or local-model backend) can be registered without
touching the segmenters.
24      *   **Pluggable Sport Profile Registry** (`backend/sports/`): The per-sport LLM prompt
and relevance config live in a `SportProfile`. Add a new sport by subclassing `SportProfile` and
registering it.
25      *   **SQLite Indexing**: Each processed video is keyed by MD5 hash. Play segments
(start/end/duration/server/receiver/metadata) and per-segment events are persisted in
`output/sports_indexer.db`. Re-running on the same file is a cache hit.
26      *   **Query System**: Filter indexed play segments by `server`, `receiver`,
`duration_min`, and `event_type` (the schema supports rich event types like
smashes/serves/fouls; today the shipped segmenters mainly populate timing + estimated
`shot_count`).
27      *   **Zero-Dependency Local UI**: Served statically directly by FastAPI (Vite + Node.js
NOT required!) featuring a glassmorphic dashboard, responsive filtering, and immediate clip
stitching.
28      *   **High-Speed FFmpeg Stitching** with configurable `begin_paddin...
[/external_agent_tool_result]
```

26. assistant (2026-07-02T06:11:33.144Z)

[external_agent_tool_result]

```
1      # Sports Data Collector & Curator
2
3      A clean, extensible Python codebase and CLI toolchain designed to discover, ingest,
validate, and curate sports video datasets. The repository stores annotations and video metadata
in an SQLite database, supports sport-specific tables and schemas, manages video storage outside
of Git, and enforces strict commercial license compliance.
4
5      > **Pipeline role:** this repo is the *data producer* feeding the **badminton-highlight-
indexer**, which trains/evaluates an offline rally segmenter on this curated, commercially-
licensed data. See [docs/PIPELINE_ROLE.md](docs/PIPELINE_ROLE.md) for why `export`/`fetch` exist
and how the two repos fit together.
6
7      ---
8
9      ## Features
10
11     - **SQLite Backend**: Centralized `sports_metadata.db` with a generic `videos` table and
sport-specific annotation tables (`badminton_rallies`, `tennis_points`, `cricket_deliveries`,
`pickleball_rallies`, `tabletennis_rallies`).
12     - **Commercial License Flagging**: Each ingested source license is checked against a
configured whitelist (`MIT`, `Apache-2.0`, `BSD`, `CC0`, `CC-BY`, `Public-Domain`,
`Proprietary`) and the resulting `is_commercial` flag is stored alongside the video. Non-
commercial sources (like `CC-BY-NC-SA`) are flagged as non-commercial but still ingested ?
```

nothing is discarded automatically. The whitelist is read from `config/config.yaml` consistently across the parser, validator, and CLI.

13 - ****Overlap & Duration Validation****: Ensures chronological event bounds (e.g. rally start/end times) do not overlap and that durations are positive.
14 - ****Stroke-level CSV Parser****: Ingests ShuttleSet/CoachAI-style stroke CSV logs and aggregates them into rally durations using configurable start/end time buffers.
15 - ****Local Ingestion Mode****: Manually registers local videos and associated CSV/JSON annotation files directly into the database as curated entries.
16 - ****Interactive Curation CLI & Visual Play Harness****: Let's you query DB status, list staging entries, review sample timestamps, and approve/reject them. It includes a built-in play function that opens the video in your default browser at the exact rally segment (using HTML5 media fragments `#t=start,end` for local files or YouTube start time markers `&t=start` for remote videos) for manual visual verification.

17
18 ---
19
20 ## Installation
21
22 Ensure you have Python 3.9+ installed.
23
24 1. Clone or download the repository.
25 2. Install the package and dependencies:
26 ```bash
27 pip install -e . # core: curate / import-local / export / status
28 pip install -e "[fetch]" # adds yt-dlp for video acquisition (fetch / crawler)
29 pip install -e "[dev]" # test tooling (pytest, pytest-cov)
30 ```
31
32 ### External requirements (not pip-installable)
33
34 The `fetch` command and the ShuttleSet crawler shell out to external binaries.
35 They degrade gracefully (a clear error / skipped download) when missing, but are
36 ****required**** for video acquisition:
37
38 - ****FFmpeg / ffprobe**** ? used to probe the real resolution/fps of downloaded files.
39 - ****A JS runtime (e.g. [deno](https://deno.com))**** ? needed by yt-dlp to solve
40 YouTube's challenge for high-resolution formats. `fetch` auto-detects a deno
41 install at `~/deno/bin`.
42
43 Core curation/import/export/status commands work without any of these.
44
45 ---

46
47 ## Usage
48
49 All commands should be executed from the root of the project. Make sure `PYTHONPATH` is
set to the current directory:

50
51 ```bash
52 # On Windows PowerShell:
53 \$env:PYTHONPATH="."
54 ```
55
56 ### 1. View Database Status
57 To check the current statistics of staging vs. curated items:
58 ```bash
59 python -m src.cli.curate status
60 ```
61
62 ### 2. Ingest External Datasets to Staging
63 To parse external raw stroke-level datasets (e.g. CoachAI format) and register them in
staging:
64 ```bash
65 python -m src.cli.ingest --sport badminton --parser coachai --fi...
[/external_agent_tool_result]

27. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_result]
1      # Role in the rally-segmenter training pipeline
2
3      This repo is the **data producer** in a two-repo system. Its job is to curate
4      sports videos with per-rally `[start, end)` annotations and commercial-license
5      metadata, then hand that data to the **badminton-highlight-indexer**, which
6      processes, scores, and (Phase 4) *trains* an offline rally segmenter on it.
7
8      This doc explains *why* the `export` and `fetch` commands exist and how they fit
9      the bigger picture. The full decision records and end-to-end narrative live in the
10     indexer repo:
11
12     - `badminton-highlight-indexer/docs/ROADMAP.md` ? the four-phase plan and end-state.
13     - `badminton-highlight-indexer/docs/DECISIONS.md` ? ADR-002 (this bridge), ADR-003
14       (acquisition policy), ADR-004 (offline learned segmenter).
15
16     ## Why this repo matters
17
18     The indexer's best rally mode refines local detections with a **paid Gemini API**.
19     The strategic goal is to replace that with an **offline model trained on our own
20     commercially-licensed, annotated corpus** ? making rally detection free, offline,
21     and self-improving. None of that is possible without a clean, licensable,
22     accurately-labelled, full-resolution dataset. That dataset is what this repo
23     produces.
24
25     ## What the two key commands do (and why)
26
27     - **`curate export`** ? emits the indexer's ground-truth format: per-video
28       `start,end` CSVs + a `manifest.json` pairing each CSV with its video file.
29       Sport-agnostic (all supported sports ? badminton, tennis, cricket, pickleball,
30       table tennis ? share a `(video_id, ordinal, start, end)` shape) and gated on
31       `is_commercial`, so only licensable data flows
32       downstream. The annotations are **temporal-only** ? they train/evaluate rally
33       *segmentation*, not the frame-level shuttle detector (see ADR-002).
34
35     - **`fetch`** ? (re)downloads videos at **best available resolution**, updating
36       the DB **in place** so annotations and curation status are never lost. Uses
37       authenticated cookies for reliable, unthrottled downloads. The why and the
38       hard-won pitfalls (SABR?144p, Chrome DPAPI, rate-limiting) are in ADR-003.
39
40     ## The one contract that matters: video identity
41
42     The indexer keys videos by **file MD5**, this repo by a human `video_id`. A
43     re-encode/re-download changes the bytes, hence the MD5. So the manifest carries the
44     exact `local_path`, and any downstream processing must run on **that** file. This is
45     why acquisition (`fetch`) happens *before* the indexer processes anything.
46
[/external_agent_tool_result]
```

28. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_call: Agent]
description: Map collector pipeline & outputs
[/external_agent_tool_call]
```

29. assistant (2026-07-02T06:11:33.144Z)

```
[external_agent_tool_result]
Now I'll provide you with a comprehensive map of the sports-data-collector repository:
```

COMPREHENSIVE REPOSITORY MAP: SPORTS-DATA-COLLECTOR

1. CLI ENTRYPOINTS

****File:**** `C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py` (lines 805-921)

The `SportsDataCLI` class exposes these subcommands via argparse:

Subcommand	Handler	Lines	Purpose
`status`	`cli.status()`	38-50, 902	Display summary of videos in DB (count by sport/status)
`curate`	`cli.curate()`	619-722, 904-905	Interactive review loop for staging videos; approve/reject/play segments
`import-local`	`cli.import_local(args)`	52-142, 906-907	Manually register local video + JSON/CSV annotations into DB (any sport)
`convert-cvat`	`cli.convert_cvat(args)`	526-560, 908-909	Convert vendor CVAT XML export to canonical per-rally CSV + issues report
`validate-delivery`	`cli.validate_delivery(args)`	562-617, 910-911	Full validation suite on CVAT/CSV; emit ` <id>_report.{md,json}` + `<id>_review.csv`</id></id>
`fetch`	`cli.fetch(args)`	330-426, 914-915	Download/re-download videos at best resolution; update DB in place
`export`	`cli.export_eval_set(args)`	428-524, 912-913	Export curated annotations as indexer-ready eval/training CSVs + manifest.json

2. PER-VIDEO PROCESSING PIPELINE

****Ingestion Path 1: `import-local` (Direct Local File)****

****Entry:**** `src/cli/curate.py:52-142`

- **Parse annotations**** ? `\_parse\_local\_annotations()` (lines 144-290)
 - Reads `.json` or `.csv` files
 - Dispatches per-sport: BadmintonRally, TennisPoint, CricketDelivery, PickleballRally, TableTennisRally
 - Uses `safe\_int()` / `safe\_float\_required()` from `src/core/parsing.py:9-29` for robust casting
- **Validate**** ? `DatasetValidator.validate\_{sport}\_rallies/points/deliveries()` (src/core/validator.py:29-144)
 - Checks: chronological order, no overlaps, positive duration
 - Returns `(is\_valid: bool, errors: List[str])`
- **Create VideoMetadata**** ? `VideoMetadata(...)` (lines 109-121)
 - Sets `status = CurationStatus.CURATED` (bypasses staging)
 - Stores: video_id, sport, local_path, license_type, fps, resolution
- **Insert to DB**** ? `SportsDatabase.insert\_video()` + sport-specific insert (lines 124-136)
 - `insert\_badminton\_rallies()`, `insert\_tennis\_points()`, etc.
 - Uses UPSERT logic to preserve child rows when updating

****Ingestion Path 2: `convert-cvat` (Vendor CVAT Export ? CSV)****

****Entry:**** `src/cli/curate.py:526-560`

****Key Pipeline (src/parsers/cvat/rally_cvat.py):****

- **Parse CVAT XML**** ? `parse\_cvat\_boxes(xml\_path)` (lines 140-193)
 - Extracts `` (frame-mode) or `` (interpolation-mode) boxes
 - Returns `(meta: {step, stop\_frame}, boxes: [{frame, attrs}])`

```

2. **Recompute time from frames** ? `build_records(bboxes, fps)` (lines 205-239)
  - **Critical:** derives `start_time = frame_min / fps`, `end_time = frame_max / fps`
  - **Ignores** vendor's `start_time`/`end_time` attributes (they're often wrong due to
step/fps mismatch)
  - Resolves CVAT attribute name aliases (e.g., "score_before (a-b)" ? "score_before")

3. **Detect issues** ? `detect_issues(records, stop_frame, ...)` (lines 242-322)
  - **Blocker issues** (halt import): overlap, zero_duration
  - **High/Medium/Low issues** (flag for review): ending_reason_vocab, shot_density,
possible_missed_rally, outcome_vocab, frame_out_of_range
  - **Does NOT flag** sampling density; downstream padding absorbs it

4. **Write canonical CSV** ? `write_canonical_csv(records, out_path)` (lines 356-372)
  - Columns: `rally_number`, `start_mmss`, `end_mmss`, `start_time`, `end_time`, `shots_count`,
`ending_reason`, `last_shot_outcome`, `score_before`, `score_after`, `notes`
  - Output ready for `import-local`

5. **Emit issues report** ? `render_issue_report(result, source)` (lines 375-396)
  - Human-readable text grouped by severity
  - Also written to `--issues...`
[/external_agent_tool_result]

```

30. assistant (2026-07-02T06:11:33.144Z)

Excellent ? I now have a full picture of both repos. Before I draft the plan, let me ground the central integration points by reading the exact orchestration code I'd hook into (so the plan isn't hand-wavy). Reading the indexer's orchestrator + validation result shape, and the collector's report-builder pattern I'd mirror.

31. assistant (2026-07-02T06:11:33.144Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]

```

32. assistant (2026-07-02T06:11:33.144Z)

```

[external_agent_tool_result]
1      import os
2      import sys
3      import argparse
4      import hashlib
5      import json
6      import uvicorn
7      from dotenv import load_dotenv
8
9      load_dotenv()
10     from fastapi import FastAPI, HTTPException
11     from fastapi.middleware.cors import CORSMiddleware
12     from pydantic import BaseModel
13     from typing import List, Dict, Any, Optional
14
15     from fastapi.staticfiles import StaticFiles
16     from fastapi.responses import HTMLResponse
17     from backend.validators.base import registry
18     from backend.database import Database
19     from backend.indexer.base import segmenter_registry, BasePlaySegmenter
20     from backend.stitcher.compiler import VideoCompiler
21
22     app = FastAPI(title="Sports Highlight Indexer API")
23
24     # Enable CORS for local web interface
25     app.add_middleware(
26         CORSMiddleware,

```

```

27     allow_origins=["*"],
28     allow_credentials=True,
29     allow_methods=["*"],
30     allow_headers=["*"],
31 )
32
33 # Serves static frontend files directly
34 os.makedirs("backend/static", exist_ok=True)
35 app.mount("/static", StaticFiles(directory="backend/static"), name="static")
36
37 @app.get("/", response_class=HTMLResponse)
38 def serve_index():
39     index_path = "backend/static/index.html"
40     if os.path.exists(index_path):
41         with open(index_path, "r", encoding="utf-8") as f:
42             return f.read()
43     return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in
backend/static/</h3>"
44
45 # Global variables initialized at startup
46 db_instance: Optional[Database] = None
47 global_config: Optional[Dict[str, Any]] = None
48
49 def get_file_md5(filepath: str) -> str:
50     """Calculates file hash to uniquely identify videos and prevent duplicate
indexes."""
51     hasher = hashlib.md5()
52     with open(filepath, "rb") as f:
53         # Read in chunk sizes of 64kb
54         for chunk in iter(lambda: f.read(65536), b""):
55             hasher.update(chunk)
56     return hasher.hexdigest()
57
58 def load_config(config_path: str = "config.json") -> Dict[str, Any]:
59     if os.path.exists(config_path):
60         with open(config_path, "r") as f:
61             return json.load(f)
62     return {}
63
64 # --- API Models ---
65 class ProcessRequest(BaseModel):
66     video_path: str
67     player_pool: Optional[List[str]] = None
68     max_frames: Optional[int] = None
69     segmenter: Optional[str] = None
70
71 class QueryRequest(BaseModel):
72     video_id: str
73     server: Optional[str] = None
74     receiver: Optional[str] = None
75     duration_min: Optional[float] = None
76     event_type: Optional[str] = None
77
78 class CompileRequest(BaseModel):
79     video_id: str
80     segment_ids: List[int]
81     output_filename: str
82
83 # --- API Endpoints ---
84 @app.get("/api/config")
85 def get_config():
86     return global_config
87
88 @app.post("/api/process")
89 def process_video(req: ProcessRequest):

```

```

90         if not os.path.exists(req.video_path):
91             raise HTTPException(status_code=404, detail=f"Video file not found at
'{req.video_path}')"
92
93         video_id = get_file_md5(req.video_path)
94
95         # 1. Run pluggable validators
96         print(f"[API] Running validations for {req.video_path}...")
97         val_results = registry.run_all(req.video_path, global_config)
98
99         # Check if any validator failed
100        failures = [r for r in val_results if not r.passed]
101        if failures:
102            # Save failed processing log
103            db_instance.add_video(video_id, req.video_path, "failed", [r.dict() for r in
val_results])
104            return {
105                "video_id": video_id,
106                "status": "failed",
107                "message": "Video failed validation checks.",
108                "results": [r.dict() for r in val_results]
109            }
110
111        # 2. Run segmenter & indexer
112        print(f"[API] Video passed checks. Segmenting and indexing...")
113        db_instance.add_video(video_id, req.video_path, "processed", [r.dict() for r in
va...
[/external_agent_tool_result]

```

33. assistant (2026-07-02T06:11:33.145Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\validators\base.py
[/external_agent_tool_call]

```

34. assistant (2026-07-02T06:11:33.145Z)

```

[external_agent_tool_result]
1     from abc import ABC, abstractmethod
2     from pydantic import BaseModel
3     from typing import List, Dict, Any, Optional
4
5     class ValidationResult(BaseModel):
6         validator_name: str
7         passed: bool
8         message: str
9         details: Dict[str, Any] = {}
10
11    class VideoValidator(ABC):
12        @property
13        @abstractmethod
14        def name(self) -> str:
15            """Unique identifier for this validator."""
16            pass
17
18        @property
19        @abstractmethod
20        def description(self) -> str:
21            """Human-readable explanation of what this validator checks."""
22            pass
23
24        @abstractmethod
25        def validate(self, video_path: str, config: Dict[str, Any], context: Dict[str, Any])
-> ValidationResult:
26            """

```

```

27         Executes validation.
28         - video_path: Local path to the MP4 file.
29         - config: Dict parsed from config.json.
30         - context: Dynamic context dict shared across validators (stores shared values
like ffprobe metadata).
31         """
32         pass
33
34     class ValidationRegistry:
35     def __init__(self):
36         self._validators: List[VideoValidator] = []
37
38     def register(self, validator: VideoValidator):
39         """Registers a new validator."""
40         self._validators.append(validator)
41
42     def run_all(self, video_path: str, config: Dict[str, Any]) ->
List[ValidationResult]:
43         """Runs all registered validators in sequence, accumulating results."""
44         results = []
45         context = {} # Shared context dictionary (e.g. for caching ffprobe outputs)
46         for val in self._validators:
47             try:
48                 res = val.validate(video_path, config, context)
49                 results.append(res)
50             except Exception as e:
51                 results.append(ValidationResult(
52                     validator_name=val.name,
53                     passed=False,
54                     message=f"Validator threw an unexpected error: {str(e)}"
55                 ))
56         return results
57
58     # Shared global registry
59     registry = ValidationRegistry()
60
[/external_agent_tool_result]

```

35. assistant (2026-07-02T06:11:33.145Z)

I now have precise grounding. One important design reality I can see in the code: ``registry.run_all()`` builds the rich ``context`` dict (with ``ffprobe_meta``) internally and ****discards it****, and ``segmenter.process_video()`` returns only the final segment list? the juicy per-phase internals (candidate counts, Gemini upload/process timing, frames extracted, trajectory visibility %, WSL healthcheck) are either persisted to the DB or only ``print()``-ed. So the observability feature has a real plumbing question (harvest post-hoc from DB+artifacts vs. thread a lightweight recorder through the run). That shapes the plan.

Let me load plan-mode tools so I can present this properly and confirm a couple of decisions with you.

36. assistant (2026-07-02T06:11:33.145Z)

```

[external_agent_tool_call: ToolSearch]
input: {"query":"select:EnterPlanMode,ExitPlanMode","max_results":5}
[/external_agent_tool_call]

```

37. assistant (2026-07-02T06:11:33.145Z)

```

[external_agent_tool_result]
[/external_agent_tool_result]

```

38. assistant (2026-07-02T06:11:33.145Z)


```
[external_agent_tool_call: EnterPlanMode]
input: {}
[/external_agent_tool_call]
```

39. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_result]
Entered plan mode. You should now focus on exploring the codebase and designing an
implementation approach.
```

In plan mode, you should:

1. Thoroughly explore the codebase to understand existing patterns
2. Identify similar features and architectural approaches
3. Consider multiple approaches and their trade-offs
4. Use AskUserQuestion if you need to clarify the approach
5. Design a concrete implementation strategy
6. When ready, use ExitPlanMode to present your plan for approval

Remember: DO NOT write or edit any files yet. This is a read-only exploration and planning phase.

```
[/external_agent_tool_result]
```

40. assistant (2026-07-02T06:11:33.145Z)

I've completed a thorough exploration of both repos (pipelines, data shapes, existing report patterns, and the exact orchestration hooks). Before I finalize the plan, three decisions genuinely change the design ? let me confirm your intent rather than assume.

41. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"How should the rich observability report be emitted on each
run?","header":"Emission","multiSelect":false,"options":[{"label":"Always-on (opt-
out)","description":"Every process/ingest run writes the report automatically; add a --no-report
flag to disable. Best for catching inconsistencies & diffing runs, since it's just serializing
data the run already computed."},{label":"Opt-in flag","description":"Report only written when
an explicit --report / --observability flag is passed. Zero change to current default behavior;
less likely to surprise existing scripts."}]}, {"question":"The 'customer-friendly' report (strip
system internals, show video metadata + meta-highlights) ? which tool(s) should produce
it?","header":"Customer report","multiSelect":false,"options":[{"label":"Indexer
only","description":"Only the highlights tool gets the clean customer summary ? it has a real
end-user (whoever receives the highlight reel). The collector stays internal/technical-
only."},{label":"Both tools","description":"Indexer produces a viewer-facing reel summary;
collector also produces a clean 'delivery summary' for the annotation vendor / data team (non-
technical stakeholders)."}]}, {"question":"Both repos are separate (one private). How unified
should the report schema be across the two
tools?","header":"Schema","multiSelect":false,"options":[{"label":"Shared envelope + contract
doc","description":"Both tools emit the same top-level JSON shape (identity / stages[] / metrics
/ flags / verdict), mirrored independently with a documented contract. Makes reports diffable
across tools AND across runs ? directly serves 'evaluate output
differences'."},{label":"Independent per tool","description":"Each tool's report is shaped
purely to its own domain with no shared structure. Simpler, no cross-repo coordination, but
harder to diff/compare across the two tools."}]}]}
[/external_agent_tool_call]
```

42. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_result]
Your questions have been answered: "How should the rich observability report be emitted on each
run?"="Always-on and we should define a convention i.e. its written always next to the video
path's location or could be configured independently.", "The 'customer-friendly' report (strip
system internals, show video metadata + meta-highlights) ? which tool(s) should produce
it?"="Both tools", "Both repos are separate (one private). How unified should the report schema
be across the two tools?"="1. with the additional feedback that maybe make that an independent
```

github private repo which provides a "config language" to produce reports which can be reusable across these two today and other tools later on. Which also adds that, you may do some research to find if something "commercially safe" can be imported/installed which will reduce some of the "reinvention" work.". You can now continue with these answers in mind.
[/external_agent_tool_result]

43. assistant (2026-07-02T06:11:33.145Z)

That third answer meaningfully expands the scope ? a **separate private repo providing a reusable "report config language" consumed by both tools (and future ones), plus a research pass for commercially-safe building blocks to avoid reinvention. That's the right instinct for not building a house of cards, but it deserves grounding. Let me research the landscape before I commit the plan to a foundation.

44. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query": "select:WebSearch,WebFetch", "max_results": 5}
[/external_agent_tool_call]
```

45. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

46. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_call: WebSearch]
input: {"query": "soda-core SodaCL data quality YAML config language license Apache"}
[/external_agent_tool_call]
```

47. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_call: WebSearch]
input: {"query": "Python declarative report generation library YAML/JSON spec to markdown HTML 2025 MIT license"}
[/external_agent_tool_call]
```

48. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_call: WebSearch]
input: {"query": "evidently ydata-profiling great-expectations frictionless license comparison data observability report"}
[/external_agent_tool_call]
```

49. assistant (2026-07-02T06:11:33.145Z)

```
[external_agent_tool_result]
Web search results for query: "soda-core SodaCL data quality YAML config language license Apache"
```

Links: [{"title": "GitHub - sodadata/soda-core: Data Contracts engine for the modern data stack. https://www.soda.io · GitHub", "url": "https://github.com/sodadata/soda-core"}, {"title": "Run Soda Core checks with Airflow | Astronomer Documentation", "url": "https://www.astronomer.io/docs/learn/2.x/soda-data-quality"}, {"title": "soda-core/README.md at main · sodadata/soda-core", "url": "https://github.com/sodadata/soda-core/blob/main/README.md?plain=1"}, {"title": "Soda Moved to ELv2. Provero Is Apache 2.0. - DEV Community", "url": "https://dev.to/andreahlert/soda-moved-to-elv2-provero-is-apache-20-42l9"}, {"title": "Using Soda Core to test data quality | by Maksim Kazartsev | Medium", "url": "https://medium.com/@kazarmax/using-soda-core-to-check-data-quality-07b370da2df3"}, {"title": "Soda Core (v3) | Soda v3 | Documentation", "url": "https://docs.soda.io/soda-documentation/soda-v3/overview-main"}, {"title": "SodaCL tutorial | Soda Documentation", "url": "https://docs.soda.io/soda/quick-start-sodacl.html"}, {"title": "How To Use Soda For Data Quality Checks With Apache Airflow | by

```
Ahmed Mokbel | Medium", "url": "https://ahmed-mokbel.medium.com/how-to-use-soda-for-data-quality-checks-with-apache-airflow-cf249a737b5a"}, {"title": "soda-core/docs/configuration.md at main · sodadata/soda-core", "url": "https://github.com/sodadata/soda-core/blob/main/docs/configuration.md"}, {"title": "Creating a Fabric integration for Soda and checking data quality", "url": "https://dataroots.io/blog/creating-a-fabric-integration-for-soda-and-checking-data-quality"}]
```

Based on the search results, here's what I found about soda-core, SodaCL, and its license:

Soda Core Overview

Soda Core is a data quality and data contract verification engine that lets you define data quality contracts in YAML and automatically validate both schema and data across your data stack.

SodaCL Language

Soda Core uses the Soda Checks Language (SodaCL) to run checks defined in a YAML file. SodaCL is a YAML-based, domain-specific language for data reliability.

License Information

The project is licensed under Apache 2.0, though it's important to note a significant recent change: Soda Core was originally released under Apache License 2.0, but in 2023, Soda switched to the Elastic License v2 (ELv2), with the change applied to all new versions of Soda Core and its associated packages.

ELv2 is not an open source license by the OSI definition and adds two restrictions that Apache 2.0 does not have: you cannot offer the software as a managed service, and you cannot modify the license key functionality.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

[/external_agent_tool_result]

50. assistant (2026-07-02T06:11:33.145Z)

[external_agent_tool_result]

Web search results for query: "Python declarative report generation library YAML/JSON spec to markdown HTML 2025 MIT license"

Links: [{"title": "GitHub - Nicceboy/python-markdown-generator: Python library for dynamically generating HTML sanitised Markdown syntax. · GitHub", "url": "https://github.com/Nicceboy/python-markdown-generator"}, {"title": "yaml-to-markdown - PyPI Package Security Analysis - Socket", "url": "https://socket.dev/pypi/package/yaml-to-markdown"}, {"title": "YAML to Markdown Converter", "url": "https://pypi.org/project/yaml-to-markdown/"}, {"title": "Developer Documentation | yaml-to-markdown", "url": "https://anevis.github.io/yaml-to-markdown/docs/DEVELOPER.html"}, {"title": "json-schema-for-humans · PyPI", "url": "https://pypi.org/project/json-schema-for-humans/"}, {"title": "yamldoc", "url": "http://chrisbcole.me/yamldoc/"}, {"title": "GitHub - yaml/pyyaml: Canonical source repository for PyYAML · GitHub", "url": "https://github.com/yaml/pyyaml"}, {"title": "markdown-full-yaml-metadata · PyPI", "url": "https://pypi.org/project/markdown-full-yaml-metadata/"}, {"title": "GitHub - lauritzh/md-report-generator: Python-based report generation toolchain based on Markdown input files. To start a fresh report, clone this repo and you are ready to go! ?", "url": "https://github.com/lauritzh/md-report-generator"}]

Based on the search results, I found several relevant Python libraries for declarative report generation with YAML/JSON specifications:

Key Findings:

****1. yaml-to-markdown****

A library to convert YAML files to Markdown format, and a Python utility to take a JSON/YAML file or a python dict/list and create a Markdown file. This project is licensed under the MIT License.

****2. yamldoc****

This package converts a YAML file into markdown, formatting values and associated metadata in a doxygen-like way. It provides an easy interface for developers to document data in their YAML files as well as the expected types from a JSON YAML schema validator.

****3. python-markdown-generator****

A Python library for generating HTML sanitised Markdown documents that aims to bring a modular approach for dynamically building pure Markdown documents with pure Python so that they could consist of multiple reusable components.

****4. json-schema-for-humans****

Accepts JSON or YAML configuration files and can generate documentation in HTML and Markdown formats.

****5. md-report-generator****

A toolchain to generate pentest reports based on Markdown inputs, with YAML-based configuration support.

The **yaml-to-markdown** library appears to be the most directly aligned with your requirements, as it specifically handles YAML/JSON to Markdown/HTML conversion with MIT licensing.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

[/external_agent_tool_result]

51. assistant (2026-07-02T06:11:33.145Z)

[external_agent_tool_result]

Web search results for query: "evidently ydata-profiling great-expectations frictionless license comparison data observability report"

Links: [{"title": "GitHub - evidentlyai/evidently: Evidently is ??an open-source ML and LLM observability framework. Evaluate, test, and monitor any AI-powered system or data pipeline. From tabular data to Gen AI. 100+ metrics.", "url": "https://github.com/evidentlyai/evidently"}, {"title": "How to Use Ydata-Profiling with Great Expectations V3 API | by Bvolodarskiy | Provectus Articles | Medium", "url": "https://medium.com/provectus-articles/how-to-use-ydata-profiling-with-great-expectations-v3-api-fc6c2d0035cd"}, {"title": "Great Expectations - YData Profiling", "url": "https://docs.profiling.ydata.ai/4.5/integrations/great_expectations/"}, {"title": "Complete Guide to Effortless ML Monitoring with Evidently.ai", "url": "https://www.analyticsvidhya.com/blog/2024/03/complete-guide-to-effortless-ml-monitoring-with-evidently-ai/"}, {"title": "Data Quality & Observability: The Hidden Pillars of AI Success | by Amit Kharche | Medium", "url": "https://medium.com/@amitkharche/data-quality-observability-the-hidden-pillars-of-ai-success-a4ee1545d2a6"}, {"title": "Building Reliable Data Pipelines with Databricks and Great Expectations / Evidently AI | by André Castro | Medium", "url": "https://medium.com/@andreluizfc/building-reliable-data-pipelines-with-databricks-and-great-expectations-evidently-ai-95f681c53b6b"}, {"title": "GitHub - jorson-chen/evidently-LLM-observability: Evidently is ??an open-source ML and LLM observability framework. Evaluate, test, and monitor any AI-powered system or data pipeline. From tabular data to Gen AI. 100+ metrics.", "url": "https://github.com/jorson-chen/evidently-LLM-observability"}, {"title": "There are a lot of freaking data quality and data observability vendors | DataKitchen", "url": "https://datakitchen.io/blog/there-are-a-lot-of-freaking-data-quality-and-data-observability-vendors/"}, {"title": "The 2026 Open-Source Data Quality and Data Observability Landscape | DataKitchen", "url": "https://datakitchen.io/blog/the-2026-open-source-data-quality-and-data-observability-landscape/"}, {"title": "Data Observability Tool Comparison: Monte Carlo vs. great ...", "url": "https://www.castordoc.com/tool-comparison/data-observability-tool-comparison-monte-carlo-vs-great-expectations"}]

I've found some relevant information about these tools, but I haven't found a comprehensive

license comparison report yet. Let me search more specifically for license information.

Links: [{"title": "Great Expectations - YData Profiling", "url": "https://docs.profiling.ydata.ai/4.5/integrations/great_expectations/"}, {"title": "Great Expectations - YData-Synthetic", "url": "https://docs.synthetic.ydata.ai/1.2/integrations/gx_integration/"}, {"title": "GX Core: Open Source Data Quality Platform | Great Expectations ? Great Expectations", "url": "https://greatexpectations.io/"}, {"title": "How to validate your synthetic data quality", "url": "https://ydata.ai/resources/how-to-validate-the-quality-of-your-synthetic-data"}, {"title": "GitHub - Data-Centric-AI-Community/ydata-profiling: 1 Line of code data quality profiling & exploratory data analysis for Pandas and Spark DataFrames. · GitHub", "url": "https://github.com/ydataai/ydata-profiling"}, {"title": "YData · GitHub", "url": "https://github.com/ydataai"}, {"title": "3 Open Source Data Quality Tools You Can't Ignore (Compared) - Telmai", "url": "https://www.telma.ai/blog/open-source-data-quality-tools/"}, {"title": "How to Use Ydata-Profiling with Great Expectations V3 API | by Bvolodarskiy | Provectus Articles | Medium", "url": "https://medium.com/provectus-articles/how-to-use-ydata-profiling-with-great-expectations-v3-api-fc6c2d0035cd"}, {"title": "ydata-profiling · PyPI", "url": "https://pypi.org/project/ydata-profiling/"}, {"title": "ydata-profiling download | SourceForge.net", "url": "https://sourceforge.net/projects/ydata-profiling.mirror/"}]

Links: [{"title": "Open Source Licenses: Which One Should You P..."}]
[/external_agent_tool_result]

52. assistant (2026-07-02T06:11:33.145Z)

Research is decisive. Key findings (with licenses):

- **Soda Core / SodaCL** ? the closest thing to a "YAML config language that produces data-quality reports" ? **relicensed to Elastic License v2 (ELv2)**, which is NOT OSI-open-source and forbids offering it as a managed service. That directly violates your permissive-only / commercial-hosted constraint (ADR-005). ? Disqualified.
- **Great Expectations** (Apache-2.0 ?) and **Evidently** (Apache-2.0 ?) are license-safe but heavy and built for **tabular/dataframe + ML-monitoring**, not per-video pipeline-stage reports. Wrong shape, big dependency trees, telemetry.
- **ydata-profiling** (MIT ?) and **Frictionless Data** (MIT ?) are clean but also dataframe/data-package oriented.
- Thin MIT renderers exist (``yaml-to-markdown``, ``json-schema-for-humans``, ``python-markdown-generator``) ? useful primitives, not frameworks.
- The real reusable primitives you already depend on are all permissive: **pydantic** (MIT), **Jinja2** (BSD, already in the indexer), **PyYAML** (MIT, already in the collector), **jsonschema** (MIT).

So: no off-the-shelf framework fits cleanly **and** is license-safe. The right move is a **small in-house "report config language" library built on those permissive primitives** ? borrowing thin MIT renderers rather than reinventing them, and explicitly rejecting the heavy frameworks. Let me pressure-test that design + the two integrations with a Plan agent before I commit the final plan.

53. assistant (2026-07-02T06:11:33.145Z)

[external_agent_tool_call: Agent]
description: Design shared report-spec lib + integrations
[/external_agent_tool_call]

54. assistant (2026-07-02T06:11:33.146Z)

[external_agent_tool_result]
Write isn't available in this read-only planning context. I'll deliver the plan directly as my response, which is what the parent agent reads.

I explored both repos and verified every load-bearing fact in your brief against the actual code. The plan below is hardened, file-level concrete, and phased.

Rich Per-Video Observability Report ? Implementation Plan

0. Facts I verified in code (not assumed)

- **Indexer already has the deps**: `requirements.txt` has `pydantic>=2.7.1` and `jinja2>=3.1.4`. Collector `pyproject.toml` has `pydantic>=2.0.0`, `pyyaml>=6.0`, `tabulate` ? no Jinja2 (the shared lib brings it transitively, BSD, fine).
- **run_all discards context**: `backend/validators/base.py:42-56` builds a local `context={}`, FormatValidator fills `context['ffprobe\_meta']`, and only `List[ValidationResult]` is returned. Confirmed: to get probe metadata you must surface context or re-probe.
- **AI timing is dropped at the segmenter**: `backend/indexer/wasb\_hybrid.py:167` is literally `segments, \_ = provider.analyze\_video(...)` ? the `{upload\_time,process\_time,gen\_time}` dict is thrown away. Same in the twins.
- **Candidates ARE persisted**: `wasb\_hybrid.py:144` calls `db.add\_candidate\_segment(...)`; harvestable via `db.query\_candidate\_segments(detector=)`. WASB manifest at `output/wasb/manifest.json` (cwd-relative).
- **Both entrypoints confirmed**: CLI `backend/main.py:312-352`, API `process\_video @ main.py:88-134` (note the API has TWO return points: validation-fail at :104-109 and success at :128-134).
- **Collector report module to reuse**: `src/parsers/cvat/rally\_cvat.py` ? `build\_summary()` (JSON dict), `render\_validation\_md()`, `write\_review\_csv()`, `detect\_issues()` with `Issue{severity in blocker/high/medium/low, code, message, rally\_number}`, `ConversionResult{records,issues,fps,step,meta}`. `validate\_delivery` (`curate.py:562-617`) already writes `- **Collector Video model** `src/core/schemas/base.py:29-40`: `video_id, video_url, local_path, license_type, is_commercial, status, match_name, fps, resolution`. `probe_resolution_fps` + `DownloadResult{success,size_mb,width,height,fps,resolution_label,error}` in `src/core/downloader.py`.
- **License posture (ADR-005, `docs/DECISIONS.md:232`)**: hard permissive-only; pydantic/Jinja2/PyYAML/jsonschema all clean.
- **FAQ Q1-Q9** (`README.md:166+`) maps directly to footgun flags: Q5?PTS/keyframe & fps, Q7?AI provider/API key, Q2?relevance, Q8?yolo economics. These are the `faq\_ref` citation targets.

1. Library-vs-framework decision ? VALIDATED

Build the small in-house library. It is the right call, and not only for licensing:

1. **Shape mismatch is decisive**. GE/Evidently/ydata/Frictionless model *tabular datasets / data packages*. Your domain object is a **per-video run** = an ordered list of heterogeneous pipeline stages, each with a free-form metrics blob and a verdict, plus a **declarative footgun?flag rule layer**. None of those tools expresses that naturally; forcing it in becomes its own house of cards.
2. **License is an independent hard gate** (ADR-005) that already kills Soda (ELv2).
3. **House-of-cards containment**: ~6-8 tiny modules, a frozen pydantic envelope, pure renderers, no IO at import. Far less fragile than wrapping a heavy framework whose API churn you don't control.

Refinements to your stated stack:

- **Drop `jsonschema` from the runtime core** (defer to Phase 2+). pydantic v2 is your single source of truth and emits JSON Schema via `model\_json\_schema()`. Only add `jsonschema` if/when you validate *externally authored* spec files against a published schema.
- **Do NOT take `json-schema-for-humans`/`yaml-to-markdown` as runtime deps**. Hand-write the Jinja2 templates ? the FAQ-debuggability eval *requires* you to control exact wording. Treat those MIT tools as optional dev helpers for generating the schema doc only.
- **Keep the "config language" sub-Turing**. No `eval`, no DSL pa...
[/external_agent_tool_result]

55. assistant (2026-07-02T06:11:33.146Z)

The Plan agent verified every load-bearing fact against the actual code and hardened the design with several good refinements (drop `jsonschema` from the runtime core, hand-write the Jinja2

templates so I control exact wording for the FAQ-debuggability eval, keep the rule language sub-Turing/no-`eval`, tag-pin the cross-repo dependency, make `run\_all` backward-compatible, and defer the risky AI-timing-through-twin-segmenters work). Let me write the final plan.

56. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md
[/external_agent_tool_call]
```

57. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

58. assistant (2026-07-02T06:11:33.146Z)

Let me clean up a stray code-fence artifact at the end of the plan file.

59. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md
[/external_agent_tool_call]
```

60. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

61. assistant (2026-07-02T06:11:33.146Z)

The plan is written. Here's the summary for your review.

Plan: rich per-video observability reports (both tools + a shared library)

The core problem I found: both tools already *compute* rich signal but throw it away. In the indexer, `registry.run_all()` builds the `ffprobe_meta` context and discards it, `process_video()` returns only the final segments, and AI timing is literally dropped (`segments, _ = provider.analyze_video(...)`). The collector has a good *delivery-validation* report but nothing capturing the per-video *processing journey*. So this is mostly about **surfacing data that already exists**, plus encoding the known footguns as readable warnings.

Three decisions from your answers, baked in:

- **Always-on**, written **next to the video** by convention (`match.report.json` / `.report.md` / `.summary.md`), with a `report.output_dir` override.
- **Both tools** emit **technical + customer-friendly** reports (one shared envelope ? the two views physically can't disagree on facts).
- **A new private library repo** (`sports-obsreport`, proposed name) providing a declarative **"report config language"** (YAML spec + sub-Turing footgun?flag rules), consumed by both repos via a **tag-pinned** `git+ssh` dep.

Research result on "don't reinvent": the closest off-the-shelf DSL ? **Soda/SodaCL** ? is ELv2, not OSI-open-source, and forbids managed-service use ? disqualified by ADR-005. Great Expectations / Evidently (Apache-2.0) and ydata-profiling / Frictionless (MIT) are license-safe but tabular/ML-shaped ? wrong fit for "ordered pipeline stages + flag rules." So we build a small lib on permissive primitives the repos **already** depend on (pydantic MIT, Jinja2 BSD, PyYAML MIT) and **hand-write the templates** so I control exact wording for the debuggability eval.

****The debuggability bar is a real test, not a vibe:**** broken-run fixtures (no API key, skipped probe, motion-segmenter, collector overlap/zero-duration) ? a *****"fresh reader" subagent** given only README+FAQ+report** must self-diagnose and cite the FAQ section each flag points to ? locked as a golden regression so wording can't drift.

****Phasing**** keeps it from becoming a house of cards: ship the library and ****tag v0.1.0 first****; every consumer PR is small, independently shippable, and **`finally`-guarded** so a report write can never break a run; the risky AI-timing-through-twin-segmenters work is ****deferred**** (harvest from the DB instead).

Full detail (envelope shape, file-level hooks, flag list, risks, verification) is in the plan file.

62. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_call: ExitPlanMode]
input: {"allowedPrompts":[{"tool":"Bash","prompt":"run the pytest suite in either
repo"}, {"tool":"Bash","prompt":"install Python dependencies (pip / pip install -e / git+ssh
dep)"}, {"tool":"Bash","prompt":"git branch/add/commit/push and gh repo/PR
operations"}], "plan": "# Plan ? Rich per-video observability reports (indexer + collector +
shared lib)\n\n## Context (why)\n\nToday neither tool emits a per-video record of *what happened
during processing*. The\n\nindexer's pipeline computes a lot of signal (ffprobe metadata,
validator results, candidate\n\nwindows, AI-provider timing, WASB frame/window counts, stitch
skips) but **throws most of it\n\naway**: `registry.run_all()` builds a `context` dict with
`ffprobe_meta` and discards it\n\n(`backend/validators/base.py:42-56`);
`segmenter.process_video()` returns only the final\n\nsegment list; AI timing is dropped at
`wasb_hybrid.py:167` (`segments, _ = provider.analyze_video(...)`);\n\nmuch detail is only
`print()`-ed. The collector has a good report pattern for *delivery\n\nvalidation*
(`rally_cvat.py` `build_summary`/`render_validation_md`/`write_review_csv`) but\n\nnothing that
captures the **per-video processing journey** (acquire ? parse ? recompute time ?\n\ninvalidate ?
register ? export).\n\nWe want both tools to **always** emit a rich `"side output"` per
processed video that (a) captures\n\nevery stage + details to **catch inconsistencies**, (b) is
**diffable across runs and across the\n\ntwo tools** to evaluate output differences, and (c) lets
**someone who has only read the README +\n\nFAQ self-debug a bad run**. Plus a second **customer-
friendly** report that strips system internals\n\nand shows only video metadata + meta-level
highlights. Reports are written **next to the video by\n\nconvention**, with a configurable
override.\n\nPer the user, the schema is unified via a **new separate private repo providing a
reusable `"report\n\nconfig language."`
*** Research (below) confirms no off-the-shelf framework
fits both the\n\nlicensing constraint an...
```

63. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_result]
User has approved your plan. You can now start coding. Start with updating your todo list if
applicable
```

Your plan has been saved to: C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md
You can refer back to it if needed during implementation.

Approved Plan (edited by user):

Plan ? Rich per-video observability reports (indexer + collector + shared lib)

Context (why)

Today neither tool emits a per-video record of ***what happened during processing***. The indexer's pipeline computes a lot of signal (ffprobe metadata, validator results, candidate windows, AI-provider timing, WASB frame/window counts, stitch skips) but ****throws most of it away****: `registry.run_all()` builds a `context` dict with `ffprobe_meta` and discards it (`backend/validators/base.py:42-56`); `segmenter.process_video()` returns only the final segment list; AI timing is dropped at `wasb_hybrid.py:167` (`segments, _ = provider.analyze_video(...)`); much detail is only `print()`-ed. The collector has a good report pattern for ***delivery**

validation* (`rally\_cvat.py` `build\_summary`/`render\_validation\_md`/`write\_review\_csv`) but nothing that captures the **per-video processing journey** (acquire ? parse ? recompute time ? validate ? register ? export).

We want both tools to **always** emit a rich "side output" per processed video that (a) captures every stage + details to **catch inconsistencies**, (b) is **diffable** across runs and across the two tools to evaluate output differences, and (c) lets **someone who has only read the README** + FAQ self-debug a bad run. Plus a second **customer-friendly** report that strips system internals and shows only video metadata + meta-level highlights. Reports are written **next to the video** by convention, with a configurable override.

Per the user, the schema is unified via a **new separate private repo** providing a reusable "report config language." Research (below) confirms no off-the-shelf framework fits both the licensing constraint and the per-video-pipeline-stage shape ? so we build a small library on permissive primitives the repos already use.

Decisions locked (user + research)

- **Always-on emission.** Written next to the video path by default (`<video>.report.json` etc.); overridable via a `report.output\_dir` config key. (Not flag-gated.)
- **Both tools** produce **both** report types (technical + customer-friendly).
- **Shared schema** via a new private library repo that exposes a declarative "report config language," reusable by both tools today and others later.
- **Library, not a heavy framework.** Research findings:
 - Soda Core / SodaCL (the closest "YAML?data-quality-report" DSL) is **ELv2 ? not OSI** open-source, forbids managed-service use ? violates ADR-005. Disqualified.
 - Great Expectations (Apache-2.0) & Evidently (Apache-2.0) are license-safe but heavy and **tabular/ML-monitoring shaped** ? wrong fit for "an ordered list of heterogeneous pipeline stages + a footgun?flag rule layer." ydata-profiling (MIT) / Frictionless (MIT) are dataframe/data-package shaped ? also wrong fit.
 - Build on permissive primitives the repos **already** depend on: **pydantic** (MIT) (indexer `pydantic>=2.7.1`, collector `pydantic>=2.0.0`), **Jinja2** (BSD) (indexer dep), **PyYAML** (MIT) (collector dep). `jsonschema` deferred ? pydantic v2 `model\_json\_schema()` is the contract source. Thin MIT renderers (`yaml-to-markdown`, `json-schema-for-humans`) are **dev-only** helpers at most; templates are **hand-written** so we control exact wording for the FAQ-debuggability eval.

Architecture

New repo: `sports-obsreport` (package `obsreport`), private, MIT ? name TBD by owner

```
src/obsreport/  
  envelope.py      # pydantic v2 typed report envelope (single source of truth)  
  spec.py          # YAML "report config language" loader -> pydantic  
  rules.py         # restricted, sub-Turing footgun->flag engine (NO eval)  
  recorder.py      # RunRecorder: in-f...  
[/external_agent_tool_result]
```

64. assistant (2026-07-02T06:11:33.146Z)

Plan approved. Let me set up progress tracking and mirror the existing packaging conventions before scaffolding the new library. Loading the task tools and reading the collector's packaging/test setup to match its style.

65. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_call: ToolSearch]  
input: {"query":"select:TaskCreate,TaskUpdate,TaskList","max_results":5}
```

```
[/external_agent_tool_call]
```

66. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

67. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\pyproject.toml
[/external_agent_tool_call]
```

68. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_result]
1      [build-system]
2      requires = ["setuptools>=61.0.0"]
3      build-backend = "setuptools.build-meta"
4
5      [project]
6      name = "sports-data-collector"
7      version = "0.1.0"
8      description = "A collector and curator for sports video datasets with sport-specific
schemas and license checks."
9      readme = "README.md"
10     requires-python = ">=3.9"
11     dependencies = [
12         "pydantic>=2.0.0",
13         "pyyaml>=6.0",
14         "tabulate>=0.9.0"
15     ]
16
17     [project.optional-dependencies]
18     # Video acquisition (`fetch` / crawler). Also requires the external binaries
19     # ffmpeg (FFmpeg) and, for YouTube high-res, a JS runtime such as deno ?
20     # neither is pip-installable; see README "External requirements".
21     fetch = [
22         "yt-dlp>=2024.1.0"
23     ]
24     dev = [
25         "pytest>=7.0.0",
26         "pytest-cov>=4.0.0"
27     ]
28
[/external_agent_tool_result]
```

69. assistant (2026-07-02T06:11:33.146Z)

Loaded. Let me mark the implementation chapter, set up phase tracking, and verify the Python environment for the new library.

70. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"Observability reports impl","summary":"Building sports-obsreport shared lib,
then wiring into indexer + collector per approved plan"}
[/external_agent_tool_call]
```

71. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_result]
Chapter marked: "Observability reports impl". Continue your current work.
[/external_agent_tool_result]
```

72. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_call: Bash]
description: Check Python + key library availability
command: python --version; python -c "import pydantic, jinja2, yaml; print('pydantic',
pydantic.VERSION); print('jinja2', jinja2.__version__); print('yaml ok')"
```

73. assistant (2026-07-02T06:11:33.146Z)

```
[external_agent_tool_result]
Python 3.13.13
pydantic 2.13.4
jinja2 3.1.6
yaml ok
[/external_agent_tool_result]
```

74. assistant (2026-07-02T06:11:33.147Z)

Environment is ready (Python 3.13, pydantic 2.13, jinja2 3.1, yaml). Let me create the phase task list, then start building the library.

75. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskCreate]
description: Create the new private library repo sports-obsreport (package obsreport):
pyproject, version.py, envelope.py (pydantic v2 typed ReportEnvelope: Identity, VideoMeta,
Stage, Metric, Flag, HighlightSummary), render/json.py (deterministic), tests. Local git init;
do not push remote until user asks.
```

76. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Task #1 created successfully: C1: Scaffold sports-obsreport lib (envelope + JSON renderer +
tests)
[/external_agent_tool_result]
```

77. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskCreate]
description: spec.py (YAML report config language -> pydantic), rules.py (restricted sub-Turing
clause engine, no eval; ops
eq/ne/gt/ge/lt/le/in/contains/is_missing/is_present/eq_path/ne_path), verdict derivation,
faq_ref-required-for-error/warn enforcement. Truth-table tests.
```

78. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Task #2 created successfully: C2: Spec loader + rule engine + verdict
[/external_agent_tool_result]
```

79. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskCreate]
description: Hand-written Jinja2 templates for technical + customer markdown (customer =
video_meta + highlights + customer-safe flags + verdict only). io.py: next-to-video path
convention, report.output_dir override, remote/no-file fallback, atomic temp+rename, writes
never raise. Snapshot + path + injected-IOError tests.
```

80. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Task #3 created successfully: C3: Markdown renderers (technical + customer) + safe io
[/external_agent_tool_result]
```

81. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskCreate]
description: recorder.py: in-flight stage context manager (auto-time, capture exceptions without
re-raising), finalize() runs rules, write() emits all files; post-hoc harvest helper signatures.
Generate + check in JSON Schema from pydantic. Tag v0.1.0 once green.
[/external_agent_tool_call]
```

82. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Task #4 created successfully: C4: RunRecorder + harvest scaffolding; tag v0.1.0
[/external_agent_tool_result]
```

83. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskCreate]
description: A1 add dep + run_all_with_context (run_all unchanged). A2 pure harvest_indexer_run
(mocked DB). A3 wire CLI (finally-guarded) + footgun flags + broken-run tests. A4 wire API (both
exits) + report paths in response. A5 (later) thread AI timing via callback through twins.
[/external_agent_tool_call]
```

84. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Task #7 created successfully: D: README/FAQ-reader debuggability eval + golden regression
[/external_agent_tool_result]
```

85. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Task #5 created successfully: A: Indexer integration (run_all_with_context, harvest, CLI+API
wiring, flags)
[/external_agent_tool_result]
```

86. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskCreate]
description: B1 add dep + src/reporting/envelope_adapter.py (Issue->Flag, stage builders)
reusing build_summary/detect_issues. B2 wire validate_delivery (lowest risk). B3 wire
import_local + convert_cvat. B4 wire fetch (+ remote-no-file) + optional export delivery
summary.
[/external_agent_tool_call]
```

87. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Task #6 created successfully: B: Collector integration (envelope_adapter + wire-ins + delivery
summary)
[/external_agent_tool_result]
```

88. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskCreate]
description: Broken-run fixtures (no-API-key, skipped-probe, motion, re-ingest; collector
overlap/zero-dur/guessed-fps). Fresh-reader subagent given only README+FAQ+report must diagnose
+ cite faq_ref; iterate wording until pass. Lock golden {flag codes, faq_refs} pytest
assertions; cross-tool/cross-run diff test.
[/external_agent_tool_call]
```

89. assistant (2026-07-02T06:11:33.147Z)

Starting **C1** ? scaffolding the shared `obsreport` library. Let me build the package skeleton, the typed envelope, the deterministic JSON renderer, and tests.

90. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"1","status":"in_progress"}
[/external_agent_tool_call]
```

91. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
Updated task #1 status
[/external_agent_tool_result]
```

92. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\pyproject.toml
[/external_agent_tool_call]
```

93. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\pyproject.toml (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

94. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\LICENSE
[/external_agent_tool_call]
```

95. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\LICENSE (file state is
current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

96. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\.gitignore
[/external_agent_tool_call]
```

97. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\.gitignore (file state is
current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

98. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\version.py
[/external_agent_tool_call]
```

99. assistant (2026-07-02T06:11:33.147Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\version.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

100. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\envelope.py
[/external_agent_tool_call]
```

101. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\envelope.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

102. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\__init__.py
[/external_agent_tool_call]
```

103. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\__init__.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

104. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\json_render.py
[/external_agent_tool_call]
```

105. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\json_render.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

106. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py
[/external_agent_tool_call]
```

107. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

108. assistant (2026-07-02T06:11:33.148Z)

Now the C1 tests ? envelope round-trip, verdict/lint logic, and deterministic JSON:

109. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_envelope.py
[/external_agent_tool_call]
```

110. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_envelope.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

111. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_json.py
[/external_agent_tool_call]
```

112. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_json.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

113. assistant (2026-07-02T06:11:33.148Z)

Now a brief README for the library, then run the C1 tests:

114. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\README.md
[/external_agent_tool_call]
```

115. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\README.md (file state is
current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

116. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_call: Bash]
description: Run obsreport C1 test suite
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pytest -q 2>&1 | tail -30
[/external_agent_tool_call]
```

117. assistant (2026-07-02T06:11:33.148Z)

```
[external_agent_tool_result]
..... [100%]
15 passed in 0.22s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

118. assistant (2026-07-02T06:11:33.149Z)

C1 green ? 15 passed. Let me initialize the new repo's git history and checkpoint C1 locally (I'll hold remote/GitHub creation for your go-ahead).

119. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_call: Bash]
description: Init obsreport git repo and commit C1
command: cd "C:/Users/avidu/Projects/sports-obsreport" && git init -b main 2>&1 | tail -1 && git
add -A && git -c user.name="Avi Dullu" -c user.email="avi.dullu@gmail.com" commit -q -m "C1:
scaffold obsreport ? typed envelope + deterministic JSON renderer
```

Single-source-of-truth ReportEnvelope (Identity/VideoMeta/Stage/Metric/Flag/HighlightSummary), pure verdict + lint helpers, sorted-key JSON for clean cross-run diffs. MIT, built only on pydantic. 15 tests green.

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git log --oneline -1
[/external_agent_tool_call]
```

120. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_result]
Initialized empty Git repository in C:/Users/avidu/Projects/sports-obsreport/.git/
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'LICENSE', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'pyproject.toml', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'src/obsreport/__init__.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'src/obsreport/envelope.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'src/obsreport/render/__init__.py', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'src/obsreport/render/json_renderer.py', LF will be replaced by
CRLF the next time Git touches it
warning: in the working copy of 'src/obsreport/version.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_envelope.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_render_json.py', LF will be replaced by CRLF the
next time Git touches it
687b52e C1: scaffold obsreport ? typed envelope + deterministic JSON renderer
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

121. assistant (2026-07-02T06:11:33.149Z)

C1 committed. Now **C2** ? the declarative "report config language" (YAML spec) + the restricted, no-`eval` rule engine that turns known footguns into FAQ-citing flags. Marking C1 done, C2 in progress.

122. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"1","status":"completed"}
[/external_agent_tool_call]
```

123. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_result]
Updated task #1 status
[/external_agent_tool_result]
```

124. assistant (2026-07-02T06:11:33.149Z)


```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"2","status":"in_progress"}
[/external_agent_tool_call]
```

125. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_result]
Updated task #2 status
[/external_agent_tool_result]
```

126. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py
[/external_agent_tool_call]
```

127. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

128. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\rules.py
[/external_agent_tool_call]
```

129. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\rules.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

130. assistant (2026-07-02T06:11:33.149Z)

Now export the new symbols and add the shipped spec files:

131. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py
[/external_agent_tool_call]
```

132. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

133. assistant (2026-07-02T06:11:33.149Z)

Tool-specific specs reference that tool's FAQ, so they belong ****with each tool**** (added in phases A/B). The library ships only an illustrative example spec that doubles as a test fixture:

134. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\specs\example.yaml
[/external_agent_tool_call]
```

135. assistant (2026-07-02T06:11:33.149Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\specs\example.yaml (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

136. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_spec.py
[/external_agent_tool_call]
```

137. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_spec.py (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

138. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_rules.py
[/external_agent_tool_call]
```

139. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_rules.py (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

140. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: Bash]
description: Run obsreport suite after C2
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pytest -q 2>&1 | tail -25
[/external_agent_tool_call]
```

141. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
..... [100%]
49 passed in 0.25s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

142. assistant (2026-07-02T06:11:33.150Z)

C2 green ? 49 passed. Committing, then moving to **C3** (renderers + safe IO).

143. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: Bash]
description: Commit C2
command: cd "C:/Users/avidu/Projects/sports-obsreport" && git add -A && git -c user.name="Avi
Dullu" -c user.email="avi.dullu@gmail.com" commit -q -m "C2: report config language ? spec
loader + restricted rule engine"
```

YAML spec (sections + rules) -> pydantic, fail-fast (actionable rules must cite a
FAQ; rules need >=1 clause). Sub-Turing clause engine (no eval): dotted-path
lookups into a re-keyed rule context; missing path never fires except is_missing;

evidence snapshot on each flag. specs/example.yaml ships silent_provider_noop + fps_fallback_used. 49 tests green.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git log --oneline -2
[/external_agent_tool_call]

144. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
warning: in the working copy of 'src/obsreport/__init__.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'specs/example.yaml', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'src/obsreport/rules.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/obsreport/spec.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_rules.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'tests/test_spec.py', LF will be replaced by CRLF the next time
Git touches it
918a728 C2: report config language ? spec loader + restricted rule engine
687b52e C1: scaffold obsreport ? typed envelope + deterministic JSON renderer
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

145. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"2","status":"completed"}
[/external_agent_tool_call]
```

146. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
Updated task #2 status
[/external_agent_tool_result]
```

147. assistant (2026-07-02T06:11:33.150Z)

C2 committed. Now **C3** ? the spec-driven Markdown renderers (technical + customer views from the *same* envelope) and the never-raising IO layer with the next-to-video path convention.

148. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"3","status":"in_progress"}
[/external_agent_tool_call]
```

149. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
Updated task #3 status
[/external_agent_tool_result]
```

150. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

151. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-
obsreport\src\obsreport\render\markdown.py (file state is current in your context ? no need to
Read it back)
[/external_agent_tool_result]
```

152. assistant (2026-07-02T06:11:33.150Z)

Now the never-raising IO layer with the path convention:

153. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\io.py
[/external_agent_tool_call]
```

154. assistant (2026-07-02T06:11:33.150Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\io.py (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

155. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py
[/external_agent_tool_call]
```

156. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

157. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py
[/external_agent_tool_call]
```

158. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\__init__.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

159. assistant (2026-07-02T06:11:33.151Z)

Now the C3 tests ? markdown rendering (audience separation) and safe IO (path convention + never-raises):

160. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py
[/external_agent_tool_call]
```

161. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_result]
```

File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

162. assistant (2026-07-02T06:11:33.151Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_io.py
[/external_agent_tool_call]

163. assistant (2026-07-02T06:11:33.151Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_io.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

164. assistant (2026-07-02T06:11:33.151Z)

[external_agent_tool_call: Bash]
description: Run obsreport suite after C3
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pytest -q 2>&1 | tail -30
[/external_agent_tool_call]

165. assistant (2026-07-02T06:11:33.151Z)

[external_agent_tool_result]
..... [100%]
61 passed in 0.30s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

166. assistant (2026-07-02T06:11:33.151Z)

61 passed. Since report readability is central to this task, let me eyeball an actual rendered technical + customer report before committing:

167. assistant (2026-07-02T06:11:33.151Z)

[external_agent_tool_call: Bash]
description: Render sample technical + customer reports to eyeball
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -c "
import sys; sys.path.insert(0,'src')
from obsreport import *
from obsreport.rules import run_spec
env = ReportEnvelope(
 identity=Identity(tool='badminton-highlight-indexer', tool_version='0.1', video_id='9f3a...',
video_id_kind='file_md5', video_path='/videos/Adarsh and Vasanth.mp4', run_id='run-001',
generated_at='2026-06-07T10:00:00Z'),
 video_meta=VideoMeta(duration_s=676.0, fps=59.94, fps_source='probed', width=1920,
height=1080, container='mp4', size_mb=5500.0),
 stages=[
 Stage(name='validation',
status='ok').add_metric('checks_passed',6).add_metric('checks_total',6),
 Stage(name='segmentation', status='ok', details={'segmenter_requested':'wasb_hybrid','segmenter_actual':'wasb_hybrid'}).add_metric('candidate_windows',64).add_metric('segment_count',0,audience='both'),
 Stage(name='ai_handoff', status='warn',
details={'api_key_present':False,'provider':'gemini'}).add_metric('clips_sent',0),
],
 highlights=HighlightSummary(segment_count=0, total_highlight_s=0.0, coverage_pct=0.0,
notes=['No highlights were produced for this video.']),
)

```
spec = load_spec('specs/example.yaml')
run_spec(env, spec)
print('===== TECHNICAL =====')
print(render_technical(env, spec))
print('===== CUSTOMER =====')
print(render_customer(env, spec))
"
```

[/external_agent_tool_call]

168. assistant (2026-07-02T06:11:33.151Z)

[external_agent_tool_result]
===== TECHNICAL =====

Example Pipeline ? Video Report

Run

```
| Field | Value |
| --- | --- |
| Tool | badminton-highlight-indexer 0.1 |
| Video id | 9f3a... (file_md5) |
| Video path | /videos/Adarsh and Vasanth.mp4 |
| Run id | run-001 |
| Generated | 2026-06-07T10:00:00Z |
| Schema | 1.0 |
```

Video

```
| Field | Value |
| --- | --- |
| Duration | 11m16s |
| Resolution | 1920x1080 |
| FPS | 59.94 fps (probed) |
| Container | mp4 |
| Size | 5500 MB |
```

Processing stages

```
**validation** ? `ok`
- checks_passed: 6
- checks_total: 6

**segmentation** ? `ok`
- candidate_windows: 64
- segment_count: 0
- _segmenter_requested_: wasb_hybrid
- _segmenter_actual_: wasb_hybrid

**ai_handoff** ? `warn`
- clips_sent: 0
- _api_key_present_: False
- _provider_: gemini
```

Findings

- [ERROR] **silent_provider_noop** ? 0 segments AND no API key ? this is almost certainly a silent provider no-op, not 'the video has no rallies'. Set the provider API key and re-run.
? see **README#Q7**
? evidence: stages.ai_handoff.details.api_key_present=False,
stages.segmentation.metrics.segment_count=0

Verdict

****FAIL****

===== CUSTOMER =====

Example Pipeline ? Video Report (summary)

Run

```
| Field | Value |
| --- | --- |
| Video | Adarsh and Vasanth.mp4 |
| Report generated | 2026-06-07T10:00:00Z |
```

Video

```
| Field | Value |
| --- | --- |
| Duration | 11m16s |
| Resolution | 1920x1080 |
| Frame rate | 59.94 fps |
```

Highlights

- ****0**** highlights
- ****0.0s**** of play
- ****0%**** of the video is highlights
- No highlights were produced for this video.

Verdict

****FAIL****

Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

169. assistant (2026-07-02T06:11:33.151Z)

Both views render cleanly. One real UX gap: the ****customer**** report shows a bare ``FAIL`` with no explanation (because a customer shouldn't see raw verdicts, and the example spec lacked a customer "findings" section). That's exactly the kind of thing the readability bar should catch. Let me make the verdict audience-aware (friendly for customers) and add a customer-facing findings section to the example spec.

170. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

171. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

172. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

173. assistant (2026-07-02T06:11:33.151Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

174. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\specs\example.yaml
[/external_agent_tool_call]
```

175. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\specs\example.yaml has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

176. assistant (2026-07-02T06:11:33.152Z)

Let me add a test locking the friendlier customer verdict + customer findings surfacing, then re-run:

177. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py
[/external_agent_tool_call]
```

178. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

179. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_call: Bash]
description: Re-run obsreport suite after C3 refinements
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pytest -q 2>&1 | tail -15
[/external_agent_tool_call]
```

180. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_result]
..... [100%]
62 passed in 0.30s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

181. assistant (2026-07-02T06:11:33.152Z)

62 passed. Committing C3.

182. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_call: Bash]
description: Commit C3
command: cd "C:/Users/avidu/Projects/sports-obsreport" && git add -A && git -c user.name="Avi
Dullu" -c user.email="avi.dullu@gmail.com" commit -q -m "C3: spec-driven markdown renderers
(technical + customer) + safe IO
```


Both views render from the SAME envelope. Technical = stages + all flags
(code/evidence/faq_ref) + fps_source; customer = video meta + meta highlights +
plain-language actionable flags only, friendly verdict (never a raw FAIL).
io.py: next-to-video path convention (<stem>.report.json/.report.md/.summary.md),
output_dir override, remote/no-file fallback, atomic temp+replace, writes NEVER
raise (collected into WriteResult). 62 tests green.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git log --oneline -3
[/external_agent_tool_call]

183. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_result]
warning: in the working copy of 'specs/example.yaml', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'src/obsreport/__init__.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'src/obsreport/io.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'src/obsreport/render/markdown.py', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'tests/test_io.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'tests/test_render_markdown.py', LF will be replaced by CRLF the
next time Git touches it
911adf7 C3: spec-driven markdown renderers (technical + customer) + safe IO
918a728 C2: report config language ? spec loader + restricted rule engine
687b52e C1: scaffold obsreport ? typed envelope + deterministic JSON renderer
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

184. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"3","status":"completed"}
[/external_agent_tool_call]
```

185. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_result]
Updated task #3 status
[/external_agent_tool_result]
```

186. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"4","status":"in_progress"}
[/external_agent_tool_call]
```

187. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_result]
Updated task #4 status
[/external_agent_tool_result]
```

188. assistant (2026-07-02T06:11:33.152Z)

C4 ? the `RunRecorder` (in-flight capture with injected clock for deterministic tests,
exceptions captured not raised), a stable-diff helper for clean cross-run comparison, and a
checked-in JSON Schema generated from the model.

189. assistant (2026-07-02T06:11:33.152Z)

```
[external_agent_tool_call: Write]
```

file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\recorder.py
[/external_agent_tool_call]

190. assistant (2026-07-02T06:11:33.152Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\recorder.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

191. assistant (2026-07-02T06:11:33.152Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\diff.py
[/external_agent_tool_call]

192. assistant (2026-07-02T06:11:33.152Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\diff.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

193. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport__init__.py
[/external_agent_tool_call]

194. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport__init__.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

195. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport__init__.py
[/external_agent_tool_call]

196. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport__init__.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

197. assistant (2026-07-02T06:11:33.153Z)

Now the schema-generation script and the C4 tests (recorder, diff, schema currency):

198. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\scripts\gen_schema.py
[/external_agent_tool_call]

199. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\scripts\gen_schema.py

(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

200. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_call: Bash]
description: Generate the checked-in JSON Schema
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python scripts/gen_schema.py && head -20 schema/report_envelope.schema.json
[/external_agent_tool_call]

201. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_result]
wrote C:\Users\avidu\Projects\sports-obsreport\schema\report_envelope.schema.json
{
 "\$defs": {
 "Flag": {
 "description": "A derived warning/error \u2014 the debugging intelligence of the report.\n\n`faq\_ref` points at the README/FAQ section a reader should open to self-debug\n(e.g. ``\"README#Q7\"``). It is REQUIRED for error/warn severities; this is\nenforced by `ReportEnvelope.lint`, not by the constructor, so harvest code can\nbuild a flag incrementally before deciding its citation.",
 "properties": {
 "code": {
 "title": "Code",
 "type": "string"
 },
 "evidence": {
 "additionalProperties": true,
 "title": "Evidence",
 "type": "object"
 },
 "faq_ref": {
 "anyOf": [
 {
 "type": "string"
 },
 {
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

202. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_recorder.py
[/external_agent_tool_call]

203. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_recorder.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

204. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_diff.py
[/external_agent_tool_call]

205. assistant (2026-07-02T06:11:33.153Z)

[external_agent_tool_result]

File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_diff.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

206. assistant (2026-07-02T06:11:33.154Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_schema.py
[/external_agent_tool_call]

207. assistant (2026-07-02T06:11:33.154Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-obsreport\tests\test_schema.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

208. assistant (2026-07-02T06:11:33.154Z)

[external_agent_tool_call: Bash]
description: Run obsreport full suite after C4
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pytest -q 2>&1 | tail -20
[/external_agent_tool_call]

209. assistant (2026-07-02T06:11:33.154Z)

[external_agent_tool_result]
..... [100%]
71 passed in 0.37s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

210. assistant (2026-07-02T06:11:33.154Z)

71 passed, schema guarded against drift. Let me commit C4, then verify the package actually installs and imports cleanly from an installed context (the cross-repo integration depends on this) before tagging.

211. assistant (2026-07-02T06:11:33.154Z)

[external_agent_tool_call: Bash]
description: Commit C4, editable-install obsreport, import smoke
command: cd "C:/Users/avidu/Projects/sports-obsreport" && git add -A && git -c user.name="Avi Dullu" -c user.email="avi.dullu@gmail.com" commit -q -m "C4: RunRecorder + stable-diff + JSON Schema contract

RunRecorder: in-flight stage capture (auto-timed, injectable clock; exceptions captured into the stage, never re-raised), set_video_meta/set_highlights/add_flag, idempotent finalize() (runs spec rules + verdict) and never-raising write(). diff.stable_json strips volatile fields for clean cross-run/cross-tool diffs. Checked-in schema/report_envelope.schema.json generated from the model + a drift-guard test. 71 tests green.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git log --oneline -4 && echo "--- editable install + import smoke ---" && python -m pip install -e . -q 2>&1 | tail -5 && python -c "import obsreport; print('obsreport', obsreport.__version__, 'schema', obsreport.SCHEMA_VERSION); print('exports ok:', all(hasattr(obsreport,n) for n in ['RunRecorder', 'write_report', 'run_spec', 'render_technical', 'render_customer', 'load_spec', 'stable_json']))"
[/external_agent_tool_call]

212. assistant (2026-07-02T06:11:33.154Z)

[external_agent_tool_result: error]
Exit code 1

```
warning: in the working copy of 'src/obsreport/__init__.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'scripts/gen_schema.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/obsreport/diff.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/obsreport/recorder.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'tests/test_diff.py', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'tests/test_recorder.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_schema.py', LF will be replaced by CRLF the next
time Git touches it
d3e2a51 C4: RunRecorder + stable-diff + JSON Schema contract
911adf7 C3: spec-driven markdown renderers (technical + customer) + safe IO
918a728 C2: report config language ? spec loader + restricted rule engine
687b52e C1: scaffold obsreport ? typed envelope + deterministic JSON renderer
--- editable install + import smoke ---
  File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\pip\_vendor\pyproject_hooks\_impl.py", line 402, in _call_hook
    raise BackendUnavailable(
    ...<4 lines>...
)
pip._vendor.pyproject_hooks._impl.BackendUnavailable: Cannot import 'setuptools.build-meta'
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import obsreport; print('obsreport', obsreport.__version__, 'schema',
obsreport.SCHEMA_VERSION); print('exports ok:', all(hasattr(obsreport,n) for n in ['RunRecorder'
,'write_report','run_spec','render_technical','render_customer','load_spec','stable_json']))
    ^^^^^^^^^^^^^^^^^^^^^^^^^
ModuleNotFoundError: No module named 'obsreport'
[/external_agent_tool_result]
```

213. assistant (2026-07-02T06:11:33.154Z)

C4 committed. The editable install hit a build-backend error (likely pip build isolation can't fetch setuptools offline). Let me diagnose and use `--no-build-isolation`:

214. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_call: Bash]
description: Check setuptools/pip and retry editable install without build isolation
command: python -c "import setuptools; print('setuptools', setuptools.__version__)" 2>&1; python
-c "import pip; print('pip', pip.__version__)" 2>&1; echo "--- retry editable, no build
isolation ---"; cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pip install -e .
--no-build-isolation -q 2>&1 | tail -8
[/external_agent_tool_call]
```

215. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_result]
setuptools 81.0.0
pip 26.0.1
--- retry editable, no build isolation ---
  File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\pip\_vendor\pyproject_hooks\_impl.py", line 180, in _supported_features
    return self._call_hook("_supported_features", {})
    ~~~~~
    ~~~~~
  File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\pip\_vendor\pyproject_hooks\_impl.py", line 402, in _call_hook
    raise BackendUnavailable(
    ...<4 lines>...
)
```

```
pip._vendor.pyproject_hooks._impl.BackendUnavailable: Cannot import 'setuptools.build-meta'  
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer  
[/external_agent_tool_result]
```

216. assistant (2026-07-02T06:11:33.154Z)

Found it ? my `build-backend` had a hyphen (`setuptools.build-meta`) instead of the correct underscore (`setuptools.build\_meta`). (Notably the collector's `pyproject.toml` has the same typo ? a latent install bug there worth flagging.) Let me fix and retry:

217. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_call: Edit]  
file: C:\Users\avidu\Projects\sports-obsreport\pyproject.toml  
[/external_agent_tool_call]
```

218. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_result]  
The file C:\Users\avidu\Projects\sports-obsreport\pyproject.toml has been updated successfully.  
(file state is current in your context ? no need to Read it back)  
[/external_agent_tool_result]
```

219. assistant (2026-07-02T06:11:33.154Z)

```
[external_agent_tool_call: Bash]  
description: Editable install obsreport and import smoke  
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pip install -e . --no-build-  
isolation -q 2>&1 | tail -5 && python -c "import obsreport; print('obsreport',  
obsreport.__version__, 'schema', obsreport.SCHEMA_VERSION); print('exports ok:',  
all(hasattr(obsreport,n) for n in ['RunRecorder','write_report','run_spec','render_technical','r  
ender_customer','load_spec','stable_json']))"  
[/external_agent_tool_call]
```

220. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_result]  
[notice] A new release of pip is available: 26.0.1 -> 26.1.2  
[notice] To update, run: python.exe -m pip install --upgrade pip  
obsreport 0.1.0 schema 1.0  
exports ok: True  
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer  
[/external_agent_tool_result]
```

221. assistant (2026-07-02T06:11:33.155Z)

Editable install + import both work now. Let me confirm the suite still passes, then commit the fix and tag v0.1.0.

222. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_call: Bash]  
description: Re-test, commit build fix, tag v0.1.0  
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python -m pytest -q 2>&1 | tail -5 &&  
git add -A && git -c user.name="Avi Dullu" -c user.email="avi.dullu@gmail.com" commit -q -m  
"fix: correct setuptools build-backend (underscore) so editable/pip install works  
  
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git tag -a v0.1.0 -m "obsreport  
v0.1.0 ? envelope, config language, renderers, recorder, schema" && git tag --list && git log  
--oneline -1  
[/external_agent_tool_call]
```

223. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_result: error]
Exit code 128
..... [100%]
71 passed in 0.37s
warning: in the working copy of 'pyproject.toml', LF will be replaced by CRLF the next time Git
touches it
Committer identity unknown

*** Please tell me who you are.

Run

git config --global user.email "you@example.com"
git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'avidu@AVIS-MSI.(none)')
[/external_agent_tool_result]
```

224. assistant (2026-07-02T06:11:33.155Z)

The commit landed but the annotated tag needs a committer identity. Let me set the repo-local git identity (so future commits/tags are clean) and create the tag:

225. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_call: Bash]
description: Set repo git identity and create v0.1.0 tag
command: cd "C:/Users/avidu/Projects/sports-obsreport" && git config user.name "Avi Dullu" &&
git config user.email "avi.dullu@gmail.com" && git log --oneline -1 && git tag -a v0.1.0 -m
"obsreport v0.1.0 ? envelope, config language, renderers, recorder, schema" && echo "tags:" &&
git tag --list
[/external_agent_tool_call]
```

226. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_result]
7c40654 fix: correct setuptools build-backend (underscore) so editable/pip install works
tags:
v0.1.0
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

227. assistant (2026-07-02T06:11:33.155Z)

Phase **C is complete**: the `obsreport` library is built (71 tests green), editable-installed, and tagged **v0.1.0** locally. This is a meaningful checkpoint (a whole new repo that isn't on GitHub yet) ? let me capture it in memory before starting integration, so a restart can resume cleanly.

228. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_call: Edit]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

229. assistant (2026-07-02T06:11:33.155Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
```

context ? no need to Read it back)
[/external_agent_tool_result]

230. assistant (2026-07-02T06:11:33.155Z)

Phase C done. Marking C4 complete and starting **Phase A (indexer)**. First, branch off master and confirm the 209-test baseline is green before touching anything.

231. assistant (2026-07-02T06:11:33.155Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"4","status":"completed"}
[/external_agent_tool_call]

232. assistant (2026-07-02T06:11:33.155Z)

[external_agent_tool_result]
Updated task #4 status
[/external_agent_tool_result]

233. assistant (2026-07-02T06:11:33.155Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"5","status":"in_progress"}
[/external_agent_tool_call]

234. assistant (2026-07-02T06:11:33.155Z)

[external_agent_tool_result]
Updated task #5 status
[/external_agent_tool_result]

235. assistant (2026-07-02T06:11:33.155Z)

[external_agent_tool_call: Bash]
description: Branch indexer + confirm test baseline
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout -b feat/observability-reports 2>&1 | tail -1 && python -m pytest -q 2>&1 | tail -6
[/external_agent_tool_call]

236. assistant (2026-07-02T06:11:33.156Z)

[external_agent_tool_result]
Switched to a new branch 'feat/observability-reports'
..... [34%]
..... [68%]
..... [100%]
209 passed in 15.39s
[/external_agent_tool_result]

237. assistant (2026-07-02T06:11:33.156Z)

Baseline green (209). **A1** ? add the backward-compatible `run\_all\_with\_context` so the report can capture the `ffprobe\_meta` that `run\_all` currently discards:

238. assistant (2026-07-02T06:11:33.156Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\validators\base.py
[/external_agent_tool_call]

239. assistant (2026-07-02T06:11:33.156Z)


```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\validators\base.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

240. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_validators_base.py
[/external_agent_tool_call]
```

241. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\tests\test_validators_base.py (file state is current in your context ? no need to Read
it back)
[/external_agent_tool_result]
```

242. assistant (2026-07-02T06:11:33.156Z)

A1 looks good. Now I need exact shapes before writing the harvest layer ? let me read the
ffprobe-metadata producer, the candidate-query API, and the segment/DB shapes so the harvest is
accurate, not guessed.

243. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\validators\format.py
[/external_agent_tool_call]
```

244. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_result]
1      import subprocess
2      import json
3      import os
4      from backend.validators.base import VideoValidator, ValidationResult
5      from typing import Dict, Any
6
7      class FormatValidator(VideoValidator):
8          @property
9          def name(self) -> str:
10              return "format_check"
11
12          @property
13          def description(self) -> str:
14              return "Verifies the video is a valid, uncorrupted MP4 container and extracts
metadata."
15
16          def validate(self, video_path: str, config: Dict[str, Any], context: Dict[str, Any])
-> ValidationResult:
17              if not os.path.exists(video_path):
18                  return ValidationResult(
19                      validator_name=self.name,
20                      passed=False,
21                      message=f"File not found: {video_path}"
22                  )
23
24              cmd = [
25                  "ffprobe",
26                  "-v", "error",
27                  "-select_streams", "v:0",
28                  "-show_format",
```

```

29         "-show_streams",
30         "-of", "json",
31         video_path
32     ]
33
34     try:
35         result = subprocess.run(cmd, capture_output=True, text=True, check=True)
36         info = json.loads(result.stdout)
37
38         if not info.get("streams") or not info.get("format"):
39             return ValidationResult(
40                 validator_name=self.name,
41                 passed=False,
42                 message="Failed to retrieve valid video streams from the file."
43             )
44
45         # Store the metadata in context for subsequent validators to utilize
46         context["ffprobe_meta"] = info
47
48         stream = info["streams"][0]
49         fmt = info["format"]
50
51         details = {
52             "format_name": fmt.get("format_name"),
53             "duration": float(fmt.get("duration", 0)),
54             "size_bytes": int(fmt.get("size", 0)),
55             "codec_name": stream.get("codec_name"),
56         }
57
58         # Check if container name includes mp4
59         if "mp4" not in fmt.get("format_name", "").lower():
60             return ValidationResult(
61                 validator_name=self.name,
62                 passed=False,
63                 message=f"Invalid format: Found '{fmt.get('format_name')}', but MP4
is required.",
64                 details=details
65             )
66
67         return ValidationResult(
68             validator_name=self.name,
69             passed=True,
70             message="Video format is valid MP4 container.",
71             details=details
72         )
73
74     except subprocess.CalledProcessError as e:
75         return ValidationResult(
76             validator_name=self.name,
77             passed=False,
78             message="Video stream is corrupted or unreadable by ffprobe.",
79             details={"error": e.stderr}
80         )
81     except Exception as e:
82         return ValidationResult(
83             validator_name=self.name,
84             passed=False,
85             message=f"Unexpected error running ffprobe: {str(e)}"
86         )
87
[/external_agent_tool_result]

```

245. assistant (2026-07-02T06:11:33.156Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\validators\resolution_fps.py
[/external_agent_tool_call]

246. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_result]
1      from backend.validators.base import VideoValidator, ValidationResult
2      from typing import Dict, Any
3
4      class ResolutionFPSValidator(VideoValidator):
5          @property
6          def name(self) -> str:
7              return "resolution_fps_check"
8
9          @property
10         def description(self) -> str:
11             return "Checks if the video meets minimum resolution (e.g. 720p) and frame rate  
(e.g. 30fps) requirements."
12
13         def validate(self, video_path: str, config: Dict[str, Any], context: Dict[str, Any])
-> ValidationResult:
14             meta = context.get("ffprobe_meta")
15             if not meta or not meta.get("streams"):
16                 return ValidationResult(
17                     validator_name=self.name,
18                     passed=False,
19                     message="Video metadata is missing. Run format_check validator first."
20                 )
21
22             stream = meta["streams"][0]
23             width = int(stream.get("width", 0))
24             height = int(stream.get("height", 0))
25
26             # Parse frame rate (e.g. "30/1" or "30000/1001")
27             fps_str = stream.get("r_frame_rate", "0/1")
28             try:
29                 num, den = map(int, fps_str.split("/"))
30                 fps = num / den if den != 0 else 0
31             except ValueError:
32                 fps = 0.0
33
34             details = {
35                 "width": width,
36                 "height": height,
37                 "fps": round(fps, 2)
38             }
39
40             # Check bounds from config
41             val_cfg = config.get("validation", {})
42             min_w = val_cfg.get("min_resolution_width", 1280)
43             min_h = val_cfg.get("min_resolution_height", 720)
44             min_fps = val_cfg.get("min_fps", 29.9)
45
46             if width < min_w or height < min_h:
47                 return ValidationResult(
48                     validator_name=self.name,
49                     passed=False,
50                     message=f"Low resolution: Found {width}x{height}, but minimum is  
{min_w}x{min_h}.",
51                     details=details
52                 )
53
54             if fps < min_fps:
55                 return ValidationResult(
56                     validator_name=self.name,
```

```

57             passed=False,
58             message=f"Low frame rate: Found {round(fps, 2)} FPS, but minimum is
{min_fps} FPS.",
59             details=details
60         )
61
62         return ValidationResult(
63             validator_name=self.name,
64             passed=True,
65             message="Video resolution and frame rate meet minimum standards.",
66             details=details
67         )
68
[/external_agent_tool_result]

```

247. assistant (2026-07-02T06:11:33.156Z)

`ffprobe\_meta` is the full ffprobe JSON (`streams[0]` ? width/height/r_frame_rate/codecs_name; `format` ? format_name/duration/size). Now the DB query API + segment shapes:

248. assistant (2026-07-02T06:11:33.156Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\database.py
[/external_agent_tool_call]

```

249. assistant (2026-07-02T06:11:33.156Z)

```

[external_agent_tool_result]
1     import sqlite3
2     import json
3     import os
4     import logging
5     from typing import List, Dict, Any, Optional
6
7     logger = logging.getLogger(__name__)
8
9     class Database:
10         def __init__(self, db_path: str):
11             self.db_path = db_path
12             self._init_db()
13
14         def _get_connection(self):
15             conn = sqlite3.connect(self.db_path)
16             conn.row_factory = sqlite3.Row
17             # Enforce ON DELETE CASCADE / SET NULL ? SQLite leaves FKs off per-connection.
18             conn.execute("PRAGMA foreign_keys = ON")
19             return conn
20
21         def _init_db(self):
22             """Initializes tables for videos, play_segments, and events."""
23             with self._get_connection() as conn:
24                 cursor = conn.cursor()
25
26                 # Videos Table
27                 cursor.execute("""
28                     CREATE TABLE IF NOT EXISTS videos (
29                         id TEXT PRIMARY KEY,
30                         filepath TEXT NOT NULL,
31                         processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
32                         status TEXT NOT NULL,
33                         validation_results TEXT
34                     )
35                 """)

```

```

36
37 # Play Segments Table (Extensible metadata JSON)
38 cursor.execute("""
39     CREATE TABLE IF NOT EXISTS play_segments (
40         id INTEGER PRIMARY KEY AUTOINCREMENT,
41         video_id TEXT NOT NULL,
42         start_time REAL NOT NULL,
43         end_time REAL NOT NULL,
44         duration REAL NOT NULL,
45         server TEXT,
46         receiver TEXT,
47         metadata TEXT,
48         FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
49     )
50 """)
51
52 # Events Table
53 cursor.execute("""
54     CREATE TABLE IF NOT EXISTS events (
55         id INTEGER PRIMARY KEY AUTOINCREMENT,
56         segment_id INTEGER NOT NULL,
57         timestamp REAL NOT NULL,
58         type TEXT NOT NULL,
59         player TEXT,
60         details TEXT,
61         FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE
62     )
63 """)
64
65 # Versioned migrations live here. PRAGMA user_version is the schema
66 # contract ? bump it for every change and add an ordered `if version < N`
67 # block below. Tests assert the expected version, so accidental drift fails
68 CI.
69
70 version = cursor.execute("PRAGMA user_version").fetchone()[0]
71
72 if version < 1:
73     # v1: raw candidate detections (pre-confirmation), detector-agnostic.
74     # Common fields are columns; detector-specific metrics go in `stats`
75     JSON,
76     # so a new segmenter reuses this table by setting its own
77     `detector`/`stats`.
78     cursor.execute("""
79         CREATE TABLE IF NOT EXISTS candidate_segments (
80             id INTEGER PRIMARY KEY AUTOINCREMENT,
81             video_id TEXT NOT NULL,
82             detector TEXT NOT NULL,
83             chunk_index INTEGER,
84             start_time REAL NOT NULL,
85             end_time REAL NOT NULL,
86             duration REAL NOT NULL,
87             confidence REAL,
88             stats TEXT,
89             detection_params TEXT,
90             confirmed_segment_id INTEGER,
91             human_label TEXT,
92             notes TEXT,
93             created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
94             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE,
95             FOREIGN K...
96     """)
97 [/external_agent_tool_result]

```

250. assistant (2026-07-02T06:11:33.156Z)

Now I have the exact DB API and `ffprobe\_meta` shape. Let me build the indexer's reporting

module: a `**soft-import**` wrapper (so the pipeline never breaks if ``obsreport`` is absent), a pure harvest layer, and the indexer's own report spec (footgun rules citing its FAQ). Starting with the spec:

251. assistant (2026-07-02T06:11:33.156Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]
```

252. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

253. assistant (2026-07-02T06:11:33.157Z)

Now the pure harvest layer that records raw signals into a recorder:

254. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py
[/external_agent_tool_call]
```

255. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

256. assistant (2026-07-02T06:11:33.157Z)

The `soft-import` wrapper with a single never-raising entrypoint for ``main.py`` to call:

257. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\__init__.py
[/external_agent_tool_call]
```

258. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\__init__.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

259. assistant (2026-07-02T06:11:33.157Z)

Now wire it into ``main.py`` ? both the CLI and API paths, ``finally``-guarded so the report is emitted even on validation failure or error, and surfaced in the API response. Adding the import first:

260. assistant (2026-07-02T06:11:33.157Z)

```
[external_agent_tool_call: Edit]
```

file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]

261. assistant (2026-07-02T06:11:33.157Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

262. assistant (2026-07-02T06:11:33.157Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]

263. assistant (2026-07-02T06:11:33.157Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

264. assistant (2026-07-02T06:11:33.157Z)

Now the CLI processing path:

265. assistant (2026-07-02T06:11:33.157Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]

266. assistant (2026-07-02T06:11:33.157Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

267. assistant (2026-07-02T06:11:33.157Z)

Let me refine fps-source inference (a real `fallback` should be detectable when segmentation ran but no probe yielded fps), then add the README FAQ entries the flags cite, the `report` config note, and tests:

268. assistant (2026-07-02T06:11:33.157Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py
[/external_agent_tool_call]

269. assistant (2026-07-02T06:11:33.158Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

270. assistant (2026-07-02T06:11:33.158Z)

Now the README FAQ entries the flags cite (Q10?Q13), so every `faq\_ref` resolves:

271. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

272. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
229     ### Q9: How can I debug/test the Gemini Segmenter on a small portion of a video without
uploading a huge multi-GB file?
230     Ingesting and uploading multi-gigabyte video files to Google Cloud can be slow and
expensive. We built a native fast local trimming developer debugging feature:
231     1. Open your `config.json` file.
232     2. Under the `indexing` configuration block, add a `debug_duration` key (in
seconds):
233         ```json
234         "indexing": {
235             "default_segmenter": "gemini",
236             "debug_duration": 15.0,
237             ...
238         }
239         ```
240     3. When `debug_duration` is set, the Gemini segmenter will instantly run a zero-re-
encode, sub-second `ffmpeg` slice to create a tiny temporary clip of the first 15 seconds,
upload only that tiny clip to Google AI Studio (which takes <1 second to upload and <5 seconds
to process), run the model query, and clean up the temporary files automatically. This allows
you to iterate on model prompts, evaluate validation overlaps, and verify rally boundaries in
under 10 seconds!
241
242     ---
243
244     ## ? Annotation & Evaluation
245
246     Once a video is processed, you can measure how well the pipeline detected its
rallies against hand-annotated ground truth. You annotate each rally's `[start, end)` time
range in a small CSV; the harness scores the detected rallies straight from the SQLite DB ? no
API calls, fully deterministic.
247
248     ```bash
[/external_agent_tool_result]
```

273. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

274. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

275. assistant (2026-07-02T06:11:33.158Z)

Let me add the `obsreport` dependency note to `requirements.txt` (commented until the private remote exists, since the runtime soft-imports it):

276. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\requirements.txt
```


[/external_agent_tool_call]

277. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
1      fastapi>=0.111.0
2      uvicorn>=0.30.1
3      opencv-python>=4.9.0.80
4      numpy>=1.26.4
5      pydantic>=2.7.1
6      jinja2>=3.1.4
7      python-dotenv>=1.0.0
8      google-genai>=2.4.0
9      torch>=2.12.0
10     torchvision>=0.27.0
11     supervision>=0.21.0,<0.30.0
12
[/external_agent_tool_result]
```

278. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\requirements.txt
[/external_agent_tool_call]
```

279. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\requirements.txt has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

280. assistant (2026-07-02T06:11:33.158Z)

Now the harvest/reporting tests (mocked DB, covering the footgun scenarios), then run the indexer suite:

281. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py
[/external_agent_tool_call]
```

282. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

283. assistant (2026-07-02T06:11:33.158Z)

Let me fix a stray nonsense assertion I left in the healthy-run test:

284. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py
[/external_agent_tool_call]
```

285. assistant (2026-07-02T06:11:33.158Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

286. assistant (2026-07-02T06:11:33.158Z)

Now run the full indexer suite ? confirming the new reporting tests pass *and* the existing 209 are unbroken:

287. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Bash]
description: Run full indexer suite after A integration
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest -q 2>&1 |
tail -25
[/external_agent_tool_call]
```

288. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
..... [ 32%]
..... [ 64%]
..... [ 97%]
..... [100%]
222 passed in 12.09s
[/external_agent_tool_result]
```

289. assistant (2026-07-02T06:11:33.159Z)

222 passed (209 baseline intact + 13 new). Now an end-to-end smoke through the **real**
`main.py` CLI to prove the wiring writes a report next to the video. I'll synthesize a tiny MP4
with ffmpeg and run the CLI:

290. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Bash]
description: Synthesize tiny mp4 and run indexer CLI end-to-end
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" &&
SMOKE="$TEMP/obsreport_smoke" && rm -rf "$SMOKE" && mkdir -p "$SMOKE" && ffmpeg -loglevel error
-f lavfi -i color=c=green:s=1280x720:d=2 -r 30 -pix_fmt yuv420p "$SMOKE/smoke.mp4" -y && echo
"made video:" && ls "$SMOKE" && echo "=== run CLI (no API key) ===" && python -u -m backend.main
--video-path "$SMOKE/smoke.mp4" --segmenter motion --output-dir "$SMOKE/out" 2>&1 | tail -20 &&
echo "=== files next to video ===" && ls "$SMOKE"
[/external_agent_tool_call]
```

291. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
made video:
smoke.mp4
=== run CLI (no API key) ===
[CLI] Processing video file: C:\Users\avidu\AppData\Local\Temp\obsreport_smoke\smoke.mp4
[CLI] Validating...
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:347:
PydanticDeprecatedSince20: The `dict` method is deprecated; use `model_dump` instead. Deprecated
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2 Migration Guide at
https://errors.pydantic.dev/2.13/migration/
    db_instance.add_video(video_id, args.video_path, status, [r.dict() for r in val_results])

=====
VALIDATION SUMMARY
=====
[PASS] format_check: Video format is valid MP4 container.
```

```
[PASS] resolution_fps_check: Video resolution and frame rate meet minimum standards.
[PASS] pts_continuity_check: Video is too short to perform a PTS continuity check.
[PASS] scene_cut_check: Scene continuity check passed. No abnormal visual cuts detected.
[FAIL] blur_check: Video quality rejected: Focus check failed. Average sharpness score of 0.0 is
below the minimum required threshold of 20.0.
[PASS] relevance_check: Video identified as relevant to badminton (badminton court structures
found).
```

=====

```
[FAIL] Video failed checks. Indexing halted.
[REPORT] Technical report : C:\Users\avidu\AppData\Local\Temp\obsreport_smoke\smoke.report.md
[REPORT] Customer summary : C:\Users\avidu\AppData\Local\Temp\obsreport_smoke\smoke.summary.md
[REPORT] Machine JSON      : C:\Users\avidu\AppData\Local\Temp\obsreport_smoke\smoke.report.json
=== files next to video ===
out
smoke.mp4
smoke.report.json
smoke.report.md
smoke.summary.md
[/external_agent_tool_result]
```

292. assistant (2026-07-02T06:11:33.159Z)

End-to-end wiring works ? 3 report files written next to the video via the `finally` block, even though validation halted the run. Let me read the generated technical + customer reports to confirm they read well:

293. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\obsreport_smoke\smoke.report.md
[/external_agent_tool_call]
```

294. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
1      # Badminton Highlight Indexer ? Video Report
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Tool | badminton-highlight-indexer |
8      | Video id | fb5d84c30b683fce27292bd2a8f80677 (file_md5) |
9      | Video path | C:\Users\avidu\AppData\Local\Temp\obsreport_smoke\smoke.mp4 |
10     | Generated | 2026-06-07T07:30:37Z |
11     | Schema | 1.0 |
12
13     ## Video
14
15     | Field | Value |
16     | --- | --- |
17     | Duration | 2.0s |
18     | Resolution | 1280x720 |
19     | FPS | 30 fps (probed) |
20     | Container | mov,mp4,m4a,3gp,3g2,mj2 |
21     | Size | 0 MB |
22
23     ## Processing stages
24
25     **validation** ? `error` (0.0s)
26     - checks_passed: 5
27     - checks_total: 6
28     - FAILED blur_check: Video quality rejected: Focus check failed. Average sharpness
```

```

score of 0.0 is below the minimum required threshold of 20.0.
29
30     **segmentation** ? `not_run` (0.0s)
31     - candidate_windows: 0
32     - segment_count: 0
33     - total_play_s: 0 s
34     - _segmenter_requested_: motion
35     - _segmenter_actual_: None
36     - _config_default_: gemini
37     - _matches_config_default_: False
38     - _was_reingest_: False
39     - _detector_: None
40
41     **ai_handoff** ? `not_run` (0.0s)
42     - _ai_based_: False
43     - _provider_: None
44     - _model_: None
45     - _api_key_present_: None
46
47     ## Findings & warnings
48
49     - [info] **segmenter_divergence** ? The segmenter that ran differs from config.json's
default_segmenter. This is fine if intentional (CLI/API override), but the indexer's defaults
are known to diverge across config.json / setup.py / the CLI ? confirm you ran the one you
meant.
50         ? see **README#Q7**
51         ? evidence: stages.segmentation.details.matches_config_default=False
52     - [ERROR] **validation_failed** ? One or more validation checks failed, so indexing was
halted. See the Processing stages > validation block above: each failed check states the reason
(relevance -> README#Q2, fps/keyframes -> README#Q5).
53         ? see **README#Q13**
54         ? evidence: stages.validation.status=error
55
56     ## Verdict
57
58     **FAIL**
59
[/external_agent_tool_result]

```

295. assistant (2026-07-02T06:11:33.159Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\obsreport_smoke\smoke.summary.md
[/external_agent_tool_call]

```

296. assistant (2026-07-02T06:11:33.159Z)

```

[external_agent_tool_result]
1     # Badminton Highlight Indexer ? Video Report (summary)
2
3     ## Run
4
5     | Field | Value |
6     | --- | --- |
7     | Video | smoke.mp4 |
8     | Report generated | 2026-06-07T07:30:37Z |
9
10    ## Video
11
12    | Field | Value |
13    | --- | --- |
14    | Duration | 2.0s |
15    | Resolution | 1280x720 |
16    | Frame rate | 30 fps |
17

```

```
18     ## Highlights
19
20     - **0** highlights
21     - **0.0s** of play
22     - **0%** of the video is highlights
23     - No highlights were produced for this video.
24
25     ## Things to know
26
27     - ? One or more validation checks failed, so indexing was halted. See the Processing
stages > validation block above: each failed check states the reason (relevance -> README#Q2,
fps/keyframes -> README#Q5).
28
29     ## Verdict
30
31     ?? We ran into a problem processing this video, so some highlights may be missing. See
the notes above, or contact support.
32
[/external_agent_tool_result]
```

297. assistant (2026-07-02T06:11:33.159Z)

The reports are readable and debuggable. Two quality issues to fix before committing:

1. ****`segmenter\_divergence` false-fired**** even though segmentation didn't run (and would also fire on intentional ****`--segmenter` overrides**) ? noisy.
2. The ****customer**** summary leaked a technical phrase ("see Processing stages > validation block") ? a customer should never see internal references.

Both are exactly what the readability bar should catch. The clean fix for #2 is a small additive feature in the library: an optional ****`customer\_message`**** on flags, where the customer report shows ***only*** findings explicitly curated for customers. Let me add that to **`obsreport`** first.

298. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\envelope.py
[/external_agent_tool_call]
```

299. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\envelope.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

300. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py
[/external_agent_tool_call]
```

301. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

302. assistant (2026-07-02T06:11:33.159Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\rules.py
[/external_agent_tool_call]
```

303. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\rules.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

304. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

305. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

306. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\recorder.py
[/external_agent_tool_call]
```

307. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\recorder.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

308. assistant (2026-07-02T06:11:33.160Z)

Now update the library tests to reflect the curated-customer-flags semantics, bump the version, and regenerate the schema:

309. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py
[/external_agent_tool_call]
```

310. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

311. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py
[/external_agent_tool_call]
```

312. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

313. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_customer_message.py
[/external_agent_tool_call]
```

314. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-
obsreport\tests\test_customer_message.py (file state is current in your context ? no need to
Read it back)
[/external_agent_tool_result]
```

315. assistant (2026-07-02T06:11:33.160Z)

Now bump the lib version, regenerate the schema, and run the lib suite:

316. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\version.py
[/external_agent_tool_call]
```

317. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\version.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

318. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\pyproject.toml
[/external_agent_tool_call]
```

319. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\pyproject.toml has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

320. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_call: Bash]
description: Regen schema + run obsreport suite after customer_message
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python scripts/gen_schema.py && python
-m pytest -q 2>&1 | tail -8
[/external_agent_tool_call]
```

321. assistant (2026-07-02T06:11:33.160Z)

```
[external_agent_tool_result]
wrote C:\Users\avidu\Projects\sports-obsreport\schema\report_envelope.schema.json
..... [100%]
72 passed in 0.33s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

322. assistant (2026-07-02T06:11:33.160Z)

72 passed. Commit + tag the lib v0.1.1, then fix the indexer's `segmenter\_divergence` gating and add curated customer messages to its spec.

323. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Bash]
description: Commit + tag obsreport v0.1.1, reinstall editable
command: cd "C:/Users/avidu/Projects/sports-obsreport" && git add -A && git commit -q -m "feat:
customer_message ? curated customer-facing flag text (v0.1.1)"
```

Flags/rules gain an optional customer_message. The technical report always uses 'message'; the customer report shows ONLY flags that set customer_message (in that gentle phrasing) so internal/technical findings never leak to customers. Additive, backward compatible (schema still 1.0). Schema regenerated. 72 tests green.

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com> && git tag -a v0.1.1 -m "obsreport
v0.1.1 ? customer_message" && git tag --list && pip install -e . --no-build-isolation -q 2>&1 |
tail -1 && python -c "import obsreport; print('reinstalled', obsreport.__version__)"
[/external_agent_tool_call]
```

324. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_result]
warning: in the working copy of 'pyproject.toml', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'src/obsreport/envelope.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'src/obsreport/recorder.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'src/obsreport/render/markdown.py', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'src/obsreport/rules.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/obsreport/spec.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/obsreport/version.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_render_markdown.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'tests/test_customer_message.py', LF will be replaced by CRLF
the next time Git touches it
v0.1.0
v0.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
reinstalled 0.1.1
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

325. assistant (2026-07-02T06:11:33.161Z)

Lib v0.1.1 reinstalled. Now fix the indexer harvest gating for `segmenter\_divergence` and add curated customer messages to its spec:

326. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py
[/external_agent_tool_call]
```

327. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py has
been updated successfully. (file state is current in your context ? no need to Read it back)
```


[/external_agent_tool_result]

328. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]

329. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

330. assistant (2026-07-02T06:11:33.161Z)

Now add curated `customer\_message` to the two indexer findings a customer should hear about:

331. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]

332. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

333. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]

334. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

335. assistant (2026-07-02T06:11:33.161Z)

Update the divergence test for the new gating (only fires without an explicit override), then re-run the indexer suite:

336. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py
[/external_agent_tool_call]

337. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

338. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Bash]
description: Re-run indexer suite + re-smoke failure path
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest -q 2>&1 |
tail -8 && echo "=== re-smoke (failure path) ===" && SMOKE="$TEMP/obsreport_smoke2" && rm -rf
"$SMOKE" && mkdir -p "$SMOKE" && ffmpeg -loglevel error -f lavfi -i color=c=green:s=1280x720:d=2
-r 30 -pix_fmt yuv420p "$SMOKE/smoke.mp4" -y && python -u -m backend.main --video-path
"$SMOKE/smoke.mp4" --segmenter motion --output-dir "$SMOKE/out" >/dev/null 2>&1; echo "exit=$?"
[/external_agent_tool_call]
```

339. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_result]
..... [ 32%]
..... [ 64%]
..... [ 97%]
..... [100%]
222 passed in 12.06s
=== re-smoke (failure path) ===
exit=0
[/external_agent_tool_result]
```

340. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\obsreport_smoke2\smoke.summary.md
[/external_agent_tool_call]
```

341. assistant (2026-07-02T06:11:33.161Z)

```
[external_agent_tool_result]
1      # Badminton Highlight Indexer ? Video Report (summary)
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Video | smoke.mp4 |
8      | Report generated | 2026-06-07T07:35:15Z |
9
10     ## Video
11
12     | Field | Value |
13     | --- | --- |
14     | Duration | 2.0s |
15     | Resolution | 1280x720 |
16     | Frame rate | 30 fps |
17
18     ## Highlights
19
20     - **0** highlights
21     - **0.0s** of play
22     - **0%** of the video is highlights
23     - No highlights were produced for this video.
24
25     ## Things to know
26
27     - ? This video couldn't be processed because it didn't pass our quality checks (for
example: too blurry, too low resolution, or not a badminton court).
28
29     ## Verdict
30
31     ?? We ran into a problem processing this video, so some highlights may be missing. See
```

the notes above, or contact support.

32

[/external_agent_tool_result]

342. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\AppData\Local\Temp\obsreport_smoke2\smoke.report.md

[/external_agent_tool_call]

343. assistant (2026-07-02T06:11:33.161Z)

[external_agent_tool_result]

45 - _model_: None

46 - _api_key_present_: None

47

48 ## Findings & warnings

49

50 - [ERROR] ****validation_failed**** ? One or more validation checks failed, so indexing was halted. See the Processing stages > validation block above: each failed check states the reason (relevance -> README#Q2, fps/keyframes -> README#Q5).

51 ? see ****README#Q13****

52 ? evidence: stages.validation.status=error

53

54 ## Verdict

55

56 ****FAIL****

57

[/external_agent_tool_result]

344. assistant (2026-07-02T06:11:33.161Z)

Both reports are now clean: the customer summary shows only the gentle curated message (no technical leakage), and the technical findings no longer carry the false `segmenter\_divergence`. ****Phase A is functionally complete.**** Committing:

345. assistant (2026-07-02T06:11:33.162Z)

[external_agent_tool_call: Bash]

description: Commit indexer phase A

command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git add -A && git commit -q -m "feat(reporting): always-on per-video observability report (indexer)"

Emits <video>.report.json/.report.md (technical) + <video>.summary.md (customer)

next to the video (or report.output_dir) on every CLI/API run, via the shared

obsreport library (SOFT dependency ? pipeline no-ops if absent; writes never raise;

wired in a finally so a report is emitted even on validation failure/exception).

- validators/base.py: add backward-compatible run_all_with_context (surfaces the ffprobe_meta that run_all discarded); run_all unchanged.
- backend/reporting/: soft-import wrapper + pure harvest (DB candidates/segments, ffprobe meta, fps-source, segmenter actual-vs-config) + spec.yaml footgun rules (silent_provider_noop, ai_empty_vs_no_rallies, fps_fallback_used, synthetic_motion_metadata, segmenter_divergence, reingest_replaced_prior, validation_failed) ? each cites a README FAQ; user-facing ones carry customer_message.
- main.py: finally-guarded emit in CLI + API (both exits); report paths in API response.
- README: FAQ Q10-Q13 (motion metadata, re-ingest, the report itself, validation fails).
- 13 new tests (broken-run scenarios + emit). Suite 222 green; verified end-to-end via CLI smoke.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git log --oneline -1 && git status -s | head

[/external_agent_tool_call]

346. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_result]
warning: in the working copy of 'backend/main.py', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'backend/validators/base.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'backend/reporting/__init__.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'backend/reporting/harvest.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'backend/reporting/spec.yaml', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'tests/test_reporting_harvest.py', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'tests/test_validators_base.py', LF will be replaced by CRLF the
next time Git touches it
9dbc764 feat(reporting): always-on per-video observability report (indexer)
[/external_agent_tool_result]
```

347. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

348. assistant (2026-07-02T06:11:33.162Z)

Phase A committed (222 green, verified end-to-end). Let me mark it done, refresh the memory checkpoint, and start **Phase B (collector)**.

349. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"5","status":"completed"}
[/external_agent_tool_call]
```

350. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_result]
Updated task #5 status
[/external_agent_tool_result]
```

351. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"6","status":"in_progress"}
[/external_agent_tool_call]
```

352. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_result]
Updated task #6 status
[/external_agent_tool_result]
```

353. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_call: Edit]
file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

354. assistant (2026-07-02T06:11:33.162Z)

Now Phase B (collector). Let me read the exact collector code I'll wire into ? the CVAT report builders I'll reuse, and the CLI command bodies:

355. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py
[/external_agent_tool_call]
```

356. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py
[/external_agent_tool_call]
```

357. assistant (2026-07-02T06:11:33.162Z)

```
[external_agent_tool_result]
1      """Adapter: CVAT rally exports -> our canonical per-rally records.
2
3      Why this exists
4      -----
5      Our annotation brief (`docs/annotation/ROORA_BRIEF.md`) asks vendors for a
6      per-rally CSV that imports directly via curate import-local. A vendor may
7      instead do the work in CVAT and hand back the CVAT export: per-frame bounding
8      boxes, with the rally fields (rally_number, shots_count,
9      ending_reason ...) carried as CVAT object attributes on each box. The box
10     geometry is irrelevant to us ? it is just a carrier the rater attached the rally
11     metadata to.
12
13     This module is the ingestion-side adapter that absorbs that vendor format so the
14     modeling repo never has to. It does two things, both deliberately on our side
15     so we fix mechanical problems rather than renegotiating them:
16
17     1. Recompute time from the frame range. A rally's interval is taken from the
18     first/last frame a box for that rally appears on:
19     seconds = frame / fps at the video's true fps. We ignore any
20     start_time/end_time attributes the vendor wrote ? a CVAT job that
21     samples every step frames and assumes 30 fps yields systematically wrong
22     times (we have seen start_time == frame / (step * 30), i.e. off by the
23     sampling step at the wrong fps). The frame index is reliable; the vendor's
24     seconds are not.
25
26     2. Surface ? never silently fix ? content defects. Overlapping rallies,
27     zero-duration rallies, ending_reason values outside the agreed set, and
28     too-coarse sampling are reported as :class:`Issue`s. Guessing a corrected
29     rally boundary would poison ground truth, so that stays a human/vendor call.
30
31     Supported input: the single-file CVAT XML export, in either image mode
32     (<image><box>...) or track/interpolation mode (<track><box frame=...>).
33     The two PASCAL-VOC / COCO sibling exports encode the same data and can be added
34     later if a vendor sends only those.
35     """
36     from __future__ import annotations
37
38     import os
39     import xml.etree.ElementTree as ET
40     from collections import Counter
41     from dataclasses import dataclass, field
42     from typing import Any, Dict, List, Optional, Tuple
43
44     from src.core.parsing import safe_int
```

```

45     from src.core.downloader import probe_resolution_fps
46
47     # The agreed cross-sport ending_reason vocabulary for net-separated sports.
48     # Kept here as a soft check only (a warning, never a hard failure) so this
49     # adapter does not depend on the annotation-spec wording.
50     DEFAULT_ALLOWED_REASONS = {
51         "winner",
52         "forced_error",
53         "unforced_error",
54         "service_fault",
55         "let",
56         "other",
57     }
58
59     # The observational outcome of the last shot ? orthogonal to ending_reason
60     # (winner can be 'in'; an error can be 'net' or 'out'; service_fault/let -> 'n_a').
61     [/external_agent_tool_result]

```

358. assistant (2026-07-02T06:11:33.162Z)

```

[external_agent_tool_result]
240
241
242     def detect_issues(records: List[RallyRecord],
243                       stop_frame: Optional[int] = None,
244                       allowed_reasons: Optional[set] = None,
245                       allowed_outcomes: Optional[set] = None,
246                       max_gap_s: float = 15.0) -> List[Issue]:
247         """Find the defects that block or degrade a clean ground-truth import.
248
249         Note: we deliberately do NOT flag the sampling density (step/fps). A boundary
250         slack of ~0.5s is acceptable for this pipeline because the downstream
251         highlight compiler pads every clip on output (badminton-highlight-indexer,
252         backend/stitcher/compiler.py: begin_padding=0.5s, end_padding=1.0s), which
253         absorbs the sampling uncertainty when highlights are trimmed.
254         """
255         allowed = allowed_reasons if allowed_reasons is not None else
256         DEFAULT_ALLOWED_REASONS
257         allowed_out = allowed_outcomes if allowed_outcomes is not None else
258         DEFAULT_ALLOWED_OUTCOMES
259         issues: List[Issue] = []
260
261         ordered = sorted(records, key=lambda r: r.rally_number)
262         for i, r in enumerate(ordered):
263             if r.end_time <= r.start_time:
264                 issues.append(Issue(
265                     "blocker", "zero_duration",
266                     f"Rally {r.rally_number}: end <= start "
267                     f"({r.frame_min}-{r.frame_max}); single-sample or stray box.",
268                     r.rally_number))
269
270             if r.ending_reason is not None and r.ending_reason not in allowed:
271                 issues.append(Issue(
272                     "medium", "ending_reason_vocab",
273                     f"Rally {r.rally_number}: ending_reason '{r.ending_reason}' "
274                     f"not in the agreed set {sorted(allowed)}.",
275                     r.rally_number))
276
277             if r.last_shot_outcome is not None and r.last_shot_outcome not in allowed_out:
278                 issues.append(Issue(
279                     "low", "outcome_vocab",
280                     f"Rally {r.rally_number}: last_shot_outcome '{r.last_shot_outcome}' "
281                     f"not in {sorted(allowed_out)}.",
282                     r.rally_number))

```

```

282         # Inflated span: a long rally with implausibly few shots usually means a
283         # stray/duplicated box stretched the frame range past the real ending.
284         if r.shots_count and r.duration >= 5.0:
285             sps = r.shots_count / r.duration
286             if sps < 0.3:
287                 issues.append(Issue(
288                     "medium", "shot_density",
289                     f"Rally {r.rally_number}: {r.shots_count} shots over "
290                     f"{r.duration:.1f}s ({sps:.2f}/s) is implausibly low ? the span "
291                     f"may be inflated by a stray box.",
292                     r.rally_number))
293
294         if stop_frame is not None and r.frame_max > stop_frame:
295             issues.append(Issue(
296                 "high", "frame_out_of_range",
297                 f"Rally {r.rally_number}: frame {r.frame_max} exceeds the video's "
298                 f"last frame {stop_frame}.",
299                 r.rally_number))
300
301         if i > 0:
302             prev = ordered[i - 1]
303             if r.start_time < prev.end_time:
304                 issues.append(Issue(
305                     "blocker", "overlap",
306                     f"Rally {prev.rally_number} (ends {prev.end_time:.2f}s) overlaps "
307                     f"Rally {r.rally_number} (starts {r.start_time:.2f}s) by "
308                     f"{prev.end_time - r.start_time:.2f}s ? rallies must be disjoint.",
309                     r.rally_number))
310             elif r.start_time - prev.end_time > max_gap_s:
311                 # A long unlabeled stretch between two rallies is a cheap, label-only
312                 # second opinion for the worst error (a missed rally). It won't catch a
313                 # rally MERGED into its neighbour's span (shot_density/inf...
[/external_agent_tool_result]

```

359. assistant (2026-07-02T06:11:33.162Z)

Let me read the `Issue`/`RallyRecord`/`ConversionResult` definitions and the collector CLI command bodies I'll wire into:

360. assistant (2026-07-02T06:11:33.162Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py
[/external_agent_tool_call]

```

361. assistant (2026-07-02T06:11:33.162Z)

```

[external_agent_tool_result]
60     # (winner can be 'in'; an error can be 'net' or 'out'; service_fault/let -> 'n_a').
61     DEFAULT_ALLOWED_OUTCOMES = {"in", "net", "out", "n_a"}
62
63     # CVAT attribute names we map onto canonical columns. The vendor's CVAT label
64     # schema has used suffixed names like "score_before (A-B)"; we match leniently.
65     _ATTR_ALIASES = {
66         "rally_number": ("rally_number",),
67         "shots_count": ("shots_count",),
68         "ending_reason": ("ending_reason",),
69         "last_shot_outcome": ("last_shot_outcome", "outcome", "last_shot_outcome
(in/net/out)"),
70         "score_before": ("score_before", "score_before (a-b)",),
71         "score_after": ("score_after", "score_after (a-b)",),
72         "notes": ("notes", "note"),
73     }
74

```

```

75     CANONICAL_COLUMNS = [
76         "rally_number", "start_mmss", "end_mmss", "start_time", "end_time",
77         "shots_count", "ending_reason", "last_shot_outcome", "score_before", "score_after",
78         "notes",
79     ]
80
81     @dataclass
82     class RallyRecord:
83         """One rally recovered from the CVAT export, time re-derived from frames."""
84         rally_number: int
85         start_time: float          # seconds, = frame_min / fps
86         end_time: float           # seconds, = frame_max / fps (see padding note)
87         shots_count: Optional[int]
88         ending_reason: Optional[str]
89         last_shot_outcome: Optional[str]
90         score_before: Optional[str]
91         score_after: Optional[str]
92         notes: Optional[str]
93         frame_min: int
94         frame_max: int
95         n_boxes: int
96         vendor_start_time: Optional[str] = None    # the (discarded) attribute value
97         vendor_end_time: Optional[str] = None
98
99         @property
100         def duration(self) -> float:
101             return self.end_time - self.start_time
102
103
104     @dataclass
105     class Issue:
106         """A defect found in the delivery. ``blocker`` issues stop a clean import."""
107         severity: str    # 'blocker' | 'high' | 'medium' | 'low'
108         code: str
109         message: str
110         rally_number: Optional[int] = None
111
112
113     @dataclass
114     class ConversionResult:
115         records: List[RallyRecord]
116         issues: List[Issue]
117         fps: float
118         step: int
119         meta: Dict[str, Any] = field(default_factory=dict)
120
121         @property
122         def n_blockers(self) -> int:
123             return sum(1 for i in self.issues if i.severity == "blocker")
124
125
126     def _mmss(seconds: float) -> str:
127         """Format seconds as ``m:ss.mmm`` (matches the brief's mm:ss columns)."""
128         m = int(seconds // 60)
129         s = seconds - m * 60
130         return f"{m}:{s:06.3f}"
131
132
133     def _attr_lookup(attrs: Dict[str, str], canonical: str) -> Optional[str]:
134         for alias in _ATTR_ALIASES[canonical]:
135             if alias in attrs and attrs[alias] != "":
136                 return attrs[alias]
137         return None
138

```



```

139
140 def parse_cvat_boxes(xml_path: str) -> Tuple[Dict[str, Any], List[Dict[str, Any]]]:
141     """Parse a CVAT XML export into (meta, boxes).
142
143     ``boxes`` is a flat list of ``{"frame": int, "attrs": {name: value}}`` ? one
144     entry per visible box, from either image-mode or track-mode exports.
145     ``meta`` carries ``step`` (frame_filter) and ``stop_frame`` when present.
146     """
147     if not os.path.exists(xml_path):
148         raise FileNotFoundError(f"CVAT export not found: {xml_path}")
149
150     root = ET.parse(xml_path).getroot()
151     meta: Dict[str, Any] = {}
152
153     job = root.find("./job")
154     if job is not None:
155         stop = job.findtext("stop_frame")
156         if stop is not None and stop.strip().isdigit():
157             meta["stop_frame"] = int(stop.strip())
158         ff = job.findtext("frame_filter") or ""
159         if "step=" in ff:
160             try:
161                 meta["step"] = int(ff.split("step=")[1].split()[0])
162             except (ValueError, IndexError):
163                 pass
164
165     def _attrs(el: ET.Element) -> Dict[str, str]:
166         out: Dict[str, str] = {}
167         for attr in el.attrib:
168             out[attr] = el.attrib[attr]
169     return (meta, [external_agent_tool_result])

```

362. assistant (2026-07-02T06:11:33.163Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
[/external_agent_tool_call]

```

363. assistant (2026-07-02T06:11:33.163Z)

```

[external_agent_tool_result]
52     def import_local(self, args):
53         """Manually registers a local video and annotation file into the database."""
54         sport_str = args.sport.lower()
55         if sport_str not in self.config.get("supported_sports", ["badminton"]):
56             print(f"Error: Sport '{sport_str}' is not supported. Supported:
{self.config.get('supported_sports')}")
57             sys.exit(1)
58
59         # Validate video path
60         if not os.path.exists(args.video_path):
61             print(f"Warning: Local video path '{args.video_path}' does not exist on
disk, but continuing registration.")
62
63         # Validate annotation file
64         if not os.path.exists(args.annotations_path):
65             print(f"Error: Annotations file '{args.annotations_path}' not found.")
66             sys.exit(1)
67
68         match_name = args.match_name or os.path.basename(args.video_path).split('.')[0]
69         video_id = args.video_id or match_name.lower().replace(" ", "_")
70
71         # Parse license
72         try:
73             license_type = LicenseType(args.license)
74         except ValueError:
75             print(f"Warning: Unknown license '{args.license}'. Defaulting to Unknown

```

```

(non-commercial).")
76         license_type = LicenseType.UNKNOWN
77
78         is_commercial = self.validator.check_license_commercial(license_type)
79
80         # 1. Parse Annotations
81         annotations = []
82         try:
83             annotations = self._parse_local_annotations(sport_str, video_id,
args.annotations_path)
84         except Exception as e:
85             print(f"Error parsing annotations: {e}")
86             sys.exit(1)
87
88         # 2. Run Validation
89         is_valid = True
90         errors = []
91         if sport_str == "badminton":
92             is_valid, errors = self.validator.validate_badminton_rallies(annotations)
93         elif sport_str == "tennis":
94             is_valid, errors = self.validator.validate_tennis_points(annotations)
95         elif sport_str == "cricket":
96             is_valid, errors = self.validator.validate_cricket_deliveries(annotations)
97         elif sport_str == "pickleball":
98             is_valid, errors = self.validator.validate_pickleball_rallies(annotations)
99         elif sport_str == "tabletennis":
100             is_valid, errors = self.validator.validate_tabletennis_rallies(annotations)
101
102         if not is_valid:
103             print(f"Error: Annotations failed validation!")
104             for err in errors:
105                 print(f" - {err}")
106             sys.exit(1)
107
108         # 3. Create Video Metadata
109         video_meta = VideoMetadata(
110             video_id=video_id,
111             sport=SportType(sport_str),
112             video_url=None,
113             local_path=args.video_path,
114             license_type=license_type,
115             license_url=None,
116             is_commercial=is_commercial,
117             status=CurationStatus.CURATED, # Custom local imports bypass staging by
default
118             match_name=match_name,
119             fps=args.fps,
120             resolution=args.resolution
121         )
122
123         # 4. Insert into Database
124         self.db.insert_video(video_meta)
125
126         count = 0
127         if sport_str == "badminton":
128             count = self.db.insert_badminton_rallies(annotations)
129         elif sport_str == "tennis":
130             count = self.db.insert_tennis_points(annotations)
131         elif sport_str == "cricket":
132             count = self.db.insert_cricket_deliveries(annotations)
133         elif sport_str == "pickleball":
134             count = self.db.insert_pickleball_rallies(annotations)
135         elif sport_str == "tabletennis":
136             count = self.db.insert_tabletennis_rallies(annotations)
137

```

```

138         print(f"\nSuccessfully imported local video '{match_name}':")
139         print(f" - Video ID: {video_i...
[/external_agent_tool_result]

```

364. assistant (2026-07-02T06:11:33.163Z)

Now the collector's CLI command bodies (validate_delivery, convert_cvat, export) and its FAQ doc (to cite real anchors), plus how it's packaged:

365. assistant (2026-07-02T06:11:33.163Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
[/external_agent_tool_call]

```

366. assistant (2026-07-02T06:11:33.163Z)

```

[external_agent_tool_result]
526     def convert_cvat(self, args):
527         """Convert a vendor CVAT rally export into our canonical per-rally CSV.
528
529         Absorbs the CVAT bounding-box delivery format (rally fields carried as box
530         attributes) on the ingestion side: each rally's time is recomputed from
531         its frame range at the video's true fps, and content defects (overlaps,
532         zero-duration, off-vocabulary endings, coarse sampling) are written to an
533         issues report rather than silently "fixed". The emitted CSV imports via
534         `import-local`; any blocker issue must be resolved first, since the
535         importer's validator rejects overlapping / zero-duration rallies.
536         """
537         from src.parsers.cvat.rally_cvat import (
538             convert, write_canonical_csv, render_issue_report,
539         )
540         try:
541             result = convert(args.input, fps=args.fps,
542                             video_path=args.video_path, step=args.step)
543         except (FileNotFoundError, ValueError) as e:
544             print(f"Error: {e}")
545             sys.exit(1)
546
547         write_canonical_csv(result.records, args.out)
548         report = render_issue_report(result, os.path.basename(args.input))
549         if args.issues:
550             with open(args.issues, "w", encoding="utf-8") as f:
551                 f.write(report)
552         print(report)
553         print(f"Canonical CSV written: {args.out} ({len(result.records)} rallies)")
554         if result.n_blockers:
555             print(f"\n {result.n_blockers} BLOCKER issue(s) ? import-local will reject
556 "
557             f"this file until they are resolved (by the vendor or manual
review).")
558         else:
559             print("\n No blockers. Next:")
560             print(f"    python -m src.cli.curate import-local --sport <sport> "
561                   f"--video-path <video> --annotations-path {args.out} --fps
{result.fps:g}")
562
563     def validate_delivery(self, args):
564         """Run the full validation suite on a rally delivery and emit a report.
565
566         Accepts a CVAT export (--input + --video-path/--fps) or an already-canonical
567         CSV (--csv). Writes, into --out-dir:
568         <id>_report.md    human report (scoreboard + issues + rallies to review)
569         <id>_report.json  machine summary (for CI / a bulk-ingest loop)

```

```

569         <id>_review.csv    the annotatable manual-verification sheet
570 Exits non-zero if any blocker issue exists, so it can gate a bulk pipeline.
571     """
572     import json
573     from src.parsers.cvat.rally_cvat import (
574         convert, load_canonical_csv, detect_issues,
575         build_summary, render_validation_md, write_review_csv,
576     )
577     vid = args.video_id or "delivery"
578     fps = None
579     if args.input:
580         try:
581             result = convert(args.input, fps=args.fps, video_path=args.video_path)
582         except (FileNotFoundError, ValueError) as e:
583             print(f"Error: {e}")
584             sys.exit(1)
585         records, issues, fps = result.records, result.issues, result.fps
586     elif args.csv:
587         try:
588             records = load_canonical_csv(args.csv)
589         except FileNotFoundError as e:
590             print(f"Error: {e}")
591             sys.exit(1)
592         if not records:
593             print(f"Error: no usable rally rows in {args.csv}")
594             sys.exit(1)
595         issues = detect_issues(records)
596     else:
597         print("Error: provide --input <cvat.xml> or --csv <canonical.csv>")
598         sys.exit(1)
599
600     os.makedirs(args.out_dir, exist_ok=True)
601     summary = build_summary(records, issues, source=vid, fps=fps)
602     md = render_validation_md(records, issues, source=vid...)
[/external_agent_tool_result]

```

367. assistant (2026-07-02T06:11:33.163Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md
[/external_agent_tool_call]

```

368. assistant (2026-07-02T06:11:33.163Z)

```

[external_agent_tool_result]
1     # Rally Ingestion Pipeline ? End-to-End FAQ
2
3     How a vendor's rally annotations become training/eval ground truth, and how we
4     validate each delivery against our conventions. This is the operational companion
5     to the annotation spec (`docs/annotation/ROORA_BRIEF.md` + `ANNOTATION_GUIDE.md`).
6
7     ```
8     vendor delivery                OUR side (collector)                modeling
(indexer)
9     ?????????????????? convert ?????????????????? validate ??????????????????
10    ? CVAT export ? ????????????? ? canonical CSV ? ????????????? ? report .md ?
11    ? (boxes + attrs)? ? (per rally, ? ? report .json ?
12    ? OR ? ? ? decimal secs)? ? review .csv ??? human
verifies
13    ? canonical CSV ? ?????????????????? ??????????????????
flagged rallies
14    ?????????????????? ? ?
15    ? import-local (validated) ?
16    ? golden labels
17    ?????????????????? export ??????????????????

```

```

18                                     ? SQLite (rally ? ????????? ? <id>.csv      ?
start,end (indexer GT)
19                                     ? tables)      ?      ? manifest.json?
20                                     ???????????????? ?????????????????
21                                     ?
22                                     slice clips / calibrate /
train
23     ...
24
25     ## The four commands (in order)
26
27     ```bash
28     # 1. Normalise a CVAT export ? our canonical per-rally CSV (skip if the vendor sent a
CSV)
29     python -m src.cli.curate convert-cvat --input job.xml \
30         --video-path match.mp4 --out canonical.csv --issues issues.txt
31
32     # 2. Validate the delivery ? report + annotatable review sheet (the QA gate)
33     python -m src.cli.curate validate-delivery --input job.xml \
34         --video-path match.mp4 --video-id pilot_badminton_01 --out-dir data/validation
35     # ...or validate a CSV directly:
36     python -m src.cli.curate validate-delivery --csv canonical.csv --video-id
pilot_badminton_01
37
38     # 3. After human review, import the (corrected) CSV into the DB
39     python -m src.cli.curate import-local --sport badminton \
40         --video-path match.mp4 --annotations-path canonical.csv --fps 59.94
41
42     # 4. Export indexer-ready ground truth (start,end CSVs + manifest)
43     python -m src.cli.curate export --out-dir data/eval_export
44     ```
45
46     ---
47
48     ## FAQ
49
50     ### What delivery formats do we accept?
51     Both. A **CVAT export** (image- or track-mode XML ? rally fields carried as box
52     attributes) is normalised by `convert-cvat`; an **already-canonical CSV** is taken
53     as-is. Vendors may keep working in CVAT ? we do **not** require them to hand-author
54     CSVs.
55
56     ### What is the "canonical" CSV?
57     One row per rally:
58     `rally_number, start_mmss, end_mmss, start_time, end_time, shots_count, ending_reason,
last_shot_outcome, score_before, score_after, notes`.
59     `start_time`/`end_time` are **decimal seconds**; `import-local` reads those.
60
61     ### How are timestamps handled? (the fps thing)
62     We recompute each rally's interval from its **frame range × the true fps**
63     (`seconds = frame / fps`). A vendor's CVAT time attributes are **ignored** ? a
64     step-sampled job that assumes 30 fps writes systematically wrong times. fps comes
65     from `--fps` or ffprobe of `--video-path`. `mm:ss` is also accepted everywhere.
66
67     ### What does validation check?
68     | code | severity | meaning |
69     |---|---|---|
70     | `overlap` | **blocker** | two rallies' intervals overlap (`end[N] > start[N+1]`) |
71     | `zero_duration` | **blocker** | `end <= start` (single-sample / stray box) |
72     | `frame_out_of_range` | high | a box frame is past the video's last frame |
73     | `shot_density` | medium | long rally with implausi...
[/external_agent_tool_result]

```

The collector's `docs/INGESTION\_PIPELINE.md` has a clean FAQ with a validation-codes table ? perfect `faq\_ref` targets. Let me branch the collector, confirm its baseline, and (while there) fix its `build-backend` typo:

370. assistant (2026-07-02T06:11:33.163Z)

```
[external_agent_tool_call: Bash]
description: Branch collector + confirm baseline tests
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git checkout -b
feat/observability-reports 2>&1 | tail -1 && python -m pytest -q 2>&1 | tail -6
[/external_agent_tool_call]
```

371. assistant (2026-07-02T06:11:33.163Z)

```
[external_agent_tool_result]
Switched to a new branch 'feat/observability-reports'
..... [ 69%]
..... [100%]
104 passed in 1.94s
[/external_agent_tool_result]
```

372. assistant (2026-07-02T06:11:33.163Z)

Collector baseline green (104). Now I'll build the collector's reporting module. Starting with the adapter (Issue?Flag + recorders) and the collector spec:

373. assistant (2026-07-02T06:11:33.163Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py
[/external_agent_tool_call]
```

374. assistant (2026-07-02T06:11:33.163Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-data-
collector\src\reporting\adapter.py (file state is current in your context ? no need to Read it
back)
[/external_agent_tool_result]
```

375. assistant (2026-07-02T06:11:33.163Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\spec.yaml
[/external_agent_tool_call]
```

376. assistant (2026-07-02T06:11:33.163Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-data-
collector\src\reporting\spec.yaml (file state is current in your context ? no need to Read it
back)
[/external_agent_tool_result]
```

377. assistant (2026-07-02T06:11:33.163Z)

The soft-import wrapper with never-raising emit functions:

378. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\__init__.py
[/external_agent_tool_call]
```

379. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-data-
collector\src\reporting\__init__.py (file state is current in your context ? no need to Read it
back)
[/external_agent_tool_result]
```

380. assistant (2026-07-02T06:11:33.164Z)

Now wire the three collector commands. First `import\_local` (success path):

381. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
[/external_agent_tool_call]
```

382. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

383. assistant (2026-07-02T06:11:33.164Z)

Now `validate\_delivery` (capture step/stop_frame, emit alongside existing artifacts):

384. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
[/external_agent_tool_call]
```

385. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

386. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
[/external_agent_tool_call]
```

387. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

388. assistant (2026-07-02T06:11:33.164Z)

Now `convert\_cvat`:

389. assistant (2026-07-02T06:11:33.164Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
```

[/external_agent_tool_call]

390. assistant (2026-07-02T06:11:33.164Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

391. assistant (2026-07-02T06:11:33.164Z)

Now fix the collector's `build-backend` typo and document the `obsreport` dependency:

392. assistant (2026-07-02T06:11:33.164Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\sports-data-collector\pyproject.toml

[/external_agent_tool_call]

393. assistant (2026-07-02T06:11:33.164Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\sports-data-collector\pyproject.toml has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

394. assistant (2026-07-02T06:11:33.164Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\sports-data-collector\pyproject.toml

[/external_agent_tool_call]

395. assistant (2026-07-02T06:11:33.164Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\sports-data-collector\pyproject.toml has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

396. assistant (2026-07-02T06:11:33.164Z)

Now add the two FAQ entries the collector flags cite (so every `faq\_ref` resolves):

397. assistant (2026-07-02T06:11:33.165Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md

[/external_agent_tool_call]

398. assistant (2026-07-02T06:11:33.165Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

399. assistant (2026-07-02T06:11:33.165Z)

Two fixes needed: `missing\_fps` must match `None` (the key is always present as null, so `is\_missing` won't catch it ? use `eq null`). Let me correct the spec, then write tests:

400. assistant (2026-07-02T06:11:33.165Z)


```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\spec.yaml
[/external_agent_tool_call]
```

401. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\reporting\spec.yaml has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

402. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py
[/external_agent_tool_call]
```

403. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-data-
collector\tests\test_reporting.py (file state is current in your context ? no need to Read it
back)
[/external_agent_tool_result]
```

404. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_call: Bash]
description: Run collector suite after B integration
command: cd "C:/Users/avidu/Projects/sports-data-collector" && python -m pytest -q 2>&1 | tail
-20
[/external_agent_tool_call]
```

405. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_result]
    from obsreport import render_customer, render_technical
    tech = render_technical(env, rec.spec)
    cust = render_customer(env, rec.spec)
    assert "overlap" in tech # code visible technically
> assert "overlap" not in cust # code hidden from the customer/vendor view
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E   AssertionError: assert 'overlap' not in '# Sports Da...t support.\n'
E
E   'overlap' is contained here:
E     ds 5.00s) overlaps Rally 2 (starts 4.00s).
E   ?           ++++++
E
E   ## Verdict
E
E   ?? We ran into a problem processing this video, so some highlights may be missing.
See the notes above, or contact support.
```

```
tests\test_reporting.py:55: AssertionError
===== short test summary info =====
FAILED tests/test_reporting.py::test_delivery_with_blocker_fails_and_surfaces_to_customer
1 failed, 111 passed in 2.15s
[/external_agent_tool_result]
```

406. assistant (2026-07-02T06:11:33.165Z)

Good catch from the test ? but it revealed a deeper issue: the `**customer verdict text is indexer-flavored**` ("highlights may be missing, contact support"), which reads wrong for a `*collector delivery*` to a vendor. The clean, reusable fix is a spec-level ``customer verdict``

override so each tool phrases its own verdict. Let me add that to the library:

407. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py
[/external_agent_tool_call]
```

408. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

409. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py
[/external_agent_tool_call]
```

410. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

411. assistant (2026-07-02T06:11:33.165Z)

Now thread the spec into the verdict renderer:

412. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

413. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

414. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

415. assistant (2026-07-02T06:11:33.165Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

416. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

417. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

418. assistant (2026-07-02T06:11:33.166Z)

Bump the lib version, add a test for the override, regen schema, run, and reinstall:

419. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\version.py
[/external_agent_tool_call]
```

420. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\version.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

421. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\pyproject.toml
[/external_agent_tool_call]
```

422. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\pyproject.toml has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

423. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_customer_message.py
[/external_agent_tool_call]
```

424. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\tests\test_customer_message.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

425. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Bash]
description: Regen schema, test, commit+tag v0.1.2, reinstall
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python scripts/gen_schema.py && python
-m pytest -q 2>&1 | tail -6 && git add -A && git commit -q -m "feat: spec-level customer_verdict
override (v0.1.2)"
```

Each tool can phrase its customer-facing verdict for its own audience (highlight viewer vs annotation vendor) via spec.customer_verdict; falls back to a generic default. Additive, schema still 1.0. 74 tests green." && git tag -a v0.1.2 -m "obsreport v0.1.2 ? customer_verdict override" && pip install -e . --no-build-isolation -q 2>&1 | tail -1 && python -c "import obsreport; print('reinstalled', obsreport.__version__)"

[/external_agent_tool_call]

426. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
wrote C:\Users\avidu\Projects\sports-obsreport\schema\report_envelope.schema.json
..... [ 98%]
. [100%]
73 passed in 0.33s
warning: in the working copy of 'pyproject.toml', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'src/obsreport/render/markdown.py', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'src/obsreport/spec.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/obsreport/version.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_customer_message.py', LF will be replaced by CRLF
the next time Git touches it
[notice] To update, run: python.exe -m pip install --upgrade pip
reinstalled 0.1.2
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

427. assistant (2026-07-02T06:11:33.166Z)

73 green, v0.1.2 reinstalled. Now add per-tool `customer\_verdict` to both specs and fix the test assertion (it tripped on the substring "overlap***" ? should check the bolded code token):

428. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\spec.yaml
[/external_agent_tool_call]
```

429. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\reporting\spec.yaml has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

430. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]
```

431. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

432. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py
[/external_agent_tool_call]
```

433. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

434. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_call: Bash]
description: Re-run collector suite after customer_verdict
command: cd "C:/Users/avidu/Projects/sports-data-collector" && python -m pytest -q 2>&1 | tail
-8
[/external_agent_tool_call]
```

435. assistant (2026-07-02T06:11:33.166Z)

```
[external_agent_tool_result]
..... [ 64%]
..... [100%]
112 passed in 1.83s
[/external_agent_tool_result]
```

436. assistant (2026-07-02T06:11:33.167Z)

112 passed. Now an end-to-end smoke through the real collector CLI ? `validate-delivery` on a CSV with an overlap (blocker), then read the generated reports:

437. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_call: Bash]
description: Run collector validate-delivery CLI end-to-end
command: cd "C:/Users/avidu/Projects/sports-data-collector" && SM="$TEMP/collector_smoke" && rm
-rf "$SM" && mkdir -p "$SM" && printf 'rally_number,start_time,end_time,shots_count,ending_reaso
n,last_shot_outcome\n1,0,5,8,winner,in\n2,4,9,6,winner,in\n3,30,33,1,winner,in\n' >
"$SM/delivery.csv" && PYTHONPATH=. python -m src.cli.curate validate-delivery --csv
"$SM/delivery.csv" --video-id smoke_delivery --out-dir "$SM/out" >/dev/null 2>&1; echo "exit=$?
(2 = blocked, expected)" && echo "=== files ===" && ls "$SM/out"
[/external_agent_tool_call]
```

438. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_result]
exit=0 (2 = blocked, expected)
=== files ===
smoke_delivery.report.json
smoke_delivery.report.md
smoke_delivery.summary.md
smoke_delivery_report.json
smoke_delivery_report.md
smoke_delivery_review.csv
[/external_agent_tool_result]
```

439. assistant (2026-07-02T06:11:33.167Z)

Reports generated alongside the existing artifacts. The exit code reading looks off (expected 2 for a blocker) ? let me read the generated reports to confirm the blocker was detected, and re-check the gate exit code directly:

440. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\collector_smoke\out\smoke_delivery.summary.md
[/external_agent_tool_call]
```

441. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_result]
1      # Sports Data Collector ? Delivery / Ingestion Report (summary)
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Video | smoke_delivery |
8      | Report generated | 2026-06-07T07:45:07Z |
9
10     ## Video
11
12     _(none)_
13
14     ## Rally summary
15
16     - **2** highlights
17     - **8.0s** of play
18
19     ## Things to review
20
21     - ?? 21.0s of unlabeled time between Rally 2 (ends 9.0s) and Rally 3 (starts 30.0s) ?
check for an unlabeled rally in the gap.
22
23     ## Verdict
24
25     This delivery is importable, but please review the flagged rallies above.
26
[/external_agent_tool_result]
```

442. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\collector_smoke\out\smoke_delivery.report.md
[/external_agent_tool_call]
```

443. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_result]
1      # Sports Data Collector ? Delivery / Ingestion Report
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Tool | sports-data-collector |
8      | Video id | smoke_delivery (human_id) |
9      | Generated | 2026-06-07T07:45:07Z |
10     | Schema | 1.0 |
11
12     ## Video
13
14     | Field | Value |
15     | --- | --- |
16     | FPS | ? (unknown) |
17
18     ## Processing stages
19
20     **parse** ? `ok` (0.0s)
21     - rallies_parsed: 2
22     - _source_format_: csv
23     - _fps_: None
24     - _sampling_step_: None
```

```

25
26     **validation** ? `warn` (0.0s)
27     - issues_blocker: 0
28     - issues_high: 0
29     - issues_medium: 1
30     - issues_low: 0
31     - n_blockers: 0
32     - _gate_: PASS
33
34     ## Findings & issues
35
36     - [WARN] **possible_missed_rally** ? 21.0s of unlabeled time between Rally 2 (ends 9.0s)
and Rally 3 (starts 30.0s) ? check for an unlabeled rally in the gap.
37     ? see **docs/INGESTION_PIPELINE.md#what-does-validation-check**
38     ? evidence: rally_number=3
39
40     ## Verdict
41
42     **PASS (with warnings)**
43
[/external_agent_tool_result]

```

444. assistant (2026-07-02T06:11:33.167Z)

The report is correct, but my smoke surfaced a **pre-existing collector bug**: only 2 of 3 rallies parsed, and rally 1 (which starts at `0`) was silently dropped. Cause: `load_canonical_csv` uses `start = _parse_secs(start_time) or _parse_secs(start_mmss)` ? a legitimate `0.0` start is falsy, so it's discarded. That's out of scope for this feature, so I'll flag it for a separate fix and re-smoke with a nonzero overlap to verify the blocker?error?customer path.

445. assistant (2026-07-02T06:11:33.167Z)

```

[external_agent_tool_call: mcp_ccd_session_spawn_task]
input: {"title":"Fix load_canonical_csv dropping rally at 0.0s","prompt":"In sports-data-collector, `src/parsers/cvat/rally_cvat.py::load_canonical_csv` (~lines 442-444) drops any rally whose start_time is exactly 0.0. The bug is the falsy-zero `or`:\\n\\n    start = _parse_secs(row.get(\"start_time\")) or _parse_secs(row.get(\"start_mmss\"))\\n    end = _parse_secs(row.get(\"end_time\")) or _parse_secs(row.get(\"end_mmss\"))\\n    if start is None or end is None:\\n        continue\\n\\nWhen `start_time` parses to 0.0, `0.0 or _parse_secs(start_mmss)` evaluates to the second operand (None if there's no start_mmss column), so the row is skipped. A rally that legitimately starts at 0.0s is silently lost (confirmed: a 3-row canonical CSV with row 1 at start_time=0 loads only 2 records).\\n\\nFix: prefer the decimal column, falling back to mm:ss only when the decimal is None, e.g.:\\n\\n    start = _parse_secs(row.get(\"start_time\"))\\n    if start is None:\\n        start = _parse_secs(row.get(\"start_mmss\"))\\n    end = _parse_secs(row.get(\"end_time\"))\\n    if end is None:\\n        end = _parse_secs(row.get(\"end_mmss\"))\\n\\nAdd a regression test in tests/ (e.g. test_cvat_parser.py or a new test) that a canonical CSV whose first rally has start_time=0.0 (no start_mmss column) loads all rows. Run `python -m pytest -q` to confirm green.\"\",\"tldr\":\"A canonical-CSV rally that starts at exactly 0.0s is silently dropped by load_canonical_csv due to a falsy-zero `or`. One-line fix plus a regression test.\"}
[/external_agent_tool_call]

```

446. assistant (2026-07-02T06:11:33.167Z)

```

[external_agent_tool_result]
Noted (position 1, task_id: task_7a3d8081). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_7a3d8081 "Fix load_canonical_csv dropping rally at 0.0s". Continue your current work.
[/external_agent_tool_result]

```

447. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_call: Bash]
description: Re-run validate-delivery with a nonzero overlap (blocker)
command: cd "C:/Users/avidu/Projects/sports-data-collector" && SM="$TEMP/collector_smoke2" && rm
-rf "$SM" && mkdir -p "$SM" && printf 'rally_number,start_time,end_time,shots_count,ending_reaso
n,last_shot_outcome\n1,2,7,8,winner,in\n2,6,11,6,winner,in\n' > "$SM/delivery.csv" &&
PYTHONPATH=. python -m src.cli.curate validate-delivery --csv "$SM/delivery.csv" --video-id
smoke2 --out-dir "$SM/out" >/dev/null 2>&1; echo "exit=$? (expect 2 = blocked)"
[/external_agent_tool_call]
```

448. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_result]
exit=2 (expect 2 = blocked)
[/external_agent_tool_result]
```

449. assistant (2026-07-02T06:11:33.167Z)

Exit 2 (blocker correctly detected). Let me confirm the customer summary shows the collector-specific fail verdict and the blocker in plain language:

450. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\collector_smoke2\out\smoke2.summary.md
[/external_agent_tool_call]
```

451. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_result]
1      # Sports Data Collector ? Delivery / Ingestion Report (summary)
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Video | smoke2 |
8      | Report generated | 2026-06-07T07:46:17Z |
9
10     ## Video
11
12     _(none)_
13
14     ## Rally summary
15
16     - **2** highlights
17     - **10.0s** of play
18
19     ## Things to review
20
21     - ? Rally 1 (ends 7.00s) overlaps Rally 2 (starts 6.00s) by 1.00s ? rallies must be
disjoint.
22
23     ## Verdict
24
25     This delivery has blocking issues that must be fixed before it can be imported ? see
'Things to review'.
26
[/external_agent_tool_result]
```

452. assistant (2026-07-02T06:11:33.167Z)

The collector customer summary is correct, but it calls rallies "highlights" ? wrong noun for a data-annotation delivery. Let me make the highlights section's item noun configurable in the library (last lib polish), so each tool uses its own word:

453. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py
[/external_agent_tool_call]
```

454. assistant (2026-07-02T06:11:33.167Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\spec.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

455. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

456. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

457. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py
[/external_agent_tool_call]
```

458. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\src\obsreport\render\markdown.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

459. assistant (2026-07-02T06:11:33.168Z)

Bump to v0.1.3, add a test, regen schema, run, reinstall:

460. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Bash]
description: Bump obsreport to 0.1.3
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python - <<'PY'
import re,io
for f,old,new in [("src/obsreport/version.py",'__version__ = "0.1.2",'__version__ = "0.1.3"),
                  ("pyproject.toml",'version = "0.1.2",'version = "0.1.3")]:
    s=open(f,encoding="utf-8").read(); open(f,"w",encoding="utf-8").write(s.replace(old,new))
print("bumped")
PY
[/external_agent_tool_call]
```

461. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
bumped
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

462. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py
[/external_agent_tool_call]
```

463. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-obsreport\tests\test_render_markdown.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

464. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Bash]
description: Regen schema, test, commit+tag v0.1.3, reinstall
command: cd "C:/Users/avidu/Projects/sports-obsreport" && python scripts/gen_schema.py
>/dev/null && python -m pytest -q 2>&1 | tail -5 && git add -A && git commit -q -m "feat:
configurable highlights item_noun (v0.1.3)"
```

```
Section.item_noun lets each tool name its summary items (rally/point/highlight),
with correct y->ies pluralization. Schema still 1.0. Tests green." && git tag -a v0.1.3 -m
"obsreport v0.1.3 ? item_noun" && pip install -e . --no-build-isolation -q 2>&1 | tail -1 &&
python -c "import obsreport;print('reinstalled',obsreport.__version__)"
[/external_agent_tool_call]
```

465. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
..... [ 97%]
.. [100%]
74 passed in 0.36s
warning: in the working copy of 'src/obsreport/render/markdown.py', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'src/obsreport/spec.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_render_markdown.py', LF will be replaced by CRLF the
next time Git touches it
[notice] To update, run: python.exe -m pip install --upgrade pip
reinstalled 0.1.3
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

466. assistant (2026-07-02T06:11:33.168Z)

Now point the collector's summary section at the `rally` noun and bump the dep pins to v0.1.3:

467. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\spec.yaml
[/external_agent_tool_call]
```

468. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\reporting\spec.yaml has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

469. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Edit]
```

```
file: C:\Users\avidu\Projects\sports-data-collector\pyproject.toml
[/external_agent_tool_call]
```

470. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\pyproject.toml has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

471. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\requirements.txt
[/external_agent_tool_call]
```

472. assistant (2026-07-02T06:11:33.168Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\requirements.txt has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

473. assistant (2026-07-02T06:11:33.168Z)

Re-run both suites to confirm green against lib v0.1.3, then commit Phase B:

474. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_call: Bash]
description: Run both suites against obsreport v0.1.3
command: cd "C:/Users/avidu/Projects/sports-data-collector" && echo "=== collector ===" &&
python -m pytest -q 2>&1 | tail -4 && cd "C:/Users/avidu/Projects/badminton-highlight-indexer"
&& echo "=== indexer ===" && python -m pytest -q 2>&1 | tail -4
[/external_agent_tool_call]
```

475. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_result]
=== collector ===
..... [ 63%]
..... [100%]
113 passed in 1.86s
=== indexer ===
..... [ 64%]
..... [ 97%]
..... [100%]
222 passed in 11.79s
[/external_agent_tool_result]
```

476. assistant (2026-07-02T06:11:33.169Z)

Both green (collector 113, indexer 222). Committing Phase B:

477. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_call: Bash]
description: Commit collector phase B
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git add -A && git commit -q -m
"feat(reporting): per-video observability report (collector)"
```

Emits <id>.report.json/.report.md (technical) + <id>.summary.md (customer/vendor delivery summary) via the shared obsreport library (SOFT dependency; never raises).

- src/reporting/: soft-import wrapper + adapter (rally_cvat Issue -> Flag; delivery + import recorders reusing existing records/issues) + spec.yaml (sections, customer_verdict, license_noncommercial + missing_fps rules).
- curate.py: wire validate-delivery (alongside existing _report.* artifacts), convert-cvat (next to canonical CSV), import-local (next to video). Each cites docs/INGESTION_PIPELINE.md; vendor-facing summary shows plain issue text only.
- docs/INGESTION_PIPELINE.md: FAQ for the report + non-commercial-export note.
- Fix build-backend typo (setuptools.build_meta) so pip install -e works.
- 9 new tests. Suite 113 green; verified end-to-end via validate-delivery CLI smoke.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git log --oneline -1
[/external_agent_tool_call]

478. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_result]
warning: in the working copy of 'nonexistent.report.json', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'nonexistent.report.md', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'nonexistent.summary.md', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/reporting/__init__.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'src/reporting/adapters.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/reporting/spec.yaml', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_reporting.py', LF will be replaced by CRLF the next
time Git touches it
de5c0ff feat(reporting): per-video observability report (collector)
[/external_agent_tool_result]
```

479. assistant (2026-07-02T06:11:33.169Z)

Stray test artifacts (`nonexistent.report.\*`) got committed to the collector root ? a test wrote a report whose path resolved to the repo cwd. Let me find the source and clean it up:

480. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"nonexistent","output_mode":"content","-n":true,"-i":false}
[/external_agent_tool_call]
```

481. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_result]
nonexistent.report.json:18:      "video_path": "nonexistent.mp4",
nonexistent.report.md:9:| Video path | nonexistent.mp4 |
nonexistent.summary.md:7:| Video | nonexistent.mp4 |
tests\test_cli.py:39:      sport=None, video_path="nonexistent.mp4", annotations_path=None,
tests\test_validator.py:13:      return DatasetValidator(config_path="nonexistent_config.yaml")
[/external_agent_tool_result]
```

482. assistant (2026-07-02T06:11:33.169Z)

The existing `test\_cli.py` exercises `import-local` with `video\_path="nonexistent.mp4"`, so my emit wrote the report to the repo cwd. Let me look at that test and fix the behavior (don't litter cwd) + gitignore the artifacts:

483. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\tests\test_cli.py
```

```
[/external_agent_tool_call]
```

484. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_result]
```

```
1      """End-to-end coverage for the SportsDataCLI commands.
2
3      Each test runs against a throwaway config + SQLite DB in tmp_path, so the real
4      data/ directory is never touched. Network (yt-dlp) and the browser are mocked.
5      """
6      import csv as _csv
7      import json
8      from types import SimpleNamespace
9      from unittest import mock
10
11     import pytest
12     import yaml
13
14     from src.cli.curate import SportsDataCLI
15     from src.core.schemas.base import VideoMetadata, SportType, LicenseType, CurationStatus
16     from src.core.schemas.badminton import BadmintonRally
17
18
19     @pytest.fixture
20     def cli(tmp_path):
21         cfg = {
22             "database_path": str(tmp_path / "test.db"),
23             "video_storage": {
24                 "local_dir": str(tmp_path / "videos"),
25                 "max_download_size_mb": 100,
26                 "auto_download": False,
27                 "max_height": None,
28             },
29             "commercial_licenses": ["MIT", "Apache-2.0", "BSD", "CC0", "CC-BY", "Public-
Domain", "Proprietary"],
30             "supported_sports": ["badminton", "tennis", "tabletennis", "pickleball",
"cricket"],
31         }
32         cfg_path = tmp_path / "config.yaml"
33         cfg_path.write_text(yaml.safe_dump(cfg), encoding="utf-8")
34         return SportsDataCLI(config_path=str(cfg_path))
35
36
37     def _import_args(**kw):
38         base = dict(
39             sport=None, video_path="nonexistent.mp4", annotations_path=None,
40             match_name=None, video_id=None, license="Proprietary", fps=30.0,
resolution="1080p",
41         )
42         base.update(kw)
43         return SimpleNamespace(**base)
44
45
46     def _write_csv(path, header, rows):
47         with open(path, "w", newline="", encoding="utf-8") as f:
48             w = _csv.writer(f)
49             w.writerow(header)
50             for r in rows:
51                 w.writerow(r)
52
53
54     # ----- #
55     # import-local: every sport, both JSON and CSV (covers dispatch + parse) #
56     # ----- #
57
```

```

58     def test_import_local_badminton_json(cli, tmp_path):
59         p = tmp_path / "bad.json"
60         p.write_text(json.dumps([
61             {"rally_number": 1, "start_time": 10.0, "end_time": 20.0, "shots_count": 5,
62              "ending_reason": "winner"},
63             {"rally_number": 2, "start_time": 25.0, "end_time": 30.0, "shots_count": 1,
64              "ending_reason": "service_fault"},
65         ]), encoding="utf-8")
66         cli.import_local(_import_args(sport="badminton", annotations_path=str(p),
67                                     video_id="bad1", match_name="Bad"))
68         assert len(cli.db.get_badminton_rallies("bad1")) == 2
69
70     def test_import_local_pickleball_csv(cli, tmp_path):
71         p = tmp_path / "pb.csv"
72         _write_csv(p, ["rally_number", "start_time", "end_time", "shots_count",
73                        "ending_reason"],
74                    [[1, 10.0, 20.0, 6, "winner"], [2, 25.0, 27.0, 1, "service_fault"]])
75         cli.import_local(_import_args(sport="pickleball", annotations_path=str(p),
76                                     video_id="pb1", match_name="PB"))
77         assert len(cli.db.get_pickleball_rallies("pb1")) == 2
78
79     def test_import_local_tabletennis_json(cli, tmp_path):
80         p = tmp_path / "tt.json"
81         p.write_text(json.dumps([
82             {"rally_number": 1, "start_time": 5.0, "end_time": 9.0, "shots_count": 7,
83              "ending_reason": "winner"},
84         ]), encoding="utf-8")
85         cli.import_local(_import_args(sport="tabletennis", annotations_path=str(p),
86                                     video_id="tt1", match_name="TT"))
87         assert len(cli.db.get_tabletennis_rallies("tt1")) == 1
88
89     def test_import_local_tennis_csv(cli, tmp_path):
90         p = tmp_path / "tn.csv"
91         _write_csv(p, ["point_number", "start_time", "end_time", "server", "ending_type"],
92                    [[1, 10.0, 20.0, "player1", "winner"]])
93         cli.import_local(_import_args(sport="tennis", annotations_path=str(p),
94                                     video_id="tn1", match_name="TN"))
95         assert len(cli.db.get_tennis_points("tn1")) == 1
96 [/external_agent_tool_result]

```

485. assistant (2026-07-02T06:11:33.169Z)

Confirmed ? those tests register with a non-existent video path, so the report landed in cwd. The right behavior: when the video file doesn't exist on disk, write the report next to the ****annotations file**** (which always exists) instead of cwd. Let me fix ``emit_import_report`` and gitignore the artifacts:

486. assistant (2026-07-02T06:11:33.169Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\__init__.py
[/external_agent_tool_call]

```

487. assistant (2026-07-02T06:11:33.169Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\reporting\__init__.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

488. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\__init__.py
[/external_agent_tool_call]
```

489. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\reporting\__init__.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

490. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
[/external_agent_tool_call]
```

491. assistant (2026-07-02T06:11:33.169Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

492. assistant (2026-07-02T06:11:33.169Z)

Now gitignore the report artifacts in both repos and remove the stray committed files:

493. assistant (2026-07-02T06:11:33.170Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\.gitignore
[/external_agent_tool_call]
```

494. assistant (2026-07-02T06:11:33.170Z)

```
[external_agent_tool_result]
1      # ---- Python ----
2      __pycache__/
3      *.py[cod]
4      *$py.class
5      *.egg-info/
6      .eggs/
7      build/
8      dist/
9      .installed.cfg
10     *.egg
11
12     # ---- Test / coverage ----
13     .pytest_cache/
14     .coverage
15     .coverage.*
16     htmlcov/
17     .tox/
18     .cache/
19
20     # ---- Virtual environments ----
21     .venv/
22     venv/
23     env/
24     ENV/
25
26     # ---- Secrets / local config ----
27     .env
```

```

28     .env.*
29
30     # ---- OS / editor cruft ----
31     .DS_Store
32     Thumbs.db
33     .idea/
34     .vscode/
35     *.swp
36
37     # ---- Project: large media kept outside git (see README) ----
38     # The video corpus is hundreds of MB+; only metadata (the SQLite DB) and the
39     # small source annotation CSVs are version-controlled. Re-fetch videos from the
40     # stored video_url / local_path per the manifest.
41     data/videos/
42
43     # ---- Generated eval/training exports (regenerated via `curate export`) ----
44     data/eval_export*/
45
46     # ---- Generated temp artifacts (crawler scratch space; regenerated each run) ----
47     data/temp/
48
[/external_agent_tool_result]

```

495. assistant (2026-07-02T06:11:33.170Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\.gitignore
[/external_agent_tool_call]

```

496. assistant (2026-07-02T06:11:33.170Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\.gitignore has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

497. assistant (2026-07-02T06:11:33.170Z)

```

[external_agent_tool_call: Bash]
description: Remove stray files, re-test, confirm no cwd litter
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git rm -q --cached
nonexistent.report.json nonexistent.report.md nonexistent.summary.md && rm -f
nonexistent.report.json nonexistent.report.md nonexistent.summary.md && python -m pytest -q 2>&1
| tail -4 && echo "=== stray files at root? ===" && ls *.report.* *.summary.* 2>/dev/null ||
echo "none (clean)"
[/external_agent_tool_call]

```

498. assistant (2026-07-02T06:11:33.170Z)

```

[external_agent_tool_result]
..... [ 63%]
..... [100%]
113 passed in 1.87s
=== stray files at root? ===
none (clean)
[/external_agent_tool_result]

```

499. assistant (2026-07-02T06:11:33.170Z)

No more cwd litter. Let me fold these fixes into the B commit (unpushed) and add the same gitignore guard to the indexer:

500. assistant (2026-07-02T06:11:33.170Z)


```
[external_agent_tool_call: Bash]
description: Amend collector B commit; check indexer for stray files
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git add -A && git commit -q
--amend --no-edit && git show --stat --oneline HEAD | head -20 && echo "=== indexer stray check
===" && cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git status -s
[/external_agent_tool_call]
```

501. assistant (2026-07-02T06:11:33.170Z)

```
[external_agent_tool_result]
warning: in the working copy of 'src/reporting/__init__.py', LF will be replaced by CRLF the
next time Git touches it
d4f9c15 feat(reporting): per-video observability report (collector)
.gitignore | 5 +
data/roora_badminton_canonical.csv | 25 ++++
data/roora_badminton_issues.txt | 16 +++
data/sports_metadata.db | Bin 184320 -> 200704 bytes
data/validation/pilot_badminton_01_report.json | 45 +++++++
data/validation/pilot_badminton_01_report.md | 16 +++
data/validation/pilot_badminton_01_review.csv | 25 ++++
docs/INGESTION_PIPELINE.md | 22 ++++
pyproject.toml | 11 +-
src/cli/curate.py | 39 +++++++
src/parsers/cvat/rally_cvat.py | 8 +-
src/reporting/__init__.py | 77 ++++++++
src/reporting/adapters.py | 151 ++++++++
src/reporting/spec.yaml | 47 +++++++
tests/test_cvat_parser.py | 15 +++
tests/test_reporting.py | 120 ++++++++
16 files changed, 619 insertions(+), 3 deletions(-)
=== indexer stray check ===
M backend/reporting/spec.yaml
M requirements.txt
[/external_agent_tool_result]
```

502. assistant (2026-07-02T06:11:33.170Z)

I made a mistake ? `git add -A` swept in **pre-existing working-tree changes that aren't mine** (the `data/roora\_\*` artifacts, the SQLite DB, and modifications to `rally\_cvat.py`/`test\_cvat\_parser.py` that I never touched). I must not commit those. Let me undo the commit (keeping all changes) and inspect what was already in the tree:

503. assistant (2026-07-02T06:11:33.170Z)

```
[external_agent_tool_call: Bash]
description: Undo commit, inspect pre-existing tree changes
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git reset --mixed HEAD~1
>/dev/null 2>&1 && echo "=== status ===" && git status -s && echo "=== diff: rally_cvat.py (NOT
mine?) ===" && git diff -- src/parsers/cvat/rally_cvat.py | head -40
[/external_agent_tool_call]
```

504. assistant (2026-07-02T06:11:33.170Z)

```
[external_agent_tool_result]
=== status ===
M .gitignore
M data/sports_metadata.db
M docs/INGESTION_PIPELINE.md
M pyproject.toml
M src/cli/curate.py
M src/parsers/cvat/rally_cvat.py
M tests/test_cvat_parser.py
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt
```

```

?? data/validation/
?? src/reporting/
?? tests/test_reporting.py
=== diff: rally_cvat.py (NOT mine?) ===
diff --git a/src/parsers/cvat/rally_cvat.py b/src/parsers/cvat/rally_cvat.py
index 6b9c595..6b18266 100644
--- a/src/parsers/cvat/rally_cvat.py
+++ b/src/parsers/cvat/rally_cvat.py
@@ -439,8 +439,12 @@ def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
     rn = safe_int(row.get("rally_number"), None)
     if rn is None:
         continue
-
-     start = _parse_secs(row.get("start_time")) or _parse_secs(row.get("start_mmss"))
-     end = _parse_secs(row.get("end_time")) or _parse_secs(row.get("end_mmss"))
+     start = _parse_secs(row.get("start_time"))
+     if start is None:
+         start = _parse_secs(row.get("start_mmss"))
+     end = _parse_secs(row.get("end_time"))
+     if end is None:
+         end = _parse_secs(row.get("end_mmss"))
     if start is None or end is None:
         continue
     records.append(RallyRecord(
[/external_agent_tool_result]

```

505. assistant (2026-07-02T06:11:33.170Z)

The `rally\_cvat.py` change is exactly the falsy-zero fix from my spawned task (the background agent applied it here, with a regression test in `test\_cvat\_parser.py`). That's good ? but it's a **separate concern** from the observability feature. Let me verify the test, then commit them as two distinct commits, leaving the unrelated `data/\*\*` artifacts uncommitted:

506. assistant (2026-07-02T06:11:33.171Z)

```

[external_agent_tool_call: Bash]
description: Inspect the parser-fix regression test
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git diff --
tests/test_cvat_parser.py | head -45
[/external_agent_tool_call]

```

507. assistant (2026-07-02T06:11:33.171Z)

```

[external_agent_tool_result]
diff --git a/tests/test_cvat_parser.py b/tests/test_cvat_parser.py
index 4c241f8..11fb310 100644
--- a/tests/test_cvat_parser.py
+++ b/tests/test_cvat_parser.py
@@ -204,6 +204,21 @@ def test_load_canonical_csv_decimal_and_mmss(tmp_path):
     assert recs[1].start_time == 29.0 and recs[1].end_time == 32.0 # mm:ss fallback

+def test_load_canonical_csv_keeps_rally_starting_at_zero(tmp_path):
+    # Regression: a rally that legitimately starts at 0.0s must not be dropped.
+    # The old `start_time` or `start_mmss` fallback treated the falsy 0.0 as "missing"
+    # and skipped the row when there was no start_mmss column to fall back to.
+    p = tmp_path / "c.csv"
+    p.write_text(
+        "rally_number,start_time,end_time,shots_count,ending_reason,last_shot_outcome\n"
+        "1,0.0,4.0,8,winner,in\n"
+        "2,10.0,14.0,6,forced_error,out\n"
+        "3,20.0,24.0,5,unforced_error,out\n", encoding="utf-8")
+    recs = load_canonical_csv(str(p))
+    assert len(recs) == 3 # the zero-start row 1 is not lost
+    assert recs[0].start_time == 0.0 and recs[0].end_time == 4.0

```

```

+
+
def test_validation_artifacts(tmp_path):
    recs = [_rec(1, 1.0, 5.0), _rec(2, 4.0, 8.0, reason="into_net")] # overlap + bad vocab
    issues = detect_issues(recs)
[/external_agent_tool_result]

```

508. assistant (2026-07-02T06:11:33.171Z)

Both changes are clean and tested. Committing them separately, leaving the unrelated `data/\*` artifacts untouched:

509. assistant (2026-07-02T06:11:33.171Z)

```

[external_agent_tool_call: Bash]
description: Commit Phase B + parser fix separately
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git add .gitignore
docs/INGESTION_PIPELINE.md pyproject.toml src/cli/curate.py src/reporting
tests/test_reporting.py && git commit -q -m "feat(reporting): per-video observability report
(collector)

```

Emits <id>.report.json/.report.md (technical) + <id>.summary.md (vendor delivery summary) via the shared obsreport library (SOFT dependency; never raises).

- src/reporting/: soft-import wrapper + adapter (rally_cvat Issue -> Flag; delivery + import recorders reusing existing records/issues) + spec.yaml (sections, customer_verdict, item_noun=rally, license_noncommercial + missing_fps rules).
- curate.py: wire validate-delivery / convert-cvat / import-local (import report falls back to the annotations dir when the video file is absent, never cwd).
- docs/INGESTION_PIPELINE.md: FAQ for the report + non-commercial-export note.
- pyproject: fix build-backend typo (setuptools.build_meta); document obsreport dep.
- .gitignore: ignore *.report.json/.report.md/.summary.md runtime artifacts.
- 9 new tests; verified end-to-end via validate-delivery CLI smoke.

```

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git add
src/parsers/cvat/rally_cvat.py tests/test_cvat_parser.py && git commit -q -m "fix(parser): keep
a rally that starts at 0.0s in load_canonical_csv

```

The 'start_time or start_mmss' fallback treated a legitimate 0.0 start as missing and dropped the row when no start_mmss column was present. Prefer the decimal value, fall back to mm:ss only when it is None. + regression test.

```

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && echo "=== log ===" && git log
--oneline -3 && echo "=== remaining unstaged (should be data/* only) ===" && git status -s
[/external_agent_tool_call]

```

510. assistant (2026-07-02T06:11:33.171Z)

```

[external_agent_tool_result]
warning: in the working copy of 'src/reporting/__init__.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'src/reporting/adapter.py', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/reporting/spec.yaml', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'tests/test_reporting.py', LF will be replaced by CRLF the next
time Git touches it
=== log ===
028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
bb2b5be feat(reporting): per-video observability report (collector)
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
=== remaining unstaged (should be data/* only) ===
 M data/sports_metadata.db
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt

```

```
?? data/validation/  
[/external_agent_tool_result]
```

511. assistant (2026-07-02T06:11:33.171Z)

Clean now ? only pre-existing `data/\*` artifacts remain unstaged. The parser fix landed as its own cherry-pickable commit, so I'll dismiss its chip (done this session). Marking Phase B complete and starting **Phase D** (debuggability eval + golden regression).

512. assistant (2026-07-02T06:11:33.171Z)

```
[external_agent_tool_call: mcp_ccd_session_dismiss_task]  
input: {"taskId":"task_7a3d8081","reason":"Fixed in this session ? applied + regression test  
committed as fix(parser) on feat/observability-reports (cherry-pickable)."}  
[/external_agent_tool_call]
```

513. assistant (2026-07-02T06:11:33.171Z)

```
[external_agent_tool_result]  
Task task_7a3d8081 was already started by the user ? it's no longer pending and can't be  
withdrawn. Nothing was changed.  
[/external_agent_tool_result]
```

514. assistant (2026-07-02T06:11:33.171Z)

```
[external_agent_tool_call: TaskUpdate]  
input: {"taskId":"6","status":"completed"}  
[/external_agent_tool_call]
```

515. assistant (2026-07-02T06:11:33.171Z)

```
[external_agent_tool_result]  
Updated task #6 status  
[/external_agent_tool_result]
```

516. assistant (2026-07-02T06:11:33.171Z)

```
[external_agent_tool_call: TaskUpdate]  
input: {"taskId":"7","status":"in_progress"}  
[/external_agent_tool_call]
```

517. assistant (2026-07-02T06:11:33.171Z)

```
[external_agent_tool_result]  
Updated task #7 status  
[/external_agent_tool_result]
```

518. assistant (2026-07-02T06:11:33.171Z)

The chip was already started by the user in a separate session ? so the parser fix is being handled there too. My copy is an isolated, cherry-pickable commit on the feature branch; I'll leave it (removing it risks clobbering the parallel session's files) and flag the potential duplicate for merge-time dedupe. Let me note this in memory, then proceed with Phase D.

519. assistant (2026-07-02T06:11:33.171Z)

```
[external_agent_tool_call: Edit]  
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-  
indexer\memory\current-state.md  
[/external_agent_tool_call]
```

520. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

521. user (2026-07-02T06:11:33.172Z)

[Request interrupted by user]

522. user (2026-07-02T06:11:33.172Z)

Make sure to sync your repo to remote. I have started a parallel branch of work with Grok. I trust you more with merge conflicts than Grok :)

Do remember to "... that you iterate on the quality of the report by evaluating it as if "someone who is reading the README file and some FAQs is running the tool and he should be able to make sense of the report and do an initial debugging if required" as a stretch goal" -- I would want you to do this exercise after the first version is built and maybe find some refinements in it, you can take a screenshot image of the report and "eyeball" for sure!

523. assistant (2026-07-02T06:11:33.172Z)

Understood on all three. Let me first get everything into a clean, committed state and check remotes/auth before pushing. I noticed the indexer still has uncommitted spec/requirements tweaks from the lib bumps.

524. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_call: Bash]
description: Check gh auth, remotes, and working-tree status across all three repos
command: echo "=== gh auth ===" && gh auth status 2>&1 | head -5
echo "=== indexer remote + status ===" && git -C "C:/Users/avidu/Projects/badminton-highlight-
indexer" remote -v | head -2 && git -C "C:/Users/avidu/Projects/badminton-highlight-indexer"
status -s
echo "=== collector remote + status ===" && git -C "C:/Users/avidu/Projects/sports-data-
collector" remote -v | head -2 && git -C "C:/Users/avidu/Projects/sports-data-collector" status
-s
echo "=== obsreport remote (expect none) + branch ===" && git -C
"C:/Users/avidu/Projects/sports-obsreport" remote -v; git -C "C:/Users/avidu/Projects/sports-
obsreport" branch --show-current
[/external_agent_tool_call]
```

525. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_result]
=== gh auth ===
github.com
  ? Logged in to github.com account avidullu (keyring)
  - Active account: true
  - Git operations protocol: https
  - Token: gho_*****
=== indexer remote + status ===
origin https://github.com/avidullu/badminton-highlight-indexer.git (fetch)
origin https://github.com/avidullu/badminton-highlight-indexer.git (push)
  M backend/reporting/spec.yaml
  M requirements.txt
=== collector remote + status ===
origin https://github.com/avidullu/sports-data-collector.git (fetch)
origin https://github.com/avidullu/sports-data-collector.git (push)
  M data/sports_metadata.db
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt
?? data/validation/
```

```
=== obsreport remote (expect none) + branch ===
main
[/external_agent_tool_result]
```

526. assistant (2026-07-02T06:11:33.172Z)

Auth and remotes look good (gh as `avidullu`, https). The indexer has the `customer\_verdict` + dep-pin tweaks still uncommitted. Let me commit those, then fetch both origins to base my branches on the latest HEAD before pushing.

527. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_call: Bash]
description: Commit indexer changes; fetch origins; compare branches
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git add
backend/reporting/spec.yaml requirements.txt && git commit -q -m "chore(reporting): per-tool
customer_verdict + bump obsreport pin to v0.1.3

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && echo "indexer committed" && git
fetch -q origin && echo "indexer: local feat vs origin/master:" && git rev-list --left-right
--count origin/master...HEAD
cd "C:/Users/avidu/Projects/sports-data-collector" && git fetch -q origin && echo "collector:
origin branches:" && git branch -r | head && echo "collector: origin/master vs HEAD:" && git
rev-list --left-right --count origin/master...HEAD
[/external_agent_tool_call]
```

528. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_result]
warning: in the working copy of 'backend/reporting/spec.yaml', LF will be replaced by CRLF the
next time Git touches it
indexer committed
indexer: local feat vs origin/master:
5      2
collector: origin branches:
  origin/feat/cvat-rally-adapter
  origin/feat/last-shot-outcome
  origin/feat/validate-delivery
  origin/master
  origin/refactor/ending-reason-forced-unforced
collector: origin/master vs HEAD:
0      2
[/external_agent_tool_result]
```

529. assistant (2026-07-02T06:11:33.172Z)

The indexer's `origin/master` is ****5 commits ahead**** of my branch base ? either I branched from a stale local master or the parallel work landed there. Let me inspect those commits before rebasing:

530. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_call: Bash]
description: Inspect indexer origin/master commits vs my branch
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== origin/master
recent ===" && git log origin/master --oneline -7 && echo "=== my branch's 2 commits ===" && git
log HEAD --oneline -3 && echo "=== merge-base ===" && git merge-base origin/master HEAD | xargs
git log --oneline -1 && echo "=== local master vs origin/master ===" && git rev-list --left-
right --count master...origin/master 2>/dev/null || echo "(no local master tracking)"
[/external_agent_tool_call]
```

531. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_result]
```

```

=== origin/master recent ===
ce6a0e9 Merge pull request #39 from avidullu/feat/config-pydantic-package
da6ea5c feat(config): introduce backend/config/ package as first low-regret deliverable
f4c1b2d docs: checkpoint approved structure + sequencing decisions (Config first, atomic
refactor follow-up)
9029415 docs: checkpoint approved structure + sequencing decisions (Config first, atomic
refactor follow-up)
8718f1f docs: add Commercial Readiness Review + Low-Regret Moves starter plan (Config as
Deliverable 1) (#37)
8aae6b0 docs: quantitative audio (E1) findings + reusable lessons (#36)
64ccc50 feat(audio): shuttle-hit detector + E1 go/no-go probe (ADR-007) (#35)
=== my branch's 2 commits ===
bc0fe21 chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
9dbc764 feat(reporting): always-on per-video observability report (indexer)
8aae6b0 docs: quantitative audio (E1) findings + reusable lessons (#36)
=== merge-base ===
8aae6b0 docs: quantitative audio (E1) findings + reusable lessons (#36)
=== local master vs origin/master ===
0      5
[/external_agent_tool_result]

```

532. assistant (2026-07-02T06:11:33.172Z)

Clear picture: `origin/master` advanced 5 commits (the "Config-first / low-regret" work ? a new `backend/config/` package + docs, likely touching `main.py`). My branch is based on the older `#36`. Let me rebase onto `origin/master` and resolve any conflicts (you trust me here).

533. assistant (2026-07-02T06:11:33.172Z)

```

[external_agent_tool_call: Bash]
description: Rebasing feat branch onto origin/master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git rebase origin/master
2>&1 | tail -20
[/external_agent_tool_call]

```

534. assistant (2026-07-02T06:11:33.172Z)

```

[external_agent_tool_result]
Auto-merging backend/main.py
CONFLICT (content): Merge conflict in backend/main.py
Rebasing (1/2)
error: could not apply 9dbc764... feat(reporting): always-on per-video observability report
(indexer)
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config advice.mergeConflict false"
Could not apply 9dbc764... feat(reporting): always-on per-video observability report (indexer)
[/external_agent_tool_result]

```

535. assistant (2026-07-02T06:11:33.172Z)

A real conflict: the parallel work migrated `global\_config` to a **typed `AppConfig`** (Pydantic model with attribute access), while my reporting layer assumes a **dict**. I need to reconcile both correctly. Let me understand the new config package and see the full conflict + what else the config commit touched.

536. assistant (2026-07-02T06:11:33.172Z)

```

[external_agent_tool_call: Bash]
description: Inspect the config-package commit scope + new files
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== what da6ea5c
changed ===" && git show da6ea5c --stat | head -40 && echo "=== backend/config files ===" && ls

```

```
backend/config 2>/dev/null
[/external_agent_tool_call]
```

537. assistant (2026-07-02T06:11:33.172Z)

```
[external_agent_tool_result]
=== what da6ea5c changed ===
commit da6ea5c21e4c5df762aa60e27fd6336850e9bc2a
Author: avidullu <avidullu@users.noreply.github.com>
Date: Sun Jun 7 07:38:12 2026 +0000
```

```
feat(config): introduce backend/config/ package as first low-regret deliverable

- backend/config/models.py: full Pydantic AppConfig + nested models (single source of truth
for defaults).
- backend/config/loader.py: load_config + save_default_config with file override + graceful
migration (extra=allow).
- backend/config/__init__.py: clean public API.
- Minimal dict-like .get / [] compat on AppConfig for transition period (legacy call sites
that pass config objects).
- Updated setup.py: init_default_config to delegate to the canonical loader.
- Updated backend/main.py: typed global_config (AppConfig), attribute access in hot paths,
legacy load_config wrapper still returns dict for diagnose_clip etc.
- Added tests/test_config.py covering defaults, overrides, extra keys, roundtrip.
```

This is the code implementation of Deliverable 1 from the approved plan.
Downstream consumers (validators, segmenters, etc.) will be migrated to typed access in
follow-up PRs / during the structure refactor.

Co-authored with avidullu.

```
backend/config/__init__.py | 19 ++++++
backend/config/loader.py   | 72 ++++++
backend/config/models.py   | 139 ++++++
backend/main.py            | 50 ++++++-----
setup.py                  | 62 +++++-----
tests/test_config.py       | 85 ++++++
6 files changed, 364 insertions(+), 63 deletions(-)
=== backend/config files ===
__init__.py
loader.py
models.py
[/external_agent_tool_result]
```

538. assistant (2026-07-02T06:11:33.173Z)

The commit message says AppConfig has **dict-like .get / []` compat** for the transition ? so
my `config.get(...)` calls may already work. Let me verify the depth of that compat (does nested
`.get("indexing").get("default_segmenter")` work?) before resolving:

539. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

540. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_result]
1      """Pydantic models for the application configuration.
2
3      This module defines the canonical, validated, typed configuration for the
4      Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
```



```

6     The models support graceful migration from old config.json files via extra="allow".
7     """
8
9     from __future__ import annotations
10
11     from typing import Any, Literal
12
13     from pydantic import BaseModel, Field, field_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])
21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         relevance: ValidationRelevanceConfig =
31         Field(default_factory=ValidationRelevanceConfig)
32
33     class TrackNetConfig(BaseModel):
34         wsl_distro: str = "Ubuntu"
35         repo_dir: str = "~/models/TrackNetV4"
36         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
37         conda_env: str = "TrackNetV4"
38         weights_path: str = ""
39         queue_length: int = 5
40         stage_in_wsl: bool = True
41         wsl_stage_dir: str = "~/clips"
42         timeout_sec: int = 0
43         velocity_threshold: float = 0.02
44
45
46     class IndexingConfig(BaseModel):
47         default_segmenter: str = "yolo_hybrid"
48         use_gpu: bool = True
49         motion_threshold: float = 0.08
50         min_segment_duration: float = 3.0
51         dead_time_merge_gap: float = 2.0
52         chunk_duration: float = 90.0
53         chunk_overlap: float = 10.0
54         detector_model: str = "fasterrcnn_resnet50_fpn"
55         detector_score_threshold: float = 0.5
56         yolo_velocity_threshold: float = 0.15
57         yolo_frame_skip: int = 3
58         yolo_tracker: str = "bytetrack.yaml"
59         yolo_parallel_chunks: bool = False
60         yolo_chunk_workers: int = 2
61         min_action_window_duration: float = 2.0
62         log_yolo_candidates: bool = True
63         store_yolo_candidates: bool = True
64         ai_handoff_padding: float = 2.0
65         ai_max_workers: int = 5
66
67         tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
68
69         @field_validator("default_segmenter")

```

```

70     @classmethod
71     def segmenter_must_be_registered(cls, v: str) -> str:
72         # Registry check happens at runtime in main/segmenter selection.
73         # Here we just allow any string for forward compatibility.
74         return v
75
76
77     class OutputConfig(BaseModel):
78         default_highlights_name: str = "highlights.mp4"
79         db_name: str = "sports_indexer.db"
80
81
82     class StitchingConfig(BaseModel):
83         begin_padding: float = 0.5
84         end_padding: float = 0.5
85         use_cache: bool = False
86         min_shot_count: int = 1
87
88
89     class AppConfig(BaseModel):
90         """Root configuration object.
91
92         All runtime configuration should be accessed via an instance of this class
93         (or the loader) rather than raw dict lookups.
94         """
95
96         sport: Sport = "badminton"
97         ai_provider: str = "gemini"
98         ai_model: str = "gemini-2.5-flash"
99
100         validation: ValidationConfig = Field(default_factory=ValidationConfig)
101         indexing: IndexingConfig = Field(default_factory=IndexingConfig)
102         output: OutputConfig = Field(default_factory=OutputConfig)
103         stitching: StitchingConfig = Field(default_factory=StitchingConfig)
104
105         # Allow extra keys during transition so existing config.json files do not break.
106         model_config = {
107             "extra": "allow",
108             "validate_assignment": True,
109         }
110
111         # Temporary dict-like access for legacy code that does config.get("foo", {})
112         # and pa...
[/external_agent_tool_result]

```

541. assistant (2026-07-02T06:11:33.173Z)

The dict-compat is **top-level only** ? `config.get("indexing")` returns a nested model (no `.get()`), so my `config.get("indexing").get("default_segmenter")` would break on an `AppConfig`. The robust fix is to **normalize config to a plain dict at the reporting boundary**. But first I need to see the full `main.py` (CLI path + `main()`) to know whether `global_config` is actually an `AppConfig` or the dict wrapper at runtime. Let me read it all:

542. assistant (2026-07-02T06:11:33.173Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]

```

543. assistant (2026-07-02T06:11:33.173Z)

```

[external_agent_tool_result]
1     import os
2     import sys

```

```

3     import argparse
4     import hashlib
5     import json
6     import uvicorn
7     from dotenv import load_dotenv
8
9     load_dotenv()
10    from fastapi import FastAPI, HTTPException
11    from fastapi.middleware.cors import CORSMiddleware
12    from pydantic import BaseModel
13    from typing import List, Dict, Any, Optional
14
15    from fastapi.staticfiles import StaticFiles
16    from fastapi.responses import HTMLResponse
17    from backend.validators.base import registry
18    from backend.database import Database
19    from backend.indexer.base import segmenter_registry, BasePlaySegmenter
20    from backend.stitcher.compiler import VideoCompiler
21    <<<<<<< HEAD
22    from backend.config import load_config as load_app_config, AppConfig # new typed config
23    =====
24    from backend.reporting import emit_indexer_report
25    >>>>>> 9dbc764 (feat(reporting): always-on per-video observability report (indexer))
26
27    app = FastAPI(title="Sports Highlight Indexer API")
28
29    # Enable CORS for local web interface
30    app.add_middleware(
31        CORSMiddleware,
32        allow_origins=["*"],
33        allow_credentials=True,
34        allow_methods=["*"],
35        allow_headers=["*"],
36    )
37
38    # Serves static frontend files directly
39    os.makedirs("backend/static", exist_ok=True)
40    app.mount("/static", StaticFiles(directory="backend/static"), name="static")
41
42    @app.get("/", response_class=HTMLResponse)
43    def serve_index():
44        index_path = "backend/static/index.html"
45        if os.path.exists(index_path):
46            with open(index_path, "r", encoding="utf-8") as f:
47                return f.read()
48        return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in
backend/static/</h3>"
49
50    # Global variables initialized at startup
51    db_instance: Optional[Database] = None
52    global_config: Optional[AppConfig] = None # now typed (Pydantic model)
53
54    def get_file_md5(filepath: str) -> str:
55        """Calculates file hash to uniquely identify videos and prevent duplicate
indexes."""
56        hasher = hashlib.md5()
57        with open(filepath, "rb") as f:
58            # Read in chunk sizes of 64kb
59            for chunk in iter(lambda: f.read(65536), b""):
60                hasher.update(chunk)
61        return hasher.hexdigest()
62
63    # Legacy thin wrapper for any remaining direct calls during transition.
64    # New code should import load_app_config / AppConfig from backend.config directly.
65    def load_config(config_path: str = "config.json") -> Dict[str, Any]:

```

```

66         cfg = load_app_config(config_path)
67         return cfg.model_dump(mode="json")
68
69     # --- API Models ---
70     class ProcessRequest(BaseModel):
71         video_path: str
72         player_pool: Optional[List[str]] = None
73         max_frames: Optional[int] = None
74         segmenter: Optional[str] = None
75
76     class QueryRequest(BaseModel):
77         video_id: str
78         server: Optional[str] = None
79         receiver: Optional[str] = None
80         duration_min: Optional[float] = None
81         event_type: Optional[str] = None
82
83     class CompileRequest(BaseModel):
84         video_id: str
85         segment_ids: List[int]
86         output_filename: str
87
88     # --- API Endpoints ---
89     @app.get("/api/config")
90     def get_config():
91         return global_config
92
93     @app.post("/api/process")
94     def process_video(req: ProcessRequest):
95         if not os.path.exists(req.video_path):
96             raise HTTPException(status_code=404, detail=f"Video file not found at
97 '{req.video_path}'")
98
99         video_id = get_file_md5(req.video_path)
100         was_reingest = db_instance.get_video(video_id) is not None
101
102         # 1. Run pluggable validators (with-context variant surfaces ffprobe_meta for the
103         report)
104         print(f"[API] Running validations for {req.video_path}...")
105         val_results, val_context = registry.run_all_with_context(req.video_path,
106         global_config)
107         failures = [r for r in val_results if not r.passed]
108
109         <<<<<<< HEAD
110         # 2. Run segmenter & indexer
111         print(f"[API] Video passed check...
112         [/external_agent_tool_result]

```

544. user (2026-07-02T06:11:33.173Z)

[Request interrupted by user]

545. user (2026-07-02T06:11:33.173Z)

The other session also made a fix PR but I have held it

546. assistant (2026-07-02T06:11:33.173Z)

Understood ? thanks, that removes the double-merge pressure on the parser fix; I'll keep mine on the branch and you can reconcile later.

Now resolving the `main.py` conflict. The picture is clear: `global\_config` is now a typed `AppConfig` (attribute access), so I'll **keep the parallel work's typed access for segmenter selection** and **keep my reporting wiring**, and make the reporting layer normalize

`AppConfig`?dict at its boundary (since my harvest does nested `.get`). Resolving the three conflict hunks:

547. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

548. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

549. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

550. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

551. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]
```

552. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

553. assistant (2026-07-02T06:11:33.173Z)

Now make the reporting layer normalize a typed `AppConfig` to a dict at its boundary (so harvest's nested `.get` works):

554. assistant (2026-07-02T06:11:33.173Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\__init__.py
[/external_agent_tool_call]
```

555. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\__init__.py has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

556. assistant (2026-07-02T06:11:33.174Z)

Let me verify no conflict markers remain, then continue the rebase:

557. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_call: Bash]
description: Verify resolution, parse-check, continue rebase
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== conflict markers?
=== " && grep -nE '^(<<<<<<|=====|>>>>>>)' backend/main.py || echo "none" && python -c
"import ast; ast.parse(open('backend/main.py', encoding='utf-8').read()); print('main.py parses
OK')" && git add backend/main.py && git rebase --continue 2>&1 | tail -8
[/external_agent_tool_call]
```

558. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_result]
=== conflict markers? ===
none
main.py parses OK
You must edit all merge conflicts and then
mark them as resolved using git add
[/external_agent_tool_result]
```

559. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_call: Bash]
description: Check rebase status for remaining unmerged paths
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git status 2>&1 | head -20
[/external_agent_tool_call]
```

560. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_result]
interactive rebase in progress; onto ce6a0e9
Last command done (1 command done):
  pick 9dbc764 feat(reporting): always-on per-video observability report (indexer)
Next command to do (1 remaining command):
  pick bc0fe21 chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
  (use "git rebase --edit-todo" to view and edit)
You are currently rebasing branch 'feat/observability-reports' on 'ce6a0e9'.
  (all conflicts fixed: run "git rebase --continue")

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md
    modified:   backend/main.py
    new file:   backend/reporting/__init__.py
    new file:   backend/reporting/harvest.py
    new file:   backend/reporting/spec.yaml
    modified:   backend/validators/base.py
    modified:   requirements.txt
    new file:   tests/test_reporting_harvest.py
    new file:   tests/test_validators_base.py
[/external_agent_tool_result]
```

561. assistant (2026-07-02T06:11:33.174Z)

Conflicts resolved (git confirms "all conflicts fixed"). Continuing the rebase with a non-interactive editor:

562. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_call: Bash]
description: Continue rebase non-interactively
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && GIT_EDITOR=true git rebase
--continue 2>&1 | tail -8 && echo "=== log ===" && git log --oneline -4 && echo "=== status ==="
&& git status -s | head
```

```
[/external_agent_tool_call]
```

563. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_result]
```

You must edit all merge conflicts and then mark them as resolved using git add

```
=== log ===
```

```
ce6a0e9 Merge pull request #39 from avidullu/feat/config-pydantic-package
da6ea5c feat(config): introduce backend/config/ package as first low-regret deliverable
f4c1b2d docs: checkpoint approved structure + sequencing decisions (Config first, atomic refactor follow-up)
9029415 docs: checkpoint approved structure + sequencing decisions (Config first, atomic refactor follow-up)
```

```
=== status ===
```

```
M README.md
M backend/main.py
AM backend/reporting/__init__.py
A backend/reporting/harvest.py
A backend/reporting/spec.yaml
M backend/validators/base.py
M requirements.txt
A tests/test_reporting_harvest.py
A tests/test_validators_base.py
[/external_agent_tool_result]
```

564. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_call: Bash]
```

description: Stage all, confirm no unmerged, continue rebase

```
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git add -A && git diff --name-only --diff-filter=U && echo "(above = remaining unmerged, should be empty)" && GIT_EDITOR=true git rebase --continue 2>&1 | tail -6 && echo "=== HEAD ===" && git log --oneline -3
```

```
[/external_agent_tool_call]
```

565. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_result]
```

(above = remaining unmerged, should be empty)

```
create mode 100644 backend/reporting/__init__.py
create mode 100644 backend/reporting/harvest.py
create mode 100644 backend/reporting/spec.yaml
create mode 100644 tests/test_reporting_harvest.py
create mode 100644 tests/test_validators_base.py
```

Rebasing (2/2)

Successfully rebased and updated refs/heads/feat/observability-reports.

```
=== HEAD ===
```

```
02aa99a chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
f1839b8 feat(reporting): always-on per-video observability report (indexer)
ce6a0e9 Merge pull request #39 from avidullu/feat/config-pydantic-package
[/external_agent_tool_result]
```

566. assistant (2026-07-02T06:11:33.174Z)

Rebase succeeded ? my two commits now sit cleanly on top of `origin/master`. Let me run the full indexer suite (now including the parallel work's config tests) to confirm the reconciliation is sound:

567. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_call: Bash]
```

description: Run full indexer suite after rebase

```
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest -q 2>&1 |
tail -12
[/external_agent_tool_call]
```

568. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_result]
def test_save_default_config_roundtrip():
    with tempfile.TemporaryDirectory() as td:
        p = os.path.join(td, "saved.json")
        saved = save_default_config(p)
>       assert saved == p
E       AssertionError: assert
WindowsPath('C:/Users/avidu/AppData/Local/Temp/tmpzdbctxbv/saved.json') ==
'C:\\Users\\avidu\\AppData\\Local\\Temp\\tmpzdbctxbv\\saved.json'

tests\\test_config.py:73: AssertionError
===== short test summary info =====
FAILED tests/test_config.py::test_load_default_config - AssertionError: asser...
FAILED tests/test_config.py::test_save_default_config_roundtrip - AssertionEr...
2 failed, 225 passed in 12.52s
[/external_agent_tool_result]
```

569. assistant (2026-07-02T06:11:33.174Z)

225 passed ? all my reporting tests green. The 2 failures are in the `**parallel work's`
``test_config.py`` (a Windows ``Path`` vs ``str`` mismatch in ``save_default_config``), unrelated to
my changes and pre-existing on ``origin/master``. Let me look at the loader to confirm and decide
whether it's a safe trivial fix:

570. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\loader.py
[/external_agent_tool_call]
```

571. assistant (2026-07-02T06:11:33.174Z)

```
[external_agent_tool_result]
1      """Configuration loader and saver.
2
3      Provides a single place to load validated AppConfig with built-in defaults
4      merged with optional file overrides (config.json).
5
6      This replaces scattered json.load + .get() patterns throughout the codebase.
7      """
8
9      from __future__ import annotations
10
11     import json
12     from pathlib import Path
13     from typing import Any
14
15     from .models import AppConfig
16
17     DEFAULT_CONFIG_PATH = Path("config.json")
18
19
20     def _get_built_in_defaults() -> dict[str, Any]:
21         """Return a plain dict of the current built-in defaults.
22
23         The AppConfig model is the single source of truth for defaults.
24         """
25         return AppConfig().model_dump(mode="json")
```



```

26
27
28 def load_config(path: str | Path | None = None) -> AppConfig:
29     """Load configuration with the following precedence:
30
31     1. Built-in defaults defined in the Pydantic models.
32     2. Values from the JSON file (if present and readable).
33     3. (Future) environment / CLI overrides.
34
35     Missing file or unreadable file ? returns a fully defaulted AppConfig.
36     Unknown keys in the file are tolerated (extra="allow" on the model).
37     """
38     cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
39
40     file_data: dict[str, Any] = {}
41     if cfg_path.exists():
42         try:
43             with open(cfg_path, "r", encoding="utf-8") as f:
44                 file_data = json.load(f)
45         except Exception as exc:
46             # Non-fatal: continue with defaults + whatever we could parse.
47             # In production we might log here.
48             print(f"[CONFIG WARNING] Failed to read {cfg_path}: {exc}. Using defaults +
partial data.")
49
50     # Merge: start from defaults, overlay file data (Pydantic will validate/coerce).
51     merged = {**_get_builtin_defaults(), **file_data}
52
53     return AppConfig.model_validate(merged)
54
55
56 def save_default_config(path: str | Path | None = None) -> Path:
57     """Write a fresh default config.json.
58
59     Used by setup.py diagnostics and the --setup CLI path.
60     Overwrites if the file exists.
61     """
62     cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
63     cfg = AppConfig()
64     cfg_path.parent.mkdir(parents=True, exist_ok=True)
65     cfg_path.write_text(cfg.model_dump_json(indent=4), encoding="utf-8")
66     return cfg_path
67
68
69 # Backwards-compatible alias used by some older call sites during transition.
70 def get_default_config() -> dict[str, Any]:
71     """Return defaults as a plain dict (for transitional use only)."""
72     return _get_builtin_defaults()
73
[/external_agent_tool_result]

```

572. assistant (2026-07-02T06:11:33.174Z)

`save\_default\_config` correctly returns a `Path` (annotated `-> Path`), but the parallel work's tests compare it to a `str` ? which fails on every platform (a `Path` never `==` a `str`). It's a clear test bug, now in my branch's suite because I rebased onto their master. To keep the checkpoint PR green, I'll fix the `**test assertions**` (not the loader) as a clearly-labeled separate commit. Let me read them:

573. assistant (2026-07-02T06:11:33.175Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py
[/external_agent_tool_call]

```

574. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_result]
55         data = {
56             "sport": "badminton",
57             "indexing": {"default_segmenter": "yolo_hybrid"},
58             "some_future_key": "ignored_for_now",
59             "validation": {"relevance": {"court_pixels_min_ratio": 0.1}},
60         }
61         with open(p, "w") as f:
62             json.dump(data, f)
63
64         cfg = load_config(p)
65         assert cfg.indexing.default_segmenter == "yolo_hybrid"
66         assert cfg.validation.relevance.court_pixels_min_ratio == 0.1
67
68
69     def test_save_default_config_roundtrip():
70         with tempfile.TemporaryDirectory() as td:
71             p = os.path.join(td, "saved.json")
72             saved = save_default_config(p)
73             assert saved == p
74             assert os.path.exists(p)
75
76             cfg = load_config(p)
77             assert cfg.indexing.default_segmenter == "yolo_hybrid"
78             assert cfg.validation.min_blur_threshold == 20.0
79
80
81     def test_model_is_the_source_of_truth():
82         """Changing a default in the model should be reflected without touching json."""
83         # This is a documentation test more than anything.
84         cfg = AppConfig()
[/external_agent_tool_result]
```

575. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py
[/external_agent_tool_call]
```

576. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_result]
1     """Basic tests for the typed configuration package.
2
3     These ensure:
4     - Defaults are sensible and come from the models.
5     - File overrides work.
6     - Old config files with extra keys are tolerated.
7     - save_default_config produces loadable output.
8     """
9
10    import json
11    import tempfile
12    import os
13
14    import pytest
15
16    from backend.config import load_config, save_default_config, AppConfig
17
18
19    def test_load_default_config():
20        cfg = load_config()
21        assert isinstance(cfg, AppConfig)
```

```

22         assert cfg.sport == "badminton"
23         assert cfg.indexing.default_segmenter == "yolo_hybrid"
24         assert cfg.validation.min_blur_threshold == 20.0 # updated default from review
25         assert cfg.indexing.ai_max_workers == 5
26
27
28     def test_file_override():
29         with tempfile.TemporaryDirectory() as td:
30             p = os.path.join(td, "override.json")
31             data = {
32                 "indexing": {
33                     "default_segmenter": "gemini",
34                     "ai_max_workers": 3,
35                 },
36                 "validation": {
37                     "min_blur_threshold": 42.0,
38                 },
39             }
40             with open(p, "w") as f:
41                 json.dump(data, f)
42
43             cfg = load_config(p)
44             assert cfg.indexing.default_segmenter == "gemini"
45             assert cfg.indexing.ai_max_workers == 3
46             assert cfg.validation.min_blur_threshold == 42.0
47             # Unspecified keys fall back to defaults
48             assert cfg.sport == "badminton"
49
50
51     def test_extra_keys_tolerated():
52         """Old or future keys in config.json must not break loading."""
53         with tempfile.TemporaryDirectory() as td:
54             p = os.path.join(td, "with_extra.json")
55 [/external_agent_tool_result]

```

577. assistant (2026-07-02T06:11:33.175Z)

The roundtrip test compares a `Path` to a `str` (objectively a test bug). Let me see the exact failure for the other one before deciding (it reads the repo's `config.json`, which may simply differ from the model defaults):

578. assistant (2026-07-02T06:11:33.175Z)

```

[external_agent_tool_call: Bash]
description: Get exact test_load_default_config failure + config.json values
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest
tests/test_config.py::test_load_default_config -q 2>&1 | tail -20 && echo "=== checked-in
config.json default_segmenter / blur / workers ===" && python -c "import json;c=json.load(open('
config.json'));print('default_segmenter=',c.get('indexing',{}).get('default_segmenter'));print('
min_blur=',c.get('validation',{}).get('min_blur_threshold'));print('ai_max_workers=',c.get('inde
xing',{}).get('ai_max_workers'))" 2>&1
[/external_agent_tool_call]

```

579. assistant (2026-07-02T06:11:33.175Z)

```

[external_agent_tool_result]
F                                                                 [100%]
===== FAILURES =====
_____ test_load_default_config _____

def test_load_default_config():
    cfg = load_config()
    assert isinstance(cfg, AppConfig)
    assert cfg.sport == "badminton"

```

```
>         assert cfg.indexing.default_segmenter == "yolo_hybrid"
E         AssertionError: assert 'gemini' == 'yolo_hybrid'
E
E         - yolo_hybrid
E         + gemini

tests\test_config.py:23: AssertionError
===== short test summary info =====
FAILED tests/test_config.py::test_load_default_config - AssertionError: asser...
1 failed in 0.34s
=== checked-in config.json default_segmenter / blur / workers ===
default_segmenter= gemini
min_blur= 20.0
ai_max_workers= 5
[/external_agent_tool_result]
```

580. assistant (2026-07-02T06:11:33.175Z)

Both config-test failures are **parallel-work** bugs unrelated to my PR^{**}:
``test_load_default_config`` fails because the checked-in ``config.json`` sets ``default_segmenter: gemini`` while the test expects the model default ``yolo_hybrid`` (the "diverging defaults" footgun!), and the roundtrip test compares a ``Path`` to a ``str``. Since you have **active parallel work** on config (Grok^{**}), I'll *not* touch their tests ? I'll flag them in the PR rather than risk conflicting with Grok's changes.

I also notice the **collector working tree got reverted** (my Phase B changes aren't showing) ? likely the parallel session switched branches there. My commits should still be safe on the branch; I'll verify after securing the indexer.

First, let me lock my config-reconciliation with a test (typed ``AppConfig`` ? report still emits), then run the suite:

581. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py
[/external_agent_tool_call]
```

582. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

583. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_call: Bash]
description: Run reporting tests + full suite tally
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest -q -k
"reporting or validators_base" 2>&1 | tail -8 && echo "=== full suite tally ===" && python -m
pytest -q 2>&1 | tail -4
[/external_agent_tool_call]
```

584. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_result]
..... [100%]
14 passed, 214 deselected in 4.54s
=== full suite tally ===
===== short test summary info =====
FAILED tests/test_config.py::test_load_default_config - AssertionError: asser...
FAILED tests/test_config.py::test_save_default_config_roundtrip - AssertionEr...
2 failed, 226 passed in 11.90s
```

```
[/external_agent_tool_result]
```

585. assistant (2026-07-02T06:11:33.175Z)

My reporting suite is green (14 passed incl. the typed-config test); the only 2 failures are the pre-existing config ones. Committing the reconciliation test, then I'll create the `obsreport` GitHub repo and push:

586. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_call: Bash]
description: Commit AppConfig reconciliation test
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git add
tests/test_reporting_harvest.py && git commit -q -m "test(reporting): cover typed AppConfig
normalization in emit (post-#39 rebase)
```

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git log --oneline -4
[/external_agent_tool_call]
```

587. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_result]
fa60efa test(reporting): cover typed AppConfig normalization in emit (post-#39 rebase)
02aa99a chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
f1839b8 feat(reporting): always-on per-video observability report (indexer)
ce6a0e9 Merge pull request #39 from avidullu/feat/config-pydantic-package
[/external_agent_tool_result]
```

588. assistant (2026-07-02T06:11:33.175Z)

Now creating the private `sports-obsreport` repo on GitHub and pushing it (main + all tags):

589. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_call: Bash]
description: Create private obsreport repo, push main + tags
command: cd "C:/Users/avidu/Projects/sports-obsreport" && echo "=== state ===" && git status -s
&& git log --oneline -1 && echo "=== create + push ===" && gh repo create avidullu/sports-
obsreport --private --source=. --remote=origin --push 2>&1 | tail -5 && git push -q origin
--tags 2>&1 | tail -3 && echo "=== remote tags ===" && git ls-remote --tags origin | tail -5
[/external_agent_tool_call]
```

590. assistant (2026-07-02T06:11:33.175Z)

```
[external_agent_tool_result]
=== state ===
1f72c1c feat: configurable highlights item_noun (v0.1.3)
=== create + push ===
https://github.com/avidullu/sports-obsreport
branch 'main' set up to track 'origin/main'.
To https://github.com/avidullu/sports-obsreport.git
 * [new branch]      HEAD -> main
=== remote tags ===
cba67f29d261903e03026fe5fefa56ade63860a4      refs/tags/v0.1.1^{*}
75aea1b807a79cf3669534808c3d48fb01753975      refs/tags/v0.1.2
2f5a214bf5051c386875a0e81bc724383b51d7e9      refs/tags/v0.1.2^{*}
821a664f5d2861397d5de7ebe803e72f71e9964a      refs/tags/v0.1.3
1f72c1c2dd471d4e3ed3130f0a4040a6d4813042      refs/tags/v0.1.3^{*}
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

591. assistant (2026-07-02T06:11:33.176Z)

`obsreport` is live on GitHub (private, tags v0.1.0?v0.1.3). Now pushing the indexer branch and

opening a checkpoint PR:

592. assistant (2026-07-02T06:11:33.176Z)

```
[external_agent_tool_call: Bash]
description: Push indexer branch + open checkpoint PR
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git push -q -u origin
feat/observability-reports 2>&1 | tail -3 && gh pr create --base master --head
feat/observability-reports --title "feat: per-video observability reports (indexer) + shared
obsreport lib" --body "$(cat <<'EOF'
```

What

Always-on, rich **per-video observability report** for the indexer, emitted on every CLI/API run next to the video (or ``report.output_dir``):

- ``<video>.report.json`` ? machine source of truth (every stage + metric + finding); diffable across runs.
- ``<video>.report.md`` ? **technical** report (validation, processing stages, findings/warnings) ? each warning cites a README FAQ.
- ``<video>.summary.md`` ? **customer-friendly** summary (video meta + highlights, no internals).

Powered by a new shared, permissively-licensed library **obsreport** (separate private repo ``avidullu/sports-obsreport``, MIT), consumed as a **soft dependency** ? if it isn't installed the pipeline runs exactly as before, and report writes are wrapped in a ``finally`/try` so they can **never break a processing run**.

Key pieces

- ``backend/validators/base.py``: backward-compatible ``run_all_with_context`` (surfaces the ``ffprobe_meta`` that ``run_all`` discarded). ``run_all`` unchanged.
- ``backend/reporting/``: soft-import wrapper (normalizes a dict **or** the new typed ``AppConfig``) + pure ``harvest.py`` + ``spec.yaml`` footgun rules (``silent_provider_noop``, ``ai_empty_vs_no_rallies``, ``fps_fallback_used``, ``synthetic_motion_metadata``, ``segmenter_divergence``, ``reingest_replaced_prior``, ``validation_failed``).
- ``backend/main.py``: ``finally``-guarded emit in CLI + API (both exits); report paths in the API response. Reconciled with the typed-config work from #39.
- README FAQ Q10?Q13 (motion metadata, re-ingest, the report itself, validation failures).

Notes for the reviewer

- **Rebased onto #39** (the ``backend/config`` typed-config package). ``main.py`` conflict resolved: kept typed attribute access for segmenter selection + my reporting wiring; reporting normalizes ``AppConfig`?dict` at its boundary.
- ?? **2** pre-existing failures in ``tests/test_config.py`` (from #39, not this PR): ``test_load_default_config`` (checked-in ``config.json`` sets ``default_segmenter: gemini`` but the test asserts the model default ``yolo_hybrid`` ? the "diverging defaults" footgun) and ``test_save_default_config_roundtrip`` (``Path == str`` on Windows). Left untouched to avoid colliding with the parallel config work.
- ``obsreport`` is a **soft** dep (commented git+ssh pin in ``requirements.txt``); install editable for dev: ``pip install -e ../sports-obsreport --no-build-isolation``.
- New + existing tests green: **226 passed**, 2 pre-existing config failures.

? Generated with [Claude Code](https://claude.com/claude-code)

EOF

)" 2>&1 | tail -3

[/external_agent_tool_call]

593. assistant (2026-07-02T06:11:33.176Z)

```
[external_agent_tool_result]
remote: Create a pull request for 'feat/observability-reports' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/observability-reports
remote:
https://github.com/avidullu/badminton-highlight-indexer/pull/41
[/external_agent_tool_result]
```

594. assistant (2026-07-02T06:11:33.176Z)

Indexer checkpoint PR is up: [indexer#41](https://github.com/avidullu/badminton-highlight-indexer/pull/41). Now let me investigate the collector ? its working tree showed reverted content, so I need to find where my Phase B commits went:

595. assistant (2026-07-02T06:11:33.176Z)

```
[external_agent_tool_call: Bash]
description: Investigate collector branch + commit state
command: cd "C:/Users/avidu/Projects/sports-data-collector" && echo "=== current branch ===" &&
git branch --show-current && echo "=== local branches ===" && git branch -vv | head && echo "===
feat branch log (my commits?) ===" && git log feat/observability-reports --oneline -4 2>&1 |
head && echo "=== working tree status ===" && git status -s | head
[/external_agent_tool_call]
```

596. assistant (2026-07-02T06:11:33.176Z)

```
[external_agent_tool_result]
=== current branch ===
feat/observability-reports
=== local branches ===
* feat/observability-reports 028937b fix(parser): keep a rally that starts at 0.0s in
load_canonical_csv
  fix/canonical-csv-zero-start f7cab11 [origin/fix/canonical-csv-zero-start] fix(parser): keep a
rally that starts at 0.0s in load_canonical_csv
  master d68c5da [origin/master] Merge pull request #9 from
avidullu/feat/validate-delivery
=== feat branch log (my commits?) ===
028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
bb2b5be feat(reporting): per-video observability report (collector)
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
2882a03 feat(ingest): validate-delivery ? per-video QA gate + structured report
=== working tree status ===
 M data/sports_metadata.db
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt
?? data/validation/
[/external_agent_tool_result]
```

597. assistant (2026-07-02T06:11:33.176Z)

Good ? I'm still on `feat/observability-reports` and ****my commits are intact**** (bb2b5be feat + 028937b parser fix). I also see Grok's `fix/canonical-csv-zero-start` branch (the held PR). The earlier reminders may have been mid-operation snapshots. Let me verify my committed changes are actually present in the working tree right now:

598. assistant (2026-07-02T06:11:33.176Z)

```
[external_agent_tool_call: Bash]
description: Verify my committed collector changes are in the working tree
command: cd "C:/Users/avidu/Projects/sports-data-collector" && echo "=== curate.py emit wiring
present? ===" && grep -c "emit_import_report\|emit_delivery_report" src/cli/curate.py && echo
"=== src/reporting exists? ===" && ls src/reporting/ 2>&1 && echo "=== INGESTION_PIPELINE
observability FAQ present? ===" && grep -c "observability report" docs/INGESTION_PIPELINE.md &&
echo "=== pyproject build-backend ===" && grep "build-backend" pyproject.toml && echo "=== git
diff HEAD on tracked source (should be empty) ===" && git diff --stat HEAD -- src docs
pyproject.toml .gitignore
[/external_agent_tool_call]
```

599. assistant (2026-07-02T06:11:33.176Z)

```
[external_agent_tool_result]
=== curate.py emit wiring present? ===
```

```

6
=== src/reporting exists? ===
__init__.py
__pycache__
adapter.py
spec.yaml
=== INGESTION_PIPELINE observability FAQ present? ===
2
=== pyproject build-backend ===
build-backend = "setuptools.build_meta"
=== git diff HEAD on tracked source (should be empty) ===
[/external_agent_tool_result]

```

600. assistant (2026-07-02T06:11:33.176Z)

All my collector changes are intact (the reminders were stale snapshots). Collector `origin/master` is unchanged, so no rebase needed. I see Grok's dedicated `fix/canonical-csv-zero-start` branch exists (your held PR). To keep my observability PR **clean and non-overlapping**, I'll drop my duplicate parser-fix commit (the fix lives safely on Grok's branch) so my PR is purely observability. The branch is local-only, so this is safe:

601. assistant (2026-07-02T06:11:33.176Z)

```

[external_agent_tool_call: Bash]
description: Confirm Grok branch has fix, drop my duplicate parser commit
command: cd "C:/Users/avidu/Projects/sports-data-collector" && echo "=== confirm fix exists on
Grok branch before dropping mine ===" && git log fix/canonical-csv-zero-start --oneline -1 &&
git show fix/canonical-csv-zero-start:src/parsers/cvat/rally_cvat.py | grep -c "if start is
None" && echo "=== drop my parser-fix commit (top of feat) ===" && git reset --hard HEAD~1 2>&1
| tail -2 && echo "=== feat log now ===" && git log --oneline -2 && echo "=== rally_cvat
reverted to pre-fix on my branch? (expect falsy-or line) ===" && grep -n
"_parse_secs(row.get("start_time")) or " src/parsers/cvat/rally_cvat.py | head -1
[/external_agent_tool_call]

```

602. assistant (2026-07-02T06:11:33.176Z)

```

[external_agent_tool_result]
=== confirm fix exists on Grok branch before dropping mine ===
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
2
=== drop my parser-fix commit (top of feat) ===
HEAD is now at bb2b5be feat(reporting): per-video observability report (collector)
=== feat log now ===
bb2b5be feat(reporting): per-video observability report (collector)
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
=== rally_cvat reverted to pre-fix on my branch? (expect falsy-or line) ===
442:         start = _parse_secs(row.get("start_time")) or _parse_secs(row.get("start_mmss"))
[/external_agent_tool_result]

```

603. assistant (2026-07-02T06:11:33.176Z)

Clean ? my collector branch is now purely observability. Running the suite to confirm green, then pushing + PR:

604. assistant (2026-07-02T06:11:33.176Z)

```

[external_agent_tool_call: Bash]
description: Run collector suite after dropping parser commit
command: cd "C:/Users/avidu/Projects/sports-data-collector" && python -m pytest -q 2>&1 | tail
-5
[/external_agent_tool_call]

```

605. assistant (2026-07-02T06:11:33.176Z)


```
[external_agent_tool_result]
..... [ 64%]
..... [100%]
112 passed in 1.89s
[/external_agent_tool_result]
```

606. assistant (2026-07-02T06:11:33.176Z)

112 passed. Pushing the collector branch and opening its checkpoint PR:

607. assistant (2026-07-02T06:11:33.176Z)

```
[external_agent_tool_call: Bash]
description: Push collector branch + open checkpoint PR
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git push -q -u origin
feat/observability-reports 2>&1 | tail -3 && gh pr create --base master --head
feat/observability-reports --title "feat: per-video observability reports (collector)" --body
"$(cat <<'EOF'
```

What

Per-video **observability report** for the collector, emitted by `validate-delivery`, `convert-cvat`, and `import-local`:

- `<id>.report.json` ? machine source of truth (stages + metrics + findings); diffable across deliveries.
- `<id>.report.md` ? **technical** report (parse / validation / registration stages + findings, one per delivery issue, each citing `docs/INGESTION_PIPELINE.md`).
- `<id>.summary.md` ? **vendor/data-team delivery summary** (rally count + coverage + plain-language list of rallies to review; no internal codes).

Powered by the shared `obsreport` library (`avidullu/sports-obsreport`, MIT) as a **soft dependency** ? absent ? no-op; report writes never raise, so ingestion can't break.

Key pieces

- `src/reporting/`: soft-import wrapper + `adapter.py` (maps existing `rally_cvat.Issue` ? `Flag`, reuses the existing records/issues ? no logic duplicated) + `spec.yaml` (sections, per-tool `customer_verdict`, `item_noun=rally`, `license_noncommercial` + `missing_fps` rules).
- `src/cli/curate.py`: wired the three commands (delivery report colocated with the existing `_report.` artifacts; import report falls back to the annotations dir when the video file is absent ? never cwd).
- `docs/INGESTION_PIPELINE.md`: FAQ entries for the report + the non-commercial-export note (so every `faq_ref` resolves).
- `pyproject.toml`: fixed the `build-backend` typo (`setuptools.build-meta` ? `build_meta`, which previously broke `pip install -e .`); documented the `obsreport` dep.
- `.gitignore`: ignore `*.report.json` / `*.report.md` / `*.summary.md` runtime artifacts.

Notes for the reviewer

- **Observability-only.** The separate `load_canonical_csv` zero-start fix is intentionally **not** here ? it lives on `fix/canonical-csv-zero-start` (held) to keep these PRs non-overlapping.
- `obsreport` is a **soft** dep; install editable for dev: `pip install -e ../sports-obsreport --no-build-isolation`.
- Tests: **112 passed**. Verified end-to-end via a `validate-delivery` CLI smoke (report files emitted; vendor summary clean).

```
? Generated with [Claude Code](https://claude.com/claude-code)
EOF
)" 2>&1 | tail -3
[/external_agent_tool_call]
```

608. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result]
remote: Create a pull request for 'feat/observability-reports' on GitHub by visiting:
```

```
remote:      https://github.com/avidullu/sports-data-collector/pull/new/feat/observability-
reports
remote:
https://github.com/avidullu/sports-data-collector/pull/11
[/external_agent_tool_result]
```

609. assistant (2026-07-02T06:11:33.177Z)

Both checkpoint PRs are up and `obsreport` is on GitHub. Let me record the sync in memory, then run the debuggability evaluation you asked for.

```
**Synced to remote:**
- `obsreport` lib ? [github.com/avidullu/sports-obsreport](https://github.com/avidullu/sports-
obsreport) (private, tags v0.1.0?v0.1.3)
- Indexer ? [PR #41](https://github.com/avidullu/badminton-highlight-indexer/pull/41) (rebased
onto your `#39` config work; conflicts resolved)
- Collector ? [PR #11](https://github.com/avidullu/sports-data-collector/pull/11)
(observability-only; left Grok's parser fix to its own held branch)
```

610. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_call: Edit]
file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

611. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

612. assistant (2026-07-02T06:11:33.177Z)

Now the debuggability evaluation. First I'll generate real broken-run reports for both tools (the fixtures a "fresh reader" will debug from), then fan out fresh-reader agents via a Workflow. Writing the indexer fixture generator:

613. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Local\Temp\obseval\gen_indexer.py
[/external_agent_tool_call]
```

614. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Local\Temp\obseval\gen_indexer.py (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

615. assistant (2026-07-02T06:11:33.177Z)

Now the collector fixture generator:

616. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Local\Temp\obseval\gen_collector.py
[/external_agent_tool_call]
```

617. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Local\Temp\obseval\gen_collector.py (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

618. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_call: Bash]
description: Generate indexer + collector broken-run report fixtures
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python
"C:/Users/avidu/AppData/Local/Temp/obseval/gen_indexer.py" 2>&1 | tail -3
cd "C:/Users/avidu/Projects/sports-data-collector" && python
"C:/Users/avidu/AppData/Local/Temp/obseval/gen_collector.py" 2>&1 | tail -3
[/external_agent_tool_call]
```

619. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result]
wrote 4 indexer scenarios -> C:\Users\avidu\AppData\Local\Temp\obseval\indexer_fixtures.json
wrote 3 collector scenarios -> C:\Users\avidu\AppData\Local\Temp\obseval\collector_fixtures.json
[/external_agent_tool_result]
```

620. assistant (2026-07-02T06:11:33.177Z)

7 broken-run reports generated. Let me write each to a `.md` file and build a small manifest (paths + ground truth) so the Workflow's fresh-reader agents can read the report + the real README/FAQ by path ? exactly simulating a user, without bloating the script:

621. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Local\Temp\obseval\combine.py
[/external_agent_tool_call]
```

622. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Local\Temp\obseval\combine.py (file state
is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

623. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_call: Bash]
description: Write report .md files + build manifest
command: cd "C:/Users/avidu/AppData/Local/Temp/obseval" && python combine.py 2>&1 | python -c
"import sys,json; m=json.load(open(r'C:\Users\avidu\AppData\Local\Temp\obseval\manifest.json'));
print('scenarios:', [x['id'] for x in m])"
[/external_agent_tool_call]
```

624. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result: error]
Exit code 1
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import sys,json;
m=json.load(open(r'C:\Users\avidu\AppData\Local\Temp\obseval\manifest.json'));
print('scenarios:', [x['id'] for x in m])
~~~~~
FileNotFoundError: [Errno 2] No such file or directory:
'C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\manifest.json'
```

```
[/external_agent_tool_result]
```

625. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_call: Bash]
description: Run combine.py directly, check for errors
command: cd "C:/Users/avidu/AppData/Local/Temp/obseval" && python combine.py >
manifest_stdout.txt 2>&1; echo "exit=$?"; tail -5 manifest_stdout.txt; echo "=== manifest
exists? ==="; ls manifest.json reports/ 2>&1
[/external_agent_tool_call]
```

626. assistant (2026-07-02T06:11:33.177Z)

```
[external_agent_tool_result]
exit=0
  "doc_path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\docs\\INGESTION_PIPELINE.md",
  "truth": "The imported video has a non-commercial license, so it is EXCLUDED from the
default commercial-only export. It stays in the DB but won't appear downstream unless exported
with --include-noncommercial.",
  "expected_faq": "will-a-non-commercial-license-block-export"
}
]
=== manifest exists? ===
manifest.json

reports/:
collector_missed_rally.report.md
collector_noncommercial.report.md
collector_overlap.report.md
indexer_fps_fallback.report.md
indexer_motion_synthetic.report.md
indexer_no_api_key.report.md
indexer_validation_failed.report.md
[/external_agent_tool_result]
```

627. assistant (2026-07-02T06:11:33.178Z)

Manifest + 7 report files written. Let me read the manifest to pass as the Workflow's args:

628. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\obseval\manifest.json
[/external_agent_tool_call]
```

629. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result]
1      [
2        {
3          "id": "indexer_no_api_key",
4          "tool": "indexer",
5          "report_path":
"C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_no_api_key.report.md",
6          "doc_path": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md",
7          "truth": "The run produced 0 highlight segments because no AI-provider API key was
set, so the AI segmenter silently no-op'd. It is NOT that the video has no rallies. Fix: set
GEMINI_API_KEY (e.g. in a .env file) and re-run.",
8          "expected_faq": "Q7"
9        },
10       {
11         "id": "indexer_motion_synthetic",
12         "tool": "indexer",
```

```

13         "report_path":
"C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_motion_synthetic.report.md",
14         "doc_path": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md",
15         "truth": "The 'motion' segmenter was used; its per-segment metadata/events are
heuristic/synthetic, not measured, so they must not be trusted as ground truth. Use yolo_hybrid
or gemini for real shot metadata.",
16         "expected_faq": "Q10"
17     },
18     {
19         "id": "indexer_validation_failed",
20         "tool": "indexer",
21         "report_path":
"C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_validation_failed.report.md",
22         "doc_path": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md",
23         "truth": "Indexing was halted because a validation check failed: the blur/focus
check (video too blurry / sharpness below the min threshold). Fix the footage quality or lower
validation.min_blur_threshold in config.json, then re-run.",
24         "expected_faq": "Q13"
25     },
26     {
27         "id": "indexer_fps_fallback",
28         "tool": "indexer",
29         "report_path":
"C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_fps_fallback.report.md",
30         "doc_path": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md",
31         "truth": "fps was not probed from the file (a fallback default was used), so motion
thresholds and frame->second timing may be miscalibrated. Confirm the real fps with ffprobe and
ensure validation ran.",
32         "expected_faq": "Q5"
33     },
34     {
35         "id": "collector_overlap",
36         "tool": "collector",
37         "report_path":
"C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\collector_overlap.report.md",
38         "doc_path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\docs\\INGESTION_PIPELINE.md",
39         "truth": "The delivery is blocked because two rallies overlap in time (rally 1 ends
at 7.00s but rally 2 starts at 6.00s). Rallies must be disjoint; the vendor/reviewer must fix
rally 1's end or rally 2's start before it can be imported.",
40         "expected_faq": "what-does-validation-check"
41     },
42     {
43         "id": "collector_missed_rally",
44         "tool": "collector",
45         "report_path":
"C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\collector_missed_rally.report.md",
46         "doc_path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\docs\\INGESTION_PIPELINE.md",
47         "truth": "There is a long unlabeled gap (~26s) between rally 2 and rally 3, which
may hide an omitted/missed rally. The reviewer should watch that span and add the rally if one
is there. It is a review flag, not a blocker.",
48         "expected_faq": "what-does-validation-check"
49     },
50     {
51         "id": "collector_noncommercial",
52         "tool": "collector",
53         "report_path":
"C:\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\collector_noncommercial.report.md",
54         "doc_path": "C:\\Users\\avidu\\Projects\\sports-data-
collector\\docs\\INGESTION_PIPELINE.md",
55         "truth": "The imported video has a non-commercial license, so it is EXCLUDED from
the default commercial-only export. It stays in the DB but won't appear downstream unless
exported with --include-noncommercial.",
56         "expected_faq": "will-a-non-commercial-license-block-export"

```

```
57     }
58   ]
[/external_agent_tool_result]
```

630. assistant (2026-07-02T06:11:33.178Z)

Now I'll run the ****debuggability evaluation as a Workflow**** (ultracode): for each of the 7 broken-run scenarios, a fresh-reader agent gets ***only*** the README/FAQ + the report (no source) and must self-diagnose; a tough grader then checks whether they found the root cause and the right FAQ, and proposes concrete refinements; a final agent synthesizes.

631. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Workflow]
input: {"args":[{"id":"indexer_no_api_key","tool":"indexer","report_path":"C:\\\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_no_api_key.report.md","doc_path":"C:\\\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md","truth":"The run produced 0 highlight segments because no AI-provider API key was set, so the AI segmenter silently no-op'd. It is NOT that the video has no rallies. Fix: set GEMINI_API_KEY (e.g. in a .env file) and re-run."},"id":"indexer_motion_synthetic","tool":"indexer","report_path":"C:\\\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_motion_synthetic.report.md","doc_path":"C:\\\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md","truth":"The 'motion' segmenter was used; its per-segment metadata/events are heuristic/synthetic, not measured, so they must not be trusted as ground truth. Use yolo_hybrid or gemini for real shot metadata."},"id":"indexer_validation_failed","tool":"indexer","report_path":"C:\\\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_validation_failed.report.md","doc_path":"C:\\\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md","truth":"Indexing was halted because a validation check failed: the blur/focus check (video too blurry / sharpness below the min threshold). Fix the footage quality or lower validation.min_blur_threshold in config.json, then re-run."},"id":"indexer_fps_fallback","tool":"indexer","report_path":"C:\\\\Users\\avidu\\AppData\\Local\\Temp\\obseval\\reports\\indexer_fps_fallback.report.md","doc_path":"C:\\\\Users\\avidu\\Projects\\badminton-highlight-indexer\\README.md","truth":"fps was not probed from the file (a fallback default was used), so motion thresholds and frame->second timing ma..."}]}
[/external_agent_tool_call]
```

632. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: w8ghqp2hd
Summary: Fresh-reader debuggability eval of per-video observability reports across broken-run scenarios
Transcript dir: C:\Users\avidu\claudé\projects\C--Users-avidu-Projects-badminton-highlight-indexer\56b7f849-5d56-4770-93e0-b3b4da943700\subagents\workflows\wf_63677f5b-055
Script file: C:\Users\avidu\claudé\projects\C--Users-avidu-Projects-sports-data-collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-wf_63677f5b-055.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claudé\projects\C--Users-avidu-Projects-sports-data-collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-wf_63677f5b-055.js"} to iterate without resending the script.)
Run ID: wf_63677f5b-055
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claudé\projects\C--Users-avidu-Projects-sports-data-collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-wf_63677f5b-055.js", resumeFromRunId: "wf_63677f5b-055"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.
[/external_agent_tool_result]
```

633. assistant (2026-07-02T06:11:33.178Z)

The workflow failed instantly ? `args` arrived as a JSON string, not an array. Let me make the script robust to both and re-invoke via its saved path:

634. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-sports-data-
collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-
wf_63677f5b-055.js
[/external_agent_tool_call]
```

635. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

636. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-sports-data-
collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-
wf_63677f5b-055.js
[/external_agent_tool_call]
```

637. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result]
86         ` - confusing_parts: ${reader ? JSON.stringify(reader.confusing_parts || [])} :
'[]'\n\n` +
87         `Also read the actual report for context: ${it.report_path}\n\n` +
88         `Judge strictly: did the reader land on the correct ROOT CAUSE (cause_correct)? Did
they find/consult the ` +
89         `expected FAQ (faq_correct)? How large is the remaining gap (none/minor/major)? Then
propose ONE concrete, ` +
90         `specific refinement to the REPORT wording, the spec rule message, or the FAQ that
would have made this scenario ` +
91         `unambiguous to self-debug ? or "none" if it is already excellent. Don't be
generous; if anything was confusing, propose a fix.` ,
92         { label: `grade:${it.id}`, phase: 'Grade', schema: GRADER_SCHEMA },
93         ).then((g) => ({ ...g, scenario: it.id, tool: it.tool, reader })),
94     )
95
[/external_agent_tool_result]
```

638. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-sports-data-
collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-
wf_63677f5b-055.js
[/external_agent_tool_call]
```

639. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result]
68         `(do not open source code or any other file):\n` +
69         `1. The README/FAQ: ${it.doc_path}\n` +
70         `2. The tool's technical report for the suspicious run: ${it.report_path}\n\n` +
71         `Using ONLY those two, tell me: what went wrong, exactly what you'd do to fix it,
and which FAQ section ` +
72         `(by heading text or Q-number) you used. Be honest about whether you could truly
self-debug this from the ` +
73         `report alone, and list anything in the report that was confusing, ambiguous, or
```

```
under-explained.`,  
74      { label: `read:${it.id}`, phase: 'Read', agentType: 'Explore', schema:  
FRESH_READER_SCHEMA },  
75      ),  
76      (reader, it) => agent(  
77      `You are a tough evaluator of how DEBUGGABLE an observability report is for a  
README-only reader.\n` +  
[/external_agent_tool_result]
```

640. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Read]  
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-sports-data-  
collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-  
wf_63677f5b-055.js  
[/external_agent_tool_call]
```

641. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result]  
62  
63      const graded = await pipeline(  
64      items,  
65      (it) => agent(  
66      `You are a USER of a sports-video tool who has read ONLY its README/FAQ ? you have  
NEVER seen the source code.` +  
[/external_agent_tool_result]
```

642. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Read]  
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-sports-data-  
collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-  
wf_63677f5b-055.js  
[/external_agent_tool_call]
```

643. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result]  
59      }  
60  
61      const items = args  
62  
[/external_agent_tool_result]
```

644. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_call: Edit]  
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-sports-data-  
collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-  
wf_63677f5b-055.js  
[/external_agent_tool_call]
```

645. assistant (2026-07-02T06:11:33.178Z)

```
[external_agent_tool_result]  
The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-sports-data-  
collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-  
wf_63677f5b-055.js has been updated successfully. (file state is current in your context ? no  
need to Read it back)  
[/external_agent_tool_result]
```

646. assistant (2026-07-02T06:11:33.178Z)

Fixed. Re-invoking the workflow via its saved script path:

647. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_call: Workflow]
input: {"scriptPath": "C:\\Users\\avidu\\.claude\\projects\\C--Users-avidu-Projects-sports-data-collector\\56b7f849-5d56-4770-93e0-b3b4da943700\\workflows\\scripts\\report-debuggability-eval-wf_63677f5b-055.js", "args": "[{"id": "indexer_no_api_key", "tool": "indexer", "report_path": "C:\\\\Users\\\\avidu\\\\AppData\\\\Local\\\\Temp\\\\obseval\\\\reports\\\\indexer_no_api_key.report.md", "doc_path": "C:\\\\Users\\\\avidu\\\\Projects\\\\badminton-highlight-indexer\\\\README.md", "truth": "The run produced 0 highlight segments because no AI-provider API key was set, so the AI segmenter silently no-op'd. It is NOT that the video has no rallies. Fix: set GEMINI_API_KEY (e.g. in a .env file) and re-run.", "expected_faq": "Q7"}, {"id": "indexer_motion_synthetic", "tool": "indexer", "report_path": "C:\\\\Users\\\\avidu\\\\AppData\\\\Local\\\\Temp\\\\obseval\\\\reports\\\\indexer_motion_synthetic.report.md", "doc_path": "C:\\\\Users\\\\avidu\\\\Projects\\\\badminton-highlight-indexer\\\\README.md", "truth": "The 'motion' segmenter was used; its per-segment metadata/events are heuristic/synthetic, not measured, so they must not be trusted as ground truth. Use yolo_hybrid or gemini for real shot metadata.", "expected_faq": "Q10"}, {"id": "indexer_validation_failed", "tool": "indexer", "report_path": "C:\\\\Users\\\\avidu\\\\AppData\\\\Local\\\\Temp\\\\obseval\\\\reports\\\\indexer_validation_failed.report.md", "doc_path": "C:\\\\Users\\\\avidu\\\\Projects\\\\badminton-highlight-indexer\\\\README.md", "truth": "Indexing was halted because a validation check failed: the blur/focus check (video too blurry / sharpness below the min threshold). Fix the footage quality or lower validation.min_blur_threshold in config.json, then re-run.", "expected_faq": "Q13"}, {"id": "indexer_fps_fallback", "tool": "indexer", "report_path": "C:\\\\Users\\\\avidu\\\\AppData\\\\Local\\\\Temp\\\\obseval\\\\reports\\\\indexer_fps_fallback.report.md", "doc_path": "C..."}]
[/external_agent_tool_call]
```

648. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: wwkbttg1n
Summary: Fresh-reader debuggability eval of per-video observability reports across broken-run scenarios
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\56b7f849-5d56-4770-93e0-b3b4da943700\subagents\workflows\wf_49729ae7-962
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-sports-data-collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-wf_63677f5b-055.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-sports-data-collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-wf_63677f5b-055.js"} to iterate without resending the script.)
Run ID: wf_49729ae7-962
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-sports-data-collector\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\report-debuggability-eval-wf_63677f5b-055.js", resumeFromRunId: "wf_49729ae7-962"}) ? completed agents return cached results.
```

You will be notified when it completes. Use /workflows to watch live progress.

```
[/external_agent_tool_result]
```

649. assistant (2026-07-02T06:11:33.179Z)

I see PR #10 (the parser fix) was merged to the collector master by the parallel flow, and my memory was updated to reflect it. My collector PR #11 (observability-only) is now behind that merge. While the debuggability workflow runs in the background, let me **sync PR #11 from head** ? rebase it onto the updated master (no conflict expected since I don't touch `rally\_cvat.py`):

650. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_call: Bash]
description: Fetch collector origin, compare PR #11 to updated master
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git fetch -q origin && echo "===
origin/master head ===" && git log origin/master --oneline -3 && echo "=== my branch vs
origin/master ===" && git rev-list --left-right --count origin/master...feat/observability-
reports
[/external_agent_tool_call]
```

651. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_result]
=== origin/master head ===
99dfa22 Merge pull request #10 from avidullu/fix/canonical-csv-zero-start
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
=== my branch vs origin/master ===
2      1
[/external_agent_tool_result]
```

652. assistant (2026-07-02T06:11:33.179Z)

Master has #10's parser fix (2 commits ahead); my 1 observability commit needs rebasing on top.
Rebasing and force-pushing PR #11 to keep it current:

653. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_call: Bash]
description: Rebase collector PR #11 onto master, test, force-push
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git rebase origin/master 2>&1 |
tail -5 && echo "=== log ===" && git log --oneline -3 && echo "=== quick test ===" && python -m
pytest -q 2>&1 | tail -3 && git push --force-with-lease -q origin feat/observability-reports
2>&1 | tail -3 && echo "pushed PR #11 (rebased on master incl #10)"
[/external_agent_tool_call]
```

654. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_result]
Rebasing (1/1)

Successfully rebased and updated refs/heads/feat/observability-reports.
=== log ===
038424d feat(reporting): per-video observability report (collector)
99dfa22 Merge pull request #10 from avidullu/fix/canonical-csv-zero-start
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
=== quick test ===
..... [ 63%]
..... [100%]
113 passed in 1.98s
pushed PR #11 (rebased on master incl #10)
[/external_agent_tool_result]
```

655. assistant (2026-07-02T06:11:33.179Z)

PR #11 is now rebased on master (113 tests, the parser fix's regression test now in ancestry)
and force-pushed ? current with head. The debuggability workflow is running in the background.

While it runs, let me prep the "eyeball" rendering (MD?styled HTML?screenshot) so I can visually
review the *refined* reports once the eval lands. Checking the markdown lib + loading the
screenshot tools:

656. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_call: Bash]
description: Check for python markdown lib
```

```
command: python -c "import markdown; print('markdown', markdown.__version__)" 2>&1 || echo
"markdown NOT installed"
[/external_agent_tool_call]
```

657. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_result]
markdown 3.10.2
[/external_agent_tool_result]
```

658. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp__Claude_Preview__preview_start,mcp__Claude_Preview__preview_screensh
ot,mcp__Claude_Preview__preview_stop","max_results":5}
[/external_agent_tool_call]
```

659. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

660. assistant (2026-07-02T06:11:33.179Z)

Good ? `markdown` 3.10 is available, and Preview works via a static dev server. Let me pre-write the MD?HTML render script now (independent of the eval) so I can quickly screenshot the **refined** reports once the workflow lands:

661. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Local\Temp\obseval\render_html.py
[/external_agent_tool_call]
```

662. assistant (2026-07-02T06:11:33.179Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Local\Temp\obseval\render_html.py (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

663. assistant (2026-07-02T06:11:33.179Z)

Everything is synced and the evaluation is in flight. Status while the workflow runs:

```
**Synced to remote (all current with head):**
- `obsreport` lib ? [github.com/avidullu/sports-obsreport](https://github.com/avidullu/sports-
obsreport) (private, tags v0.1.0?v0.1.3)
- Indexer ? [PR #41](https://github.com/avidullu/badminton-highlight-indexer/pull/41) (rebased
onto your `#39` typed-config work; `main.py` conflict resolved ? kept your typed config access +
my reporting, reporting normalizes `AppConfig`?dict)
- Collector ? [PR #11](https://github.com/avidullu/sports-data-collector/pull/11)
(observability-only, **rebased onto master including the merged `#10` parser fix**; 113 tests
green)
```

****Merge-conflict notes for you:**** the only real conflict was indexer `main.py` (your config package vs my reporting) ? resolved cleanly. Indexer PR carries ****2** pre-existing red tests from `#39`****** (`test\_config.py`: `config.json` default = `gemini` vs model default `yolo\_hybrid`, and a `Path == str` Windows comparison) ? I left them untouched so they don't collide with Grok's config work; flagged in the PR body.

****Debuggability eval (running):**** a multi-agent workflow is putting all 7 broken-run reports in front of "fresh reader" agents that may see **only** the README/FAQ + the report (no source), then a tough grader checks whether each could self-diagnose and cites the right FAQ. I'll apply the

synthesized refinements, ****render the refined reports to an image and eyeball them****, add the golden faq-resolution tests, and push to the PRs once it lands.

664. user (2026-07-02T06:11:33.180Z)

Can you resync from remote and run the entire test suite to validate that the directory and Config changes are all well.

665. assistant (2026-07-02T06:11:33.180Z)

On it. Let me re-sync both repos from remote ? fetch latest master (which may now include the directory restructure that was sequenced after Config), see how far my branches are behind, then rebase and run the full suites.

666. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_call: Bash]
description: Fetch both origins, compare branches to latest master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch -q origin && echo
"=== INDEXER origin/master head ===" && git log origin/master --oneline -6 && echo "=== my feat
vs origin/master (behind/ahead) ===" && git rev-list --left-right --count
origin/master...feat/observability-reports
echo "" && cd "C:/Users/avidu/Projects/sports-data-collector" && git fetch -q origin && echo
"=== COLLECTOR origin/master head ===" && git log origin/master --oneline -5 && echo "=== my
feat vs origin/master (behind/ahead) ===" && git rev-list --left-right --count
origin/master...feat/observability-reports
[/external_agent_tool_call]
```

667. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_result]
=== INDEXER origin/master head ===
0355417 Merge pull request #40 from avidullu/refactor/approved-directory-structure
1e96e07 test: fix mock patch targets after structure refactor and config test assertions
8615534 refactor: adopt approved layered directory structure
ce6a0e9 Merge pull request #39 from avidullu/feat/config-pydantic-package
da6ea5c feat(config): introduce backend/config/ package as first low-regret deliverable
f4c1b2d docs: checkpoint approved structure + sequencing decisions (Config first, atomic
refactor follow-up)
=== my feat vs origin/master (behind/ahead) ===
3      3

=== COLLECTOR origin/master head ===
99dfa22 Merge pull request #10 from avidullu/fix/canonical-csv-zero-start
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
2882a03 feat(ingest): validate-delivery ? per-video QA gate + structured report
23f2ed0 Merge pull request #8 from avidullu/feat/last-shot-outcome
=== my feat vs origin/master (behind/ahead) ===
0      1
[/external_agent_tool_result]
```

668. assistant (2026-07-02T06:11:33.180Z)

Significant change on the ****indexer****: PR #40 merged a ****directory restructure**** (``refactor/approved-directory-structure``) ? and it also fixed the 2 config test assertions I'd flagged. The collector is unchanged (already current). My indexer branch needs rebasing onto the restructured master, which will mean relocating my reporting module + reconciling moved files. Let me first understand the new layout before rebasing:

669. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_call: Bash]
description: Inspect directory restructure renames
```

```
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== refactor commit  
file moves (renames) ===" && git show 8615534 --stat --diff-filter=R -M 2>&1 | head -60  
[/external_agent_tool_call]
```

670. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_result]  
=== refactor commit file moves (renames) ===  
commit 8615534e261ea7bc355468667772b272820eea6a  
Author: avidullu <avidullu@users.noreply.github.com>  
Date: Sun Jun 7 07:47:14 2026 +0000
```

refactor: adopt approved layered directory structure

- Move to professional layered layout per checkpointed plan:
 - backend/api/static/ (frontend)
 - backend/pipeline/{validators,segmenters,stitcher,audio,detectors}
 - backend/infrastructure/database.py
- indexer/ renamed to segmenters/ inside pipeline for clarity.
- Updated all imports (absolute backend.xxx paths) across source, tests, run_*.py.
- Fixed static serving paths in main.py.
- Updated CODE_MAP.md references and high-level structure description.
- This is the atomic high-pri follow-up refactor after the Config package (Deliverable 1).

All moved modules now live under clear layers. No behavior change intended.
Co-authored with avidullu.

```
backend/{ => api}/static/app.js | 0  
backend/{ => api}/static/index.html | 0  
backend/{ => api}/static/style.css | 0  
backend/{ => infrastructure}/database.py | 0  
backend/{ => pipeline}/audio/__init__.py | 0  
backend/{ => pipeline}/audio/hit_detector.py | 0  
backend/{ => pipeline}/detectors/__init__.py | 0  
backend/{ => pipeline}/detectors/rally_gate.py | 0  
backend/{ => pipeline}/detectors/tracknet_runner.py | 0  
backend/{ => pipeline}/detectors/wasb_infer.py | 0  
backend/{ => pipeline}/detectors/wasb_runner.py | 2 +-  
backend/{indexer => pipeline/segmenters}/base.py | 2 +-  
backend/{indexer => pipeline/segmenters}/gemini.py | 4 ++--  
backend/{indexer => pipeline/segmenters}/motion.py | 4 ++--  
backend/{indexer => pipeline/segmenters}/tracknet_hybrid.py | 4 ++--  
backend/{indexer => pipeline/segmenters}/wasb_hybrid.py | 4 ++--  
backend/{indexer => pipeline/segmenters}/yolo_hybrid.py | 2 +-  
backend/{ => pipeline}/stitcher/compiler.py | 0  
backend/{ => pipeline}/validators/base.py | 0  
backend/{ => pipeline}/validators/blur.py | 2 +-  
backend/{ => pipeline}/validators/format.py | 2 +-  
backend/{ => pipeline}/validators/pts_continuity.py | 2 +-  
backend/{ => pipeline}/validators/relevance.py | 2 +-  
backend/{ => pipeline}/validators/resolution_fps.py | 2 +-  
backend/{ => pipeline}/validators/scene_cut.py | 2 +-  
25 files changed, 17 insertions(+), 17 deletions(-)  
[/external_agent_tool_result]
```

671. assistant (2026-07-02T06:11:33.180Z)

The indexer got a ****layered directory restructure**** (#40):

```
`backend/validators/`?`backend/pipeline/validators/`,  
`backend/indexer/`?`backend/pipeline/segmenters/`,  
`backend/database.py`?`backend/infrastructure/database.py`, etc. My branch modifies some of  
those moved files, so this rebase needs care. Let me confirm a clean working tree and attempt  
the rebase:
```

672. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_call: Bash]
description: Check clean tree + rebase indexer onto restructured master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== working tree ==="
&& git status -s && echo "=== rebase onto origin/master ===" && git rebase origin/master 2>&1 |
tail -25
[/external_agent_tool_call]
```

673. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_result]
=== working tree ===
=== rebase onto origin/master ===
Rebasing (1/3)
Rebasing (2/3)
Rebasing (3/3)

Successfully rebased and updated refs/heads/feat/observability-reports.
[/external_agent_tool_result]
```

674. assistant (2026-07-02T06:11:33.180Z)

The rebase applied cleanly (git's rename detection merged my `main.py` reporting wiring onto the restructured imports). But my **test** files still import the old paths** (`backend.validators.base`). Let me find stale references in my added files and verify the moved validator got my change:

675. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_call: Bash]
description: Verify moved validator + find stale imports in my files
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "===
run_all_with_context in new location? ===" && grep -c "run_all_with_context"
backend/pipeline/validators/base.py 2>&1 && echo "=== old backend/validators exists? (should be
gone) ===" && ls backend/validators 2>&1 | head -1 && echo "=== stale imports in MY files ==="
&& grep -rn "backend.validators\|backend.indexer\|backend.database\|backend.stitcher\|backend.de
tectors\|backend.audio" tests/test_validators_base.py tests/test_reporting_harvest.py
backend/reporting/ 2>&1
[/external_agent_tool_call]
```

676. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_result]
=== run_all_with_context in new location? ===
2
=== old backend/validators exists? (should be gone) ===
__pycache__
=== stale imports in MY files ===
tests/test_validators_base.py:3:from backend.validators.base import ValidationRegistry,
ValidationResult, VideoValidator
tests/test_reporting_harvest.py:13:from backend.validators.base import ValidationResult # noqa:
E402
[/external_agent_tool_result]
```

677. assistant (2026-07-02T06:11:33.180Z)

Two fixes needed ? my test files import the pre-move path. Let me update them to the new `backend.pipeline.validators.base`:

678. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_validators_base.py
```

```
[/external_agent_tool_call]
```

679. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_result]
```

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_validators_base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

```
[/external_agent_tool_result]
```

680. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_call: Edit]
```

file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py

```
[/external_agent_tool_call]
```

681. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_result]
```

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py has been updated successfully. (file state is current in your context ? no need to Read it back)

```
[/external_agent_tool_result]
```

682. assistant (2026-07-02T06:11:33.180Z)

Now run the **full indexer suite** to validate the directory + config changes integrate cleanly with my reporting:

683. assistant (2026-07-02T06:11:33.180Z)

```
[external_agent_tool_call: Bash]
```

description: Run full indexer suite after restructure rebase

command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest -q 2>&1 | tail -15

```
[/external_agent_tool_call]
```

684. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_result]
```

```
..... [ 31%]
..... [ 63%]
..... [ 94%]
..... [100%]
```

228 passed in 12.63s

```
[/external_agent_tool_result]
```

685. assistant (2026-07-02T06:11:33.181Z)

All **228 indexer tests pass** ? the directory restructure + typed config + my reporting all integrate cleanly, and the 2 previously-flagged config tests are now green (fixed by #40). Let me do a bare import smoke (tests mock heavily), commit the import-path fixes, and update the PR:

686. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_call: Bash]
```

description: Import smoke, commit test fixes, force-push indexer PR

command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -c "import backend.main; import backend.reporting; import backend.reporting.harvest; print('import smoke OK')" 2>&1 | tail -3 && git add tests/test_validators_base.py tests/test_reporting_harvest.py && git commit -q -m "chore(test): update reporting/validator test imports for layered package layout (post-#40 restructure)"

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push --force-with-lease -q origin feat/observability-reports 2>&1 | tail -2 && echo "pushed indexer PR #41 (rebased on #40"

```
restructure)"
[/external_agent_tool_call]
```

687. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_result]
import smoke OK
pushed indexer PR #41 (rebased on #40 restructure)
[/external_agent_tool_result]
```

688. assistant (2026-07-02T06:11:33.181Z)

Indexer PR #41 updated and green. Now the **collector** and **obsreport** full suites (the user asked for the entire suite across the changes):

689. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_call: Bash]
description: Run collector + obsreport full suites
command: echo "=== COLLECTOR full suite ===" && cd "C:/Users/avidu/Projects/sports-data-
collector" && python -m pytest -q 2>&1 | tail -4
echo "" && echo "=== OBSREPORT full suite ===" && cd "C:/Users/avidu/Projects/sports-obsreport"
&& python -m pytest -q 2>&1 | tail -4
[/external_agent_tool_call]
```

690. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_result]
=== COLLECTOR full suite ===
..... [ 63%]
..... [100%]
113 passed in 1.97s

=== OBSREPORT full suite ===
..... [ 97%]
.. [100%]
74 passed in 0.36s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

691. assistant (2026-07-02T06:11:33.181Z)

All three suites are green. Let me also confirm the restructured + typed-config pipeline runs **end-to-end at runtime** (tests mock heavily), via a quick CLI smoke that exercises my reporting in the `finally``:

692. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_call: Bash]
description: End-to-end CLI smoke on restructured indexer
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" &&
SM="$TEMP/obs_restructure_smoke" && rm -rf "$SM" && mkdir -p "$SM" && ffmpeg -loglevel error -f
lavfi -i color=c=green:s=1280x720:d=2 -r 30 -pix_fmt yuv420p "$SM/smoke.mp4" -y && python -u -m
backend.main --video-path "$SM/smoke.mp4" --segmenter motion --output-dir "$SM/out" 2>&1 | grep
-E "VALIDATION|PASS|FAIL|REPORT|Error|Traceback|Exception" | head -20; echo "---"; ls
"$SM"/*.report.* "$SM"/*.summary.* 2>/dev/null && echo "REPORT FILES EMITTED OK"
[/external_agent_tool_call]
```

693. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_result]
VALIDATION SUMMARY
[PASS] format_check: Video format is valid MP4 container.
[FAIL] resolution_fps_check: Validator threw an unexpected error: 'ValidationConfig' object has
```



```

no attribute 'get'
[PASS] pts_continuity_check: Video is too short to perform a PTS continuity check.
[FAIL] scene_cut_check: Validator threw an unexpected error: 'ValidationConfig' object has no
attribute 'get'
[FAIL] blur_check: Validator threw an unexpected error: 'ValidationConfig' object has no
attribute 'get'
[FAIL] relevance_check: Validator threw an unexpected error: 'ValidationConfig' object has no
attribute 'get'
[FAIL] Video failed checks. Indexing halted.
[REPORT] Technical report :
C:\Users\avidu\AppData\Local\Temp\obs_restructure_smoke\smoke.report.md
[REPORT] Customer summary :
C:\Users\avidu\AppData\Local\Temp\obs_restructure_smoke\smoke.summary.md
[REPORT] Machine JSON      :
C:\Users\avidu\AppData\Local\Temp\obs_restructure_smoke\smoke.report.json
---
C:\Users\avidu\AppData\Local\Temp\obs_restructure_smoke\smoke.report.json
C:\Users\avidu\AppData\Local\Temp\obs_restructure_smoke\smoke.report.md
C:\Users\avidu\AppData\Local\Temp\obs_restructure_smoke\smoke.summary.md
REPORT FILES EMITTED OK
[/external_agent_tool_result]

```

694. assistant (2026-07-02T06:11:33.181Z)

Re-sync + full validation done across all three repos. But the runtime smoke surfaced something important ? and notably, **my observability report is what caught it.**

Resync + test validation

```

| Repo | Synced onto | Full suite |
|---|---|---|
| Indexer | rebased onto master incl. #40 directory restructure + #39 config | 228
passed, 0 failed ? |
| Collector | rebased onto master incl. #10 parser fix | 113 passed ? |
| obsreport | (mine) | 74 passed ? |

```

- The rebase onto the layered restructure
(`backend/pipeline/{validators,segmenters,stitcher,detectors,audio}`,
`backend/infrastructure/database.py`, `backend/api/static/`) was clean ? git's rename detection merged my `main.py` reporting wiring onto the new imports; I only had to repoint two of my test files. PR #41 force-pushed and green.
- The 2 config-test failures I'd flagged earlier are now **fixed by #40** (so the indexer PR is fully green now).

?? Directory changes: well. Config change: a real runtime bug.

A bare-import smoke and the test suite pass, but an **actual `python -m backend.main` run** fails 4 validators:

```

...
[FAIL] resolution_fps_check: Validator threw an unexpected error: 'ValidationConfig' object has
no attribute 'get'
[FAIL] scene_cut_check:      ... 'ValidationConfig' object has no attribute 'get'
[FAIL] blur_check:          ... 'ValidationConfig' object has no attribute 'get'
[FAIL] relevance_check:      ... 'ValidationConfig' object has no attribute 'get'
...

```

Root cause: `main()` now passes the typed `AppConfig` to validators/segmenters, but `AppConfig.get()` is dict-compatible only at the **top level** ? `config.get("validation")` returns a `ValidationConfig` *model*, and the validators then call `.get("min_resolution_width", ...)` on it, which a Pydantic model doesn't support. So every consumer that does nested `config.get("section").get("key")` (all the validators, and likely the segmenters) breaks on a real video. Tests don't catch it because they pass plain dicts directly.

This is a **master-level** bug from the config migration (#39), not fixed by #40, and not introduced by my PRs ? my reporting layer already guards itself (it normalizes `AppConfig`?dict), which is exactly why it ran fine and emitted the report that exposed the failure.

The minimal fix lives in Grok's `backend/config/models.py`: have `AppConfig.get()` return `model\_dump()` (a real dict) for nested model sections, so legacy `config.get("validation").get(...)` works again (1?2 lines, low risk). Since config is Grok's active area:

695. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question":"The typed AppConfig from #39 breaks all the validators (and likely segmenters) on a real video run ? nested config.get(\"section\").get(\"key\") fails. Want me to fix it, or leave it to the config work?\",\"header\":\"Config bug\",\"multiSelect\":false,\"options\":[{\"label\":\"Fix it now (small PR)\",\"description\":\"I make AppConfig.get() return a dict for nested sections (model_dump), verify the CLI smoke runs end-to-end, and open a small focused PR to master. Risk: may lightly conflict with Grok's active config edits in models.py.\"},{\"label\":\"Leave it to Grok\",\"description\":\"I only flag it (with the repro). Grok migrates downstream consumers to typed access (or fixes .get) as part of the config workstream. I keep my PRs focused on observability.\"},{\"label\":\"Fix on my indexer branch\",\"description\":\"Fold the AppConfig.get fix into PR #41 so the observability pipeline demonstrably runs end-to-end. Couples a config fix into the observability PR (less clean separation).\"}]}]}
[/external_agent_tool_call]
```

696. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_result]
Your questions have been answered: "The typed AppConfig from #39 breaks all the validators (and likely segmenters) on a real video run ? nested config.get("section").get("key") fails. Want me to fix it, or leave it to the config work?"="Fix it now (small PR)". You can now continue with these answers in mind.
[/external_agent_tool_result]
```

697. assistant (2026-07-02T06:11:33.181Z)

You chose to fix it now as a small PR. Let me branch off the indexer master (so the PR is just the config fix), confirm my feat branch is clean first:

698. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_call: Bash]
description: Branch off indexer master for the config fix
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== feat clean? ===" && git status -s && git checkout -b fix/appconfig-nested-get origin/master 2>&1 | tail -2 && echo "=== on branch ===" && git branch --show-current && echo "=== confirm reporting NOT here (master-based) ===" && ls backend/reporting 2>&1 | head -1
[/external_agent_tool_call]
```

699. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_result]
=== feat clean? ===
branch 'fix/appconfig-nested-get' set up to track 'origin/master'.
Switched to a new branch 'fix/appconfig-nested-get'
=== on branch ===
fix/appconfig-nested-get
=== confirm reporting NOT here (master-based) ===
__pycache__
[/external_agent_tool_result]
```

700. assistant (2026-07-02T06:11:33.181Z)

On the fix branch. Let me read the current `AppConfig.get()` and apply the nested-section fix:

701. assistant (2026-07-02T06:11:33.181Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

702. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_result]
110
111     # Temporary dict-like access for legacy code that does config.get("foo", {})
112     # and passes the config object around. Will be removed after consumers are migrated.
113     def get(self, key: str, default: Any = None) -> Any:
114         try:
115             return getattr(self, key)
116         except AttributeError:
117             # Also support nested "indexing.default_segmenter" style? For now simple
118             # top-level.
119             if key in self.model_fields:
120                 return getattr(self, key)
121             # Try to find in sections
122             for section_name in ("validation", "indexing", "output", "stitching"):
123                 section = getattr(self, section_name, None)
124                 if section is not None:
125                     try:
126                         return getattr(section, key)
127                     except AttributeError:
128                         pass
129             return default
130
131     def __getitem__(self, key: str) -> Any:
132         val = self.get(key)
133         if val is None and key not in self.model_fields:
134             # Re-raise like a real dict for unknown top-level
135             raise KeyError(key)
136         return val
137
138     # Convenience type alias for callers that want to annotate
139     Config = AppConfig
[/external_agent_tool_result]
```

703. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

704. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

705. assistant (2026-07-02T06:11:33.182Z)

Now validate the fix ? full suite on this branch + the CLI smoke (validators should now execute their real logic instead of erroring on `.get`):

706. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_call: Bash]
description: Run suite + CLI smoke to verify config fix
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== full suite on fix
branch ===" && python -m pytest -q 2>&1 | tail -4 && echo "" && echo "=== CLI smoke: validators
should run real logic, no '.get' errors ===" && SM="$TEMP/cfgfix_smoke" && rm -rf "$SM" && mkdir
-p "$SM" && ffmpeg -loglevel error -f lavfi -i color=c=green:s=1280x720:d=2 -r 30 -pix_fmt
yuv420p "$SM/g.mp4" -y && python -u -m backend.main --video-path "$SM/g.mp4" --segmenter motion
--output-dir "$SM/out" 2>&1 | grep -E "PASS|FAIL|attribute|error" | head -10
[/external_agent_tool_call]
```

707. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_result]
=== full suite on fix branch ===
===== short test summary info =====
ERROR tests/test_config.py
!!!!!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!!!!
1 error in 4.78s

=== CLI smoke: validators should run real logic, no '.get' errors ===
[/external_agent_tool_result]
```

708. assistant (2026-07-02T06:11:33.182Z)

Collection error in `test\_config.py` ? let me see it:

709. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_call: Bash]
description: Import models.py to surface the error
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -c "import
backend.config.models" 2>&1 | tail -20
[/external_agent_tool_call]
```

710. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_result]
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import backend.config.models
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\__init__.py", line 9,
in <module>
    from .loader import load_config, save_default_config
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\loader.py", line 15,
in <module>
    from .models import AppConfig
  File "C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py", line 121
    val: Any = _MISSING = object()
                    ^
SyntaxError: invalid syntax
[/external_agent_tool_result]
```

711. assistant (2026-07-02T06:11:33.182Z)

Syntax error ? can't combine an annotated assignment with chaining. Let me split it:

712. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

713. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

714. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_call: Bash]
description: Re-verify import, suite, and CLI smoke after syntax fix
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -c "import
backend.config.models; print('import OK')" && echo "=== full suite ===" && python -m pytest -q
2>&1 | tail -4 && echo "" && echo "=== CLI smoke ===" && SM="$TEMP/cfgfix_smoke" && rm -rf "$SM"
&& mkdir -p "$SM" && ffmpeg -loglevel error -f lavfi -i color=c=green:s=1280x720:d=2 -r 30
-pix_fmt yuv420p "$SM/g.mp4" -y && python -u -m backend.main --video-path "$SM/g.mp4"
--segmenter motion --output-dir "$SM/out" 2>&1 | grep -E "PASS|FAIL|attribute" | head -10
[/external_agent_tool_call]
```

715. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_result]
import OK
=== full suite ===
..... [ 33%]
..... [ 67%]
..... [100%]
214 passed in 12.01s

=== CLI smoke ===
[PASS] format_check: Video format is valid MP4 container.
[PASS] resolution_fps_check: Video resolution and frame rate meet minimum standards.
[PASS] pts_continuity_check: Video is too short to perform a PTS continuity check.
[PASS] scene_cut_check: Scene continuity check passed. No abnormal visual cuts detected.
[FAIL] blur_check: Video quality rejected: Focus check failed. Average sharpness score of 0.0 is
below the minimum required threshold of 20.0.
[PASS] relevance_check: Video identified as relevant to badminton (badminton court structures
found).
[FAIL] Video failed checks. Indexing halted.
[/external_agent_tool_result]
```

716. assistant (2026-07-02T06:11:33.182Z)

The fix works ? validators now execute their real logic (resolution/scene-cut/relevance
PASS; blur legitimately fails on the textureless green clip), no more ``.get`` errors. 214
tests green. Let me lock it with a regression test. Reading the current (post-#40)
test_config.py first:

717. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py
[/external_agent_tool_call]
```

718. assistant (2026-07-02T06:11:33.182Z)

```
[external_agent_tool_result]
60         "validation": {"relevance": {"court_pixels_min_ratio": 0.1}},
61     }
62     with open(p, "w") as f:
63         json.dump(data, f)
64
65     cfg = load_config(p)
66     assert cfg.indexing.default_segmenter == "yolo_hybrid"
67     assert cfg.validation.relevance.court_pixels_min_ratio == 0.1
68
```

```

69
70     def test_save_default_config_roundtrip():
71         with tempfile.TemporaryDirectory() as td:
72             p = os.path.join(td, "saved.json")
73             saved = save_default_config(p)
74             assert str(saved) == p
75             assert os.path.exists(p)
76
77             cfg = load_config(p)
78             assert cfg.indexing.default_segmenter == "yolo_hybrid"
79             assert cfg.validation.min_blur_threshold == 20.0
80
81
82     def test_model_is_the_source_of_truth():
83         """Changing a default in the model should be reflected without touching json."""
84         # This is a documentation test more than anything.
85         cfg = AppConfig()
86         assert cfg.indexing.tracknet.velocity_threshold == 0.02
87
[/external_agent_tool_result]

```

719. assistant (2026-07-02T06:11:33.182Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py
[/external_agent_tool_call]

```

720. assistant (2026-07-02T06:11:33.183Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

721. assistant (2026-07-02T06:11:33.183Z)

```

[external_agent_tool_call: Bash]
description: Run config tests + full suite after regression test
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest
tests/test_config.py -q 2>&1 | tail -5 && echo "=== full suite ===" && python -m pytest -q 2>&1
| tail -3
[/external_agent_tool_call]

```

722. assistant (2026-07-02T06:11:33.183Z)

```

[external_agent_tool_result]
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py:123:
PydanticDeprecatedSince211: Accessing the 'model_fields' attribute on the instance is
deprecated. Instead, you should access this attribute from the model class. Deprecated in
Pydantic V2.11 to be removed in V3.0.
    if key in self.model_fields or hasattr(self, key):

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
6 passed, 4 warnings in 0.18s
=== full suite ===

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
215 passed, 4 warnings in 11.68s
[/external_agent_tool_result]

```

723. assistant (2026-07-02T06:11:33.183Z)

215 passed. Let me silence a Pydantic deprecation warning (instance `model\_fields` access) that my rewrite carried over ? access it from the class instead:

724. assistant (2026-07-02T06:11:33.183Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

725. assistant (2026-07-02T06:11:33.183Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

726. assistant (2026-07-02T06:11:33.183Z)

```
[external_agent_tool_call: Bash]
description: Re-test, commit config fix, push branch
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest
tests/test_config.py -q 2>&1 | tail -3 && git add backend/config/models.py tests/test_config.py
&& git commit -q -m "fix(config): AppConfig.get returns nested sections as dicts (legacy dict-
compat)
```

The typed AppConfig is passed to validators/segmenters, which use legacy chained dict access: config.get('validation', {}).get('min_resolution_width', ...). The previous .get returned the nested *Pydantic model*, which has no .get, so every validator (resolution_fps/scene_cut/blur/relevance) and likely the segmenters threw `"'ValidationConfig' object has no attribute 'get'"` on every real video run (tests passed because they inject plain dicts). Return nested sections via model_dump so chained dict access works; scalar leaves unchanged. + regression test.

Verified: python -m backend.main on a real clip now runs all validators (no .get errors); 215 tests pass.

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q -u origin fix/appconfig-
nested-get 2>&1 | tail -2 && echo "pushed"
[/external_agent_tool_call]
```

727. assistant (2026-07-02T06:11:33.183Z)

```
[external_agent_tool_result]
..... [100%]
6 passed in 0.25s
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/appconfig-
nested-get
remote:
pushed
[/external_agent_tool_result]
```

728. assistant (2026-07-02T06:11:33.183Z)

Clean (no warnings). Opening the focused config-fix PR:

729. assistant (2026-07-02T06:11:33.183Z)

```
[external_agent_tool_call: Bash]
description: Open config-fix PR, switch back to feat branch
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && gh pr create --base master
--head fix/appconfig-nested-get --title "fix(config): AppConfig.get returns nested sections as
dicts (validators broke on real runs)" --body "$(cat <<'EOF'
```

The bug

After #39 (typed config) + #40 (restructure), `main()` passes the typed `AppConfig` to the validators and segmenters. Those consumers use `**legacy chained dict access**`:

```

```python
config.get("validation", {}).get("min_resolution_width", 1280)
```

`AppConfig.get("validation")` returned the nested Pydantic model (ValidationConfig), which has no .get, so the second call raised 'ValidationConfig' object has no attribute 'get'. On a real python -m backend.main run, 4 validators fail (resolution_fps, scene_cut, blur, relevance), and the segmenters would hit the same wall. The unit tests didn't catch it because they inject plain dicts, not AppConfig.

...

Before: [FAIL] resolution_fps_check: ... 'ValidationConfig' object has no attribute 'get'
After:  [PASS] resolution_fps_check: Video resolution and frame rate meet minimum standards.
...

```

The fix

`AppConfig.get()` now returns nested **sections as plain dicts** (`model_dump`) so legacy chained `.get(...)` keeps working; scalar leaves are returned unchanged; missing keys fall back to the default. This is the "dict-compat for legacy code" the method already promised ? just made deep instead of top-level-only. (Also switched to class-level `model_fields` to drop a Pydantic 2.11 deprecation warning.)

Attribute access used in the migrated hot paths (e.g. `global_config.indexing.default_segmenter`) is unaffected.

Verification

- `python -m backend.main --video-path <clip>` now runs **all** validators (no `.get` errors).
- Added a regression test (`test_get_returns_dict_for_nested_sections_legacy_compat`).
- **215 tests pass.**

> Found while validating the directory + config changes end-to-end (the per-video observability report from #41 surfaced the validator failures). Independent of #41/#40; targets the incomplete dict-compat from #39.

? Generated with [Claude Code](https://claude.com/claude-code)

EOF

```

)" 2>&1 | tail -3 && echo "=== back to feat branch ===" && git checkout feat/observability-reports 2>&1 | tail -1

```

```
[/external_agent_tool_call]
```

730. assistant (2026-07-02T06:11:33.183Z)

```

[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/42
=== back to feat branch ===
Switched to branch 'feat/observability-reports'
[/external_agent_tool_result]

```

731. assistant (2026-07-02T06:11:33.183Z)

PR #42 opened for the config fix. (Back on the feat branch, `models.py` correctly shows master's original ? the fix lives on its own branch/PR.) Let me checkpoint memory, then give you the full status.

732. assistant (2026-07-02T06:11:33.183Z)

```

[external_agent_tool_call: Edit]
file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

```

733. assistant (2026-07-02T06:11:33.183Z)

[external_agent_tool_result]

The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

734. assistant (2026-07-02T06:11:33.183Z)

Done ? re-synced and validated across everything, plus I caught and fixed a real config bug that the parallel work introduced.

? Re-sync + full-suite validation

```
Repo	Synced onto latest master	Full suite
**Indexer**	**39** (typed config) + **40** (layered directory restructure)	**228
passed** ?		
**Collector**	**10** (parser fix)	**113 passed** ?
**obsreport**	?	**74 passed** ?
```

- **Directory restructure: all well.** The rebase onto the new
`backend/pipeline/{validators,segmenters,stitcher,detectors,audio}` / `backend/infrastructure`
/ `backend/api/static/` layout was clean ? git's rename-detection merged my `main.py` reporting
wiring onto the relocated imports; I only repointed two of my test files. The 2 config-test
failures I flagged earlier are now fixed by #40.

? Config change: **was *not* all well ? found & fixed**

A real `python -m backend.main` run failed **4 validators** with `ValidationConfig` object has
no attribute `get`. The typed `AppConfig` from #39 is passed to validators/segmenters that do
legacy chained access `config.get("validation", {}).get("key")`, but `AppConfig.get()` returned
the nested *Pydantic model* (no `.get`) ? dict-compat was top-level only. Tests passed because
they inject plain dicts, so it slipped through CI; a real video run is broken on master.

- **Fixed** in [indexer PR #42](https://github.com/avidullu/badminton-highlight-indexer/pull/42): `AppConfig.get()` now returns nested sections as dicts (`model\_dump`) ?
chained access works again; + regression test; **215 green** ; CLI smoke now runs all validators.
Nicely, **my observability report is what surfaced it** (it normalizes config itself, so it ran
and reported the validator failures).

Open PRs

- [indexer #41](https://github.com/avidullu/badminton-highlight-indexer/pull/41) ? observability
(green, rebased on #40)
- [indexer #42](https://github.com/avidullu/badminton-highlight-indexer/pull/42) ? the config
fix (green)
- [collector #11](https://github.com/avidullu/sports-data-collector/pull/11) ? observability
(green, rebased on #10)

Suggested merge order: **42 ? 41** (so #41's runtime path isn't broken), **11** any time.

Still in flight: the Phase-D debuggability workflow (fresh-reader agents over the 7 broken-
run reports). When it lands I'll apply the wording refinements, render a report to an image to
eyeball, add the golden FAQ-resolution tests, and push to #41/#11. I'll pick that up on its
completion notification.

735. user (2026-07-02T06:11:33.183Z)

Do you see any issues with #41? Can I merge it and then you can rebase and keep going ?

736. assistant (2026-07-02T06:11:33.183Z)

Let me verify #41's actual merge state on GitHub before giving you a grounded answer, and check
whether CI exists:

737. assistant (2026-07-02T06:11:33.183Z)

```
[external_agent_tool_call: Bash]
description: Check #41 mergeability, CI, and file overlap with #42
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && gh pr view 41 --json
mergeable,mergeStateStatus,title 2>&1 | head -20 && echo "=== CI checks on #41 (if any) ===" &&
gh pr checks 41 2>&1 | head -10 && echo "=== files #41 touches vs #42 (overlap check) ===" &&
echo "--#41--" && gh pr view 41 --json files -q '.files[].path' 2>&1 | head -30 && echo
"--#42--" && gh pr view 42 --json files -q '.files[].path' 2>&1
[/external_agent_tool_call]
```

738. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE","title":"feat: per-video observability
reports (indexer) + shared obsreport lib"}
=== CI checks on #41 (if any) ===
no checks reported on the 'feat/observability-reports' branch
=== files #41 touches vs #42 (overlap check) ===
--#41--
README.md
backend/main.py
backend/pipeline/validators/base.py
backend/reporting/__init__.py
backend/reporting/harvest.py
backend/reporting/spec.yaml
requirements.txt
tests/test_reporting_harvest.py
tests/test_validators_base.py
--#42--
backend/config/models.py
tests/test_config.py
[/external_agent_tool_result]
```

739. assistant (2026-07-02T06:11:33.184Z)

Two things resolved at once. First, your question about ****#41****:

****No issues ? safe to merge.**** GitHub reports it ``MERGEABLE`` / ``CLEAN``, no CI checks are configured (nothing to go red), and ****#41** and **#42** touch disjoint files** (``backend/reporting/*`` + ``main.py`` + ``pipeline/validators/base.py`` vs. ``backend/config/models.py``) ? so they merge independently in any order. 228 tests green. One design note: the reports only **emit** when ``obsreport`` is installed (soft dep) ? it's editable-installed on your machine so it works; a fresh clone/CI no-ops gracefully. Recommend merging ****#42**** too so real video runs aren't broken by the validator bug.

Go ahead and merge **#41** ? I'll rebase **#42** and my follow-on work onto the new master.

Also, the ****debuggability workflow just finished****. Let me read the full results:

740. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\56b7f849-5d56-4770-93e0-b3b4da943700\tasks\wwkbitg1n.output
[/external_agent_tool_call]
```

741. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
1      {
2      "summary": "Fresh-reader debuggability eval of per-video observability reports across
broken-run scenarios",
3      "agentCount": 15,
```

```

4      "logs": [],
5      "result": {
6          "scores": [
7              {
8                  "cause_correct": true,
9                  "faq_correct": true,
10                 "gap": "minor",
11                 "refinement": "In the silent_provider_noop finding, name the active segmenter
and tie it to the AI handoff so the reader does not get distracted wondering whether the
segmenter name is the fault. E.g.: \"The 'wasb_hybrid' segmenter found 1 candidate window but
produced 0 confirmed segments because it hands those windows to the AI provider (gemini), and no
API key was set (api_key_present=False) ? so the provider silently did nothing. This is a
missing credential, NOT 'the video has no rallies'. Fix: set GEMINI_API_KEY in a .env file in
the project root (see README#Q7) and re-run.\"",
12                 "notes": "Reader landed exactly on the root cause (missing GEMINI_API_KEY ->
silent AI provider no-op, 0 segments, NOT a no-rallies video) and consulted the expected FAQ
(README#Q7, which exists at README.md:197). Fix proposed by reader matches ground truth.
could_self_debug=true is justified.\n\nReport quality is high: the finding explicitly rules out
the false 'no rallies' interpretation, gives the GEMINI_API_KEY/.env fix, points to Q7, and
cites evidence (api_key_present=False, segment_count=0).\n\nRemaining gap is minor and stems
from one report detail: the segmentation stage prominently shows
_segmenter_requested/_actual/_config_default_ = wasb_hybrid with
_matches_config_default_=True, but never connects wasb_hybrid to the AI handoff. This led the
reader to a side-quest ? believing 'wasb_hybrid' is undocumented/invalid and must be changed to
yolo_hybrid/gemini/motion. In fact wasb_hybrid is a real, registered segmenter
(backend/indexer/wasb_hybrid.py) that routes through the gemini provider; it is just absent from
Q7's enumeration. The reader's 'invalid segmenter' worry is therefore a false lead that did not
block self-debug but did add noise.\n\nTwo smaller nits the reader raised: (1) 'silent provider
no-op' is jargon, though the finding immediately glosses it; (2) the finding says '.env file'
but not where ? the project-root location lives only in Q7. The proposed refinement folds the
segmenter-to-handoff link and the explicit project-root path into the finding so the reader
never has to leave it.\n\nSecondary doc-gap worth a follow-up (not blocking this scenario):
README Q7 lists only yolo_hybrid/gemini/motion and omits wasb_hybrid, which is what made the
reader think it was invalid.",
13                 "scenario": "indexer_no_api_key",
14                 "tool": "indexer",
15                 "reader": {
16                     "diagnosis": "The tool ran the segmenter with AI handoff enabled but the
Gemini API key was not set (GEMINI_API_KEY missing from .env). The AI provider silently exited
without processing any segments, resulting in 0 confirmed segments and making it incorrectly
appear as if \"the video has no rallies.\" This is a silent failure caused by a missing
credential, not a video content issue.",
17                     "fix": "Create a .env file in the project root
(C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\) with the line
GEMINI_API_KEY=your_actual_key. Obtain an API key from Google AI Studio. Verify config.json has
a valid segmenter name (yolo_hybrid, gemini, or motion ? not wasb_hybrid). Re-run the processing
command.",
18                     "faq_section": "README#Q7 ? How do I configure and run an AI-powered segmenter
(Gemini / YOLO-hybrid)?",
19                     "could_self_debug": true,
20                     "confusing_parts": [
21                         "The segmenter name 'wasb_hybrid' is not documented and appears invalid ?
the report does not explain where this came from or how to correct it to a valid option
(yolo_hybrid, gemini, or motion).",
22                         "The term 'silent provider no-op' is technical jargon ? clearer phrasing for
users would be 'The AI provider e...
[/external_agent_tool_result]

```

742. assistant (2026-07-02T06:11:33.184Z)

The eval is in and the result is strong:

Debuggability eval ? verdict: *all 7 scenarios self-debuggable*** ?**

`can\_readers\_debug: true`. Every fresh reader (given **only** README/FAQ + the report) landed on

the ****correct root cause****; 6/7 also cited the exact expected FAQ. Gaps were all "minor" (one "none"). So the reports already pass your bar ? but the agents found concrete polish, two of them genuinely important:

****High-priority (real doc defects):****

1. ****`fps\_fallback\_used`** cites the wrong FAQ** ? it points at README Q5, which is about HEVC keyframes and never mentions fps. Needs its own fps FAQ. *(A faq existence-test wouldn't catch this ? Q5 exists; it's semantically wrong. Good catch by the eval.)*
2. ****README Q7 omits `wasb\_hybrid`/`tracknet\_hybrid`**** ? readers see `wasb\_hybrid` in the report and waste effort thinking it's an invalid segmenter (false lead in 3 scenarios).

****Minor polish:**** name the active segmenter + explicit `.env` path in `silent\_provider\_noop`; show the numeric fps fallback (`30 (fallback)` not `? (fallback)`); flag motion metadata as synthetic at the data level, not just prose; make the blur failure line name `validation.min\_blur\_threshold` + remedies; add frame-level numbers to the collector overlap finding; mark `possible\_missed\_rally` as review-only/non-guarantee.

Now let me honor your "screenshot and eyeball" ? generating a representative technical + customer pair and rendering them to an image:

743. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Local\Temp\obseval\gen_eyeball_indexer.py
[/external_agent_tool_call]
```

744. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Local\Temp\obseval\gen_eyeball_indexer.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

745. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\AppData\Local\Temp\obseval\gen_eyeball_collector.py
[/external_agent_tool_call]
```

746. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\AppData\Local\Temp\obseval\gen_eyeball_collector.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

747. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_call: Bash]
description: Generate eyeball reports + render combined HTML
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python
"C:/Users/avidu/AppData/Local/Temp/obseval/gen_eyeball_indexer.py" 2>&1 | tail -2
cd "C:/Users/avidu/Projects/sports-data-collector" && python
"C:/Users/avidu/AppData/Local/Temp/obseval/gen_eyeball_collector.py" 2>&1 | tail -2
cd "C:/Users/avidu/AppData/Local/Temp/obseval" && python render_html.py
"C:/Users/avidu/AppData/Local/Temp/obseval/html/index.html" "Indexer ? technical
report::html_src/indexer.report.md" "Indexer ? customer summary::html_src/indexer.summary.md"
"Collector ? technical report::html_src/collector.report.md" "Collector ? delivery summary
(customer)::html_src/collector.summary.md" 2>&1 | tail -2
[/external_agent_tool_call]
```

748. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
```

```
wrote indexer eyeball md
wrote collector eyeball md
wrote C:/Users/avidu/AppData/Local/Temp/obseval/html/index.html
[/external_agent_tool_result]
```

749. assistant (2026-07-02T06:11:33.184Z)

Now serve the HTML and screenshot it. Creating a throwaway static-server config (untracked; I'll remove it after):

750. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\launch.json
[/external_agent_tool_call]
```

751. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\launch.json (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

752. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_call: mcp_Claude_Preview__preview_start]
input: {"name":"obseval-report-preview"}
[/external_agent_tool_call]
```

753. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
{
  "serverId": "9dd33862-4c8c-4d15-b578-61bbaf6f5328",
  "port": 8731,
  "name": "obseval-report-preview",
  "reused": false
}
Server started successfully on port 8731.
[/external_agent_tool_result]
```

754. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_call: mcp_Claude_Preview__preview_screenshot]
input: {"serverId":"9dd33862-4c8c-4d15-b578-61bbaf6f5328"}
[/external_agent_tool_call]
```

755. assistant (2026-07-02T06:11:33.184Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

756. assistant (2026-07-02T06:11:33.184Z)

The render looks clean (tables, fps shown as "59.94 fps (probed)"). Let me widen the viewport to capture the full reports ? the cards are laid out side by side ? and grab the key sections (Findings + customer summaries):

757. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp_Claude_Preview__preview_resize,mcp_Claude_Preview__preview_eval",
max_results":5}
```

```
[/external_agent_tool_call]
```

758. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result]  
[/external_agent_tool_result]
```

759. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_resize]  
input: {"serverId":"9dd33862-4c8c-4d15-b578-61bbaf6f5328","width":1260,"height":1180}  
[/external_agent_tool_call]
```

760. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result]  
Viewport set to 1260x1180.  
[/external_agent_tool_result]
```

761. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]  
input: {"serverId":"9dd33862-4c8c-4d15-b578-61bbaf6f5328","expression":"window.scrollTo(0, 560);  
document.title"}  
[/external_agent_tool_call]
```

762. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result]  
""  
[/external_agent_tool_result]
```

763. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_screenshot]  
input: {"serverId":"9dd33862-4c8c-4d15-b578-61bbaf6f5328"}  
[/external_agent_tool_call]
```

764. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result: error]  
preview_screenshot timed out after 30s. The preview window may be stuck (modal dialog,  
navigation hang, or unresponsive renderer). Check preview_console_logs for errors.  
[/external_agent_tool_result]
```

765. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_screenshot]  
input: {"serverId":"9dd33862-4c8c-4d15-b578-61bbaf6f5328"}  
[/external_agent_tool_call]
```

766. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result: error]  
preview_screenshot timed out after 30s. The preview window may be stuck (modal dialog,  
navigation hang, or unresponsive renderer). Check preview_console_logs for errors.  
[/external_agent_tool_result]
```

767. assistant (2026-07-02T06:11:33.185Z)

The screenshot tool hung after resize (I already have one clean capture confirming the visual layout renders well). Rather than fight it, let me eyeball the two ****customer summaries**** (the "customer-friendly video report" you care most about) directly from the rendered source, then

clean up the preview:

768. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_stop]
input: {"serverId":"9dd33862-4c8c-4d15-b578-61bbaf6f5328"}
[/external_agent_tool_call]
```

769. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result]
Server 9dd33862-4c8c-4d15-b578-61bbaf6f5328 stopped
[/external_agent_tool_result]
```

770. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\obseval\html_src\indexer.summary.md
[/external_agent_tool_call]
```

771. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result]
1      # Badminton Highlight Indexer ? Video Report (summary)
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Video | Adarsh and Vasanth.mp4 |
8      | Report generated | 2026-06-07T08:29:42Z |
9
10     ## Video
11
12     | Field | Value |
13     | --- | --- |
14     | Duration | 11m16s |
15     | Resolution | 1920x1080 |
16     | Frame rate | 59.94 fps |
17
18     ## Highlights
19
20     - **0** highlights
21     - **0.0s** of play
22     - **0%** of the video is highlights
23     - No highlights were produced for this video.
24
25     ## Things to know
26
27     - ? We couldn't generate highlights for this video due to a configuration issue on our
28     side. Please contact support.
29
30     ## Verdict
31
32     ?? We couldn't finish generating highlights for this video. Please contact support.
[/external_agent_tool_result]
```

772. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\obseval\html_src\collector.summary.md
[/external_agent_tool_call]
```

773. assistant (2026-07-02T06:11:33.185Z)

```
[external_agent_tool_result]
1      # Sports Data Collector ? Delivery / Ingestion Report (summary)
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Video | pilot_badminton_03 |
8      | Report generated | 2026-06-07T08:29:43Z |
9
10     ## Video
11
12     | Field | Value |
13     | --- | --- |
14     | Duration | 10m00s |
15     | Frame rate | 59.94 fps |
16
17     ## Rally summary
18
19     - *** rallies
20     - **14.0s** of play
21     - **2.3%** of the video is rallies
22
23     ## Things to review
24
25     - ? Rally 1 (ends 7.00s) overlaps Rally 2 (starts 6.00s) by 1.00s ? rallies must be
disjoint.
26     - ?? 26.0s of unlabeled time between Rally 2 (ends 11.0s) and Rally 3 (starts 40.0s) ?
check for an unlabeled rally in the gap.
27     - ?? Rally 3: 2 shots over 4.0s (0.50/s) is implausibly low ? the span may be inflated
by a stray box.
28
29     ## Verdict
30
31     This delivery has blocking issues that must be fixed before it can be imported ? see
'Things to review'.
32
[/external_agent_tool_result]
```

774. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_call: Bash]
description: Remove throwaway launch.json, confirm clean tree
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && rm -f .claude/launch.json
&& rmdir .claude 2>/dev/null; git status -s | head; echo "launch.json cleaned; working tree:";
git status -s | wc -l
[/external_agent_tool_call]
```

775. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_result]
launch.json cleaned; working tree:
0
[/external_agent_tool_result]
```

776. assistant (2026-07-02T06:11:33.186Z)

Cleaned up the throwaway preview config ? working tree is clean.

Eyeball verdict (screenshot + rendered customer summaries)

The rendered reports look clean and professional: tidy tables, `Video` shows **`59.94 fps (probed)`** (the probe-vs-fallback distinction is visible), and the **customer summaries strip

all internals** ? they show only video metadata, highlight/rally counts, plain-language "things to know/review," and a gentle audience-appropriate verdict:

- **Indexer customer:** "We couldn't finish generating highlights for this video. Please contact support." (no codes, no jargon) ?
- **Collector delivery summary:** *3 rallies · 14.0s of play · "Rally 1 overlaps Rally 2 by 1.00s?" + "This delivery has blocking issues that must be fixed?" (vendor-actionable, no internal codes) ?

No layout/visual problems ? the only improvements are the content refinements the eval surfaced.

Bottom line on your question

##41 has no issues ? merge away. It's `MERGEABLE`/`CLEAN`, 228 tests green, and touches files disjoint from #42, so order doesn't matter. Recommend merging ##42 (config fix) too so real video runs work.

After you merge ? my plan (I'll rebase and keep going)

1. **Rebase** #42 (and anything else) onto the new master.
2. **Apply the eval's refinements** as a focused follow-up:
 - *High:* repoint `fps\_fallback\_used` to a new fps FAQ (not the keyframe Q5); add `wasb\_hybrid`/`tracknet\_hybrid` to README Q7's segmenter list.
 - *Minor:* numeric fps fallback (`30 (fallback)`), motion-metadata "synthetic" surfaced at the data level, self-remediating blur line, frame-level detail on the collector overlap finding, review-only framing on `possible\_missed\_rally`.
 - Push indexer refinements as a new PR; push collector refinements to ##11 (still open).
3. **Add the golden regression tests** (faq-ref-resolution: every flag's cited anchor exists in its doc; + expected flag-set per fixture) to both repos.

Go ahead and merge #41 (and #42) whenever you're ready ? ping me and I'll rebase and roll straight into the refinements + golden tests.

777. user (2026-07-02T06:11:33.186Z)

Can you review the comments on #42 and respond?

778. assistant (2026-07-02T06:11:33.186Z)

Let me pull the comments on PR #42:

779. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_call: Bash]
description: Fetch #42 comments and reviews
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== ISSUE/REVIEW
COMMENTS ===" && gh pr view 42 --json comments,reviews -q '.comments[]?, .reviews[]? | {author:
.author.login, body: .body}' 2>&1 | head -80
[/external_agent_tool_call]
```

780. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_result]
=== ISSUE/REVIEW COMMENTS ===
{"author":"avidullu","body":""}
[/external_agent_tool_result]
```

781. assistant (2026-07-02T06:11:33.186Z)

The top-level view shows little ? the comments are likely **inline review comments** on specific lines. Let me fetch those via the API:

782. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_call: Bash]
description: Fetch inline review comments + review summaries on #42
```

```
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== INLINE REVIEW
COMMENTS on #42 ===" && gh api repos/avidullu/badminton-highlight-indexer/pulls/42/comments -q
'.[ ] | "?? \(.user.login) on \(.path):\(.line // .original_line)\n\(.body)\n"' 2>&1 | head -120
&& echo "" && echo "=== REVIEW SUMMARIES ===" && gh api repos/avidullu/badminton-highlight-
indexer/pulls/42/reviews -q '.[ ] | "?? \(.user.login) [\(.state)]\n\(.body)\n"' 2>&1 | head -60
[/external_agent_tool_call]
```

783. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_result]
=== INLINE REVIEW COMMENTS on #42 ===

=== REVIEW SUMMARIES ===
?? avidullu [PENDING]
[/external_agent_tool_result]
```

784. assistant (2026-07-02T06:11:33.186Z)

A `PENDING` review by you means draft comments not yet submitted ? those aren't visible via the API until you submit the review. Let me check the other comment channels (PR conversation, commit comments) to be sure I'm not missing anything:

785. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_call: Bash]
description: Check issue comments, all reviews, inline comment count on #42
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== PR conversation
(issue) comments ===" && gh api repos/avidullu/badminton-highlight-indexer/issues/42/comments -q
'.[ ] | "?? \(.user.login)\n\(.body)\n"' 2>&1 | head -60 && echo "[end issue comments]" && echo
"" && echo "=== all reviews (id/state/body) ===" && gh api repos/avidullu/badminton-highlight-
indexer/pulls/42/reviews 2>&1 | python -c "import sys,json; d=json.load(sys.stdin); [print('id',
r['id'], 'state', r['state'], 'author', r['user']['login'], 'body=', repr(r.get('body', ''))) for r in
d]" 2>&1 | head -30 && echo "" && echo "=== review comments incl pending (via review id) ===" &&
gh api repos/avidullu/badminton-highlight-indexer/pulls/42/comments 2>&1 | python -c "import
sys,json; d=json.load(sys.stdin); print('inline comment count:', len(d))" 2>&1
[/external_agent_tool_call]
```

786. assistant (2026-07-02T06:11:33.186Z)

```
[external_agent_tool_result]
=== PR conversation (issue) comments ===
[end issue comments]

=== all reviews (id/state/body) ===
id 4444704980 state PENDING author avidullu body= ''

=== review comments incl pending (via review id) ===
inline comment count: 0
[/external_agent_tool_result]
```

787. assistant (2026-07-02T06:11:33.187Z)

There are no submitted comments, but there's a **pending review** (your draft, id `4444704980`). Since I'm authenticated as your account, let me try to read that pending review's draft comments directly:

788. assistant (2026-07-02T06:11:33.187Z)

```
[external_agent_tool_call: Bash]
description: Read pending review draft comments as the author
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== pending review
draft comments (as author) ===" && gh api repos/avidullu/badminton-highlight-
indexer/pulls/42/reviews/4444704980/comments 2>&1 | python -c "import sys,json
try:
```

```

d=json.load(sys.stdin)
print('count:', len(d))
for c in d:
    print('??', c.get('path'), ':', c.get('line') or c.get('original_line'))
    print(c.get('body'))
    print()
except Exception as e:
    print('err:', e)
" 2>&1 | head -80
[/external_agent_tool_call]

```

789. assistant (2026-07-02T06:11:33.187Z)

```

[external_agent_tool_result]
=== pending review draft comments (as author) ===
count: 5
?? backend/config/models.py : None
Good catch on the real-run breakage vs unit tests. The previous .get() only handled top-level
and didn't dump nested Pydantic models to dicts, which is exactly what broke the legacy chained
access in validators/segmenters.

```

The updated logic + docstring clearly explains the intent for legacy compat. Nice.

```

?? backend/config/models.py : None
The fallback loop for leaf keys inside sections is still limited to the four known sections. If
we ever add another top-level config section (e.g. `audio` or `jobs` in the future), keys inside
it won't be discoverable via the legacy .get() fallback.

```

Worth a small comment noting this, or we can make the section list dynamic from model_fields later.

```

?? backend/config/models.py : None
Minor: the `get()` method is accumulating more and more legacy compat logic. Consider adding a
short class or method-level comment like:

```

Temporary migration shim for legacy dict-style access.

Will be removed once all call sites (validators, segmenters, etc.) are migrated to attribute access.

This sets expectations for future cleanup.

```

?? tests/test_config.py : None
The new regression test is excellent and directly targets the exact failure mode (chained .get
on nested sections including relevance).

```

One small suggestion: you could also add a quick assertion that a top-level scalar with a default still works (you have one for the missing key at the end, but making the scalar case explicit is nice for coverage).

```

?? backend/config/models.py : None
Overall this is a solid, minimal fix that restores the promised legacy compat without changing
the happy path for attribute access.

```

The PR description is clear and links the context well. Good work surfacing this during the end-to-end validation.

No blocking issues from my side. A couple of nits above for polish.
[/external_agent_tool_result]

790. assistant (2026-07-02T06:11:33.187Z)

I can see your 5 pending-review comments. Three are actionable nits (2 praise). Since the cleanest way to "respond" to review feedback is to address it, let me apply the nits to #42's branch. Switching to it and checking master state:

791. assistant (2026-07-02T06:11:33.187Z)

```
[external_agent_tool_call: Bash]
description: Switch to fix branch, check if master advanced
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout fix/appconfig-
nested-get 2>&1 | tail -1 && git fetch -q origin && echo "=== is #41/#42 merged? origin/master
head ===" && git log origin/master --oneline -3 && echo "=== this branch vs origin/master ==="
&& git rev-list --left-right --count origin/master...HEAD
[/external_agent_tool_call]
```

792. assistant (2026-07-02T06:11:33.187Z)

```
[external_agent_tool_result]
Switched to branch 'fix/appconfig-nested-get'
=== is #41/#42 merged? origin/master head ===
0355417 Merge pull request #40 from avidullu/refactor/approved-directory-structure
1e96e07 test: fix mock patch targets after structure refactor and config test assertions
8615534 refactor: adopt approved layered directory structure
=== this branch vs origin/master ===
0          1
[/external_agent_tool_result]
```

793. assistant (2026-07-02T06:11:33.187Z)

Master hasn't moved (neither #41 nor #42 merged yet) ? no rebase needed. Let me apply the three actionable nits. First, make the section scan dynamic (nit #2) and sharpen the shim comment (nit #3):

794. assistant (2026-07-02T06:11:33.187Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

795. assistant (2026-07-02T06:11:33.187Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has been modified since read, either by the user or by a linter. Read it
again before attempting to write it.</tool_use_error>
[/external_agent_tool_result]
```

796. assistant (2026-07-02T06:11:33.187Z)

Let me re-read the current state of the file on this branch before editing:

797. assistant (2026-07-02T06:11:33.187Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]
```

798. assistant (2026-07-02T06:11:33.187Z)

```
[external_agent_tool_result]
110
111         # Temporary dict-like access for legacy code that does config.get("foo", {})
112         # and passes the config object around. Will be removed after consumers are migrated.
113         #
114         # A nested *section* (e.g. config.get("validation")) is returned as a plain dict
115         # via model_dump, so legacy callers that chain dict access ?
116         # config.get("validation", {}).get("min_resolution_width", ...) ? keep working.
117         # (Returning the Pydantic section model instead breaks them, since a model has
118         # no .get; that bit the validators on every real run.) Scalar leaves are returned
119         # as-is.
```

```

120     def get(self, key: str, default: Any = None) -> Any:
121         _MISSING = object()
122         val: Any = _MISSING
123         if key in type(self).model_fields or hasattr(self, key):
124             val = getattr(self, key, _MISSING)
125         if val is _MISSING:
126             # Fall back to a leaf key living inside a known section (legacy
convenience).
127             for section_name in ("validation", "indexing", "output", "stitching"):
128                 section = getattr(self, section_name, None)
129                 if section is not None and hasattr(section, key):
130                     val = getattr(section, key)
131                     break
132         if val is _MISSING:
133             return default
134         if isinstance(val, BaseModel):
135             return val.model_dump(mode="json")
136         return val
137
138     def __getitem__(self, key: str) -> Any:
139         val = self.get(key)
140         if val is None and key not in self.model_fields:
141             # Re-raise like a real dict for unknown top-level
142             raise KeyError(key)
143         return val
144
145
146     # Convenience type alias for callers that want to annotate
147     Config = AppConfig
[/external_agent_tool_result]

```

799. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
[/external_agent_tool_call]

```

800. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

801. assistant (2026-07-02T06:11:33.188Z)

Now nit #4 ? add the explicit scalar-with-default assertion to the regression test, plus a dynamic-section-discovery assertion to lock the nit #2 improvement:

802. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py
[/external_agent_tool_call]

```

803. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_result]
88
89     def test_get_returns_dict_for_nested_sections_legacy_compat():
90         """Regression: legacy consumers (validators/segmenters) do
91         ``config.get("validation", {}).get("min_resolution_width", ...)``. The dict-compat
92         .get must return nested SECTIONS as plain dicts (not Pydantic models), or that
93         chained access raises 'X object has no attribute get' on every real run."""

```

```

94         cfg = AppConfig()
95         validation = cfg.get("validation", {})
96         assert isinstance(validation, dict)
97         assert validation.get("min_resolution_width") == 1280
98         # nested-of-nested is also dict-accessible
99         assert isinstance(validation.get("relevance"), dict)
100        assert validation["relevance"].get("court_pixels_min_ratio") == 0.05
101        assert cfg.get("indexing", {}).get("default_segmenter") == "yolo_hybrid"
102        # scalar leaves are returned unchanged; missing keys fall back to the default
103        assert cfg.get("ai_provider") == "gemini"
104        assert cfg.get("does_not_exist", "fallback") == "fallback"
105
[/external_agent_tool_result]

```

804. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py
[/external_agent_tool_call]

```

805. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

806. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_call: Bash]
description: Run config tests + full suite after nits
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest
tests/test_config.py -q 2>&1 | tail -4 && echo "=== full suite (no regressions) ===" && python
-m pytest -q 2>&1 | tail -3
[/external_agent_tool_call]

```

807. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_result]
..... [100%]
6 passed in 0.19s
=== full suite (no regressions) ===
..... [ 66%]
..... [100%]
215 passed in 11.94s
[/external_agent_tool_result]

```

808. assistant (2026-07-02T06:11:33.188Z)

All green (215; no warnings). Let me commit and push to #42:

809. assistant (2026-07-02T06:11:33.188Z)

```

[external_agent_tool_call: Bash]
description: Commit + push review-nit fixes to #42
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git add
backend/config/models.py tests/test_config.py && git commit -q -m "review: address #42 nits ?
dynamic section scan, shim comment, scalar-default test

```

- AppConfig.get section fallback now discovers nested sections dynamically from model_fields (not a hardcoded validation/indexing/output/stitching list), so a future top-level section (e.g. audio/jobs) is covered automatically.
- Reworded the comment to mark get()/__getitem__ as a TEMPORARY migration shim to be removed once validators/segmenters move to typed access.

- Tests: assert a present scalar ignores the default arg, and that section-leaf keys resolve via the dynamic scan (validation + stitching). `__getitem__` uses class-level `model_fields` (drops the last instance-access deprecation).

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q origin fix/appconfig-nested-get 2>&1 | tail -2 && echo "pushed; #42 updated"
[/external_agent_tool_call]
```

810. assistant (2026-07-02T06:11:33.188Z)

```
[external_agent_tool_result]
pushed; #42 updated
[/external_agent_tool_result]
```

811. assistant (2026-07-02T06:11:33.188Z)

Done ? addressed all the actionable feedback and pushed to #42. Here's my point-by-point response to your 5 review comments:

****1.** "Good catch on real-run breakage vs unit tests? nice." **** ?** Thanks. The key was that the old `.get()` only handled top-level and returned the nested `*model*`; dumping sections to dicts is what restores the chained legacy access. **? (no change needed)**

****2.** "Fallback loop is limited to the four known sections? new section like ``audio`/`jobs`` won't be discoverable; make it dynamic from ``model_fields``." **** ?** ****Done**, the better option. ****** The section scan now iterates ``type(self).model_fields`` and recurses into any field whose value is a ``BaseModel``, so a future top-level section is covered automatically with no hardcoded list to maintain. Added test assertions proving leaf keys resolve from ``validation`` and ``stitching`` via the dynamic scan.

****3.** "``get()`` is accumulating legacy compat logic; add a 'temporary migration shim' comment." **** ?** ****Done**. ****** Reworded the comment on ``get()`` to: `"Temporary migration shim for legacy dict-style access? Will be removed once all call sites (validators, segmenters, etc.) move to typed attribute access."` Sets the cleanup expectation explicitly.

****4.** "Regression test is excellent; also assert a top-level scalar with a default still works." **** ?** ****Done**. ****** Added ``assert cfg.get("ai_provider", "ignored-default") == "gemini"`` (present scalar ignores the default arg), alongside the existing missing-key?default case.

****5.** "Solid minimal fix, no blocking issues, nits for polish." **** ?** Appreciated. All nits applied; also switched `__getitem__` to class-level ``model_fields`` to drop the last Pydantic instance-access deprecation warning. ****215 tests pass, no warnings.****

Pushed as commit "review: address #42 nits ?" ? your pending line comments will show as outdated since those lines changed.

Two notes:

- Your review is still `**`PENDING`**` (draft) ? the comments aren't publicly submitted yet, so I responded here rather than posting threaded replies (which aren't possible until you submit). If you'd like, I can post a single summary comment on #42 recording these resolutions ? say the word and I'll add it (it'd appear under your account, since that's the authenticated identity).
- #42 is still based on the pre-#41 master (they're disjoint, so it merges cleanly in any order). Whenever you merge #41/#42, ping me and I'll rebase and roll into the eval refinements + golden tests.

812. user (2026-07-02T06:11:33.189Z)

I submitted the comments sa well. So now we are good to merge this?

813. assistant (2026-07-02T06:11:33.189Z)

Let me verify #42's current state ? the submitted review, mergeability, and that my fixes are on the branch:

814. assistant (2026-07-02T06:11:33.189Z)

```
[external_agent_tool_call: Bash]
description: Check #42 review state, mergeability, new comments, branch HEAD
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== reviews (state)
=== " && gh api repos/avidullu/badminton-highlight-indexer/pulls/42/reviews -q '[] |
"\(.user.login): \(.state)"' 2>&1 && echo "" && echo "=== mergeable / state ===" && gh pr view
42 --json mergeable,mergeStateStatus,reviewDecision -q
'{mergeable,mergeStateStatus,reviewDecision}' 2>&1 && echo "" && echo "=== any NEW unaddressed
inline comments? ===" && gh api repos/avidullu/badminton-highlight-indexer/pulls/42/comments -q
'[] | "\(.path):\(.line // .original_line) [\(.user.login)] \(.body[0:80])"' 2>&1 | head -20 &&
echo "[end]" && echo "" && echo "=== PR branch HEAD on origin == my fix? ===" && git rev-parse
HEAD && gh api repos/avidullu/badminton-highlight-indexer/pulls/42 -q '.head.sha' 2>&1
[/external_agent_tool_call]
```

815. assistant (2026-07-02T06:11:33.189Z)

```
[external_agent_tool_result]
=== reviews (state) ===
avidullu: COMMENTED

=== mergeable / state ===
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE","reviewDecision":""}

=== any NEW unaddressed inline comments? ===
backend/config/models.py:115 [avidullu] Good catch on the real-run breakage vs unit tests. The
previous .get() only hand
backend/config/models.py:122 [avidullu] The fallback loop for leaf keys inside sections is still
limited to the four kno
backend/config/models.py:112 [avidullu] Minor: the `get()` method is accumulating more and more
legacy compat logic. Con
tests/test_config.py:100 [avidullu] The new regression test is excellent and directly targets
the exact failure mode
backend/config/models.py:1 [avidullu] Overall this is a solid, minimal fix that restores the
promised legacy compat wi
[end]

=== PR branch HEAD on origin == my fix? ===
1358521ce3587b684a1bdb8b8ff3680e9ff7b463
1358521ce3587b684a1bdb8b8ff3680e9ff7b463
[/external_agent_tool_result]
```

816. assistant (2026-07-02T06:11:33.189Z)

Yes ? **#42 is good to merge.** Verified:

- **All 5 review comments addressed** and pushed (origin HEAD = my fix commit `1358521`). The submitted review is **`COMMENTED`** (non-blocking ? your own note said "No blocking issues from my side"). The inline comments now show against changed lines, so they'll render as *outdated*. No new/unaddressed feedback.
- **`MERGEABLE` / `CLEAN`** against current master.
- **215 tests pass, no warnings.**
- The two actionable nits are genuinely improved: dynamic section discovery (future-proof) + the temporary-shim comment; plus the extra test assertions.

Merge whenever you like. Quick notes:

- **Order doesn't matter for conflicts** (#42 and #41 touch disjoint files), but #42 is the one that **unbreaks real video runs** (the validators), so merging it is pure upside.
- After you merge, I'll **rebase #41 onto the new master**, then roll into the eval refinements + golden tests as we planned.

Want me to merge it for you now (`gh pr merge 42`)? I can do #42 and, if you want, #41 too ? just say go, and tell me your preference (squash / merge commit / rebase). Otherwise click merge and ping me, and I'll pick up the rebase.

817. user (2026-07-02T06:11:33.189Z)

I merged 42. Can I merge 41 now?

818. assistant (2026-07-02T06:11:33.189Z)

Let me verify before you do ? I'll confirm #42 landed, rebase #41 onto the new master, and run the full suite so we know the *combined* state is green (not just that GitHub says mergeable):

819. assistant (2026-07-02T06:11:33.189Z)

```
[external_agent_tool_call: Bash]
description: Confirm #42 merged, rebase #41 onto new master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout
feat/observability-reports 2>&1 | tail -1 && git fetch -q origin && echo "=== master now has
#42? ===" && git log origin/master --oneline -3 && echo "=== #41 vs new master (behind/ahead)
===" && git rev-list --left-right --count origin/master...HEAD && echo "=== rebase #41 onto new
master ===" && git rebase origin/master 2>&1 | tail -6
[/external_agent_tool_call]
```

820. assistant (2026-07-02T06:11:33.189Z)

```
[external_agent_tool_result]
Switched to branch 'feat/observability-reports'
=== master now has #42? ===
ab7f895 Merge pull request #42 from avidullu/fix/appconfig-nested-get
1358521 review: address #42 nits ? dynamic section scan, shim comment, scalar-default test
d02554e fix(config): AppConfig.get returns nested sections as dicts (legacy dict-compat)
=== #41 vs new master (behind/ahead) ===
3          4
=== rebase #41 onto new master ===
Rebasing (1/4)
Rebasing (2/4)
Rebasing (3/4)
Rebasing (4/4)
```

Successfully rebased and updated refs/heads/feat/observability-reports.
[/external_agent_tool_result]

821. assistant (2026-07-02T06:11:33.189Z)

#42 is on master, and #41 rebased onto it cleanly (disjoint, no conflicts). Now the decisive check ? full suite on the combined state, then I'll make the PR current:

822. assistant (2026-07-02T06:11:33.189Z)

```
[external_agent_tool_call: Bash]
description: Full suite on combined state, push #41, check mergeability
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest -q 2>&1 |
tail -4 && git push --force-with-lease -q origin feat/observability-reports 2>&1 | tail -2 &&
echo "" && echo "=== #41 mergeable now? ===" && gh pr view 41 --json mergeable,mergeStateStatus
-q '{mergeable,mergeStateStatus}' 2>&1
[/external_agent_tool_call]
```

823. assistant (2026-07-02T06:11:33.189Z)

```
[external_agent_tool_result]
..... [ 62%]
..... [ 94%]
..... [100%]
229 passed in 12.35s

=== #41 mergeable now? ===
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE"}
```

[/external_agent_tool_result]

824. assistant (2026-07-02T06:11:33.189Z)

****Yes ? merge #41 now.**** Verified on the combined state, not just GitHub's word:

- Rebased #41 onto the new master (which now includes ****#42**** `ab7f895`) ? ****clean, no conflicts**** (disjoint files).
- ****229 tests pass**** on master + #42 + #41 together (incl. the typed-`AppConfig`?report test now running against the *fixed* config). No warnings.
- GitHub: ****`MERGEABLE` / `CLEAN`****, and the PR branch is now current (force-pushed the rebase).

So the full pipeline is sound end-to-end: validators run (thanks to #42), and every run emits the report (#41). Go ahead and merge.

Once it's in, ping me and I'll:

1. Rebase remaining work onto the post-#41 master,
2. Apply the eval's refinements (the fps-FAQ repoint, README Q7 segmenter list, + the minor wording/data-surfacing fixes) as a follow-up PR, and
3. Add the golden regression tests (faq-ref-resolution + expected flag-set) ? plus push the collector-side refinements to ****#11****.

825. user (2026-07-02T06:11:33.189Z)

oh great you know just merge it yourself if its good to go!

826. assistant (2026-07-02T06:11:33.190Z)

On it ? merging #41 myself.

827. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_call: Bash]
description: Merge #41 (merge commit, delete branch), sync local master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout master 2>&1 |
tail -1 && gh pr merge 41 --merge --delete-branch 2>&1 | tail -6 && echo "=== sync local master
===" && git pull --ff-only origin master 2>&1 | tail -3 && echo "=== master head ===" && git log
--oneline -3
[/external_agent_tool_call]
```

828. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_result]
Switched to branch 'master'
=== sync local master ===
  create mode 100644 tests/test_config.py
  create mode 100644 tests/test_reporting_harvest.py
  create mode 100644 tests/test_validators_base.py
=== master head ===
98bb1d9 Merge pull request #41 from avidullu/feat/observability-reports
22d6410 chore(test): update reporting/validator test imports for layered package layout
(post-#40 restructure)
56ca0e0 test(reporting): cover typed AppConfig normalization in emit (post-#39 rebase)
[/external_agent_tool_result]
```

829. assistant (2026-07-02T06:11:33.190Z)

#41 is merged to master (`98bb1d9`). Let me confirm it's clean and green post-merge:

830. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_call: Bash]
description: Confirm #41 merged, master green, branch cleaned
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== #41 state ===" &&
```

```
gh pr view 41 --json state,mergedAt -q '{state,mergedAt}' 2>&1 && echo "=== full suite on merged master ===" && python -m pytest -q 2>&1 | tail -3 && echo "=== branches (feat gone?) ===" && git branch | head && echo "remote feat:" && git ls-remote --heads origin feat/observability-reports 2>&1 | head -1 || echo "(remote feat deleted)"
[/external_agent_tool_call]
```

831. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_result]
=== #41 state ===
{"mergedAt":"2026-06-07T08:44:39Z","state":"MERGED"}
=== full suite on merged master ===
..... [ 94%]
..... [100%]
229 passed in 12.67s
=== branches (feat gone?) ===
  fix/appconfig-nested-get
* master
remote feat:
[/external_agent_tool_result]
```

832. assistant (2026-07-02T06:11:33.190Z)

Both PRs are in. ****#41 is `MERGED`**, master is green (**229 passed**), and the feature branch is cleaned up. Let me tidy the stale local config-fix branch and checkpoint memory:**

833. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_call: Bash]
description: Delete merged local fix branch, show state
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git branch -d fix/appconfig-nested-get 2>&1 | tail -1 && echo "=== remaining branches ===" && git branch && echo "=== recent master ===" && git log --oneline -4
[/external_agent_tool_call]
```

834. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_result]
Deleted branch fix/appconfig-nested-get (was 1358521).
=== remaining branches ===
* master
=== recent master ===
98bb1d9 Merge pull request #41 from avidullu/feat/observability-reports
22d6410 chore(test): update reporting/validator test imports for layered package layout (post-#40 restructure)
56ca0e0 test(reporting): cover typed AppConfig normalization in emit (post-#39 rebase)
b8904dc chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
[/external_agent_tool_result]
```

835. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_call: Edit]
file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]
```

836. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

837. assistant (2026-07-02T06:11:33.190Z)

Both merged, master green (229), branches cleaned. ? The indexer pipeline is now whole: validators run (via #42) and every run emits the report (via #41).

Now I'll **keep going** with the debuggability refinements the eval surfaced. Starting the indexer follow-up branch and applying the high-value fixes. First, mark a new chapter and branch:

838. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"Debuggability refinements","summary":"Applying fresh-reader eval refinements +
golden faq-resolution tests after #41/#42 merged"}
[/external_agent_tool_call]
```

839. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_result]
Chapter marked: "Debuggability refinements". Continue your current work.
[/external_agent_tool_result]
```

840. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_call: Bash]
description: Branch for refinements; locate spec + README FAQ anchors
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout -b
feat/report-debuggability-refinements 2>&1 | tail -1 && echo "=== confirm reporting spec +
README present on master ===" && ls backend/reporting/spec.yaml && grep -n "### Q9\|### Q10\|##
?" README.md | head
[/external_agent_tool_call]
```

841. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_result]
Switched to a new branch 'feat/report-debuggability-refinements'
=== confirm reporting spec + README present on master ===
backend/reporting/spec.yaml
166:## ? Frequently Asked Questions (FAQ)
229:### Q9: How can I debug/test the Gemini Segmenter on a small portion of a video without
uploading a huge multi-GB file?
242:### Q10: Can I trust the per-segment metadata from the `motion` segmenter?
[/external_agent_tool_result]
```

842. assistant (2026-07-02T06:11:33.190Z)

```
[external_agent_tool_result]
197     ### Q7: How do I configure and run an AI-powered segmenter (Gemini / YOLO-hybrid)?
198     Both `yolo_hybrid` (default) and `gemini` need an API key for the AI provider. To use
Google Gemini:
199     1. Create a local `.env` file in the project root (this is already in `.gitignore` to
prevent secret leaks).
200     2. Set your Google AI Studio API key in the file:
201         ```.env
202         GEMINI_API_KEY=your_key_here
203         ```
204     3. Run with whichever segmenter you want:
205         ```bash
206         # Default: cheap candidate filtering with YOLO, then precise boundaries via Gemini
207         python -m backend.main --video-path "C:/path/to/video.mp4" --segmenter yolo_hybrid
208
209         # Full-video Gemini analysis (more expensive, highest semantic precision per call)
210         python -m backend.main --video-path "C:/path/to/video.mp4" --segmenter gemini
211
```

```
212      # Pure CV baseline ? no API key needed
213      python -m backend.main --video-path "C:/path/to/video.mp4" --segmenter motion
214      ```
215      *Or set `"default_segmenter": "yolo_hybrid"` (or `"gemini"/`"motion"`) under
216      `"indexing"` in `config.json`.
217      The AI provider used by `yolo_hybrid` and `gemini` is controlled by `ai_provider` +
218      `ai_model` at the top level of `config.json`. Today only `"gemini"` is registered; new providers
219      can be added under `backend/providers/`.
220      ### Q8: How does the `yolo_hybrid` pipeline actually save money?
221      Sending a multi-gigabyte video to a hosted LLM is slow and expensive. `yolo_hybrid` runs
222      in four phases:
223      [/external_agent_tool_result]
```

843. assistant (2026-07-02T06:11:33.191Z)

Let me read the README FAQ region (Q7 + Q13) to apply the segmenter-list and new fps-FAQ edits precisely:

844. assistant (2026-07-02T06:11:33.191Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

845. assistant (2026-07-02T06:11:33.191Z)

Now apply the **Q7 segmenter-list fix** (the #1 false-lead from the eval) ? note
`wasb\_hybrid`/`tracknet\_hybrid` are valid:

846. assistant (2026-07-02T06:11:33.191Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

847. assistant (2026-07-02T06:11:33.191Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

848. assistant (2026-07-02T06:11:33.191Z)

Now read the Q13 region to add the new fps FAQ (Q14) after it:

849. assistant (2026-07-02T06:11:33.191Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

850. assistant (2026-07-02T06:11:33.191Z)

```
[external_agent_tool_result]
255
256      To read the technical report: scan Verdict first (PASS / PASS-with-warnings / FAIL),
257      then Findings & warnings ? each finding has a plain-English message, evidence, and a `? see
258      README#Qn` pointer. To turn reports off or relocate them, set `"report": {"enabled": false}` or
259      `"report": {"output_dir": "...}"` in `config.json`.
260
```

```

258     ### Q13: A validation check failed and indexing was halted ? what do I do?
259     Open the report's Processing stages ? validation block (or the VALIDATION SUMMARY
printed to the console). Each failed check states the exact reason. Common cases: relevance
(the court color didn't match ? see [Q2](#q2-why-does-my-video-fail-the-relevance-check));
resolution/fps (below the configured minimums ? adjust validation.min_ in config.json);
format (not an MP4 container); keyframes/PTS (HEVC/GoPro keyframe quirks ? see
[Q5](#q5-why-did-the-video-fail-with-no-keyframes--packet-timestamps-could-be-extracted)). Fix
the underlying cause or relax the relevant threshold in config.json, then re-run.
260
261     ---
262
263     ## ? Annotation & Evaluation
264
265     Once a video is processed, you can measure how well the pipeline detected its
rallies against hand-annotated ground truth. You annotate each rally's [start, end) time
range in a small CSV; the harness scores the detected rallies straight from the SQLite DB ? no
API calls, fully deterministic.
266
267     ```bash
268     python run_eval.py --scaffold --annotations rallies.csv          # make an empty
template
269     # ...fill in start,end rows, then:
270     python run_eval.py --video "C:/videos/match.mp4" --annotations rallies.csv --show-
failures
271     ```
272
273     It reports precision/recall/F1, mean IoU, and boundary error for two stages ? raw
detector candidates vs. AI-confirmed final segments ? which pinpoints whether a weak result
is a detection problem or a segmentation problem.
274
275     > Rally timestamps evaluate and tune segmentation; they do not train the shuttle
detector (that needs per-frame coordinate labels). See
[docs/EVALUATION.md](docs/EVALUATION.md) for the annotation format, CLI flags, and how to
read the output.
276
[/external_agent_tool_result]

```

851. assistant (2026-07-02T06:11:33.191Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]

```

852. assistant (2026-07-02T06:11:33.191Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

853. assistant (2026-07-02T06:11:33.191Z)

Now update the spec: repoint `fps_fallback_used` to the new Q14, and sharpen the `silent_provider_noop` + `synthetic_motion_metadata` wording. Reading the current spec:

854. assistant (2026-07-02T06:11:33.191Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]

```

855. assistant (2026-07-02T06:11:33.191Z)

```

[external_agent_tool_result]

```

```

1      # Observability report spec for the badminton-highlight-indexer.
2      # The report ENGINE (config language, renderers) lives in the shared `obsreport`
3      # library; this file is the indexer-specific layout + footgun rules. Every
4      # actionable rule cites a README/FAQ section so a reader can self-debug.
5      name: indexer
6      title: Badminton Highlight Indexer ? Video Report
7
8      # Customer here = whoever receives the highlight reel.
9      customer_verdict:
10         pass: "? Your highlights are ready."
11         pass_with_warnings: "? Your highlights are ready. (A couple of minor notes above.)"
12         fail: "?? We couldn't finish generating highlights for this video. Please contact
support."
13
14     sections:
15         - { id: run,          title: "Run",          kind: identity,  audience: both }
16         - { id: video,       title: "Video",        kind: video_meta, audience: both }
17         - { id: highlights,  title: "Highlights",    kind: highlights, audience: customer
18     }
19         - { id: cust_notes,  title: "Things to know", kind: flags,      audience: customer
20     }
21         - { id: stages,     title: "Processing stages", kind: stages,     audience:
technical }
22         - { id: findings,   title: "Findings & warnings", kind: flags,      audience:
technical }
23         - { id: verdict,    title: "Verdict",        kind: verdict,    audience: both }
24
25     rules:
26         # The classic silent failure: 0 segments because the AI provider no-op'd on a
27         # missing API key ? NOT because the video has no rallies.
28         - code: silent_provider_noop
29           severity: error
30           when:
31             - { path: "stages.ai_handoff.details.api_key_present", op: eq, value: false }
32             - { path: "stages.segmentation.metrics.segment_count", op: eq, value: 0 }
33           message: >-
34             0 segments AND no AI provider API key was set ? this is almost certainly a
35             silent provider no-op, not 'the video has no rallies'. Set the provider key
36             (e.g. GEMINI_API_KEY in a .env file) and re-run.
37           faq_ref: "README#Q7"
38           customer_message: >-
39             We couldn't generate highlights for this video due to a configuration issue
40             on our side. Please contact support.
41
42         # API key WAS present but an AI segmenter still produced nothing ? likely a
43         # provider error / quota / WSL healthcheck issue rather than a truly empty video.
44         - code: ai_empty_vs_no_rallies
45           severity: warn
46           when:
47             - { path: "stages.ai_handoff.details.ai_based", op: eq, value: true }
48             - { path: "stages.ai_handoff.details.api_key_present", op: eq, value: true }
49             - { path: "stages.segmentation.metrics.segment_count", op: eq, value: 0 }
50           message: >-
51             An AI-based segmenter returned 0 segments even though the API key is set.
52             Empty can mean a provider error, exhausted quota, or (for WASB/TrackNet) a
53             failed WSL healthcheck ? these all return [] silently. Re-run with -u for
54             unbuffered logs and check the provider/WSL output before concluding 'no rallies'.
55           faq_ref: "README#Q7"
56
57         # fps was assumed rather than probed -> velocity thresholds + timing may be wrong.
58         - code: fps_fallback_used
59           severity: warn
60           when:
61             - { path: "video_meta.fps_source", op: eq, value: fallback }
62           message: >-

```

```
61         fps was assumed (a fallback default), not probed from the file. Motion
62         thresholds and frame->second timing may be miscalibrated. Confirm the real
63         fps with `ffprobe` and ensure validation ran (it probes the file).
64         faq_ref: "README#Q5"
65
66         # The motion baseline fabricates its per-segment metadata/events ? they are not
67         # measured. Warn so nobody trusts them as ground truth.
68         - code: synthetic_motion_metadata
69           severity: warn
70           when:
71             - { path: "stages.segmentation.details.segmenter_actual", op: eq, value: motion }
72           message: >-
73             The 'motion' segmenter was used. Its per-segment metadata (intensity, speed,
74             ...
[/external_agent_tool_result]
```

856. assistant (2026-07-02T06:11:33.191Z)

Applying the spec refinements ? `silent\_provider\_noop` (tie to AI handoff + explicit `.env` path), `fps\_fallback\_used` (?Q14, reworded), and `synthetic\_motion\_metadata` (front-load "placeholder/unmeasured"):

857. assistant (2026-07-02T06:11:33.191Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]
```

858. assistant (2026-07-02T06:11:33.191Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

859. assistant (2026-07-02T06:11:33.192Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]
```

860. assistant (2026-07-02T06:11:33.192Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

861. assistant (2026-07-02T06:11:33.192Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml
[/external_agent_tool_call]
```

862. assistant (2026-07-02T06:11:33.192Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml has
been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

863. assistant (2026-07-02T06:11:33.192Z)

Now the harvest refinements ? show the numeric fps fallback value, and surface motion's synthetic metadata at the data level. Reading harvest.py:

864. assistant (2026-07-02T06:11:33.192Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py
[/external_agent_tool_call]
```

865. assistant (2026-07-02T06:11:33.192Z)

```
[external_agent_tool_result]
100     st.add_metric("total_play_s", total_play, unit="s")
101     # "diverges" only when a segmenter actually RAN, the user did NOT explicitly
102     # pick it, and it differs from config's default (the genuine surprise case).
103     # An explicit --segmenter override or a halted run is not a divergence.
104     matches_default = True
105     if ran and segmenter_actual and config_default:
106         matches_default = (segmenter_actual == config_default)
107     st.details.update({
108         "segmenter_requested": segmenter_requested,
109         "segmenter_actual": segmenter_actual,
110         "config_default": config_default,
111         "segmenter_explicitly_requested": segmenter_requested is not None,
112         "matches_config_default": matches_default,
113         "was_reingest": was_reingest,
114         "detector": detector_tag,
115     })
116
117
118     def _ai_stage(rec: "obsreport.RunRecorder", config: Dict[str, Any], segmenter_actual:
Optional[str],
119                 play_segments: List[Dict[str, Any]], ran: bool) -> None:
120         """Only meaningful for AI-based segmenters. Records whether the provider key was
121         present ? the signal the silent_provider_noop rule keys on."""
122         ai_based = segmenter_actual in _AI_SEGMENTERS
123         provider = (config.get("ai_provider") or "gemini")
124         key_env = f"{provider.upper()}_API_KEY"
125         with rec.stage("ai_handoff") as st:
126             if not ran or not ai_based:
127                 st.status = "not_run" if not ran else "skipped"
128             st.details.update({
129                 "ai_based": ai_based,
130                 "provider": provider if ai_based else None,
131                 "model": config.get("ai_model") if ai_based else None,
132                 "api_key_present": bool(os.environ.get(key_env)) if ai_based else None,
133             })
134             if ai_based:
135                 st.add_metric("confirmed_segments", len(play_segments))
136
137
138     def _highlights(rec: "obsreport.RunRecorder", play_segments: List[Dict[str, Any]],
139                   video_meta: Dict[str, Any]) -> None:
140         total = round(sum(float(s.get("duration", 0.0)) for s in play_segments), 1)
141         coverage = None
142         dur = video_meta.get("duration_s")
143         if dur:
144             coverage = round(100.0 * total / dur, 1)
145         notes = []
146         if not play_segments:
147             notes.append("No highlights were produced for this video.")
148         rec.set_highlights(segment_count=len(play_segments), total_highlight_s=total,
149                           coverage_pct=coverage, notes=notes)
150
151
```


file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py
[/external_agent_tool_call]

872. assistant (2026-07-02T06:11:33.192Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

873. assistant (2026-07-02T06:11:33.192Z)

Now the ****golden FAQ-resolution test**** ? the committed guarantee that every flag's cited anchor actually exists in the README (this is what makes the reports durably self-debuggable):

874. assistant (2026-07-02T06:11:33.192Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_spec_faq.py
[/external_agent_tool_call]

875. assistant (2026-07-02T06:11:33.192Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_spec_faq.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

876. assistant (2026-07-02T06:11:33.193Z)

Run the full suite + the new golden test:

877. assistant (2026-07-02T06:11:33.193Z)

[external_agent_tool_call: Bash]
description: Run the new faq-resolution golden test
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest tests/test_reporting_spec_faq.py -q 2>&1 | tail -15
[/external_agent_tool_call]

878. assistant (2026-07-02T06:11:33.193Z)

[external_agent_tool_result]
..
2 passed in 0.24s [100%]
[/external_agent_tool_result]

879. assistant (2026-07-02T06:11:33.193Z)

All faq_refs resolve. Now the full suite, then I'll regenerate a couple of refined reports to confirm the wording lands:

880. assistant (2026-07-02T06:11:33.193Z)

[external_agent_tool_call: Bash]
description: Full indexer suite after refinements
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python -m pytest -q 2>&1 | tail -4
[/external_agent_tool_call]

881. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_result]
..... [ 62%]
..... [ 93%]
..... [100%]
231 passed in 12.21s
[/external_agent_tool_result]
```

882. assistant (2026-07-02T06:11:33.193Z)

231 passed (229 + 2 new). Let me eyeball the refined wording on the three affected scenarios (fps numeric, motion data-level, silent-provider):

883. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_call: Bash]
description: Render refined scenarios, grep key lines
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python - <<'PY'
import os
import obsreport
from backend.reporting import harvest
from backend.pipeline.validators.base import ValidationResult
def vr(n,p,m="ok"): return ValidationResult(validator_name=n,passed=p,message=m)
FF={"streams":[{"width":1920,"height":1080,"r_frame_rate":"60000/1001","codec_name":"hevc"}],"format":{"format_name":"mov,mp4,m4a","duration":"676.0","size":"55000000000"}}

print("==== FPS FALLBACK (no ffprobe meta) ====")
rec=harvest.record_run(video_id="x",video_path="/v/m.mp4",config={"indexing":{"default_segmenter":"wasb_hybrid"},"ai_provider":"gemini"},
    val_results=[vr("format_check",True)],val_context={},play_segments=[{"duration":5.0,"metadata":{"detector":"wasb-hybrid"}}],candidates=[{"id":1}],
    segmenter_requested="wasb_hybrid",segmenter_actual="wasb_hybrid",was_reingest=False,segmentation_ran=True)
md=obsreport.render_technical(rec.envelope,rec.spec)
for ln in md.splitlines():
    if "FPS" in ln or "fps_fallback" in ln or "README#Q14" in ln: print(" ",ln.strip())

print("==== MOTION (synthetic data-level) ====")
os.environ["GEMINI_API_KEY"]="x"
rec=harvest.record_run(video_id="y",video_path="/v/m.mp4",config={"indexing":{"default_segmenter":"motion"}}),
    val_results=[vr("format_check",True)],val_context={"ffprobe_meta":FF},play_segments=[{"duration":5.0,"metadata":{"detector":"motion"}}],candidates=[],
    segmenter_requested="motion",segmenter_actual="motion",was_reingest=False,segmentation_ran=True)
md=obsreport.render_technical(rec.envelope,rec.spec)
for ln in md.splitlines():
    if "metadata_source" in ln or "PLACEHOLDER" in ln or "synthetic_motion" in ln: print(" ",ln.strip())

print("==== SILENT PROVIDER NOOP ====")
os.environ.pop("GEMINI_API_KEY",None)
rec=harvest.record_run(video_id="z",video_path="/v/m.mp4",config={"ai_provider":"gemini","indexing":{"default_segmenter":"wasb_hybrid"}},
    val_results=[vr("format_check",True)],val_context={"ffprobe_meta":FF},play_segments=[],candidates=[{"id":1}],
    segmenter_requested="wasb_hybrid",segmenter_actual="wasb_hybrid",was_reingest=False,segmentation_ran=True)
md=obsreport.render_technical(rec.envelope,rec.spec)
import re
m=re.search(r"silent_provider_noop.*?(?=\n- |\n##)", md, re.S)
print(" ", (m.group(0)[:400] if m else "NOT FOUND").replace("\n","\n  "))
PY
[/external_agent_tool_call]
```

884. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_result]
===== FPS FALLBACK (no ffprobe meta) =====
  | FPS | 30 fps (fallback) |
  - [WARN] **fps_fallback_used** ? fps was NOT probed from the file ? a fallback default (shown
as the fps value) was used. Because rally timing and motion thresholds scale with fps,
boundaries and motion sensitivity may be miscalibrated. This usually means validation didn't run
(it probes the file) or ffprobe couldn't read r_frame_rate. Confirm the real fps with ffprobe
and re-run.
  ? see **README#Q14**
===== MOTION (synthetic data-level) =====
  - _metadata_source_: synthetic/heuristic ? NOT measured
  - [WARN] **synthetic_motion_metadata** ? The 'motion' segmenter was used: its per-segment
metadata (intensity, speed, events, server/receiver) are PLACEHOLDER values (unmeasured) ?
fabricated by the heuristic, NOT derived from the video ? so a PASS with segments here does not
mean the metadata is trustworthy. Use yolo_hybrid/gemini for real shot metadata.
===== SILENT PROVIDER NOOP =====
  silent_provider_noop** ? 0 segments AND no AI provider API key was set. The segmenter (see
Processing stages > segmentation > segmenter_actual) hands its candidate windows to the AI
provider, which silently did nothing without a key ? so this is a MISSING CREDENTIAL, not 'the
video has no rallies'. Fix: put GEMINI_API_KEY=<key> in a .env file in the project root and re-
run (see README#Q7).
  ? s
[/external_agent_tool_result]
```

885. assistant (2026-07-02T06:11:33.193Z)

All three refinements landed cleanly: `**`FPS | 30 fps (fallback)`**`, `motion `metadata_source: synthetic/heuristic ? NOT measured`` at the data level, and the silent-provider finding now ties the segmenter?AI-handoff with the explicit ``.env`` path. Committing + opening the indexer refinements PR:

886. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_call: Bash]
description: Commit + push + open indexer refinements PR
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git add -A && git status -s
&& git commit -q -m "feat(reporting): debuggability refinements from fresh-reader eval + faq-
resolution golden test"
```

A 7-scenario fresh-reader eval (agents given only README/FAQ + a report) found all scenarios self-debuggable; these address the gaps it surfaced:

- README Q7: document `wasb_hybrid/tracknet_hybrid` as valid segmenters (route to the gemini provider) ? readers were mistaking them for invalid (false lead in 3 cases).
- New README Q14 (fps fallback) + repoint `fps_fallback_used` `faq_ref` off Q5 (which is about HEVC keyframes, not fps) to Q14. harvest now shows the numeric fallback `(FPS '30 (fallback)' not '?' (fallback))`.
- `silent_provider_noop`: tie segmenter -> AI handoff + explicit `.env` project-root path.
- `synthetic_motion_metadata`: front-load PLACEHOLDER/unmeasured; surface `metadata_source` at the segmentation-stage data level (not just the warning).
- README Q13: add blur remediation (`min_blur_threshold` + `re-encode-on-0.0` hint).
- NEW golden test: every rule's `faq_ref` resolves to a real README heading + actionable rules must cite a FAQ.

231 tests green; refined wording verified by rendering the affected scenarios.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com> && git push -q -u origin feat/report-debuggability-refinements 2>&1 | tail -2 && gh pr create --base master --head feat/report-debuggability-refinements --title "feat(reporting): debuggability refinements + faq-resolution golden test" --body "\$(cat <<'EOF'

Follow-up to #41. A multi-agent `**fresh-reader eval**` (7 broken-run reports, each shown to an agent that may see `*only*` the README/FAQ + the report ? no source) found `**all 7 scenarios self-debuggable**`; this PR applies the concrete gaps it surfaced.

High-impact

- **README Q7** now documents ``wasb_hybrid`/`tracknet_hybrid`` as valid, registered segmenters that route candidate windows to the ``gemini`` provider. Readers were seeing ``segmenter_actual: wasb_hybrid`` in reports and wasting effort thinking it was invalid (a false lead in 3/7 scenarios).
- **fps FAQ fixed**: ``fps_fallback_used`` cited ``README#Q5``, which is about HEVC keyframes and never mentions fps. Added a dedicated **Q14** (what "fallback" means, why it matters, how to confirm real fps) and repointed the rule. The report now prints the numeric fallback (``FPS | 30 (fallback)``).

Wording / data-surfacing

- ``silent_provider_noop``: ties the segmenter ? AI handoff and gives the explicit ``.env`` project-root path.
- ``synthetic_motion_metadata``: front-loads "PLACEHOLDER/unmeasured"; the segmentation stage now carries ``metadata_source: synthetic/heuristic ? NOT measured`` at the **data** level, not just in prose.
- README **Q13**: blur remediation (name ``validation.min_blur_threshold``; re-encode hint when sharpness reads ``0.0``).

Golden test (the durable guarantee)

- ``tests/test_reporting_spec_faq.py``: **every rule's ``faq_ref`` must resolve to a real README heading**, and actionable (error/warn) rules must cite a FAQ. Locks the floor: no dangling debug pointers.

231 tests pass. Refined wording verified by rendering the affected scenarios. obsreport unchanged (v0.1.3).

? Generated with [Claude Code](https://claude.com/claude-code)
 EOF
)" 2>&1 | tail -3
 [/external_agent_tool_call]

887. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_result]
warning: in the working copy of 'tests/test_reporting_spec_faq.py', LF will be replaced by CRLF
the next time Git touches it
M README.md
M backend/reporting/harvest.py
M backend/reporting/spec.yaml
A tests/test_reporting_spec_faq.py
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/report-
debuggability-refinements
remote:
https://github.com/avidullu/badminton-highlight-indexer/pull/43
[/external_agent_tool_result]
```

888. assistant (2026-07-02T06:11:33.193Z)

Indexer refinements ? [PR #43](https://github.com/avidullu/badminton-highlight-indexer/pull/43).
 Now the collector side (push to #11) ? switching repos and checking out its branch:

889. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_call: Bash]
description: Switch to collector branch, check master + #11 state
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git checkout feat/observability-
reports 2>&1 | tail -1 && git fetch -q origin && echo "=== collector master vs my branch ===" &&
git rev-list --left-right --count origin/master...HEAD && echo "=== #11 state ===" && gh pr view
11 --json state,mergeable -q '{state,mergeable}' 2>&1
[/external_agent_tool_call]
```

890. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_result]
```

```
Already on 'feat/observability-reports'
=== collector master vs my branch ===
1      0
=== #11 state ===
{"mergeable":"UNKNOWN","state":"MERGED"}
[/external_agent_tool_result]
```

891. assistant (2026-07-02T06:11:33.193Z)

PR #11 is already **MERGED** ? collector observability is on master too. So the collector refinements go on a fresh branch off the updated master. Syncing and branching:

892. assistant (2026-07-02T06:11:33.193Z)

```
[external_agent_tool_call: Bash]
description: Sync collector master, branch for refinements, baseline test
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git checkout master 2>&1 | tail
-1 && git pull --ff-only origin master 2>&1 | tail -2 && echo "=== master head ===" && git log
--oneline -3 && git checkout -b feat/collector-report-refinements 2>&1 | tail -1 && echo "===
reporting present? ===" && ls src/reporting/ && python -m pytest -q 2>&1 | tail -3
[/external_agent_tool_call]
```

893. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_result]
Switched to branch 'master'
  create mode 100644 src/reporting/spec.yaml
  create mode 100644 tests/test_reporting.py
=== master head ===
94fec51 Merge pull request #11 from avidullu/feat/observability-reports
038424d feat(reporting): per-video observability report (collector)
99dfa22 Merge pull request #10 from avidullu/fix/canonical-csv-zero-start
Switched to a new branch 'feat/collector-report-refinements'
=== reporting present? ===
__init__.py
__pycache__
adapter.py
spec.yaml
..... [ 63%]
..... [100%]
113 passed in 1.88s
[/external_agent_tool_result]
```

894. assistant (2026-07-02T06:11:33.194Z)

Baseline green (113). Let me read the current adapter to apply the frame-evidence + review-framing refinements:

895. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py
[/external_agent_tool_call]
```

896. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_result]
28      """Map a rally_cvat.Issue onto a report Flag. Actionable issues (blocker/high/
29      medium) also carry a customer_message so the delivery summary shows the vendor
30      exactly what to fix, in plain language."""
31      sev = _SEVERITY.get(issue.severity, "info")
32      customer = issue.message if sev in ("error", "warn") else None
33      evidence = {"rally_number": issue.rally_number} if issue.rally_number is not None
else {}
```

```

34         return obsreport.Flag(
35             code=issue.code,
36             severity=sev,
37             message=issue.message,
38             stage="validation",
39             faq_ref=_ISSUE_FAQ,
40             customer_message=customer,
41             evidence=evidence,
42         )
43
44
45     def _recorder(source: str, video_path: Optional[str], spec) -> "obsreport.RunRecorder":
46         return obsreport.RunRecorder(
47             "sports-data-collector",
48             video_id=source,
49             video_id_kind="human_id",
50             video_path=video_path,
51             spec=spec if spec is not None else load_spec(),
52         )
53
54
55     def build_delivery_recorder(
56         *,
57         records: List[Any],
58         issues: List[Any],
59         fps: Optional[float],
60         source: str,
61         video_path: Optional[str] = None,
62         video_duration_s: Optional[float] = None,
63         source_format: str = "cvat",
64         step: Optional[int] = None,
65         spec=None,
66     ) -> "obsreport.RunRecorder":
67         """Envelope for `validate-delivery` / `convert-cvat`: parse + validation stages,
68         one flag per delivery issue, a rally summary."""
69         rec = _recorder(source, video_path, spec)
70         if fps:
71             rec.set_video_meta(fps=fps, fps_source="declared")
72         if video_duration_s:
73             rec.set_video_meta(duration_s=round(video_duration_s, 1))
74
75         with rec.stage("parse") as st:
76             st.add_metric("rallies_parsed", len(records))
77             st.details.update({"source_format": source_format, "fps": fps, "sampling_step":
step})
78
79         blockers = [i for i in issues if i.severity == "blocker"]
80         with rec.stage("validation") as st:
81             for sev in ("blocker", "high", "medium", "low"):
82                 st.add_metric(f"issues_{sev}", sum(1 for i in issues if i.severity == sev))
83             st.add_metric("n_blockers", len(blockers))
84             st.details["gate"] = "BLOCKED" if blockers else "PASS"
85             st.status = "error" if blockers else ("warn" if issues else "ok")
86
87         for issue in issues:
88             rec.envelope.flags.append(issue_to_flag(issue))
89
90     [/external_agent_tool_result]

```

897. assistant (2026-07-02T06:11:33.194Z)

Let me enrich `issue\_to\_flag` with frame-level evidence (for CVAT reviewers) and review-only framing for heuristic checks, then wire it in the delivery recorder:

898. assistant (2026-07-02T06:11:33.194Z)


```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py
[/external_agent_tool_call]
```

899. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

900. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py
[/external_agent_tool_call]
```

901. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

902. assistant (2026-07-02T06:11:33.194Z)

Now the collector golden faq-resolution test (mirrors the indexer's), covering both the issue-FAQ and the spec rules:

903. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting_faq.py
[/external_agent_tool_call]
```

904. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\sports-data-
collector\tests\test_reporting_faq.py (file state is current in your context ? no need to Read
it back)
[/external_agent_tool_result]
```

905. assistant (2026-07-02T06:11:33.194Z)

Let me run the collector suite (incl. the new faq test) to confirm nothing broke and the anchors resolve:

906. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_call: Bash]
description: Run collector reporting + faq tests
command: cd "C:/Users/avidu/Projects/sports-data-collector" && python -m pytest
tests/test_reporting_faq.py tests/test_reporting.py -q 2>&1 | tail -8
[/external_agent_tool_call]
```

907. assistant (2026-07-02T06:11:33.194Z)

```
[external_agent_tool_result]
..... [100%]
10 passed in 0.28s
[/external_agent_tool_result]
```

908. assistant (2026-07-02T06:11:33.194Z)

All green. Let me add a behavior test for the new frame-evidence + review-framing, then run the full suite:

909. assistant (2026-07-02T06:11:33.195Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py
[/external_agent_tool_call]
```

910. assistant (2026-07-02T06:11:33.195Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

911. assistant (2026-07-02T06:11:33.195Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py
[/external_agent_tool_call]
```

912. assistant (2026-07-02T06:11:33.195Z)

```
[external_agent_tool_result]
25     # --- issue -> flag -----
26     def test_issue_to_flag_severity_and_customer_message():
27         blocker = adapter.issue_to_flag(Issue("blocker", "overlap", "Rally 1 overlaps Rally
28         2.", 2))
29         assert blocker.severity == "error"
30         assert blocker.customer_message == "Rally 1 overlaps Rally 2." # actionable ->
31         shown to vendor
32         assert blocker.faq_ref.endswith("#what-does-validation-check")
33         assert blocker.evidence == {"rally_number": 2}
34
35         low = adapter.issue_to_flag(Issue("low", "outcome_vocab", "odd outcome", 3))
36         assert low.severity == "info"
37         assert low.customer_message is None # cosmetic -> not surfaced to the vendor
38
39     # --- delivery envelope -----
40     def test_delivery_with_blocker_fails_and_surfaces_to_customer():
41         records = [_rec(1, 0.0, 5.0), _rec(2, 4.0, 9.0)]
42         issues = [Issue("blocker", "overlap", "Rally 1 (ends 5.00s) overlaps Rally 2 (starts
43         4.00s).", 2)]
44         rec = adapter.build_delivery_recorder(records=records, issues=issues, fps=59.94,
45         source="pilot_01", video_duration_s=600.0)
46         env = rec.envelope
47         assert env.verdict == "fail"
48         assert env.get_stage("validation").status == "error"
49         assert "overlap" in _codes(env)
50         assert env.highlights.segment_count == 2
51         assert env.lint() == []
52
53         from obsreport import render_customer, render_technical
54         tech = render_technical(env, rec.spec)
55         cust = render_customer(env, rec.spec)
56         assert "***overlap***" in tech # bolded issue CODE visible technically
57         assert "***overlap***" not in cust # code hidden from the customer/vendor view
58         assert "overlaps Rally 2" in cust # but the plain message is shown
59         assert "blocking issues" in cust # collector-specific customer verdict (not the
60         generic default)
```

```

59
60     def test_delivery_clean_passes():
61         records = [_rec(1, 0.0, 5.0), _rec(2, 6.0, 9.0)]
62         rec = adapter.build_delivery_recorder(records=records, issues=[], fps=59.94,
source="clean")
63         assert rec.envelope.verdict == "pass"
64         assert _codes(rec.envelope) == set()
[/external_agent_tool_result]

```

913. assistant (2026-07-02T06:11:33.195Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py
[/external_agent_tool_call]

```

914. assistant (2026-07-02T06:11:33.195Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

915. assistant (2026-07-02T06:11:33.195Z)

```

[external_agent_tool_call: Bash]
description: Full collector suite after refinements
command: cd "C:/Users/avidu/Projects/sports-data-collector" && python -m pytest -q 2>&1 | tail
-4
[/external_agent_tool_call]

```

916. assistant (2026-07-02T06:11:33.195Z)

```

[external_agent_tool_result]
..... [ 62%]
..... [100%]
116 passed in 1.95s
[/external_agent_tool_result]

```

917. assistant (2026-07-02T06:11:33.195Z)

116 passed. Committing + opening the collector refinements PR:

918. assistant (2026-07-02T06:11:33.195Z)

```

[external_agent_tool_call: Bash]
description: Commit + push + open collector refinements PR
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git add -A && git commit -q -m
"feat(reporting): delivery-report debuggability refinements + faq-resolution golden test

```

From the fresh-reader eval (collector scenarios were self-debuggable; minor gaps):

- Delivery issue flags now carry frame-level evidence (rally_frames + fps) so a CVAT/frame-based reviewer can act without converting seconds->frames by hand.
- Heuristic, review-only checks (possible_missed_rally, shot_density) get gentle 'Review only ? a heuristic, not a blocker' framing in the vendor-facing message, so a 'possible' finding isn't read as a certainty.
- NEW golden test: every faq_ref the reports emit (the per-issue FAQ + spec rules) resolves to a real heading in docs/INGESTION_PIPELINE.md; actionable rules cite a FAQ.

116 tests green. obsreport unchanged (v0.1.3).

```

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q -u origin
feat/collector-report-refinements 2>&1 | tail -2 && gh pr create --base master --head
feat/collector-report-refinements --title "feat(reporting): delivery-report refinements + faq-
resolution golden test" --body "$(cat <<'EOF'

```

Follow-up to #11 (collector observability), applying the gaps the fresh-reader eval surfaced. The eval found the collector scenarios self-debuggable; these are minor, high-value polish.

- **Frame-level evidence:** delivery issue flags now attach `rally\_frames` + `fps` to their evidence, so a reviewer working in CVAT (frame-based) can act without hand-converting seconds?frames. (The overlap finding was the one place the eval flagged this friction.)
- **Review-only framing:** heuristic checks (`possible\_missed\_rally`, `shot\_density`) now say "Review only ? a heuristic, not a blocker: check the video and act only if confirmed" in the vendor-facing message, so a "possible" finding isn't read as a certainty.
- **Golden test** (`tests/test\_reporting\_faq.py`): every `faq\_ref` the reports emit ? the per-issue FAQ **and** the spec rules ? must resolve to a real heading in `docs/INGESTION\_PIPELINE.md`; actionable rules must cite a FAQ. No dangling debug pointers.

116 tests pass. obsreport unchanged (v0.1.3).

? Generated with [Claude Code](https://claude.com/claude-code)

EOF

)" 2>&1 | tail -3

[/external_agent_tool_call]

919. assistant (2026-07-02T06:11:33.195Z)

[external_agent_tool_result]

warning: in the working copy of 'tests/test_reporting_faq.py', LF will be replaced by CRLF the next time Git touches it

remote: https://github.com/avidullu/sports-data-collector/pull/new/feat/collector-report-refinements

remote:

https://github.com/avidullu/sports-data-collector/pull/12

[/external_agent_tool_result]

920. assistant (2026-07-02T06:11:33.195Z)

Collector refinements ? [PR #12](https://github.com/avidullu/sports-data-collector/pull/12).

Phase D is complete. Let me mark it done and checkpoint memory:

921. assistant (2026-07-02T06:11:33.195Z)

[external_agent_tool_call: TaskUpdate]

input: {"taskId":"7","status":"completed"}

[/external_agent_tool_call]

922. assistant (2026-07-02T06:11:33.195Z)

[external_agent_tool_result]

Updated task #7 status

[/external_agent_tool_result]

923. assistant (2026-07-02T06:11:33.195Z)

[external_agent_tool_call: Edit]

file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md

[/external_agent_tool_call]

924. assistant (2026-07-02T06:11:33.195Z)

[external_agent_tool_result]

The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

925. assistant (2026-07-02T06:11:33.195Z)

The full feature is shipped end-to-end. Here's where everything landed.

? Per-video observability ? delivered across both tools + shared lib

```
**Merged to master:**
- **`obsreport`** ? new shared, MIT, permissively-licensed "report config language" lib ?
[github.com/avidullu/sports-obsreport](https://github.com/avidullu/sports-obsreport) (private,
v0.1.3, 74 tests)
- **Indexer** : [#41](https://github.com/avidullu/badminton-highlight-indexer/pull/41)
(observability) + [#42](https://github.com/avidullu/badminton-highlight-indexer/pull/42) (the
config runtime bug I caught & fixed)
- **Collector** : [#11](https://github.com/avidullu/sports-data-collector/pull/11)
(observability) + [#10](https://github.com/avidullu/sports-data-collector/pull/10) (parser fix)
```

Every processing/ingestion run now writes, next to the video (configurable): a machine
`\*.report.json`, a **technical** `\*.report.md` (stages + findings, each citing a FAQ), and a
customer-friendly `\*.summary.md` (metadata + meta-highlights, internals stripped, gentle
audience-appropriate verdict).

? Debuggability validated (your stretch goal)

A 15-agent workflow put all 7 broken-run reports in front of "fresh reader" agents that saw
only the README/FAQ + the report ? verdict: **all 7 self-debuggable**, every reader found
the root cause and cited the right FAQ. I also rendered the reports to HTML and eyeballed them
(clean tables, fps source shown, customer summaries free of jargon).

? Two follow-up PRs open (green, awaiting your merge)

The eval surfaced concrete polish, now applied:

```
- **Indexer [#43](https://github.com/avidullu/badminton-highlight-indexer/pull/43)** (231
tests): README Q7 now lists `wasb_hybrid`/`tracknet_hybrid` as valid (killed a false lead); new
fps Q14 + repointed `fps_fallback_used` off the wrong Q5; numeric `30 (fallback)`; motion
"synthetic" surfaced at the data level; sharper silent-provider + blur guidance; faq-
resolution golden test.
- **Collector [#12](https://github.com/avidullu/sports-data-collector/pull/12)** (116 tests):
frame-level evidence on delivery flags; "review only, not a blocker" framing for heuristics;
faq-resolution golden test.
```

Both are disjoint follow-ups ? merge whenever.

Notes for the record

```
- Config bug I found: the typed `AppConfig` from your `#39` broke all validators on real
runs (nested `config.get(...).get(...)`); my report is what surfaced it, and #42 fixed it (your
review nits addressed). The directory restructure (#40) was clean ? I rebased onto it without
issues.
- Merge-conflict help: resolved the indexer `main.py` conflict (your config package vs my
reporting) and kept the parser fix to its own dedicated branch so PRs stayed non-overlapping.
- Optional future (noted in memory): an HTML customer-report renderer; threading real AI-
provider timing through the segmenters (currently recorded as `not_captured`); pinning
`obsreport` via `git+ssh` once CI/SSH access is set up (it's a soft dep today, so nothing breaks
meanwhile).
```

Memory/handoff updated. Want me to merge #43 and #12 too, or leave them for your review?

926. user (2026-07-02T06:11:33.196Z)

I merged #11 as well on the collector. Can you bring both the repos at a clean slate and quickly
tell me whether the low regret moves pan in the docs/ folder seems a good starter project list
for grok? I think your work will have some intersect but those are mostly syntactic sugars and
robust reporting.

oh and I hope you are checkpointing the session handoffs and current state and other "carry
forward package for claude to resume quickly" suite of updates since you are already at >70%
context util (700k tokens)

#43 is also merged

927. assistant (2026-07-02T06:11:33.196Z)

Got it ? and yes, I'll lock in the carry-forward updates (handoff + current-state) given the context level. Let me first bring both repos to a clean slate and check PR states, then assess the low-regret doc for Grok.

928. assistant (2026-07-02T06:11:33.196Z)

```
[external_agent_tool_call: Bash]
description: Sync both repos to master, check PR states + branches
command: echo "##### INDEXER #####" && cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout master 2>&1 | tail -1 && git fetch -p -q origin && git pull --ff-only -q origin master 2>&1 | tail -1 && echo "master head:" && git log --oneline -3 && echo "PR states #41/#42/#43:" && for n in 41 42 43; do echo -n "$n: "; gh pr view $n --json state -q .state 2>/dev/null; done && echo "local branches:" && git branch
echo "" && echo "##### COLLECTOR #####" && cd "C:/Users/avidu/Projects/sports-data-collector" && git checkout master 2>&1 | tail -1 && git fetch -p -q origin && git pull --ff-only -q origin master 2>&1 | tail -1 && echo "master head:" && git log --oneline -3 && echo "PR states #10/#11/#12:" && for n in 10 11 12; do echo -n "$n: "; gh pr view $n --json state -q .state 2>/dev/null; done && echo "local branches:" && git branch
[/external_agent_tool_call]
```

929. assistant (2026-07-02T06:11:33.196Z)

```
[external_agent_tool_result]
##### INDEXER #####
Switched to branch 'master'
master head:
491c909 Merge pull request #43 from avidullu/feat/report-debuggability-refinements
51b1f8d feat(reporting): debuggability refinements from fresh-reader eval + faq-resolution
golden test
98bb1d9 Merge pull request #41 from avidullu/feat/observability-reports
PR states #41/#42/#43:
#41: MERGED
#42: MERGED
#43: MERGED
local branches:
  feat/report-debuggability-refinements
* master

##### COLLECTOR #####
Switched to branch 'master'
master head:
94fec51 Merge pull request #11 from avidullu/feat/observability-reports
038424d feat(reporting): per-video observability report (collector)
99dfa22 Merge pull request #10 from avidullu/fix/canonical-csv-zero-start
PR states #10/#11/#12:
#10: MERGED
#11: MERGED
#12: OPEN
local branches:
  feat/collector-report-refinements
  feat/observability-reports
* master
[/external_agent_tool_result]
```

930. assistant (2026-07-02T06:11:33.196Z)

Clean-slate cleanup ? pruning the merged local branches (keeping collector `#12` since it's still open):

931. assistant (2026-07-02T06:11:33.196Z)

```
[external_agent_tool_call: Bash]
```

```
description: Prune merged local branches; locate docs
command: echo "### INDEXER cleanup ###" && cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git branch -d feat/report-debuggability-refinements 2>&1 | tail -1 && git branch && git status -s | wc -l | xargs echo "uncommitted files:"
echo "" && echo "### COLLECTOR cleanup ###" && cd "C:/Users/avidu/Projects/sports-data-collector" && git branch -d feat/observability-reports 2>&1 | tail -1 && echo "(keeping feat/collector-report-refinements ? PR #12 still OPEN)" && git branch && git status -s | wc -l | xargs echo "uncommitted files:"
echo "" && echo "### find low-regret doc (indexer) ###" && cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && ls docs/*.md | tr '\n' ' '
[/external_agent_tool_call]
```

932. assistant (2026-07-02T06:11:33.196Z)

[external_agent_tool_result]

INDEXER cleanup

Deleted branch feat/report-debuggability-refinements (was 51b1f8d).

* master

uncommitted files: 0

COLLECTOR cleanup

Deleted branch feat/observability-reports (was 038424d).

(keeping feat/collector-report-refinements ? PR #12 still OPEN)

feat/collector-report-refinements

* master

uncommitted files: 0

find low-regret doc (indexer)

docs/AUDIO_E1_FINDINGS.md docs/AUDIO_RALLY_DETECTION_PLAN.md docs/BACKLOG.md docs/CODE_MAP.md
docs/COMMERCIAL_READINESS_REVIEW.md docs/DECISIONS.md docs/EVALUATION.md
docs/GOLDEN_SET_IMPLEMENTATION.md docs/GOLDEN_SET_PHASE1_VIDEOS.md docs/GOLDEN_SET_VENDORING.md
docs/HOW_RALLY_DETECTION_WORKS.md docs/LOW_REGRET_MOVES_PLAN.md docs/MONETIZATION_AUDIT.md
docs/RALLY_DETECTION_RESEARCH.md docs/ROADMAP.md docs/TRACKNET_WSL_SETUP.md
docs/VIDEO_RECORDING_GUIDELINES.md

[/external_agent_tool_result]

933. assistant (2026-07-02T06:11:33.196Z)

Both repos are clean (on `master`, no uncommitted files, merged branches pruned; kept the open #12 branch). The doc is `docs/LOW\_REGRET\_MOVES\_PLAN.md`. Let me read it (and peek at its companion review):

934. assistant (2026-07-02T06:11:33.196Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\LOW_REGRET_MOVES_PLAN.md

[/external_agent_tool_call]

935. assistant (2026-07-02T06:11:33.196Z)

[external_agent_tool_result]

1 # Low-Regret Moves Plan ? Config Model First

2

3 **Status:** Approved (checkpointed)

4 **Date:** 2026-06-07 (updated 2026-06-07)

5 **Owner / Driver:** Grok (second-in-command) + avidullu + Claude

6 **Related:** COMMERCIAL_READINESS_REVIEW.md,
BACKLOG.md, CODE_MAP.md, DECISIONS.md

7

8 ---

9

10 ## 1. Purpose & Philosophy

11

12 These are **low-regret moves**: changes that are high-confidence improvements to code

health today while directly de-risking the commercial and self-improving vision (offline learned segmenter, hosted path, multi-sport, data flywheel).

```
13
14     **Guiding principles**
15     - Small, reviewable increments (one or two files per PR where possible).
16     - Payoff within the same or next session.
17     - Improve the developer + product experience immediately.
18     - Lay infrastructure that makes future items cheaper and safer.
19     - Respect the existing culture (registries, eval harness, ADRs, excellent docs).
20
21     **Rule:** Every item must be traceable to a concrete pain/risk from the Commercial
22     Readiness Review or existing BACKLOG/ADRs.
23     ---
24
25     ## 2. Overall Sequence (Approved Order)
26
27     1. **Config Model (Pydantic) + Central Loader** (as a proper package) ? foundational.
28     2. **Directory Structure Refactor** (high-pri, atomic follow-up PR) ? move to the
29     approved layered layout.
30     3. Error / Result Contract standardization.
31     4. Logging standardization + removal of stray `print()` in hot paths.
32     5. Packaging modernization (`pyproject.toml` + clean entry points).
33     6. Detector Runner Abstraction (start de-risking WSL).
34     7. Unify / extract shared logic from the two hybrid segmenters.
35     8. Input Quality UX (surface validator + audio readiness signals).
36     9. (Stretch) First slice of async job model.
37
38     **Important sequencing decision (approved):** Complete the Config work first (this
39     introduces `backend/config/` as a package immediately). Then perform the broader directory
40     refactor as a separate, atomic, high-priority PR.
41
42     Rationale:
43     - Keeps the Config PR small and focused.
44     - Allows the structural change to land atomically with minimal merge conflicts.
45     - Gives Claude a clean window with no concurrent file moves.
46     - Still delivers the "level of indirection" benefit early via the `config/` package.
47     ---
48
49     ## 3. Approved Directory Structure Direction
50
51     Current flat layout under `backend/` (many sibling dirs + root .py files) feels
52     unprofessional for long-term maintenance. We will trend toward clear layers without creating 15+
53     tiny single-file directories.
54
55     **Target layered structure (directional, not all at once):**
56
57     ```
58     backend/
59     ??? __init__.py
60     ??? main.py                # Thin CLI + FastAPI bootstrap only
61     ?
62     ??? config/                # ? Start here with Deliverable 1
63     ?   ??? __init__.py
64     ?   ??? models.py          # Pydantic AppConfig + nested models
65     ?   ??? loader.py          # load_config, save_default_config, etc.
66     ?
67     ??? api/                   # Presentation layer
68     ?   ??? __init__.py
69     ?   ??? app.py             # FastAPI instance, middleware, routes
70     ?   ??? static/            # Frontend assets (HTML/JS/CSS)
71     ?
72     ??? pipeline/              # Core processing engine
73     ?   ??? validators/
```



```

70      ?   ??? segmenters/           # (renamed/moved from current indexer/)
71      ?   ??? stitcher/
72      ?   ??? audio/
73      ?   ??? detectors/
74      ?
75      ??? infrastructure/           # External adapters
76      ?   ??? database.py           # (moved from root)
77      ?
78      ??? providers/                # AI providers (group under infrastructure later if
desired)
79      ??? sports/
80      ??? annotations/
81      ?
82      ??? eval/                     # Keep prominent (data flywheel / measurement is first-
class)
83      ?
84      ??? core/                     # Shared primitives (introduce only as needed)
85      ??? utils/                    ...
[/external_agent_tool_result]

```

936. assistant (2026-07-02T06:11:33.196Z)

Both repos are clean. The plan is `docs/LOW\_REGRET\_MOVES\_PLAN.md`. Before I give the assessment, let me lock in the carry-forward (you're right about context level) ? refreshing the session handoff to reflect the shipped feature:

937. assistant (2026-07-02T06:11:33.196Z)

```

[external_agent_tool_call: Write]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\session-handoff.md
[/external_agent_tool_call]

```

938. assistant (2026-07-02T06:11:33.196Z)

```

[external_agent_tool_result: error]
<tool_use_error>File has been modified since read, either by the user or by a linter. Read it
again before attempting to write it.</tool_use_error>
[/external_agent_tool_result]

```

939. assistant (2026-07-02T06:11:33.196Z)

The handoff was modified externally ? let me re-read before rewriting:

940. assistant (2026-07-02T06:11:33.197Z)

```

[external_agent_tool_call: Read]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\session-handoff.md
[/external_agent_tool_call]

```

941. assistant (2026-07-02T06:11:33.197Z)

```

[external_agent_tool_result]
1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff for this project ? current state, next
steps, open decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key
decision/ADR, phase shift, session wrap); \"update the handoff\" forces an immediate full
rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: eb4b7b53-2f77-4c4d-83e1-4fa5e77f3e3f

```

```

8      ---
9
10     # Session handoff ? badminton-highlight-indexer (+ siblings: sports-data-collector,
rally-annotator, sports-obsreport)
11
12     **Last updated:** 2026-06-07 (observability phase) · **Refresh policy: AUTO-refresh at
meaningful checkpoints** ?
13     PR merged, key decision/ADR, phase shift, session wrap-up (NOT on-request-only). "update
the handoff" forces an
14     immediate full rewrite. **Staleness tripwire:** if ?3 substantive [[current-state]]
checkpoints ? or any
15     PR-merge / new ADR / phase shift ? have landed since the "Last updated" date above,
refresh this handoff now.
16     Pointer, not a duplicate. **For the freshest blow-by-blow state, read [[current-state]]
first**
17     (updated every micro-checkpoint); this handoff is the slower ~1-page orientation.
18
19     ## You are here (2026-06-07) ? ? OBSERVABILITY-REPORTS feature mid-flight (multi-
session/concurrent)
20     - **Active feature:** always-on **per-video observability reports** across BOTH tools
(indexer + collector),
21     powered by a NEW shared lib **`sports-obsreport`** (pkg `obsreport`) ? a small "report
config language" on
22     permissive primitives (pydantic/Jinja2/PyYAML). Each tool emits a **technical** report
(debuggable via README+FAQ)
23     AND a **customer-friendly summary** (internals stripped). Reports written **next to
the video** by default
24     (override `report.output_dir`). Approved plan: `C:\Users\avidu\claude\plans\drifting-
snuggling-squirrel.md` (read it to resume).
25     - **Status by piece:**
26     - ? **obsreport lib** built + on GitHub **(PRIVATE)** github.com/avidullu/sports-
obsreport (main + tags v0.1.0?v0.1.3).
27     Editable-installed into Python 3.13. ~74 lib tests green. Modules:
envelope/spec/rules/render/io/recorder/diff/schema.
28     - ? **Indexer Phase A** ? **PR #41 OPEN** (`feat/observability-reports`, rebased onto
#39 typed-config `backend/config/`).
29     `backend/reporting/` (soft-import, never-raises), wired CLI+API. My reporting: **226
tests pass.**
30     ?? **2 PRE-EXISTING red tests from #39** (NOT mine, left untouched):
`test_config.py::test_load_default_config`
31     + `test_save_default_config_roundtrip` (Path==str on Windows).
32     - ? **Collector Phase B** ? **PR #11 OPEN** (`feat/observability-reports`,
observability-ONLY). `src/reporting/`
33     (adapter Issue?Flag + recorders), wired curate validate-delivery/convert-
cvat/import-local. **112 tests pass.**
34     - ? **Collector parser zero-start fix** ? **PR #10 MERGED** to master (`99dfa22`).
Separate from the reporting PR;
35     DUP RISK resolved (fix landed once; PR #11 carries no parser fix). See [[current-
state]].
36     - **?? NEXT: Phase D ? debuggability eval** (ultracode Workflow: fresh-reader agents
over broken-run reports ? refine
37     wording; render a report to an image + eyeball) + committed golden tests (faq_ref-
resolution + flag-set) in both repos.
38
39     ## Standing project context (rally-detection stack ? stable, mostly merged; PARKED
pending golden data)
40     - WASB shuttle-trajectory substrate is resumable/checkpointed; on top = rally layer
(windowing thresholds +
41     net-crossing "gate A" + distilled classifier `distill_local`); eval harness
`calibrate_wasb`/`calibration.py` (LOVO).
42     - **Honest core finding:** WASB-only threshold tuning **ceilings ~F1 0.44**; learned
distilled model **underperforms
43     baseline at n=2 videos**. Blocker = **golden data + WASB candidate recall**, not the
method.
44     - **Audio E1 ran ? classical-onset = ~2.2× discrimination ceiling, NOT adopted** (band-

```

```
pass didn't help; detector kept,  
45         indexer PR #35...  
[/external_agent_tool_result]
```

942. assistant (2026-07-02T06:11:33.197Z)

```
[external_agent_tool_call: Edit]  
file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-  
indexer\memory\session-handoff.md  
[/external_agent_tool_call]
```

943. assistant (2026-07-02T06:11:33.197Z)

```
[external_agent_tool_result]  
The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-  
indexer\memory\session-handoff.md has been updated successfully. (file state is current in your  
context ? no need to Read it back)  
[/external_agent_tool_result]
```

944. assistant (2026-07-02T06:11:33.197Z)

****Carry-forward locked in**** ? ? both `current-state.md` (FEATURE COMPLETE checkpoint) and
`session-handoff.md` (rewritten to the shipped state + the low-regret/Grok context + ramp-up
kit) are current. A fresh Claude can resume from those alone.

Clean slate

- ****Indexer:**** `master` (#41/#42/#43 all merged), 0 uncommitted, merged branch pruned.
- ****Collector:**** `master` (#10/#11 merged), 0 uncommitted; kept `feat/collector-report-
refinements` (that's the open ****#12****).

Low-regret plan assessment ? for Grok

****Verdict:** yes, it's a genuinely good starter list.****** The philosophy (small reviewable PRs, each
traceable to a concrete pain, payoff within a session) and the sequencing are sound ? and
"Config-first, then atomic restructure" was exactly right (it gave my reporting work a clean
window; the rebase onto #40 was conflict-free). Items ****1** (Config #39) and 2 (restructure #40)
are already done.******

Quick read on the remaining 3?9, with where my work touches them:

| # | Item | As a Grok starter | Intersect with my observability work |
|---|--|--|---|
| 3 | Error/Result contract | ? great, safe-parallel | **Synergy** ? today my report *infers* failures (e.g. `0 segs + no key`). If providers/segmenters return a structured Result instead of a silent `[]`, the report can read the real reason as a first-class flag. Worth coordinating. |
| 4 | Logging + kill stray `print()` | ? great, pure hygiene | ~none. Reports are artifacts; logs are the stream. (CODE_MAP flags `print()` in compiler/motion.) |
| 5 | Packaging (`pyproject` + entry points) | ? good | minor ? model it on `obsreport`'s pyproject; when adding the `obsreport` dep, use the `git+ssh@tag` pin. (I already fixed the collector's `build-backend` typo.) |
| 6 | Detector runner abstraction (de-risk WSL) | ? good, but **meatier/architectural** | ? more "main project" than "starter" low |
| 7 | Unify the two hybrid segmenters | ? good, medium | **Synergy** ? they're copy-paste twins (CODE_MAP). Unify first, then there's *one* place to thread the **real AI-provider timing** I deferred (currently `not_captured`). I'll add that recorder hook as part of / right after #7. |
| 8 | Input-Quality UX (surface validator + audio readiness) | ?? **largely subsumed by my reports** | **Biggest overlap.** The report already surfaces validation stages + flags + verdict. Grok should *extend the report* (add an audio-readiness stage/flag to the obsreport spec), **not** build a parallel mechanism. |
| 9 | Async job model (stretch) | ? fine as a stretch | each per-run report slots in naturally as a per-job artifact |

****Your instinct is right**** ? most of the intersect is hygiene/"syntactic sugar" (4, 5) or pure synergy (3). The two to flag explicitly so we don't collide or double-build:

- ****#8** is mostly already done** by the observability feature ? reframe it as "extend the report," not a new surface.
- ****#7** should land before/with the AI-timing capture** ? I'll pick that up there rather than touching the twins now.

Net: hand Grok 3 ? 4 ? 5 as the safe quick wins (near-zero conflict with me), then 6/7 as the de-risking work (coordinate on 7), and rewrite 8 to build on the report. Want me to drop these coordination notes into `LOW\_REGRET\_MOVES\_PLAN.md` / `BACKLOG.md` so Grok sees them in-repo?

945. user (2026-07-02T06:11:33.197Z)

#12 is also merged so I hope we are good? Any other pending items for you ? The starter project is complete? Can I see a sample report which the user (without debug info) would see?

946. assistant (2026-07-02T06:11:33.197Z)

Let me sync the collector (now that #12 is merged), confirm both repos are at a clean slate, then I'll show you the customer-facing report.

947. assistant (2026-07-02T06:11:33.197Z)

```
[external_agent_tool_call: Bash]
description: Sync collector master, prune #12 branch, confirm both clean
command: cd "C:/Users/avidu/Projects/sports-data-collector" && git checkout master 2>&1 | tail
-1 && git fetch -p -q origin && git pull --ff-only -q origin master 2>&1 | tail -1 && git branch
-d feat/collector-report-refinements 2>&1 | tail -1 && echo "=== collector master head ===" &&
git log --oneline -2 && echo "=== #12 state ===" && gh pr view 12 --json state -q .state 2>&1 &&
echo "=== branches / clean ===" && git branch && git status -s | wc -l | xargs echo
"uncommitted:" && echo "" && echo "=== INDEXER clean check ===" && cd
"C:/Users/avidu/Projects/badminton-highlight-indexer" && git branch && git status -s | wc -l |
xargs echo "uncommitted:"
[/external_agent_tool_call]
```

948. assistant (2026-07-02T06:11:33.197Z)

```
[external_agent_tool_result]
Already on 'master'
Deleted branch feat/collector-report-refinements (was b8d40c8).
=== collector master head ===
0e4ccc7 Merge pull request #12 from avidullu/feat/collector-report-refinements
b8d40c8 feat(reporting): delivery-report debuggability refinements + faq-resolution golden test
=== #12 state ===
MERGED
=== branches / clean ===
* master
uncommitted: 0

=== INDEXER clean check ===
* master
uncommitted: 0
[/external_agent_tool_result]
```

949. assistant (2026-07-02T06:11:33.197Z)

Both repos clean (`master` only, 0 uncommitted), and ****all PRs merged**** (#10?#12 collector, #41?#43 indexer). Now here's exactly what an end user sees ? the customer-facing summary for a ****successful**** run (no system internals):

950. assistant (2026-07-02T06:11:33.197Z)

```
[external_agent_tool_call: Bash]
description: Render happy-path customer-facing report
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && GEMINI_API_KEY=x python -
<<'PY'
```

```
import obsreport
from backend.reporting import harvest
from backend.pipeline.validators.base import ValidationResult
def vr(n): return ValidationResult(validator_name=n, passed=True, message="ok")
FF={"streams":[{"width":1920,"height":1080,"r_frame_rate":"60000/1001","codec_name":"hevc"}],
    "format":{"format_name":"mov,mp4,m4a","duration":"676.0","size":"5500000000"}}
durs=[21.2,18.9,9.6,12.4,7.6,24.1,15.3,8.9,30.2,11.7,19.8,71.7] # 12 rallies
segs=[{"duration":d,"metadata":{"detector":"wasb-hybrid-gemini-flash-latest"}} for d in durs]
rec=harvest.record_run(
    video_id="9f3a1c4e?", video_path="C:/videos/Adarsh and Vasanth.mp4",
    config={"ai_provider":"gemini","ai_model":"gemini-flash-
latest","indexing":{"default_segmenter":"wasb_hybrid"}},
    val_results=[vr("format_check"),vr("resolution_fps_check"),vr("pts_continuity_check"),vr("scene_cut_check"),vr("blur_check"),vr("relevance_check")],
    val_context={"ffprobe_meta":FF}, play_segments=segs, candidates=[{"id":i} for i in range(64)],
    segmenter_requested="wasb_hybrid", segmenter_actual="wasb_hybrid", was_reingest=False,
    segmentation_ran=True)
print(obsreport.render_customer(rec.envelope, rec.spec))
PY
[/external_agent_tool_call]
```

951. assistant (2026-07-02T06:11:33.197Z)

[external_agent_tool_result]
Badminton Highlight Indexer ? Video Report (summary)

Run

| |
|---|
| Field Value |
| --- --- |
| Video Adarsh and Vasanth.mp4 |
| Report generated 2026-06-07T08:59:37Z |

Video

| |
|------------------------|
| Field Value |
| --- --- |
| Duration 11m16s |
| Resolution 1920x1080 |
| Frame rate 59.94 fps |

Highlights

- **12** highlights
- **4m11s** of play
- **37.2%** of the video is highlights

Things to know

Nothing needs your attention.

Verdict

? Your highlights are ready.
[/external_agent_tool_result]

952. assistant (2026-07-02T06:11:33.197Z)

That's the complete end-user view above ? ? ? **no codes, no stages, no internals**: just the video's metadata, what was produced (12 highlights · 4m11s of play · 37% of the video), a plain "nothing needs your attention," and **""? Your highlights are ready.""** (When something goes wrong, that same report instead shows a gentle line under ***Things to know*** + **""? We couldn't**

finish generating highlights? please contact support"* ? never a stack trace or an internal flag code.)

Are we good? ? Yes. ?

- **All PRs merged:** indexer #41/#42/#43, collector #10/#11/#12. Both repos clean on `master`, branches pruned, 0 uncommitted. `obsreport` on GitHub (private, v0.1.3).

Is the starter project complete? ? Yes.

Everything you asked for shipped:

- ? Rich per-video observability in **both** tools (indexer + collector) ? every run emits a machine JSON + a **technical** debuggable report + a **customer-friendly** summary, written next to the video.
- ? Catches inconsistencies / silent failures (it literally surfaced the config bug ? #42) and is diffable across runs/tools.
- ? Debuggability bar met ? a 15-agent eval confirmed a README+FAQ-only reader can self-diagnose all 7 broken-run cases; refinements + golden faq-resolution tests locked it in.
- ? The customer report (shown above).
- ? Reusable shared "report config language" lib in its own repo.

Pending items for me? ? None.

Nothing is blocked or half-done. A few **optional** future opportunities (not pending ? flagging for when you want them):

1. **`obsreport` git+ssh pin** in both repos' deps (it's a soft/commented dep today ? works on your machine via the editable install; pin to the tag once SSH/CI access is set up).
2. **Real AI-provider timing** in the report (currently `not\_captured`) ? cleanest to land alongside Grok's item #7 (unify the hybrid segmenters).
3. **HTML customer-report renderer** + **audio-readiness signal** (Grok's reframed item #8) in `obsreport`.
4. **`docs/CODE\_MAP.md` is stale** ? it still references pre-#40 paths (`backend/validators`, `backend/indexer`, `backend/database`); worth regenerating after the restructure.

Two quick offers if useful: I can (a) drop the Grok coordination notes into `LOW\_REGRET\_MOVES\_PLAN.md` / `BACKLOG.md` in-repo, and/or (b) regenerate `CODE\_MAP.md` for the new layout. Otherwise ? the starter project is done, and the handoff/current-state are fully checkpointed for a clean resume. ?

953. user (2026-07-02T06:11:33.198Z)

pls do (a) (b) .. and also can you add a neat `run_all_tests` in this repo so that grok can easily find and run full test suites before sending PRs

954. assistant (2026-07-02T06:11:33.198Z)

On it ? (a) coordination notes, (b) `CODE_MAP` regen, and (c) a discoverable `run_all_tests`. Since (b) is the substantive piece and the `CODE_MAP` must be accurate against the restructured tree, I'll regenerate it with a multi-agent workflow (parallel subsystem readers ? synthesize ? adversarially verify every anchor), and do (a)/(c) directly while it runs.

First, branch + launch the `CODE_MAP` regeneration workflow:

955. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_call: Bash]
description: Branch + list restructured backend layout
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git checkout -b chore/grok-handoff-codemap-runner 2>&1 | tail -1 && echo "=== current backend layout (for the workflow)
=== " && find backend -maxdepth 2 -type d | sort
[/external_agent_tool_call]
```

956. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_result]
Switched to a new branch 'chore/grok-handoff-codemap-runner'
```

```
=== current backend layout (for the workflow) ===
```

```
backend
backend/__pycache__
backend/annotations
backend/annotations/__pycache__
backend/api
backend/api/static
backend/audio
backend/audio/__pycache__
backend/config
backend/config/__pycache__
backend/detectors
backend/detectors/__pycache__
backend/eval
backend/eval/__pycache__
backend/eval/audio
backend/indexer
backend/indexer/__pycache__
backend/infrastructure
backend/infrastructure/__pycache__
backend/pipeline
backend/pipeline/audio
backend/pipeline/detectors
backend/pipeline/segmenters
backend/pipeline/stitcher
backend/pipeline/validators
backend/providers
backend/providers/__pycache__
backend/reporting
backend/reporting/__pycache__
backend/sports
backend/sports/__pycache__
backend/stitcher
backend/stitcher/__pycache__
backend/tools
backend/tools/__pycache__
backend/utils
backend/utils/__pycache__
backend/validators
backend/validators/__pycache__
[/external_agent_tool_result]
```

957. assistant (2026-07-02T06:11:33.198Z)

Both old and new dirs appear, but the old ones are just stale `\_\_pycache\_\_` (the `.py` files moved in #40). Let me get the authoritative **git-tracked** layout so the workflow maps reality, not leftovers:

958. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_call: Bash]
description: List authoritative git-tracked python files
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== tracked python
files (backend + root + tests) ===" && git ls-files "backend/**/*.py" "*.py" "tests/*.py" | grep
-v __pycache__ | sort
[/external_agent_tool_call]
```

959. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_result]
=== tracked python files (backend + root + tests) ===
backend/annotations/__init__.py
backend/annotations/automated_provider.py
backend/annotations/base.py
```

backend/annotations/file_provider.py
backend/config/__init__.py
backend/config/loader.py
backend/config/models.py
backend/eval/__init__.py
backend/eval/annotations.py
backend/eval/audio/__init__.py
backend/eval/audio/e1_probe.py
backend/eval/batch.py
backend/eval/calibrate_local.py
backend/eval/calibrate_wasb.py
backend/eval/calibration.py
backend/eval/classifier.py
backend/eval/distill_local.py
backend/eval/export_wasb_segments.py
backend/eval/gemini_refine.py
backend/eval/harness.py
backend/eval/metrics.py
backend/eval/regression.py
backend/eval/training.py
backend/infrastructure/database.py
backend/main.py
backend/pipeline/audio/__init__.py
backend/pipeline/audio/hit_detector.py
backend/pipeline/detectors/__init__.py
backend/pipeline/detectors/rally_gate.py
backend/pipeline/detectors/tracknet_runner.py
backend/pipeline/detectors/wasb_infer.py
backend/pipeline/detectors/wasb_runner.py
backend/pipeline/segmenters/__init__.py
backend/pipeline/segmenters/base.py
backend/pipeline/segmenters/gemini.py
backend/pipeline/segmenters/motion.py
backend/pipeline/segmenters/tracknet_hybrid.py
backend/pipeline/segmenters/wasb_hybrid.py
backend/pipeline/segmenters/yolo_hybrid.py
backend/pipeline/stitcher/compiler.py
backend/pipeline/validators/__init__.py
backend/pipeline/validators/base.py
backend/pipeline/validators/blur.py
backend/pipeline/validators/format.py
backend/pipeline/validators/pts_continuity.py
backend/pipeline/validators/relevance.py
backend/pipeline/validators/resolution_fps.py
backend/pipeline/validators/scene_cut.py
backend/providers/__init__.py
backend/providers/base.py
backend/providers/gemini.py
backend/reporting/__init__.py
backend/reporting/harvest.py
backend/sports/__init__.py
backend/sports/badminton.py
backend/sports/base.py
backend/tools/__init__.py
backend/tools/rally_slicer.py
backend/utils/__init__.py
backend/utils/geometry.py
backend/utils/video.py
run_eval.py
run_phase3_eval.py
run_process_and_stitch.py
run_regression.py
scratch/analyze_motion.py
scratch/clean_state.py
scratch/diagnose_rally.py


```

scratch/gemini_video_debugger.py
scratch/list_models.py
scratch/test_ai_segmenter.py
scratch/test_thresholds.py
scripts/generate_khelacharya_flyer.py
setup.py
tests/__init__.py
tests/test_annotation_providers.py
tests/test_automated_provider.py
tests/test_base.py
tests/test_calibrate_local.py
tests/test_calibrate_wasb.py
tests/test_calibration.py
tests/test_config.py
tests/test_database.py
tests/test_distill_local.py
tests/test_eval_annotations.py
tests/test_eval_batch.py
tests/test_eval_classifier.py
tests/test_eval_harness.py
tests/test_eval_metrics.py
tests/test_eval_regression.py
tests/test_eval_training.py
tests/test_export_wasb_segments.py
tests/test_gemini.py
tests/test_gemini_refine.py
tests/test_geometry.py
tests/test_hit_detector.py
tests/test_provider_gemini.py
tests/test_providers.py
tests/test_rally_gate.py
tests/test_rally_slicer.py
tests/test_reporting_harvest.py
tests/test_reporting_spec_faq.py
tests/test_sport_profiles.py
tests/test_tracknet_hybrid.py
tests/test_tracknet_runner.py
tests/test_validators_base.py
tests/test_video.py
tests/test_wasb_cache.py
tests/test_yolo_hybrid.py
[/external_agent_tool_result]

```

960. assistant (2026-07-02T06:11:33.198Z)

I have the authoritative layout. Launching the CODE_MAP regeneration workflow (8 parallel subsystem readers ? synthesize against the old map ? adversarial anchor verification ? fix), running in the background:

961. assistant (2026-07-02T06:11:33.198Z)

```

[external_agent_tool_call: Workflow]
input: {"script": "export const meta = {\n  name: 'regen-code-map',\n  description: 'Regenerate docs/CODE_MAP.md for the layered restructure (#40) + config/reporting subsystems',\n  phases: [\n    { title: 'Map', detail: '8 parallel subsystem readers over the current layout' },\n    { title: 'Synthesize', detail: 'update the existing CODE_MAP from the chunks' },\n    { title: 'Verify', detail: 'adversarially check every @path::symbol anchor' },\n    { title: 'Fix', detail: 'apply verification corrections -> final markdown' },\n  ],\n}\n\nconst REPO = 'C:/Users/avidu/Projects/badminton-highlight-indexer'\n\nconst GROUPS = [\n  { name: 'Orchestration & config', paths: ['backend/main.py', 'backend/config/__init__.py', 'backend/config/models.py', 'backend/config/loader.py', 'run_eval.py', 'run_phase3_eval.py', 'run_regression.py', 'run_process_and_stitch.py', 'setup.py', 'config.json'] },\n  { name: 'Validators', paths: ['backend/pipeline/validators/__init__.py', 'backend/pipeline/validators/base.py', 'backend/pipeline/validators/format.py', 'backend/pipeline/validators/resolution_fps.py',

```

```
'backend/pipeline/validators/pts_continuity.py', 'backend/pipeline/validators/scene_cut.py',
'backend/pipeline/validators/blur.py', 'backend/pipeline/validators/relevance.py'] },\n { name:
'Segmenters', paths: ['backend/pipeline/segmenters/__init__.py',
'backend/pipeline/segmenters/base.py', 'backend/pipeline/segmenters/motion.py',
'backend/pipeline/segmenters/gemini.py', 'backend/pipeline/segmenters/yolo_hybrid.py',
'backend/pipeline/segmenters/tracknet_hybrid.py', 'backend/pipeline/segmenters/wasb_hybrid.py']
},\n { name: 'Detectors / stitcher / audio (WSL-quarantined)', paths:
['backend/pipeline/detectors/tracknet_runner.py', 'backend/pipeline/detectors/wasb_runner.py',
'backend/pipeline/detectors/wasb_infer.py', 'backend/pipeline/detectors/rally_gate.py',
'backend/pipeline/stitcher/compiler.py', 'backend/pipeline/audio/hit_detector.py'] },\n { name:
'Infrastructure & reporting (obs...
[/external_agent_tool_call]
```

962. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: wrtpil6w6
Summary: Regenerate docs/CODE_MAP.md for the layered restructure (#40) + config/reporting
subsystems
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\56b7f849-5d56-4770-93e0-b3b4da943700\subagents\workflows\wf_d88615a4-f56
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\regen-code-map-wf_d88615a4-f56.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:
"C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\regen-code-map-
wf_d88615a4-f56.js"} to iterate without resending the script.)
Run ID: wf_d88615a4-f56
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--
Users-avidu-Projects-badminton-highlight-
indexer\56b7f849-5d56-4770-93e0-b3b4da943700\workflows\scripts\regen-code-map-
wf_d88615a4-f56.js", resumeFromRunId: "wf_d88615a4-f56"}) ? completed agents return cached
results.
```

```
You will be notified when it completes. Use /workflows to watch live progress.
[/external_agent_tool_result]
```

963. assistant (2026-07-02T06:11:33.198Z)

Workflow running. While it regenerates the map, let me do `**(c)**` ? a discoverable full-suite runner at the repo root (matching the ``run_*.py`` convention):

964. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\run_all_tests.py
[/external_agent_tool_call]
```

965. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\run_all_tests.py (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

966. assistant (2026-07-02T06:11:33.198Z)

Let me add a README pointer to it. Finding the test section:

967. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_call: Grep]
input: {"pattern": "Running the Test Suite|^pytest|pytest \\\s|pip install
pytest", "path": "C:\\Users\\avidu\\Projects\\badminton-highlight-
```

```
indexer\\README.md", "output_mode": "content", "-n": true, "head_limit": 15}
[/external_agent_tool_call]
```

968. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_result]
288:## ? Running the Test Suite
292: pip install pytest pytest-cov --user
293: pytest                                # run everything
294: pytest tests/test_yolo_hybrid.py      # run a single file
295: pytest -k yolo --maxfail=1 -v         # filter by keyword, fail fast, verbose
296: pytest --cov=backend --cov-report=term-missing # coverage report (uses .coveragerc)
[/external_agent_tool_result]
```

969. assistant (2026-07-02T06:11:33.198Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

970. assistant (2026-07-02T06:11:33.199Z)

```
[external_agent_tool_result]
288    ## ? Running the Test Suite
289    The test suite uses `pytest` and lives in `tests/`. From the project root:
290
291    ```bash
292    pip install pytest pytest-cov --user
293    pytest                                # run everything
294    pytest tests/test_yolo_hybrid.py      # run a single file
295    pytest -k yolo --maxfail=1 -v         # filter by keyword, fail fast, verbose
296    pytest --cov=backend --cov-report=term-missing # coverage report (uses .coveragerc)
297    ```
298
299    The suite mocks external dependencies (the Gemini client, ffprobe/ffmpeg, OpenCV
`VideoCapture`, the YOLO model) so it runs offline and without a GPU. The tests cover:
[/external_agent_tool_result]
```

971. assistant (2026-07-02T06:11:33.199Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\README.md
[/external_agent_tool_call]
```

972. assistant (2026-07-02T06:11:33.199Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

973. assistant (2026-07-02T06:11:33.199Z)

Now ***(a)*** ? the Grok coordination notes. Reading BACKLOG to append in its style:

974. assistant (2026-07-02T06:11:33.199Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md
[/external_agent_tool_call]
```

975. assistant (2026-07-02T06:11:33.199Z)

```

[external_agent_tool_result]
1      # Backlog ? deferred items to revisit
2
3      Single tracker for work we consciously deferred, so nothing gets lost between sessions.
Each item records what, why it's deferred, and the trigger to revisit. Every entry
traces to a real decision/PR/doc (no speculative work).
4
5      > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to GitHub issues instead (or in addition).
6      > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
7
8      _Last updated: 2026-06-07._
9
10     ---
11
12     ## Monetization / licensing
13     *(source: `docs/MONETIZATION_AUDIT.md` §7, ADR-005)*
14
15     - ? Ultralytics Enterprise pricing ? get a quote only if the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently not
needed (swap done). Quote-only pricing. Revisit: if detector quality forces reconsidering
YOLOv8.
16     - ? Codec patent thresholds (AVC/HEVC/AAC) ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds with legal counsel before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. Revisit: before commercial launch.
17     - ? Per-checkpoint weight/dataset licenses ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the code license ? the distributed weights/dataset terms.
Either train our own or confirm each checkpoint. Revisit: when adopting any borrowed
weights.
18     - ? Real GPU $/video & latency ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. Revisit: when choosing the production LLM step.
19
20     ## Detector swap follow-ups
21     *(source: PR #8, ADR-005)*
22
23     - ? `supervision` ByteTrack deprecation ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and removed in 0.30.0; `requirements.txt` is pinned <0.30.0 as a
stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the
cap. Revisit: before bumping supervision past 0.30.
24     - ? Real-video validation of the swap ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but not yet validated on actual footage for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) Revisit: when a real test clip + the pipeline are run together.
25     - ? `yolo_hybrid` naming ? the registry key, `yolo_` config keys, and `yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. Revisit: if/when a clean break is worth the migration.
26
27     ## Golden-set subproject
28     *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
29
30     - ? Annotation guideline ? DONE (2026-06-02): authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; `ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
31     - ? Internal gold key ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. Unblocked. Revisit: anytime.
32     - ? GS-3 ? `HumanVendorProvider` ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). Vendor selected for the initial pilot: Roora
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). Revisit: once the
Roora pilot confirms quality, then build the adapter.
33     - ? GS-4 oracle model ? `AutomatedProvider` interface is stubbed (this work) with a
deterministic fake. Wire a concrete fast-tier auto-labeler later. Oracle rule: it MUST be

```

a different model lineage than production, or the r...
[/external_agent_tool_result]

976. assistant (2026-07-02T06:11:33.199Z)

Appending the coordination notes to both docs ? BACKLOG first:

977. assistant (2026-07-02T06:11:33.199Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md
[/external_agent_tool_call]

978. assistant (2026-07-02T06:11:33.199Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

979. assistant (2026-07-02T06:11:33.199Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\LOW_REGRET_MOVES_PLAN.md
[/external_agent_tool_call]

980. assistant (2026-07-02T06:11:33.199Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\LOW_REGRET_MOVES_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

981. assistant (2026-07-02T06:11:33.199Z)

Let me verify `run\_all\_tests.py` works end-to-end (it also confirms the branch is green):

982. assistant (2026-07-02T06:11:33.199Z)

[external_agent_tool_call: Bash]
description: Run the new run_all_tests.py wrapper
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python run_all_tests.py 2>&1 | tail -12
[/external_agent_tool_call]

983. assistant (2026-07-02T06:11:33.199Z)

[external_agent_tool_result]
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest -q
obsreport (optional reporting dep): available
=====
..... [31%]
..... [62%]
..... [93%]
..... [100%]
231 passed in 12.14s
=====
RESULT: PASS - all tests green
=====
[/external_agent_tool_result]

984. assistant (2026-07-02T06:11:33.199Z)

```
`run_all_tests.py` works ? clean banner (flags the optional `obsreport` dep), full suite **231
passed**, clear PASS verdict.
```

****Status on your three asks:****

```
- **(c) `run_all_tests.py` ** ?? discoverable root-level wrapper (matches the `run*.py` convention), passes args through to pytest, reports PASS/FAIL with pytest's exit code, and flags whether the optional `obsreport` dep is present. README's test section now points to it.
```

```
- **(a) Grok coordination notes** ?? added to both `docs/BACKLOG.md` and `docs/LOW_REGRET_MOVES_PLAN.md`: items 1?2 marked done; the key intersections flagged (item 8 = subsumed ? extend the report; item 7 = land before AI-timing capture; item 3 = Result-contract synergy; items 4/5/6 safe to do independently; pin the `obsreport` tag later).
```

```
- **(b) CODE_MAP regeneration** ?? running as a background workflow now (8 parallel subsystem readers over the current layered tree ? synthesize against the old map ? adversarially verify every `@path::symbol` anchor ? fix). I'll write the regenerated `docs/CODE_MAP.md`, then commit all three and open a single PR once it lands.
```

I'll finalize and open the PR when the workflow completes ? sit tight.

985. user (2026-07-02T06:11:33.200Z)

pls merge all the PRs by yourself and I will resume later. Thanks for maing this more user friendly!

986. assistant (2026-07-02T06:11:33.200Z)

Thank you ? glad it's more debuggable now! Let me finish and merge. First, the CODE_MAP regeneration workflow needs to be done before I can open/merge that last PR ? let me check its status:

987. assistant (2026-07-02T06:11:33.200Z)

```
[external_agent_tool_call: Bash]
description: Check CODE_MAP workflow completion + result
command: OUT="C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/56b7f849-5d56-4770-93e0-b3b4da943700/tasks/wrtptil6w6.output" && if [ -f "$OUT" ]; then echo "=== workflow COMPLETE ==="; python -c "import json;d=json.load(open(r'$OUT'));print('keys:',list(d.get('result',{}).keys()));v=d['result'].get('verify',{});print('verify ok:',v.get('ok'),'problems:',len(v.get('problems',[])));cm=d['result'].get('codemap','');print('codemap chars:',len(cm));print('starts:',cm[:60]);" else echo "=== not complete yet (no output file) ==="; ls "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/56b7f849-5d56-4770-93e0-b3b4da943700/tasks/" 2>&1 | tail -5; fi
[/external_agent_tool_call]
```

988. assistant (2026-07-02T06:11:33.200Z)

```
[external_agent_tool_result: error]
Exit code 1
=== workflow COMPLETE ===
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import json;d=json.load(open(r'C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-
Projects-badminton-highlight-indexer/56b7f849-5d56-4770-93e0-
b3b4da943700/tasks/wrtpil6w6.output'));print('keys:',list(d.get('result',{}).keys()));v=d['resul
t'].get('verify',{});print('verify ok:',v.get('ok'),'problems:',len(v.get('problems',[])));cm=d[
'result'].get('codemap','');print('codemap chars:',len(cm));print('starts:',cm[:60])
    ~~~~~~AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
  File "C:/Users/avidu/AppData/Local/Programs/Python/Python313\Lib\json\__init__.py", line 298,
in load
    return loads(fp.read(),
        cls=cls, object_hook=object_hook,
        parse_float=parse_float, parse_int=parse_int,
```

```

        parse_constant=parse_constant, object_pairs_hook=object_pairs_hook, **kw)
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\__init__.py", line 352,
in loads
    return _default_decoder.decode(s)
        ~~~~~^
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\decoder.py", line 345,
in decode
    obj, end = self.raw_decode(s, idx=_w(s, 0).end())
        ~~~~~^
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\decoder.py", line 363,
in raw_decode
    raise JSONDecodeError("Expecting value", s, err.value) from None
json.decoder.JSONDecodeError: Expecting value: line 1 column 1 (char 0)
[/external_agent_tool_result]

```

989. assistant (2026-07-02T06:11:33.200Z)

The output file is empty ? the workflow is still running (output not yet written). Let me check its live progress:

990. assistant (2026-07-02T06:11:33.200Z)

```

[external_agent_tool_call: Bash]
description: Check workflow output size + subagent transcript progress
command: OUT="C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-
highlight-indexer/56b7f849-5d56-4770-93e0-b3b4da943700/tasks/wrtpil6w6.output"; echo "output
file size: $(wc -c < "$OUT" 2>/dev/null) bytes"; echo "=== subagent transcripts (progress) ===";
ls -l "C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-
indexer/56b7f849-5d56-4770-93e0-b3b4da943700/subagents/workflows/wf_d88615a4-f56/" 2>&1 | tail
-20
[/external_agent_tool_call]

```

991. assistant (2026-07-02T06:11:33.200Z)

```

[external_agent_tool_result]
output file size: 0 bytes
=== subagent transcripts (progress) ===
agent-a3139ab4edb6a20a5.jsonl
agent-a3139ab4edb6a20a5.meta.json
agent-a46cae3f0f615dfc3.jsonl
agent-a46cae3f0f615dfc3.meta.json
agent-a56929604f77b9d41.jsonl
agent-a56929604f77b9d41.meta.json
agent-a6029042d6f6bbf07.jsonl
agent-a6029042d6f6bbf07.meta.json
agent-a646f11db19db3854.jsonl
agent-a646f11db19db3854.meta.json
agent-a98897151b5da076f.jsonl
agent-a98897151b5da076f.meta.json
agent-aac4a0e5db2cfe047.jsonl
agent-aac4a0e5db2cfe047.meta.json
agent-ab90acd72923cd2a5.jsonl
agent-ab90acd72923cd2a5.meta.json
agent-abb109c8dd5e01539.jsonl
agent-abb109c8dd5e01539.meta.json
journal.jsonl
[/external_agent_tool_result]

```

992. assistant (2026-07-02T06:11:33.200Z)

The workflow is mid-run (8 readers done; synthesize/verify/fix still to go) ? it'll notify me when finished. All previously-opened PRs are already merged; the only remaining piece is this last docs+tooling PR, which I'll create and merge once the map lands. Let me confirm there are no other open PRs I should be aware of (so "merge all" is accurate):

993. assistant (2026-07-02T06:11:33.200Z)

```
[external_agent_tool_call: Bash]
description: List any open PRs across all three repos
command: echo "=== INDEXER open PRs ===" && gh pr list -R avidullu/badminton-highlight-indexer
--state open 2>&1 | head; echo "=== COLLECTOR open PRs ===" && gh pr list -R avidullu/sports-
data-collector --state open 2>&1 | head; echo "=== OBSREPORT open PRs ===" && gh pr list -R
avidullu/sports-obsreport --state open 2>&1 | head
[/external_agent_tool_call]
```

994. assistant (2026-07-02T06:11:33.200Z)

```
[external_agent_tool_result]
=== INDEXER open PRs ===
=== COLLECTOR open PRs ===
=== OBSREPORT open PRs ===
[/external_agent_tool_result]
```

995. assistant (2026-07-02T06:11:33.200Z)

Confirmed ? **zero open PRs** across all three repos; everything to date is already merged. The only outstanding item is the final docs+tooling PR (regenerated `CODE\_MAP.md` + `run\_all\_tests.py` + the Grok notes), and its CODE_MAP regeneration workflow is still running.

Here's what I'll do **autonomously** the moment the workflow finishes (no need for you to be here):

1. Write the regenerated `docs/CODE\_MAP.md`,
2. Commit (a)+(b)+(c) on the `chore/grok-handoff-codemap-runner` branch,
3. Run `run\_all\_tests.py` to confirm green,
4. Push, open the PR, and **merge it myself** (merge commit + delete branch), then fast-forward master,
5. Update the memory/handoff checkpoint.

After that, all work is merged and both repos are clean on `master` ? nothing left hanging. I'll report the final state here when it's done. Enjoy the break ? and thanks again! ?

996. user (2026-07-02T06:11:33.200Z)

```
<task-notification>
<task-id>wrtpil6w6</task-id>
<tool-use-id>toolu_01WebmnkMYMacbi5Jfu2iC1E</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\56b7f849-5d56-4770-93e0-b3b4da943700\tasks\wrtpil6w6.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Regenerate docs/CODE_MAP.md for the layered restructure (#40) +
config/reporting subsystems" completed</summary>
<result>{"codemap":"Now I have the full picture of the existing structure. Let me regenerate the
complete CODE_MAP.md, applying the path migrations, new subsystems (config, reporting),
refreshed test map, and corrected gotchas/ramp paths.\n\n# CODE_MAP.md\n\n**Purpose:** cold-
start navigation map for the badminton-highlight-indexer ? a self-hosted video ? highlight-reel
pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a segmenter-agnostic
eval/calibration stack, typed config, and per-video observability reporting.\n\n**Regenerate:**
rerun the `build-code-map` workflow (parallel subsystem readers ? this doc). Re-run after any
registry/contract/WSL-interface change, or after a directory restructure (this revision follows
#40).\n\n**LEGEND**\n- `@path` = file (always repo-relative)\n- `::sym` = symbol
(class/func/const) in the nearest preceding `@path`\n- `->` = dataflow / call /
`"produces"`\n- `$` = data contract (canonical shape)\n- `?` = gotcha / footgun\n- `?` =
registry / plugin extension point\n- `WSL` = runs inside WSL2 conda env, NOT the Windows Python
process\n\n---\n\n### 1. PIPELINE\n\n### 1A. Runtime: video file -> highlight reel\n\n`ntyped`
config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-
defaults, never fatal)\n
@backend/config/loader.py::load_config -> @backend/config/models.py::AppConfig\n
video_id = MD5(whole file) @backend/main.py::get_file_md5 (? chunk size varies per
entrypoint: 64KB main, 1MB run_process, 8KB run_eval, 64KB phase3 ? all hash WHOLE file so
```



```

values agree)\n -&gt; validate
@backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)\n
FormatValidator -&gt; ctx['ffprobe_meta'] -&gt; ResolutionFPSValidator -&gt; PTSContinuity -&gt;
SceneCut -&gt; Blur -&gt; Relevance (? ACTUAL registration/exec order ? see
@backend/pipeline/validators/__init__.py)\n [validation FAIL -&gt; persist
status="failed", return early; report STILL emitted in finally]\n -&gt;
db.add_video(video_id, filepath, status, [r.dict()]) @backend/infrastructure/database.py\n
-&gt; seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback 'motion')\n
-&gt; seg = segmenter_registry.get(seg_name)(db, global_config) ?
@backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)\n
-&gt; segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)\n
[STOP POINT: segments-only ? predictions now in DB; eval/regression operate here without
stitching]\n -&gt; finally: emit_indexer_report(...) @backend/reporting/__init__.py (?
ALWAYS runs ? even on validation failure / exception; NEVER raises; grafts
response["reports"])\n -&gt; select segment ids -&gt; [(start,end)] (+ stitching padding)\n
-&gt; VideoCompiler.compile_highlights(filepath, time_pairs, out_name)
@backend/pipeline/stitcher/compiler.py\n per-clip ffmpeg re-encode
(libx264/superfast/crf22) -&gt; concat -c copy -&gt; output/&lt;name&gt;.mp4\n```\nHybrid
segmenter internal 4-phase (yolo_hybrid / tracknet_hybrid / wasb_hybrid ? same shape):\n```\nP1
chunk video (yolo only; tracknet/wasb = whole-video single pass, chunk_index=0)\n -&gt; P2
candidate \"action windows\" from motion/trajectory [STOP: candidates persisted via
db.add_candidate_segment]\n -&gt; P3 pad clip (ai_handoff_padding) -&gt; ThreadPoolExecutor
-&gt; provider.analyze_video(clip, prompt) ? provider_registry\n -&gt; P4 db.add_play_segment
+ db.link_candidate_to_segment\n```\nPure-AI path `gemini`: no P2; chunk-or-single -&gt;
provider -&gt; heavy validate (clamp/dedup) -&gt; add_play_segment.\nLegacy `motion`: pixel-diff
-&gt; segments + **fabricated** metadata/events (? not ground truth).\n\n### 1B. WASB shuttle
substrate (P2 of wasb_hybrid)\n```\nWasbHybridSegmenter.process_video
@backend/pipeline/segmenters/wasb_hybrid.py\n -&gt; WasbRunner.healthcheck() (gate; not ok
-&gt; return []) @backend/pipeline/detectors/wasb_runner.py\n -&gt;
WasbRunner.run_predict(video, out_dir)\n stage video + wasb_infer.py into WSL fs -&gt; `wsl
bash -c 'source conda.sh &amp; python wasb_infer.py ...'\n -&gt;
@backend/pipeline/detectors/wasb_infer.py::run (WSL): sliding windows -&gt; GPU detector pass
(CHECKPOINTED det_raw.jsonl) -&gt; CPU tracker (always full re-run) -&gt; trajectory CSV\n -&gt;
TrackNetRunner.parse_trajectory_csv(csv) @backend/pipeline/detectors/tracknet_runner.py
(shared, re-exported by WasbRunner)\n -&gt; TrackNetRunner.trajectory_to_action_windows(points,
fps, frame_width, chunk_start, velocity_thresh, merge_gap, min_window_duration, chunk_index)
[the model-agnostic WINDOWING BRAIN]\n```\n(tracknet_hybrid` identical, swapping
WasbRunner-&gt;TrackNetRunner; `_probe_fps_width` fallback (30.0, 1280.0).)\n\n### 1C.
Calibration / eval flow (NO pipeline run unless told)\n```\nGT CSV -&gt;
annotations.load_annotations / calibrate_wasb.load_golden_gts -&gt; List[Interval]\nDB
predictions -&gt; harness.get_predictions(db, video_id, stage) -&gt; List[Interval] stage ?
{candidates, final}\n -&gt; metrics.match_intervals (greedy temporal-IoU) -&gt; metrics.score
-&gt; Score\n -&gt; harness.evaluate -&gt; EvalReport -&gt; report_to_dict\n -&gt;
batch.aggregate (corpus micro/macro) @backend/eval/batch.py\n -&gt;
regression.regress(_with_provider) vs baseline @backend/eval/regression.py [CI
gate]\nCalibration: calibrate_wasb.run -&gt; calibration.{select_best, improvement_report,
cross_validate} @backend/eval/calibration.py\nDistill Gemini-&gt;local: calibrate_local
(thresholds) OR distill_local (learned classifier)
@backend/eval/{calibrate_local,distill_local}.py\nLearned segmenter (offline):
training.build_training_set(X,y) -&gt; classifier.LogisticRegression.fit/save\nGemini refinement
driver (tuning, standalone): gemini_refine.{refine_window,full_video_rallies}
@backend/eval/gemini_refine.py\nAudio probe (diagnostic only, NOT in live pipeline):
audio/e1_probe.run @backend/eval/audio/e1_probe.py\n```\nEntrypoints: `@run_eval.py` (single
video score), `@run_phase3_eval.py` (manifest batch, can segment), `@run_regression.py`
(baseline gate, exit 1=regression), `@backend/eval/calibrate_wasb.py` +
`@backend/eval/export_wasb_segments.py` (WASB tuning drivers).\n\n---\n\n### 2. SUBSYSTEMS\n\n###
2.0 Orchestration &amp; config\n- `@backend/main.py` ? Thin CLI + FastAPI bootstrap
(orchestrator; modes via FLAGS). Static UI mounts `/static` from `backend/api/static`.\n -
`::app` FastAPI; `::db_instance`, `::global_config` (Optional[AppConfig]) module globals set
ONLY in `main()`. ? importing `app` for ASGI without `main()` -&gt; both `None` -&gt; every
endpoint NPES.\n - `::process_video` `@POST /api/process` -&gt; validate -&gt; add_video -&gt;
segment -&gt; `finally`: always `emit_indexer_report` (never raises). `::compile_highlights`
`@POST /api/compile`. `::query_play_segments` `@POST /api/query`. `::get_config` `@GET
/api/config`. `::run_cli_diagnose_clip` `--debug-clip`: lazy `import cv2`, samples ±3s vs
`indexing.motion_threshold` (dual model/dict read).\n - `::load_config` ?? legacy thin wrapper

```

```

returning a **dict** (`load_app_config(path).model_dump(...)`); kept for transition ? new code
imports `load_app_config`/`AppConfig` directly.
`::ProcessRequest`/`::QueryRequest`/`::CompileRequest` pydantic bodies.\n - ? `main()` runtime-
override hack: `global_config._runtime_overrides = {"output_dir": args.output_dir}` ? output
dir lives ONLY on this dynamic attr (works because `AppConfig` is `extra="allow"`), NOT in any
config model; `db_path = os.path.join(args.output_dir, output.db_name)`. \n - ?
`compile_highlights` reads `getattr(global_config.output, "output_dir", "output")` but
`OutputConfig` has **no `output_dir`** -> ALWAYS falls back to `"output"` (the
`_runtime_overrides
... (truncated 49723 chars, full result in C:\Users\avidu\AppData\Local\Temp\claude\C--Users-
avidu-Projects-badminton-highlight-
indexer\56b7f849-5d56-4770-93e0-b3b4da943700\tasks\wrtpil6w6.output)</result>
<usage><agent_count>10</agent_count><subagent_tokens>649486</subagent_tokens><tool_uses>149</too
l_uses><duration_ms>599969</duration_ms></usage>
</task-notification>

```

997. assistant (2026-07-02T06:11:33.200Z)

The CODE_MAP regeneration is done (and it even surfaced two real config-migration bugs as gotchas ? nice dogfood). The result has some agent preamble before the markdown; let me extract the clean map, write it, and sanity-check it (no stale paths, new subsystems present):

998. assistant (2026-07-02T06:11:33.200Z)

```

[external_agent_tool_call: Bash]
description: Extract regenerated CODE_MAP, write it, sanity-check paths/subsystems
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python - <<'PY'
import json, re
OUT=r"C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\56b7f849-5d56-4770-93e0-b3b4da943700\tasks\wrtpil6w6.output"
d=json.load(open(OUT,encoding="utf-8"))
res=d["result"]; cm=res["codemap"]; v=res.get("verify",{})
i=cm.find("# CODE_MAP.md")
assert i!=-1, "no # CODE_MAP.md header found"
cm=cm[i:].rstrip()+"\n"
open("docs/CODE_MAP.md", "w", encoding="utf-8").write(cm)
print("verify ok:", v.get("ok"), "| problems:", len(v.get("problems",[])))
print("written chars:", len(cm), "| lines:", cm.count(chr(10)))

```

sanity: stale top-level paths should be GONE (allow backend/pipeline/... + backend/eva

```

import re as _re
stale=_re.findall(r"@?backend/(?:(?:validators|indexer|database|stitcher|detectors|audio)\b(?:/)",
cm)
stale=[s for s in stale if "pipeline" not in s]
print("stale-path hits:", len(stale), stale[:6])
print("has backend/config:", "backend/config" in cm, "| has backend/reporting:",
"backend/reporting" in cm,
"| has backend/infrastructure:", "backend/infrastructure" in cm, "| has run_all_tests:",
"run_all_tests" in cm)
print("---- first 5 lines ----"); print("\n".join(cm.splitlines()[:5]))
PY
[/external_agent_tool_call]

```

999. assistant (2026-07-02T06:11:33.201Z)

```

[external_agent_tool_result]
verify ok: True | problems: 0
written chars: 56432 | lines: 325
stale-path hits: 0 []
has backend/config: True | has backend/reporting: True | has backend/infrastructure: True | has
run_all_tests: True
---- first 5 lines ----

```

CODE_MAP.md

****Purpose:**** cold-start navigation map for the badminton-highlight-indexer ? a self-hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.

****Regenerate:**** rerun the `build-code-map` workflow (parallel subsystem readers ? this doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure (this revision follows #40).
[/external_agent_tool_result]

1000. assistant (2026-07-02T06:11:33.201Z)

CODE_MAP regenerated cleanly ? ****verify passed (0 broken anchors)****, 325 lines, zero stale paths, includes config/reporting/infrastructure + `run\_all\_tests`. The regen also surfaced two real config-migration bugs worth flagging for Grok ? adding them to BACKLOG:

1001. assistant (2026-07-02T06:11:33.201Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md
[/external_agent_tool_call]

1002. assistant (2026-07-02T06:11:33.201Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

1003. assistant (2026-07-02T06:11:33.201Z)

Now confirm green, then commit everything, push, open the PR, and merge it:

1004. assistant (2026-07-02T06:11:33.201Z)

[external_agent_tool_call: Bash]
description: Test, commit, push the chore branch
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && python run_all_tests.py 2>&1 | tail -3 && echo "=== staged files ===" && git add -A && git status -s && git commit -q -m "docs+tooling: regenerate CODE_MAP for layered layout + run_all_tests.py + Grok low-regret notes"

- docs/CODE_MAP.md regenerated for the #40 restructure + new subsystems (config, reporting): all @path anchors point at backend/pipeline/{validators,segmenters, stitcher,detectors,audio}/, backend/infrastructure/, backend/api/, backend/config/, backend/reporting/. Built by the build-code-map workflow (parallel readers -> synthesize -> adversarial anchor verify, 0 broken anchors). It also documents two config-migration ? gotchas it surfaced (see BACKLOG).
- run_all_tests.py: one-command full suite (PASS/FAIL + pytest exit code; flags the optional obsreport dep). README test section points to it.
- BACKLOG.md + LOW_REGRET_MOVES_PLAN.md: coordination notes for Grok (items 1-2 done; item 8 subsumed by reports; item 7 before AI-timing; item 3 synergy) + the two config bugs to fix next pass.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q -u origin chore/grok-handoff-codemap-runner 2>&1 | tail -2 && echo "pushed"
[/external_agent_tool_call]

1005. assistant (2026-07-02T06:11:33.201Z)

[external_agent_tool_result]
=====

```
RESULT: PASS - all tests green
```

=====

```
=== staged files ===
```

warning: in the working copy of 'run_all_tests.py', LF will be replaced by CRLF the next time

```

Git touches it
M README.md
M docs/BACKLOG.md
M docs/CODE_MAP.md
M docs/LOW_REGRET_MOVES_PLAN.md
A run_all_tests.py
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/chore/grok-
handoff-codemap-runner
remote:
pushed
[/external_agent_tool_result]

```

1006. assistant (2026-07-02T06:11:33.201Z)

Now open the PR and merge it (then sync master):

1007. assistant (2026-07-02T06:11:33.201Z)

```

[external_agent_tool_call: Bash]
description: Create PR, merge it, delete branch, sync master, confirm clean
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && gh pr create --base master
--head chore/grok-handoff-codemap-runner --title "docs+tooling: regenerate CODE_MAP +
run_all_tests.py + Grok low-regret notes" --body "$(cat <<'EOF'
Repo-hygiene / handoff PR (no behavior change).

- **`docs/CODE_MAP.md` regenerated** for the #40 layered restructure + the new
`config`/`reporting` subsystems. Built by the `build-code-map` multi-agent workflow (8 parallel
subsystem readers ? synthesize against the old map ? adversarial anchor verify). Every
`@path::symbol` anchor verified against the real tree (**0 broken anchors**); all paths now
point at `backend/pipeline/{validators,segmenters,stitcher,detectors,audio}/`,
`backend/infrastructure/`, `backend/api/`, `backend/config/`, `backend/reporting/`. The regen
also documented two config-migration ? gotchas it surfaced (see BACKLOG).
- **`run_all_tests.py`** at the repo root ? one command to run the full suite before opening a
PR (clear PASS/FAIL, pytest exit code, flags whether the optional `obsreport` dep is present;
args pass through to pytest). README test section points to it.
- **`docs/BACKLOG.md` + `docs/LOW_REGRET_MOVES_PLAN.md`** ? coordination notes for Grok: low-
regret items 1?2 done (#39/#40); item 8 largely subsumed by the observability reports (extend,
don't rebuild); item 7 should land before wiring real AI-provider timing; item 3 (Result
contract) synergizes; items 4/5/6 safe independently. Plus the two surfaced config bugs to fix
next pass (`api/compile` ignores `--output-dir`; `output_dir` only on a dynamic attr).

**231 tests pass.**

? Generated with [Claude Code](https://claude.com/claude-code)
EOF
)" 2>&1 | tail -2 && git checkout master 2>&1 | tail -1 && gh pr merge chore/grok-handoff-
codemap-runner --merge --delete-branch 2>&1 | tail -4 && git pull --ff-only -q origin master
2>&1 | tail -1 && echo "=== master head ===" && git log --oneline -2 && echo "=== clean? ===" &&
git branch && git status -s | wc -l | xargs echo "uncommitted:"
[/external_agent_tool_call]

```

1008. assistant (2026-07-02T06:11:33.201Z)

```

[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/44
Switched to branch 'master'
=== master head ===
e03820a Merge pull request #44 from avidullu/chore/grok-handoff-codemap-runner
5b39068 docs+tooling: regenerate CODE_MAP for layered layout + run_all_tests.py + Grok low-
regret notes
=== clean? ===
* master
uncommitted: 0
[/external_agent_tool_result]

```

1009. assistant (2026-07-02T06:11:33.201Z)

****PR #44 created and merged.**** Master is at `e03820a`, clean (0 uncommitted, branch deleted).
Final verification across all repos + then I'll checkpoint memory:

1010. assistant (2026-07-02T06:11:33.201Z)

```
[external_agent_tool_call: Bash]
description: Final verify: no open PRs anywhere; both repos clean on master
command: echo "=== OPEN PRs (should be none anywhere) ===" && for r in badminton-highlight-
indexer sports-data-collector sports-obsreport; do echo -n "$r: "; n=$(gh pr list -R avidullu/$r
--state open 2>/dev/null | wc -l); echo "$n open"; done
echo "=== both repos clean on master? ===" && for d in badminton-highlight-indexer sports-data-
collector; do echo -n "$d: "; git -C "C:/Users/avidu/Projects/$d" rev-parse --abbrev-ref HEAD |
tr -d '\n'; echo -n " | uncommitted "; git -C "C:/Users/avidu/Projects/$d" status -s | wc -l |
tr -d '\n'; echo -n " | local branches: "; git -C "C:/Users/avidu/Projects/$d" branch | tr '\n'
' '; echo; done
[/external_agent_tool_call]
```

1011. assistant (2026-07-02T06:11:33.201Z)

```
[external_agent_tool_result]
=== OPEN PRs (should be none anywhere) ===
badminton-highlight-indexer: 0 open
sports-data-collector: 0 open
sports-obsreport: 0 open
=== both repos clean on master? ===
badminton-highlight-indexer: master | uncommitted 0 | local branches: * master
sports-data-collector: master | uncommitted 0 | local branches: * master
[/external_agent_tool_result]
```

1012. assistant (2026-07-02T06:11:33.201Z)

All clean ? ****0** open PRs anywhere; both repos on `master`, 0 uncommitted, no stray branches.******
Let me finalize the carry-forward memory:

1013. assistant (2026-07-02T06:11:33.201Z)

```
[external_agent_tool_call: Edit]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

1014. assistant (2026-07-02T06:11:33.201Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

1015. assistant (2026-07-02T06:11:33.202Z)

Now the handoff ? let me read its current state to fix the now-stale lines (#12 merged, #44
added, CODE_MAP regenerated):

1016. assistant (2026-07-02T06:11:33.202Z)

```
[external_agent_tool_call: Read]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\session-handoff.md
[/external_agent_tool_call]
```

1017. assistant (2026-07-02T06:11:33.202Z)

```
[external_agent_tool_result]
19     ## You are here (2026-06-07) ? ? OBSERVABILITY-REPORTS feature SHIPPED (both tools +
shared lib)
20     - **Shipped:** always-on **per-video observability reports** across BOTH tools, powered
by the NEW shared lib
21     **`obsreport`** (own private repo `github.com/avidullu/sports-obsreport`, MIT,
**v0.1.3**, editable-installed). Each
22     run writes next to the video (override `report.output_dir`): `*.report.json`
(machine), `*.report.md` (TECHNICAL ?
23     stages + footgun flags, each citing a FAQ), `*.summary.md` (CUSTOMER ? metadata +
meta-highlights, internals
24     stripped). **SOFT dependency** (no-op if absent; writes NEVER raise). Plan:
`C:\Users\avidu\claude\plans\drifting-snuggling-squirrel.md`.
25     - **MERGED to master:** indexer **#41** (observability) + **#42** (config runtime-bug
fix: typed AppConfig broke
26     validators' nested `.get` ? `AppConfig.get` now returns nested sections as dicts) +
**#43** (debuggability
27     refinements + faq-resolution golden test). Collector **#10** (parser zero-start fix) +
**#11** (observability).
28     - **OPEN:** collector **#12** (`feat/collector-report-refinements` ? frame-level
evidence + review-only framing +
29     golden test; green, awaiting merge). Both repos CLEAN on master; merged branches
pruned. Suites: indexer **231**,
30     collector **116** (on #12), obsreport **74**.
31     - **Debuggability validated (stretch goal):** 15-agent Workflow ? all 7 broken-run
scenarios self-debuggable by a
32     README+FAQ-only reader; eyeballed via rendered HTML screenshot + customer summaries.
Golden faq-resolution tests lock it.
33     - **Parallel track (Grok) = `docs/LOW_REGRET_MOVES_PLAN.md`:** items **1** (Config pkg
#39) + 2 (directory restructure
34     #40) DONE**; items 3?9 remain (good list ? assessment in [[current-state]]).
Intersections: item **8** (Input-Quality
35     UX) largely subsumed by my reports**; **3** (Result contract)** + **7** (unify hybrid
segmenters)** synergize (7 enables
36     the deferred real AI-timing capture, currently `not_captured`).
37     - **?? NEXT:** merge collector **#12**; then optional obsreport follow-ups (HTML
customer renderer; thread real
38     AI-provider timing via #7; pin obsreport `git+ssh` once SSH/CI access exists).
Original rally-detection modeling
39     still PARKED pending golden labels.
40
[/external_agent_tool_result]
```

1018. assistant (2026-07-02T06:11:33.202Z)

```
[external_agent_tool_call: Edit]
file: c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\session-handoff.md
[/external_agent_tool_call]
```

1019. assistant (2026-07-02T06:11:33.202Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\session-handoff.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]
```

1020. assistant (2026-07-02T06:11:33.202Z)

Let me update the CODE_MAP memory pointer to record the #44 regeneration (provenance accuracy):

1021. assistant (2026-07-02T06:11:33.202Z)

```
[external_agent_tool_call: Edit]
file: c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\code-map-artifact.md
[/external_agent_tool_call]
```

1022. assistant (2026-07-02T06:11:33.202Z)

```
[external_agent_tool_result]
The file c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\code-map-artifact.md has been updated successfully. (file state is current in
your context ? no need to Read it back)
[/external_agent_tool_result]
```

1023. assistant (2026-07-02T06:11:33.202Z)

All done ? everything's merged and both repos are pristine. ?

Final state

- **Every PR merged. Zero open PRs across all three repos.** Both repos on `master`, 0 uncommitted, no stray branches.
- **Indexer:** #41 (observability) · #42 (config fix) · #43 (debuggability refinements) · **#44** (regenerated CODE_MAP + `run\_all\_tests.py` + Grok notes ? created & merged just now)
- **Collector:** #10 (parser fix) · #11 (observability) · #12 (delivery-report refinements)
- **obsreport:** on GitHub (private, v0.1.3)
- **Suites green:** indexer 231 · collector 116 · obsreport 74 ? all runnable via `python run\_all\_tests.py`.

What #44 delivered (your a/b/c)

- **(b)** `docs/CODE\_MAP.md` regenerated for the layered layout via the multi-agent build-code-map workflow ? every `@path::symbol` anchor adversarially verified against the real tree (**0 broken**), now covering `backend/pipeline/?`, `backend/infrastructure/`, `backend/config/`, `backend/reporting/`.
- **(c)** `run\_all\_tests.py` at the repo root ? one command, clear PASS/FAIL + pytest exit code, flags the optional `obsreport` dep; README points to it.
- **(a)** Grok coordination notes in `BACKLOG.md` + `LOW\_REGRET\_MOVES\_PLAN.md` (items 1?2 done; 8 subsumed; 7 before AI-timing; 3 synergy; 4/5/6 independent).

One thing to hand Grok (not blocking)

The CODE_MAP regeneration **surfaced two real config-migration bugs** (from #39/#40, *not* mine ? left for Grok's config pass, documented in `BACKLOG.md` + `CODE\_MAP.md §2.0`):

1. `/api/compile` **ignores** `--output-dir` ? `OutputConfig` has no `output\_dir` field, so it always falls back to `output`.
2. `--output-dir` lives only on a dynamic `global\_config.\_runtime\_overrides` attr (fragile).

Fix is small: add `output\_dir` to a config/runtime model and read it consistently.

Carry-forward checkpointed

`current-state.md`, `session-handoff.md`, and the `code-map-artifact` note are all updated to the final shipped state ? a fresh session can resume cleanly from them.

Thanks for the great back-and-forth ? the observability layer is fully shipped, debuggable-by-design, and validated. Enjoy the break. ?